

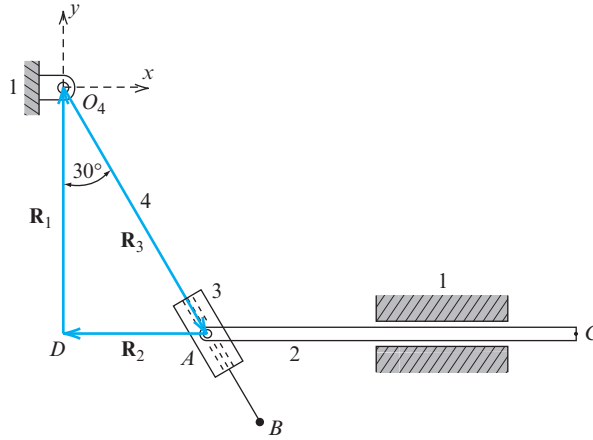


Figure 4.23 $R_{O_4D} = R_1 = 8$ ft and $R_{BO_4} = 11$ ft.

point O_4 ; thus, the input is the magnitude of vector $\mathbf{R}_2$ from point A of link 2 to point D . Also, note that the change in the input is negative; that is, the magnitude of the input is decreasing.

With the vectors shown in Fig. 4.23, the vector loop equation for the linkage is

$$\mathbf{R}_2 + \mathbf{R}_1 + \mathbf{R}_3 = \mathbf{0}, \quad (1)$$

where $\theta_1 = 90^\circ$, $\theta_2 = 180^\circ$, $\theta_3 = 300^\circ$, and, from trigonometry, $R_2 = R_1 \tan 30^\circ = 4.618$ ft. The x and y components of Eq. (1) are

$$R_2 \cos \theta_2 + R_1 \cos \theta_1 + R_3 \cos \theta_3 = 0, \quad (2a)$$

$$R_2 \sin \theta_2 + R_1 \sin \theta_1 + R_3 \sin \theta_3 = 0. \quad (2b)$$

Differentiating Eqs. (2) with respect to the input position R_2 gives

$$\cos \theta_2 - R_3 \sin \theta_3 \theta'_3 + R'_3 \cos \theta_3 = 0, \quad (3a)$$

$$\sin \theta_2 + R_3 \cos \theta_3 \theta'_3 + R'_3 \sin \theta_3 = 0. \quad (3b)$$

In matrix form, Eqs. (3) can be written as

$$\begin{bmatrix} -R_3 \sin \theta_3 & \cos \theta_3 \\ R_3 \cos \theta_3 & \sin \theta_3 \end{bmatrix} \begin{bmatrix} \theta'_3 \\ R'_3 \end{bmatrix} = \begin{bmatrix} -\cos \theta_2 \\ -\sin \theta_2 \end{bmatrix}. \quad (4)$$

The determinant of the coefficient matrix in Eq. (4) is

$$\Delta = \begin{vmatrix} -R_3 \sin \theta_3 & \cos \theta_3 \\ R_3 \cos \theta_3 & \sin \theta_3 \end{vmatrix} = -R_3 \sin^2 \theta_3 - R_3 \cos^2 \theta_3 = -R_3, \quad (5)$$

where $R_3 = (8 \text{ ft}) / \cos 30^\circ = 9.237$ ft. Therefore, $\Delta = -9.237$ ft.

Differentiating Eqs. (3) with respect to the input position, R_2 , gives

$$-R_3 \cos \theta_3 \theta_3'^2 - R_3 \sin \theta_3 \theta_3'' - 2R_3' \sin \theta_3 \theta_3' + R_3' \cos \theta_3 = 0, \quad (6a)$$

$$-R_3 \sin \theta_3 \theta_3'^2 + R_3 \cos \theta_3 \theta_3'' + 2R_3' \cos \theta_3 \theta_3' + R_3' \sin \theta_3 = 0. \quad (6b)$$

In matrix form, Eqs. (6) can be written as

$$\begin{bmatrix} -R_3 \sin \theta_3 & \cos \theta_3 \\ R_3 \cos \theta_3 & \sin \theta_3 \end{bmatrix} \begin{bmatrix} \theta_3'' \\ R_3' \end{bmatrix} = \begin{bmatrix} 2R_3' \sin \theta_3 \theta_3' + R_3 \cos \theta_3 \theta_3'^2 \\ -2R_3' \cos \theta_3 \theta_3' + R_3 \sin \theta_3 \theta_3'^2 \end{bmatrix}. \quad (7)$$

Note that the coefficient matrix in Eq. (7) is the same as the coefficient matrix in Eq. (4). The coefficient matrix does not change with differentiation of the unknown variables.

Substituting the known data into Eq. (4) and using Cramer's rule, the first-order kinematic coefficients are

$$\theta_3' = \frac{\begin{vmatrix} -\cos \theta_2 & \cos \theta_3 \\ -\sin \theta_2 & \sin \theta_3 \end{vmatrix}}{\Delta} = \frac{\sin(\theta_2 - \theta_3)}{-R_3} = +0.093 \text{ 75 rad/ft} \quad (8a)$$

and

$$R_3' = \frac{\begin{vmatrix} -R_3 \sin \theta_3 & -\cos \theta_2 \\ R_3 \cos \theta_3 & -\sin \theta_2 \end{vmatrix}}{\Delta} = \frac{R_3 \cos(\theta_2 - \theta_3)}{-R_3} = +0.500 \text{ ft/ft}. \quad (8b)$$

Substituting the known data into Eqs. (7) and using Cramer's rule, the second-order kinematic coefficients are

$$\theta_3'' = \frac{\begin{vmatrix} 2R_3' \sin \theta_3 \theta_3' + R_3 \cos \theta_3 \theta_3'^2 & \cos \theta_3 \\ -2R_3' \cos \theta_3 \theta_3' + R_3 \sin \theta_3 \theta_3'^2 & \sin \theta_3 \end{vmatrix}}{\Delta} = \frac{2R_3' \theta_3'}{-R_3} = -0.010 \text{ 1 rad/ft}^2 \quad (9a)$$

and

$$R_3'' = \frac{\begin{vmatrix} -R_3 \sin \theta_3 & 2R_3' \sin \theta_3 \theta_3' + R_3 \cos \theta_3 \theta_3'^2 \\ R_3 \cos \theta_3 & -2R_3' \cos \theta_3 \theta_3' + R_3 \sin \theta_3 \theta_3'^2 \end{vmatrix}}{\Delta} = \frac{-(R_3 \theta_3')^2}{-R_3} = +0.081 \text{ 2 ft/ft}^2. \quad (9b)$$

The angular velocity of link 3 is

$$\omega_3 = \theta_3' \dot{R}_2 = (+0.093 \text{ 75 rad/ft})(-15 \text{ ft/s}) = -1.41 \text{ rad/s}. \quad (10)$$

The negative sign indicates that the direction of the angular velocity of link 3 is clockwise. Note that $\dot{R}_2 = -15 \text{ ft/s}$, since link 2 is moving to the left (the input magnitude, R_2 , is decreasing). Also, note that link 4 is constrained to rotate with the same angular velocity as link 3. Therefore, the angular velocity of link 4 is

$$\omega_4 = \omega_3 = -1.41 \text{ rad/s (cw)}. \quad (11)$$

The angular acceleration of link 3 is

$$\begin{aligned}\alpha_3 &= \theta'_3 \ddot{R}_2 + \theta''_3 \dot{R}_2^2 \\ &= (+0.093\,75 \text{ rad/ft})(0) + (-0.010\,1 \text{ rad/ft}^2)(-15 \text{ ft/s})^2 = -2.27 \text{ rad/s}^2.\end{aligned}\quad (12)$$

The negative sign indicates that link 3 is accelerating in the clockwise direction. Since link 4 is constrained to rotate with link 3, the angular acceleration of link 4 is

$$\alpha_4 = \alpha_3 = -2.27 \text{ rad/s}^2 \text{ (cw).} \quad \text{Ans. (13)}$$

The sliding velocity of link 3 with respect to link 4 is

$$V_{A_3/4} = R'_3 \dot{R}_2 = (+0.5 \text{ ft/ft})(-15 \text{ ft/s}) = -7.5 \text{ ft/s.} \quad (14)$$

The negative sign indicates that the direction of the apparent velocity is along link 4, pointing toward the pin, O_4 ; that is, the direction is $\mathbf{V}_{A_3/4} = 7.5 \text{ ft/s} \angle 120^\circ$.

The sliding acceleration of link 3 with respect to link 4 can be written as

$$\begin{aligned}A_{A_3/4} &= R'_3 \ddot{R}_2 + R''_3 \dot{R}_2^2 \\ &= (+0.5 \text{ ft/ft})(0) + (+0.081\,2 \text{ ft/ft}^2)(-15 \text{ ft/s})^2 = +18.27 \text{ ft/s}^2.\end{aligned}\quad (15)$$

The positive sign indicates that the direction of the apparent acceleration is along link 4, pointing away from pin O_4 ; that is, the direction is $\mathbf{A}_{A_3/4} = 18.27 \text{ ft/s}^2 \angle -60^\circ$.

To determine the acceleration of point B , we draw vector $\mathbf{R}_B$, as shown in Fig. 4.24.

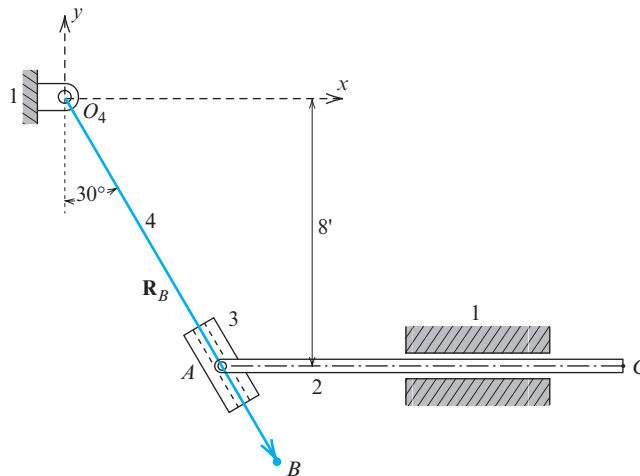


Figure 4.24 Vector for point B .

The x and y components of the position of point B are

$$x_B = R_B \cos \theta_3 = 5.50 \text{ ft}, \quad (16a)$$

$$y_B = R_B \sin \theta_3 = -9.53 \text{ ft}. \quad (16b)$$

Differentiating Eqs. (16) with respect to the input variable, R_2 , the first-order kinematic coefficients of point B are

$$x'_B = -R_B \sin \theta_3 \theta'_3 = +0.893 \text{ 09 ft/ft}, \quad (17a)$$

$$y'_B = R_B \cos \theta_3 \theta'_3 = +0.515 \text{ 63 ft/ft}. \quad (17b)$$

Then, differentiating Eqs. (17) with respect to the input, R_2 , gives the second-order kinematic coefficients of point B :

$$x''_B = -R_B \cos \theta_3 \theta_3'^2 - R_B \sin \theta_3 \theta_3'' = -0.145 \text{ ft/ft}^2, \quad (18a)$$

$$y''_B = -R_B \sin \theta_3 \theta_3'^2 + R_B \cos \theta_3 \theta_3'' = +0.028 \text{ ft/ft}^2. \quad (18b)$$

The velocity of point B can be written as

$$\begin{aligned} \mathbf{V}_B &= (x'_B \hat{\mathbf{i}} + y'_B \hat{\mathbf{j}}) \dot{R}_2 \\ &= (0.893 \text{ 09} \hat{\mathbf{i}} + 0.515 \text{ 63} \hat{\mathbf{j}}) (-15 \text{ ft/s}) = -13.40 \hat{\mathbf{i}} - 7.73 \hat{\mathbf{j}} \text{ ft/s} \\ &= 15.47 \text{ ft/s} \angle 210^\circ. \end{aligned} \quad (19)$$

As a check, the velocity of point B can also be written as

$$V_B = \omega_4 R_{BO_4} = (1.41 \text{ rad/s}) (11 \text{ ft}) = 15.51 \text{ ft/s},$$

where the difference is a result of truncation in the value of ω_4 in Eq. (10) and (11), ($\omega_4 = 1.406 \text{ 25 rad/s}$).

The acceleration of point B can be written as

$$\begin{aligned} \mathbf{A}_B &= (x'_B \hat{\mathbf{i}} + y'_B \hat{\mathbf{j}}) \ddot{R}_2 + (x''_B \hat{\mathbf{i}} + y''_B \hat{\mathbf{j}}) \dot{R}_2^2 \\ &= (0.893 \hat{\mathbf{i}} + 0.516 \hat{\mathbf{j}} \text{ ft/ft}) (0) + (-0.145 \hat{\mathbf{i}} + 0.028 \hat{\mathbf{j}} \text{ ft/ft}^2) (-15 \text{ ft/s})^2 \\ &= -32.6 \hat{\mathbf{i}} + 6.3 \hat{\mathbf{j}} \text{ ft/s}^2 = 33.2 \text{ ft/s}^2 \angle 169.1^\circ. \end{aligned} \quad \text{Ans. (20)}$$

Note that the inverse tangent function on some calculators initially indicates that the angle is -10.9° . However, this angle must be transposed into the second quadrant; that is, $\angle \mathbf{A}_B = -10.9^\circ + 180^\circ = 169.1^\circ$.

4.12 EULER-SAVARY EQUATION*

In Sec. 4.5, we developed the apparent-acceleration equation, Eq. (4.17). Then, in the examples that followed, we reported that it is very important to carefully choose a point whose apparent path with respect to another body is known so that the radius of curvature of the path, required for the normal component in Eq. (4.14), can be found. This need to know the radius of curvature of the path often dictates the method of approach to such a problem and sometimes even requires the visualization of an equivalent mechanism[†] (as was done in Example 4.7, Fig. 4.18c). It would be convenient if an arbitrary point could be chosen, and the radius of curvature of its path could be calculated. In planar mechanisms, this can be accomplished by the methods that are presented here.

When two rigid bodies move relative to each other with planar motion, an arbitrarily chosen point A of one describes a path, or locus, relative to a coordinate system fixed to the other. At a given instant, there is a point A' , attached to the other body, that is the center of curvature of the locus of A . If we take the kinematic inversion of this motion, A' also describes a locus relative to the body containing A , and it so happens that A is the center of curvature of this locus. Each point, therefore, acts as the center of curvature of the path traced by the other, and the two points are called *conjugates* of each other. The distance between these two conjugate points is the magnitude of the radius of curvature of either locus.

Figure 4.25 illustrates two circles, with centers at C and C' , commonly referred to as *osculating circles* [2]. Let us think of the circle with center C' as the fixed centrode and think of the circle with center C as the moving centrode of two bodies experiencing some particular relative planar motion. In actuality, the fixed centrode need not be fixed but is attached to the body that contains the path whose curvature is sought. Also, it is not necessary that the two centrodes be circles; we are interested only in instantaneous values, and, for convenience, we can think of the centrodes as circles matching the curvatures (osculating circles) of the two actual centrodes in the region near their point of contact, I (called the *velocity pole*). The velocity pole is an additional point, coincident with the instantaneous center of velocity.

Recall from Sec. 3.12 that the instantaneous centers of velocity are a pair of coincident points, labeled I_{jk} and I_{kj} , each attached to one of the bodies, j and k . The velocity pole, I , however, is a third point, not attached to either body, that remains coincident with the changing instantaneous centers of velocity as the motion progresses. Thus, the velocity pole can have a velocity along the two tangent centrodes that is different from the equal velocities of the two instantaneous centers of velocity.

When the bodies containing the two centrodes move relative to each other, the centrodes appear to roll against each other without slip (Sec. 3.20). Because of these properties, we can think of the two circular centrodes as actually representing the shapes of the two moving bodies, if this helps in visualizing the motion.

*Among the most important and most useful references on this subject are [2].

[†]The concept of equivalent mechanisms is an important topic. For a detailed study, consult [2].

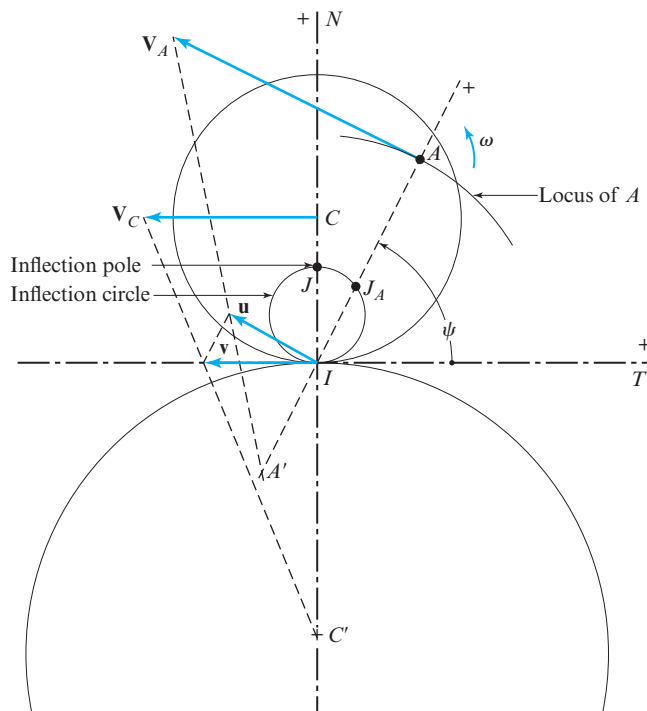


Figure 4.25 Hartmann construction.

If the moving centrode has an angular velocity ω , with respect to the fixed centrode, the velocity* of point C is

$$V_C = \omega R_{CI}. \quad (a)$$

Similarly, arbitrary point A , whose conjugate point A' we wish to find, has a velocity of

$$V_A = \omega R_{AI}. \quad (b)$$

As the motion progresses, the point of contact of the two centrodes, and therefore the location of the velocity pole, I , moves along both centrodes with some velocity (or rate) v . As shown in Fig. 4.25, the pole velocity, v , can be found by connecting a straight line from the terminus of V_C to point C' . Alternatively, the magnitude of the pole velocity can be obtained from

$$v = \frac{R_{IC'}}{R_{CC'}} V_C. \quad (c)$$

* All velocities used in this section are actually apparent velocities, with respect to the coordinate system of the fixed centrode; they are written as absolute velocities to simplify the notation.

A graphic construction for A' , the center of curvature of the locus of arbitrary point A , is shown in Fig. 4.25 and is called the *Hartmann construction*. First, we find the component, $\mathbf{u}$, of the pole velocity, $\mathbf{v}$, as that component parallel to $\mathbf{V}_A$ or perpendicular to $\mathbf{R}_{AI}$. Then, the intersection of the line AI and a line connecting the termini of the velocities $\mathbf{V}_A$ and $\mathbf{u}$ gives the location of the conjugate point, A' . The radius of curvature, ρ , of the locus of point A is $\rho = R_{AA'}$.

An analytic expression for locating point A' would also be desirable and can be derived from the Hartmann construction. The magnitude of the $\mathbf{u}$ component of the pole velocity is given by

$$u = v \sin \psi, \quad (d)$$

where ψ is the positive counterclockwise angle measured from the centrode tangent to the line of $\mathbf{R}_{AI}$. Then, noting the similar triangles in Fig. 4.25, we can write the magnitude of the $\mathbf{u}$ component of the pole velocity as

$$u = \frac{R_{IA'}}{R_{AA'}} V_A. \quad (e)$$

Now, equating of Eqs. (d) and (e), and substituting Eqs. (a), (b), and (c) into the resulting equation gives

$$u = \frac{R_{IC} R_{CI}}{R_{CC'}} \omega \sin \psi = \frac{R_{IA'} R_{AI}}{R_{AA'}} \omega. \quad (f)$$

Dividing this equation by $\omega \sin \psi$ and inverting, we obtain

$$\frac{R_{AA'}}{R_{AI} R_{IA'}} \sin \psi = \frac{R_{CC'}}{R_{CI} R_{C'I}} = \frac{\omega}{v}. \quad (g)$$

Next, upon noting that $R_{AA'} = R_{AI} - R_{A'I}$ and $R_{CC'} = R_{CI} - R_{C'I}$, we can reduce this equation to the form

$$\left(\frac{1}{R_{AI}} - \frac{1}{R_{A'I}} \right) \sin \psi = \left(\frac{1}{R_{CI}} - \frac{1}{R_{C'I}} \right). \quad (4.40)$$

This important equation is one form of the *Euler-Savary equation*. If we assume that the radii of curvature of the two centrodes, R_{CI} and $R_{C'I}$, are known, then this equation can be used to determine the position of one of the two conjugate points (A or A') from the position of the other, relative to the velocity pole, I .

Note that in using the Euler-Savary equation, we may arbitrarily choose a positive sense for the centrode tangent ($+T$); the positive centrode normal ($+N$) is then 90° counterclockwise from it. This establishes a positive direction for the line CC' in Fig. 4.25, which may be used in assigning appropriate signs to R_{CI} and $R_{C'I}$. Similarly, an arbitrary positive direction can be chosen for the line AA' . The angle ψ is then taken as positive counterclockwise from the positive centrode tangent to the positive sense of the line AA' . The sense of the line AA' also gives the appropriate signs for R_{AI} and $R_{A'I}$ for Eq. (4.40).

There is a major disadvantage to this form of the Euler-Savary equation in that the radii of curvature of both centrodes, R_{CI} and $R_{C'I}$, must be known. Usually these are not

known, any more than the curvature of the locus itself is known. However, this difficulty can be overcome by seeking another form of the equation.

Let us consider the particular point labeled J in Fig. 4.25. This point is located on the centrode normal at a location defined by

$$\frac{1}{R_{JI}} = \frac{1}{R_{CI}} - \frac{1}{R_{C'I}}. \quad (h)$$

If this particular point is chosen for A in Eq. (4.40), we find that its conjugate point, J' , must be located at infinity. The radius of curvature of the path of point J is infinite, and the locus of J therefore has an inflection point at J . The point J is called the *inflection pole*.

Let us now consider whether there are any other points, J_A , of the moving body that also have infinite radii of curvature at the instant considered. If so, then for each of these points, $R_{AA'} = \infty$. Substituting this condition into Eq. (4.40), and with the aid of Eq. (h), we obtain

$$R_{JAI} = R_{JI} \sin \psi. \quad (4.41)$$

This is the equation of a circle in polar coordinates whose diameter is R_{JI} , as shown in Fig. 4.25. This circle is called the *inflection circle*. Every point on this circle is an inflection point; it has its conjugate point at infinity, and each such point therefore has an infinite radius of curvature at the instant indicated. These points are instantaneously moving along straight lines.

Now, with the help of Eq. (4.41), the Euler-Savary equation can be written in the form

$$\frac{1}{R_{AI}} - \frac{1}{R_{A'I}} = \frac{1}{R_{JAI}}. \quad (4.42)$$

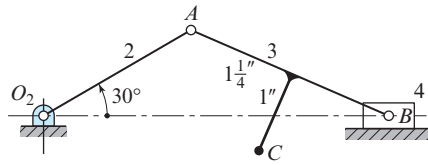
Also, after some further manipulation, the radius of curvature of the path of point A can be written as

$$\rho = R_{AA'} = \frac{R_{AI}^2}{R_{AJA}}. \quad (4.43)$$

Either of these two forms of the Euler-Savary equation, Eqs. (4.42) and (4.43), is more useful in practice than Eq. (4.40), since they do not require knowledge of the curvatures of the two centrodes. They do require finding the inflection circle, but the following example illustrates a graphic procedure for drawing the inflection circle.

EXAMPLE 4.12

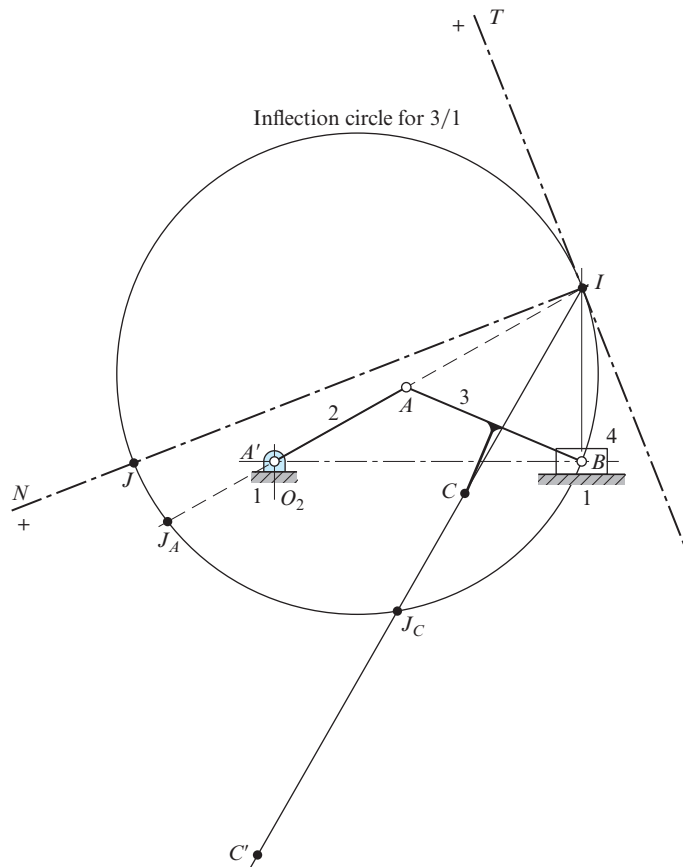
Draw the inflection circle for coupler link 3 of the slider-crank linkage in the posture shown in Fig. 4.26. Then, use the inflection circle and the Euler-Savary equation to determine the radius of curvature of the path of coupler point C .

**Figure 4.26**

$R_{AO_2} = 2$ in and $R_{BA} = 2.5$ in.

SOLUTION

We begin by locating the velocity pole, I (which is coincident with the instantaneous center of velocity), at the intersection of line O_2A and a line through B perpendicular to its direction of travel (Fig. 4.27). By definition, points B and I must both lie on the inflection circle; hence, we need only one additional point to construct the circle.

**Figure 4.27** Inflection circle.

The center of curvature of the path of point A is, of course, at O_2 , which we now call A' . Taking the positive sense of line AI as being downward and to the left, we measure $R_{AI} = 2.64$ in and $R_{AA'} = -2.00$ in. Then, substituting these values into Eq. (4.43), we obtain

$$R_{AJ_A} = \frac{R_{AI}^2}{R_{AA'}} = \frac{(2.64 \text{ in})^2}{-2.00 \text{ in}} = -3.48 \text{ in.}$$

With this, we lay off 3.48 in from A to locate J_A , which is a third point on the inflection circle. The inflection circle for the motion 3/1 can now be constructed through the three points B , I , and J_A , and the diameter of the circle is measured as

$$R_{JI} = 6.28 \text{ in.} \quad \text{Ans.}$$

The centrode normal (N) and the centrode tangent (T) can also be drawn, if desired, as shown in Fig. 4.27.

Drawing the ray R_{CI} and continuing to take the positive sense as downward and to the left, we measure $R_{CI} = 3.10$ in and $R_{CJ_C} = -1.75$ in. Substituting these values into Eq. (4.43), the radius of curvature of the path of point C is

$$\rho = R_{CC'} = \frac{R_{CI}^2}{R_{CJ_C}} = \frac{(3.10 \text{ in})^2}{-1.75 \text{ in}} = -5.49 \text{ in,} \quad \text{Ans.}$$

where the negative sign indicates that C is above C' on line $ICJ_C C'$.

4.13 BOBILLIER CONSTRUCTIONS

The Hartmann construction provides one graphic method of finding the conjugate point and the radius of curvature of the path of a moving point, but it requires knowledge of the curvature of the fixed and moving centrodes. It would be desirable to have *graphic* methods for obtaining the inflection circle and the conjugate of a given point without requiring the curvature of the centrodes. Such graphic solutions are presented in this section and are called the *Bobillier constructions*.

To understand these constructions, consider the inflection circle, the centrode normal (N), and the centrode tangent (T), as shown in Fig. 4.28. Let us select any two points, A and B , of the moving body that do not lie on a straight line through velocity pole I . Using the Euler-Savary equation, we can find the two corresponding conjugate points, A' and B' . The intersection of lines AB and $A'B'$ is labeled Q . Then, the straight line drawn through I and Q is called the *collineation axis*. This axis applies only to the two lines AA' and BB' , and so is said to belong to these two rays; also, point Q will be located differently on the collineation axis if another pair of points, A and B , is chosen on the same rays. Nevertheless, there is a unique relationship between the collineation axis and the two rays used to define it. This relationship is expressed in *Bobillier's theorem*, which states that *the angle from the collineation axis to the first ray is equal to the angle from the second ray to the centrode tangent*.

In applying the Euler-Savary equation to a planar mechanism, we can usually find two pairs of conjugate points by inspection, and from these we wish to graphically determine

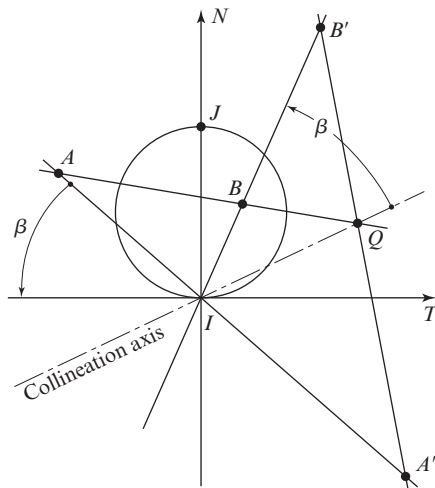


Figure 4.28 Bobillier theorem.

the inflection circle. For example, a four-bar linkage with a crank, O_2A , and a follower, O_4B , has A and O_2 as one pair of conjugate points and B and O_4 as another, when we are interested in the motion of the coupler with respect to the frame (3 with respect to 1). Given these two pairs of conjugate points, how do we use the Bobillier theorem to find the inflection circle?

In Fig. 4.29a, let A and A' and B and B' represent the known pairs of conjugate points. Rays constructed through each pair intersect at the velocity pole, I , giving one point on the inflection circle. Point Q is located next by the intersection of a ray through A and B with a ray through A' and B' . Then, the collineation axis can be drawn as line IQ .

The next step is shown in Fig. 4.29b. Drawing a straight line through I parallel to $A'B'$, we identify point W as the intersection of this line with line AB . Now, through W , we draw a second line parallel to the collineation axis. This line intersects AA' at J_A and BB' at J_B , two additional points on the inflection circle for which we are searching.

We could now construct a circle through the three points J_A, J_B , and I , but there is an easy way to do this. Remembering that a triangle inscribed in a semicircle is a right triangle having the diameter as its hypotenuse, we erect a perpendicular to AI at J_A and another perpendicular to BI at J_B . The intersection of these two perpendiculars gives the *inflection pole*, point J , as shown in Fig. 4.29c. Since IJ is the diameter, then the inflection circle, the centrode normal N , and the centrode tangent T can all be easily constructed.

To demonstrate that this construction satisfies the Bobillier theorem, note that the arc from I to J_A is inscribed by the angle that J_AI makes with the centrode tangent. But this same arc is also inscribed by angle IJB_A . Therefore, these two angles are equal. But line J_BJ_A was originally constructed parallel to the collineation axis. Therefore, line IJB also makes the same angle, β , with the collineation axis.

Our final problem is to learn how to use the Bobillier theorem to find the conjugate of another arbitrary point, say C , when the inflection circle is given. In Fig. 4.30, we draw a line through point C and the velocity pole, I , and locate the point of intersection of this line with the inflection circle (J_C). This ray serves as one of the two necessary rays to locate the collineation axis. For the other ray, we can use the centrode normal, since J and its conjugate point J' , at infinity, are both known. For these two rays, the collineation

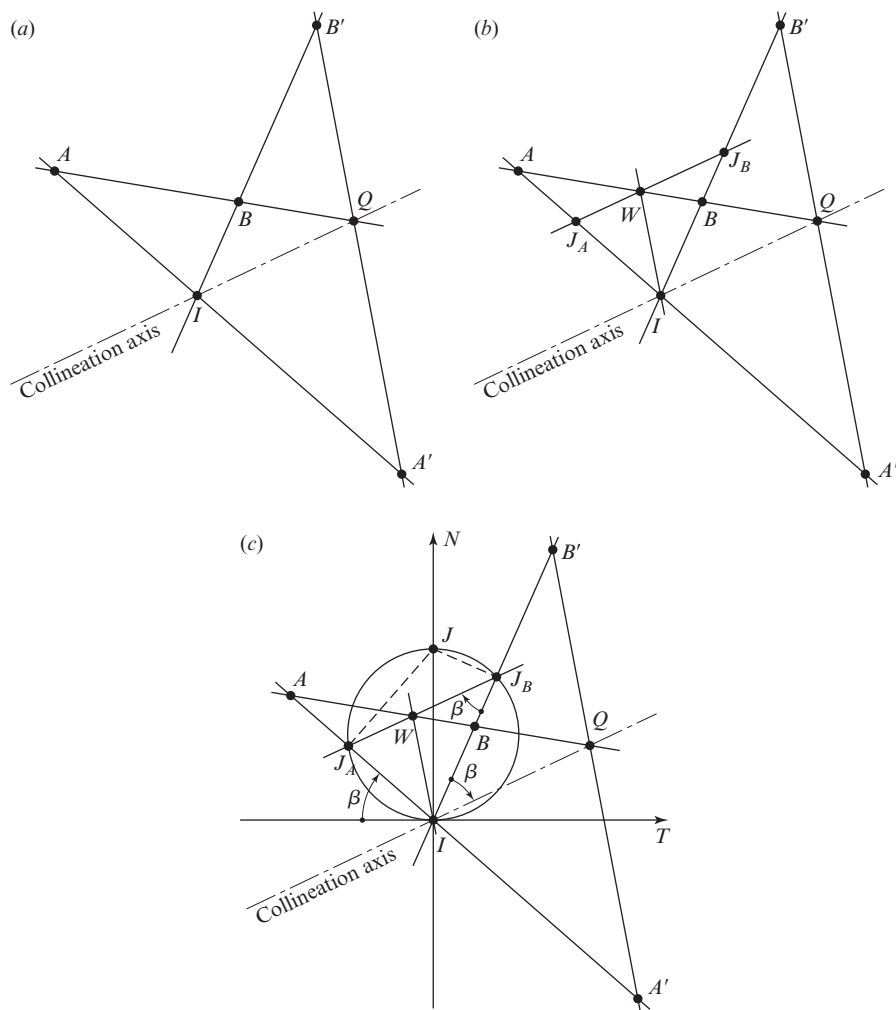


Figure 4.29 Bobillier construction for locating the inflection circle.

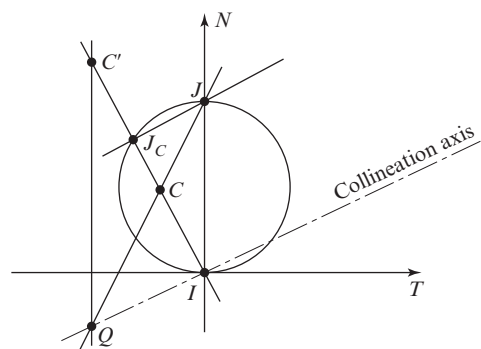


Figure 4.30 Bobillier construction for locating conjugate point C' .

axis is a line through the velocity pole, I , parallel to line $J_C J$, as we learned in Fig. 4.29c. The balance of the construction is similar to that of Fig. 4.28. Point Q is located by the intersection of a line through J and C with the collineation axis. Then a line through Q and J' at infinity intersects ray IC at C' , the conjugate point for C .

EXAMPLE 4.13

For the four-bar linkage $A'ABB'$ in the posture shown in Fig. 4.31, use the Bobillier theorem to find the center of curvature of the coupler curve of point C .

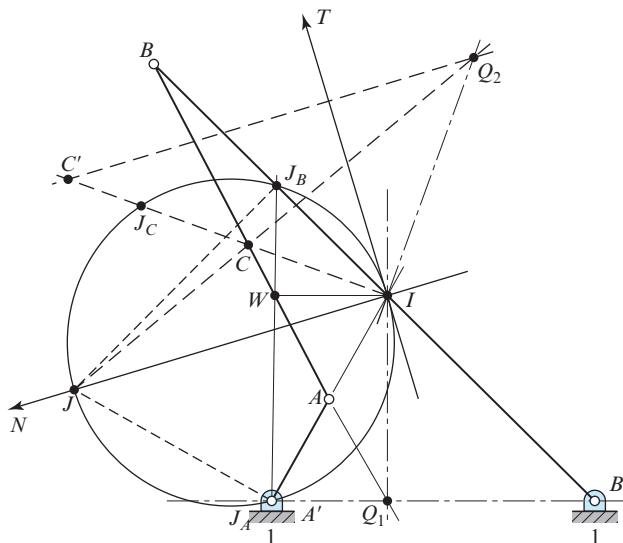


Figure 4.31 Four-bar linkage.

SOLUTION

Locate the velocity pole I at the intersection of AA' and BB' ; also locate Q_1 at the intersection of AB and $A'B'$. Line IQ_1 is the first collineation axis. Through I draw a line parallel to $A'B'$ to locate W on AB . Through W , draw a line parallel to IQ_1 to locate J_A on AA' and J_B on BB' . Then, through J_A , draw a perpendicular to AA' , and through J_B , draw a perpendicular to BB' . These perpendiculars intersect at inflection pole J and define the inflection circle, the centrode normal N , and the centrode tangent T .

To obtain the conjugate point of C , we draw ray IC and locate J_C on the inflection circle. The second collineation axis, IQ_2 , belonging to the pair of rays IC and IJ , is a line through I parallel to a line (not shown) from J to J_C . Point Q_2 is obtained as the

intersection of this collineation axis and line JC . Now, through Q_2 we draw a line parallel to the centrode normal; its intersection with ray IC yields C' , the center of curvature of the path of C .

4.14 INSTANTANEOUS CENTER OF ACCELERATION

This section defines the *instantaneous center of acceleration* for a planar mechanism. It is important to note that, in general, the instantaneous center of acceleration is not coincident with the instantaneous center of velocity. In other words, the instantaneous center of velocity has acceleration. The instantaneous center of acceleration is defined as *the instantaneous location of a pair of coincident points of two different rigid bodies where the absolute accelerations of the two bodies are identical*. If we consider a fixed and a moving body, the instantaneous center of acceleration is the point of the moving body that has zero absolute acceleration at the instant considered.

For the moving plane shown in Fig. 4.32a, assume that ω and α are known and that point A in the plane has a known acceleration, $\mathbf{A}_A$. Let Γ denote the instantaneous center of acceleration, a point of zero absolute acceleration whose location is unknown. The acceleration-difference equation can then be written as

$$\mathbf{A}_\Gamma = \mathbf{A}_A - \omega^2 \mathbf{R}_{\Gamma A} \alpha \times \mathbf{R}_{\Gamma A} = \mathbf{0}. \quad (a)$$

Solving for $\mathbf{A}_A$ gives

$$\mathbf{A}_A = \omega^2 R_{\Gamma A} \hat{\mathbf{R}}_{\Gamma A} - \alpha R_{\Gamma A} (\hat{\mathbf{k}} \times \hat{\mathbf{R}}_{\Gamma A}). \quad (b)$$

Now, recognizing that $\hat{\mathbf{R}}_{\Gamma A}$ is perpendicular to $\hat{\mathbf{k}} \times \hat{\mathbf{R}}_{\Gamma A}$, the two terms on the right-hand side of Eq. (b) are rectangular components of $\mathbf{A}_A$, as shown in Fig. 4.32b. From Fig. 4.32b, we can solve for the direction and magnitude of $\mathbf{R}_{\Gamma A}$; that is,

$$\gamma = \tan^{-1} \frac{\alpha}{\omega^2}, \quad (4.44)$$

$$R_{\Gamma A} = \frac{A_A}{\sqrt{\omega^4 + \alpha^2}} = \frac{A_A \cos \gamma}{\omega^2}. \quad (4.45)$$

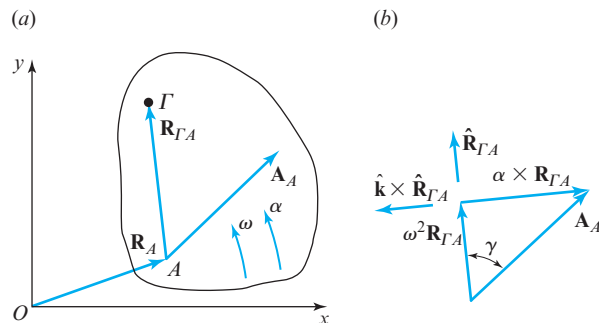


Figure 4.32 Instantaneous center of acceleration.

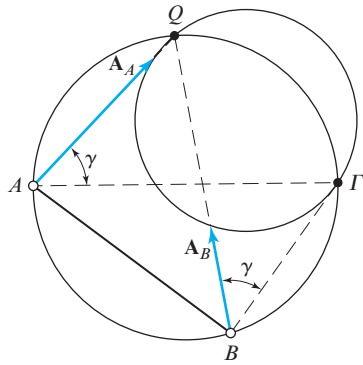


Figure 4.33 Four-circle method of locating the instantaneous center of acceleration.

Equation (4.45) states that the distance $R_{\Gamma A}$ from point A to the instantaneous center of acceleration Γ can be determined from the magnitude of the acceleration, A_A , of any point of the moving plane. Our convention is that the angle γ is defined *from* line $I\Gamma$ to the positive centrod normal.

There are many graphic methods for locating the instantaneous center of acceleration [2, pp. 148–58], [7, pp. 145–67]. Here, without proof, we present one method. In Fig. 4.33, we are given points A and B and their absolute accelerations $\mathbf{A}_A$ and $\mathbf{A}_B$. We extend $\mathbf{A}_A$ and $\mathbf{A}_B$ until they intersect at Q ; then we construct a circle through points A , B , and Q . Next, we draw another circle through the termini of $\mathbf{A}_A$ and $\mathbf{A}_B$ and point Q . The second intersection of the two circles locates point Γ , the instantaneous center of acceleration.

4.15 BRESSE CIRCLE (OR DE LA HIRE CIRCLE)

Another graphic method of locating the instantaneous center of acceleration is by drawing a circle called the *Bresse circle*. In Sec. 4.12, we defined the inflection circle as the locus of points that have their conjugate points at infinity, and each therefore has an infinite radius of curvature at the instant under consideration. Therefore, the inflection circle can also be defined as the locus of points with zero normal acceleration. The locus of points with zero tangential acceleration also defines a circle, shown in Fig. 4.34, referred to as the *Bresse circle* or the *de La Hire circle*.

The inflection circle and the Bresse circle intersect at two points, as shown in Fig. 4.34; one point is the velocity pole, I , and the other point is the instantaneous center of acceleration, Γ . Since the velocity pole, I , has acceleration, in general, then it must be discounted as a possible solution. The direction of the acceleration of the velocity pole, I , is along the positive centrod normal, N , and the magnitude can be written as

$$A_I = \omega v, \quad (4.46)$$

where v is the magnitude of the pole velocity [Eq. (b) in Sec. 4.12] and can be written as

$$v = \omega R_{II}. \quad (4.47)$$

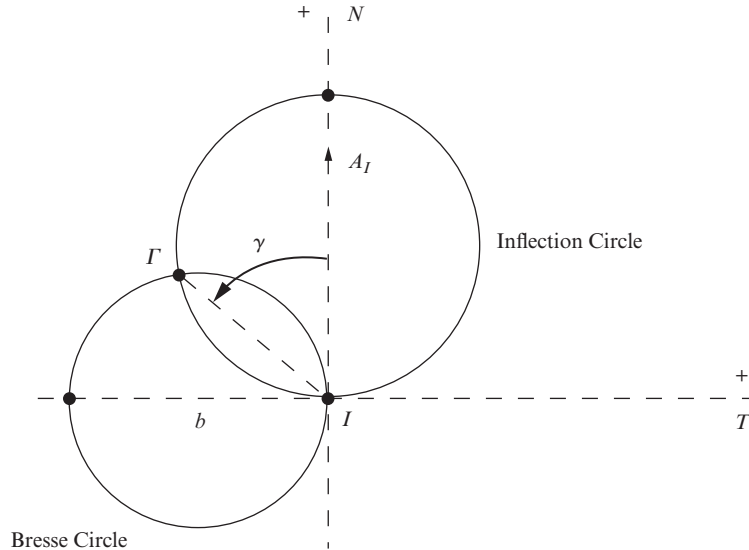


Figure 4.34 Bresse circle and the instantaneous center of acceleration.

Substituting Eq. (4.47) into Eq. (4.46), the acceleration of I can also be written as

$$A_I = \omega^2 R_{JI}.$$

The angle from the positive centrodal normal to line $I\Gamma$ (that is, the line connecting the instantaneous center of acceleration and the velocity pole) is angle γ . This angle was defined in Eq. (4.44) as

$$\gamma = \tan^{-1} \frac{\alpha}{\omega^2}.$$

The diameter of the Bresse circle can be written as

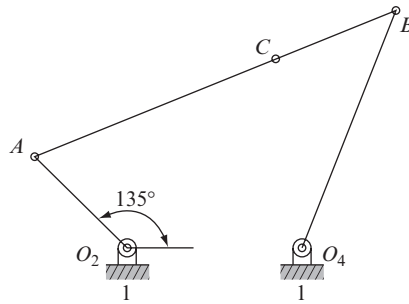
$$b = R_{JI} \frac{\omega^2}{\alpha}. \quad (4.48)$$

If the angular acceleration of the moving plane is positive (that is, counterclockwise), then the Bresse circle lies on the negative side of the centrodal tangent, T , as shown in Fig. 4.34. Alternatively, if the angular acceleration of the moving plane is negative, then the Bresse circle lies on the positive side of the centrodal tangent, T .

In the special case that the angular velocity of the moving plane is a constant (that is, $\alpha = 0$) then the diameter of the Bresse circle is infinite (the Bresse circle tends to the centrodal normal, N), and the instantaneous center of acceleration is coincident with the inflection pole, J .

EXAMPLE 4.14

Consider the four-bar linkage of Example 4.1, shown in Fig. 4.8*a* and redrawn here as Fig. 4.35. For the specified posture (that is, with the crank angle 135° ccw from the ground link, O_2O_4), the angular velocity and angular acceleration of the coupler link, AB , are $\omega_3 = 5 \text{ rad/s}$ ccw and $\alpha_3 = 20 \text{ rad/s}^2$ cw, respectively. For the instantaneous motion of the coupler link, show: (a) the velocity pole, I , centrode tangent, T , and centrode normal, N ; (b) the inflection circle and Bresse circle; and (c) the instantaneous center of acceleration of the coupler link, AB . Then, determine: (d) the radius of curvature of the trajectory of coupler point C where $R_{CB} = 6 \text{ in}$, (e) the magnitude and direction of the velocity of coupler point C , (f) the magnitude and direction of the angular velocity of the crank, (g) the magnitude and direction of the pole velocity, (h) the magnitude and direction of the acceleration of the velocity pole I , and (i) the magnitude and direction of the acceleration of C .

**Figure 4.35**

$R_{O_4O_2} = 8 \text{ in}$, $R_{AO_2} = 6 \text{ in}$,
 $R_{BA} = 18 \text{ in}$, and $R_{BO_4} = 12 \text{ in}$.

SOLUTION

- (a) The velocity pole, I , is coincident with instant center I_{13} , as shown in Fig. 4.36. The instant center I_{24} (labeled point Q) and the collineation axis, IQ , are also shown in Fig. 4.36. From Bobillier's theorem, the angle from the collineation axis to the first ray (say, link 2) is obtained from a scaled drawing as

$$\alpha = 29.08^\circ \text{ cw.} \quad \text{Ans. (1)}$$

This is equal to the angle from the second ray (say, link 4) to the centrode tangent, T . Therefore, we draw the centrode tangent, T , and the centrode normal, N , which is 90° counterclockwise from T , as shown in Fig. 4.36.

- (b) Now, by measuring ray $R_{AI} = 14.14 \text{ in}$, we can use Eq. (4.43) to locate the inflection point J_A ,

$$R_{AJ_A} = \frac{R_{AI}^2}{R_{AA'}} = \frac{(14.14 \text{ in})^2}{6.00 \text{ in}} = 33.31 \text{ in.}$$

Similarly, measuring ray $R_{BI} = 18.20 \text{ in}$, we locate the inflection point J_B ,

$$R_{BJ_B} = \frac{R_{BI}^2}{R_{BB'}} = \frac{(18.20 \text{ in})^2}{12.00 \text{ in}} = 27.60 \text{ in.}$$

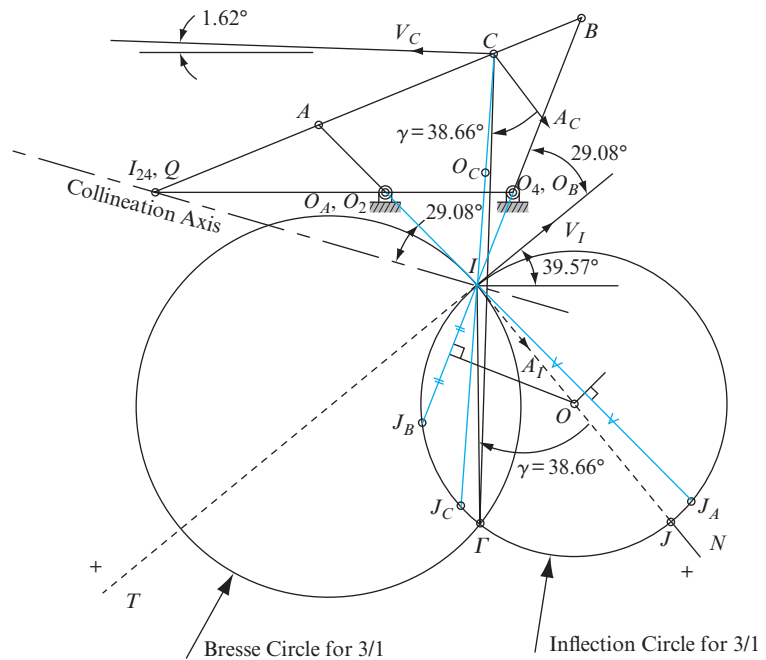


Figure 4.36 Inflection circle, Bresse circle, and instantaneous center of acceleration.

Erecting perpendiculars from these two rays, we locate the inflection pole, J , and draw the inflection circle, as shown in Fig. 4.36. The diameter of the inflection circle is

$$R_{JI} = 19.26 \text{ in.} \quad \text{Ans. (2)}$$

The diameter of the Bresse circle, from Eq. (4.48), is

$$b = R_{JI} \frac{\omega_3^2}{\alpha_3} = (19.26 \text{ in}) \frac{(5 \text{ rad/s})^2}{-20 \text{ rad/s}^2} = -24.08 \text{ in,} \quad \text{Ans. (3)}$$

where the negative sign indicates that the Bresse circle lies on the positive side of the centrod tangent.

- (c) The instantaneous center of acceleration, Γ , of the coupler link is as shown in Fig. 4.36.
- (d) The radius of curvature of the path of coupler point C (where $R_{CB} = 6$ in) is

$$\rho_C = R_{CC'} = \frac{R_{CI}^2}{R_{CJC}} = \frac{(14.67 \text{ in})^2}{28.62 \text{ in}} = 7.52 \text{ in.} \quad \text{Ans. (4)}$$

The center of curvature of the path traced by point C —that is, C' —is shown in Fig. 4.36.

- (e) The velocity of coupler point C is

$$V_C = \omega_3 R_{CI} = (5 \text{ rad/s}) (14.65 \text{ in}) = 73.25 \text{ in/s.} \quad \text{Ans. (5)}$$

The orientation of the velocity vector is as shown in Fig. 4.36.

- (f) The angular velocity of the crank can be written as

$$\omega_2 = \frac{R_{I_{23}I_{13}}}{R_{I_{23}I_{12}}} \omega_3 = \frac{14.14 \text{ in}}{6.00 \text{ in}} (5 \text{ rad/s}) = 11.78 \text{ rad/s ccw.} \quad \text{Ans. (6)}$$

- (g) The pole velocity, from Eq. (4.47), is

$$v = \omega_3 R_{JI} = (5 \text{ rad/s}) (19.26 \text{ in}) = 96.3 \text{ in/s.} \quad \text{Ans. (7)}$$

The direction of the pole velocity is as shown in Fig. 4.36.

- (h) The acceleration of the velocity pole, I , from Eq. (4.46), is

$$A_I = \omega_3 v = (5 \text{ rad/s}) (96.3 \text{ in/s}) = 481.5 \text{ in/s}^2. \quad \text{Ans. (8)}$$

The acceleration, A_I , is directed from I toward J , as shown in Fig. 4.36.

- (i) The acceleration of coupler point C is found by rearranging Eq. (4.45); that is,

$$A_C = R_{IC} \sqrt{\omega_3^4 + \alpha_3^2} = (29.64 \text{ in}) \sqrt{(5 \text{ rad/s})^4 + (20 \text{ rad/s}^2)^2} = 948.94 \text{ in/s}^2 \quad \text{Ans. (9)}$$

The direction of the acceleration of point C , from Eq. (4.44), is

$$\gamma = \tan^{-1} \frac{\alpha_3}{\omega_3^2} = \tan^{-1} \frac{-20 \text{ rad/s}^2}{(5 \text{ rad/s})^2} = -38.66^\circ, \quad \text{Ans. (10)}$$

which is a clockwise angle from $\mathbf{A}_C$ to line CI , as shown in Fig. 4.36.

4.16 RADIUS OF CURVATURE OF A POINT TRAJECTORY USING KINEMATIC COEFFICIENTS

The radius of curvature of a point trajectory (say point P), at the instant considered, can be written from Eqs. (4.2) and (4.3) or from Eq. (4.14) as

$$\rho = \frac{V_P^2}{A_P^n}. \quad (4.49)$$

From Eq. (3.34a), the speed of point P can be written as

$$V_P = r'_P \dot{\psi}. \quad (a)$$

Also, the normal component of the acceleration of point P can be written as

$$A_P^n = \mathbf{A}_P \cdot \hat{\mathbf{u}}^n, \quad (4.50)$$

where the unit normal vector to the point trajectory at the posture considered [Eq. (3.39)] is

$$\hat{\mathbf{u}}^n = \left(\frac{-y'_P}{r'_P} \right) \hat{\mathbf{i}} + \left(\frac{x'_P}{r'_P} \right) \hat{\mathbf{j}}. \quad (b)$$

Taking the time derivative of Eq. (3.33b), the acceleration of point P can be written as

$$\mathbf{A}_P = (x''_P \dot{\psi}^2 + x'_P \ddot{\psi}) \hat{\mathbf{i}} + (y''_P \dot{\psi}^2 + y'_P \ddot{\psi}) \hat{\mathbf{j}}. \quad (4.51)$$

Substituting Eqs. (b) and (4.51) into Eq. (4.50), the normal component of the acceleration of point P can be written as

$$A_P^n = \left(\frac{x'_P y''_P - y'_P x''_P}{r'_P} \right) \dot{\psi}^2. \quad (4.52)$$

Then, substituting Eqs. (a) and (4.52) into Eq. (4.49), the radius of curvature of the point trajectory at the posture considered can be written as

$$\rho = \frac{r_P'^3}{x'_P y''_P - y'_P x''_P}. \quad (4.53)$$

Sign convention: If the unit normal vector to the point trajectory $\hat{\mathbf{u}}^n$ points toward the center of curvature of the path, then the radius of curvature has a positive value. If the unit normal vector to the point trajectory points away from the center of curvature of the path, then the radius of curvature has a negative value.

The coordinates of the center of curvature of the point trajectory, at the posture under investigation, can be written as

$$x_C = x_P - \rho \left(\frac{y'_P}{r'_P} \right) \quad \text{and} \quad y_C = y_P + \rho \left(\frac{x'_P}{r'_P} \right). \quad (4.54)$$

EXAMPLE 4.15

Determine the radius of curvature and the center of curvature of the path of point B for Rapson's slide of Example 4.11.

SOLUTION

Here we continue from Example 4.11 in all respects, including equation numbers. Therefore, the unit tangent vector to the path of point B is the unit vector pointing in the direction of the velocity vector of point B . The unit tangent vector can be written as

$$\hat{\mathbf{u}}^t = \frac{x'_B \hat{\mathbf{i}} + y'_B \hat{\mathbf{j}}}{r'_B}, \quad (21)$$

where, using the data of Eqs. (17),

$$r'_B = \pm \sqrt{x_B'^2 + y_B'^2} = -\sqrt{(0.893\ 09\ \text{ft/ft})^2 + (0.515\ 63\ \text{ft/ft})^2} = -1.031\ 25\ \text{ft/ft}. \quad (22)$$

Note that the negative sign was chosen, since the input distance R_2 is becoming shorter, and, therefore, the input is negative. Substituting Eqs. (17) and (22) into Eq. (21) gives

$$\hat{\mathbf{u}}^t = \frac{x'_B \hat{\mathbf{i}} + y'_B \hat{\mathbf{j}}}{r'_B} = \frac{(0.893\ 09\ \text{ft/ft})\hat{\mathbf{i}} + (0.515\ 63\ \text{ft/ft})\hat{\mathbf{j}}}{-1.031\ 25\ \text{ft/ft}} = -0.866\ 02\hat{\mathbf{i}} - 0.500\ 00\hat{\mathbf{j}}. \quad (23)$$

As a check, using Eq. (19), the unit tangent vector can also be written as

$$\hat{\mathbf{u}}^t = \frac{\mathbf{V}_B}{V_B} = \frac{-13.40\hat{\mathbf{i}} - 7.73\hat{\mathbf{j}}\ \text{ft/s}}{15.47\ \text{ft/s}} = -0.866\ 19\hat{\mathbf{i}} - 0.499\ 68\hat{\mathbf{j}}.$$

The unit normal vector is 90° counterclockwise from the unit tangent vector; that is, using Eqs. (17) and (22),

$$\hat{\mathbf{u}}^n = \hat{\mathbf{k}} \times \hat{\mathbf{u}}^t = \hat{\mathbf{k}} \times \frac{X'_B \hat{\mathbf{i}} + Y'_B \hat{\mathbf{j}}}{r'_B} = \frac{-Y'_B \hat{\mathbf{i}} + X'_B \hat{\mathbf{j}}}{r'_B} = 0.500\ 00\hat{\mathbf{i}} - 0.866\ 02\hat{\mathbf{j}}. \quad (24)$$

The direction of the unit tangent vector and the unit normal vector are shown in Fig. 4.37.

Using Eqs. (17), (18), and (22), the radius of curvature of the path of point B can be found from Eq. (4.53) as

$$\begin{aligned} \rho_B &= \frac{r_B'^3}{x'_B y_B'' - y'_B x_B''} \\ &= \frac{(-1.031\ 25\ \text{ft/ft})^3}{(0.893\ 09\ \text{ft/ft})(0.028\ \text{ft/ft}^2) - (0.515\ 63\ \text{ft/ft})(-0.145\ \text{ft/ft}^2)} \\ &= -11.0\ \text{ft} \end{aligned} \quad \text{Ans. (25)}$$

The negative sign indicates that the unit normal vector points away from the center of curvature of the path of point B (Fig. 4.37).

The coordinates of the center of curvature of the path of point B can be found from Eqs. (4.54). Using known values give

$$x_C = x_B - \rho_B \left[\frac{y'_B}{r'_B} \right] = 5.50\ \text{ft} - (-11.0\ \text{ft}) \left[\frac{(0.515\ 63\ \text{ft/ft})}{(-1.031\ 25\ \text{ft/ft})} \right] = 0.00\ \text{ft}, \quad (26a)$$

$$y_C = y_B + \rho_B \left[\frac{x'_B}{r'_B} \right] = -9.53\ \text{ft} + (-11.0\ \text{ft}) \left[\frac{0.893\ 09\ \text{ft/ft}}{-1.031\ 25\ \text{ft/ft}} \right] = 0.00\ \text{ft}. \quad (26b)$$

Note that Eqs. (25), (26a), and (26b) make intuitive sense, since point B is located on link 4, and link 4 is pinned at the origin, O_4 , and, therefore, must rotate about the origin.

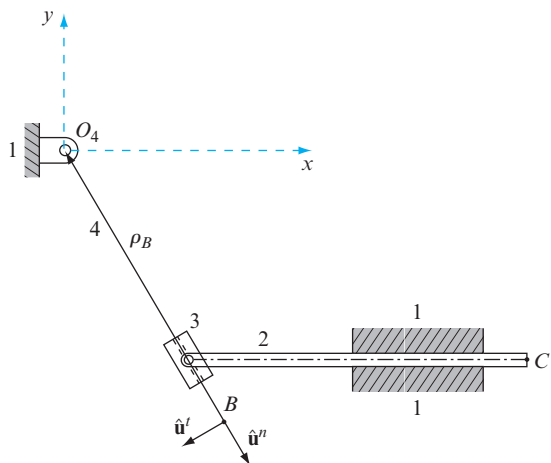


Figure 4.37 Unit tangent and unit normal vectors, and center of curvature of the path of point B .

Figure 4.37 illustrates the directions of the unit tangent vector, $\hat{\mathbf{u}}^t$, and the unit normal vector, $\hat{\mathbf{u}}^n$. Note that the unit normal vector points away from the center of curvature of the path of point B . The center of curvature of the path of point B is coincident with the ground pin O_4 . From observation, these answers are corroborated.

4.17 CUBIC OF STATIONARY CURVATURE

Now let us consider a point fixed in the coupler link of a planar four-bar linkage that generates a path relative to the frame whose radius of curvature, at the instant considered, is ρ . Since the coupler curve is, in general, a tricircular sextic [1, pp. 313–4], this radius of curvature changes continuously as the point moves. In certain situations, however, the path has stationary curvature, which means that

$$\frac{d\rho}{ds} = 0, \quad (a)$$

where s is the arc distance traveled along the path. The locus of all points in the coupler, or moving plane, which have stationary curvature at the instant considered, is called the *cubic of stationary curvature* or sometimes the *circling-point curve*. We should note that stationary curvature does not necessarily mean constant curvature, but rather that the continually varying radius of curvature is passing through a maximum or minimum.

Here, we present a quick and easy graphic method for obtaining the cubic of stationary curvature for a coupler link, as described by Hain [2, pp. 498–502]. Consider the four-bar linkage $A'ABB'$ shown in Fig. 4.38, where A' and B' are the frame pivots. Note that the loci of points A and B have stationary curvature (in fact, constant curvature about centers at A' and B' , respectively); hence, A and B must lie on the cubic.

The first step of the construction is to obtain the centrode normal and the centrode tangent. Since the inflection circle is not needed, we locate the collineation axis, IQ , as indicated and draw the centrode tangent, T , at the angle ψ from line IB' , equal to the angle from the collineation axis to line IA' . This construction follows directly from

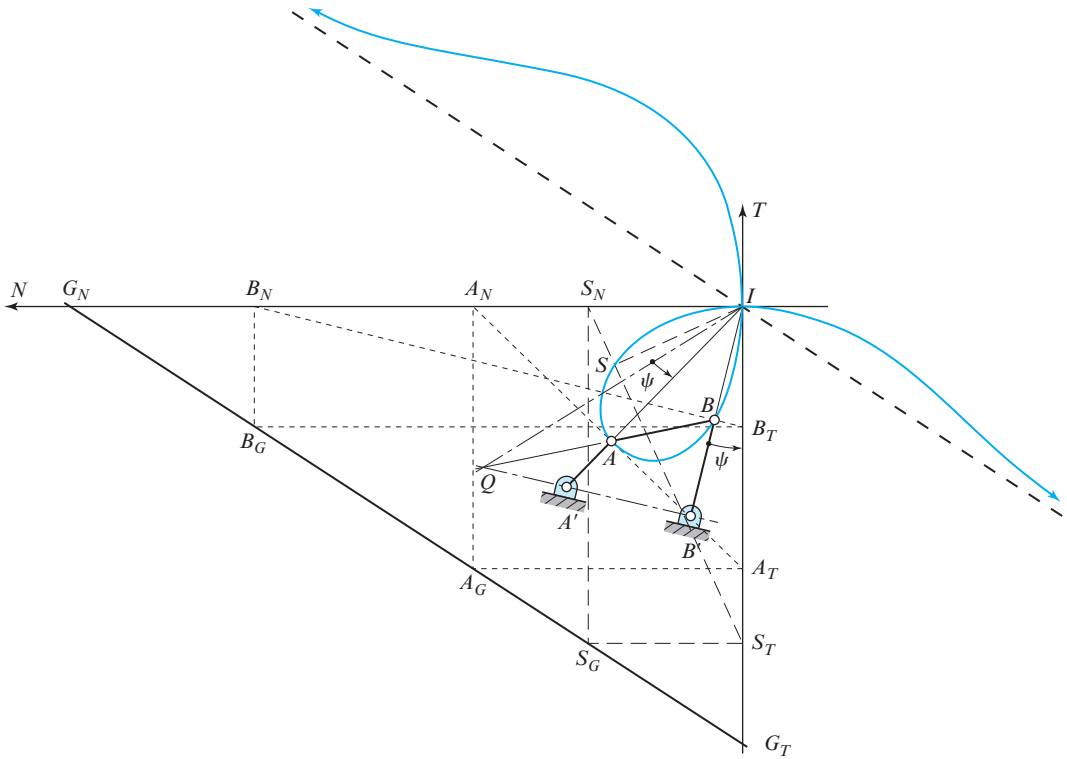


Figure 4.38 Cubic of stationary curvature.

Bobillier's theorem. We also construct the centrode normal, N . At this time it may be convenient to reorient the drawing on the working surface so that the centrode normal lies along the horizontal axis.

Next, we construct a line through A perpendicular to IA and another line through B perpendicular to IB . These lines intersect the centrode normal and centrode tangent at A_N, A_T and B_N, B_T , respectively, as shown in Fig. 4.38. Now, we draw two rectangles, $IA_N A_G A_T$ and $IB_N B_G B_T$; points A_G and B_G define an auxiliary line G that we will use to obtain other points on the cubic of stationary curvature.

Next, we choose any point, S_G , on line G . A ray parallel to N locates S_T , and another ray parallel to T locates S_N . Connecting S_T with S_N and drawing a perpendicular to this line through I locates point S , another point on the cubic of stationary curvature. We repeat this process as often as desired by choosing different points on line G , and we draw the cubic as a smooth curve through all the points S obtained.

Note that the cubic of stationary curvature has two tangents at the velocity pole I ; one is the *centrode-tangent tangent* and the other is the *centrode-normal tangent*. The radius of curvature of the cubic at these tangents is obtained as follows: Extend line G to intersect T at G_T and N at G_N , as indicated. Then, half the distance IG_T is the radius of

curvature of the cubic tangent to the centrode normal, and half the distance IG_N is the radius of curvature of the cubic tangent to the centrode tangent.

A point with interesting properties is the point of intersection of the cubic of stationary curvature with the inflection circle (other than velocity pole I); this point is called *Ball's point*. The point fixed in the coupler that is coincident with Ball's point describes a path that is practically rectilinear for a considerable distance [5, p. 357]; that is, it describes an excellent straight line near the design position, since it is located at an inflection point of its path and has stationary curvature.

The equation of the cubic of stationary curvature can be written in polar coordinates as

$$\frac{1}{r} = \frac{1}{M \sin \psi} - \frac{1}{N \cos \psi}, \quad (4.55)$$

where r is the distance from the velocity pole, I , to the point on the cubic, measured at an angle ψ from the centrode tangent.* The constant parameters M and N are determined using any two points known to lie on the cubic, such as points A and B of Fig. 4.38. Equations for M and N can be written as

$$\frac{1}{M} = \frac{1}{3} \left(\frac{1}{R_{JI}} - \frac{1}{R_{IO_M}} \right) \quad \text{and} \quad \frac{1}{N} = \frac{1}{3R_{JI}} \left(\frac{dR_{JI}}{ds} \right). \quad (4.56)$$

It so happens [8, p. 111] that M and N are, respectively, the diameters IG_T and IG_N of the circles centered on the centrode tangent and centrode normal whose radii represent the two curvatures of the cubic at velocity pole I .

The equation of the cubic of stationary curvature can also be expressed in Cartesian coordinates as

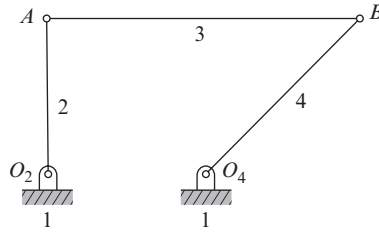
$$(x^2 + y^2) \left(\frac{x}{M} - \frac{y}{N} \right) = xy. \quad (4.57)$$

Degenerate forms. From Eq. (4.55), or Eq. (4.57), we observe that the cubic of stationary curvature degenerates to a circle and a straight line when either (a) N tends to infinity (that is, $1/N$ approaches zero) or (b) M tends to infinity (that is, $1/M$ approaches zero). Consider the following example.

EXAMPLE 4.16

For the four-bar linkage in the posture shown in Fig. 4.39, link 2 is perpendicular to the frame and the coupler link is parallel to the frame. For the absolute motion of the coupler link, determine the diameter of the inflection circle, the cubic of stationary curvature, and the location of Ball's point. Also, determine the radius of curvature and center of curvature of: (a) the moving centrode, (b) the fixed centrode, and (c) the path of coupler point C , which is midway between pins A and B .

* For a derivation of this equation see [3, p. 98] or [4, p. 206].

**Figure 4.39**

$R_{O_4O_2} = 1$ in, $R_{AO_2} = 1$ in,
 $R_{BA} = 2$ in, and $R_{BO_4} =$
 5.657 in.

SOLUTION

The velocity pole, I , is coincident with the instant center I_{13} , as shown in Fig. 4.40. The instant center I_{24} (with label Q) lies at infinity (say to the right), and the collineation axis IQ is parallel to the coupler link. From Bobillier's theorem, the angle from the collineation axis to the first ray (link 2) is $\alpha = 90^\circ$ ccw. This is equal to the angle from the second ray (link 4) to the centrode tangent, T . Therefore, the centrode tangent, T , and the centrode normal, N (which is 90° ccw from the centrode tangent, T), are as shown in Fig. 4.40. Note that link 4 lies along the centrode normal, N .

The location of the inflection point J_A for point A on the coupler link is obtained from the Euler-Savary equation, Eq. (4.43); that is,

$$R_{AJ_A} = \frac{R_{AI}^2}{R_{AA'}} = \frac{(2 \text{ in})^2}{1 \text{ in}} = 4 \text{ in.} \quad (1)$$

The inflection point, J_B , for point B on the coupler link is obtained in a similar manner; that is,

$$R_{BJ_B} = \frac{R_{BI}^2}{R_{BB'}} = \frac{(2\sqrt{2} \text{ in})^2}{\sqrt{2} \text{ in}} = 5.66 \text{ in.} \quad (2)$$

The locations of the inflection points, J_A and J_B , are shown in Fig. 4.40. Note that J_B lies on the centrode normal, N ; therefore, this inflection point is coincident with the inflection pole, J . Knowing the centrode normal, N , and the two inflection points, the inflection circle for the motion of link 3 with respect to 1 can now be drawn. The diameter of the inflection circle for the motion 3/1 is

$$R_{JI} = \frac{1}{2} R_{JB} = 2\sqrt{2} \text{ in} = 2.83 \text{ in.} \quad \text{Ans. (3)}$$

The inflection circle, the inflection pole J , and the center of the inflection circle (denoted as point O) are shown in Fig. 4.40. Note that the centrode normal, N , is directed from the velocity pole, I , toward the inflection pole, J , and the centrode tangent, T , is 90° clockwise from the centrode normal, N .

The cubic of stationary curvature equation for the coupler link, including pins A and B , can be written from Eq. (4.55) as

$$\frac{1}{R_{AI}} = \frac{1}{M \sin \psi_A} - \frac{1}{N \cos \psi_A} \quad (4a)$$

and also as

$$\frac{1}{R_{BI}} = \frac{1}{M \sin \psi_B} - \frac{1}{N \cos \psi_B}, \quad (4b)$$

where the parameters M and N are as given by Eqs. (4.56). The angle ψ_A is the counterclockwise angle from the centrode tangent, T , to the ray containing point A , and the angle ψ_B is the counterclockwise angle from the centrode tangent, T , to the ray containing point B . From the scale drawing, these angles are measured as

$$\psi_A = -45^\circ \quad \text{and} \quad \psi_B = -90^\circ. \quad (5a)$$

Note that these measurements can easily be verified from trigonometry. The distances to pin A from the velocity pole, I , and to pin B from the velocity pole, I , are measured consistently as

$$R_{AI} = 2 \text{ in} \quad \text{and} \quad R_{BI} = 2\sqrt{2} \text{ in}. \quad (5b)$$

Substituting Eqs. (5a) into Eq. (4b), gives

$$\frac{1}{R_{BI}} = \frac{1}{-M}. \quad (6)$$

Therefore, using Eq. (5b), we find the parameter

$$M = -R_{BI} = R_{JI} = -2\sqrt{2} \text{ in}. \quad \text{Ans. (7)}$$

Note that substituting $\psi_A = -45^\circ$ into Eq. (4a) indicates that $1/N = 0$, and the parameter N tends to infinity.

This is a degenerate form of the cubic of stationary curvature. The cubic of stationary curvature degenerates to a straight line (that is, the centrode normal) and a circle of diameter M (the center must lie on the centrode normal). The circle must pass through pins A and B on link 3. Therefore, the center of the circle is coincident with the ground pin O_4 .

Substituting the condition $N = \infty$ into Eq. (4.56) gives

$$\frac{1}{3R_{JI}} \left(\frac{dR_{JI}}{ds} \right) = 0. \quad (8a)$$

Since the diameter of the inflection circle is infinite, [Eq. (3)], the rate of change of the inflection circle diameter must be zero; that is,

$$\frac{dR_{JI}}{ds} = 0. \quad (8b)$$

In other words, the diameter of the inflection circle is at a maximum or a minimum as the four-bar linkage passes through this posture.

Recall that Ball's point is located at the intersection of the inflection circle and the cubic of stationary curvature. Figure 4.40 illustrates two apparent points of intersection—that is, the velocity pole, I , and the inflection pole, J . Since the velocity pole,

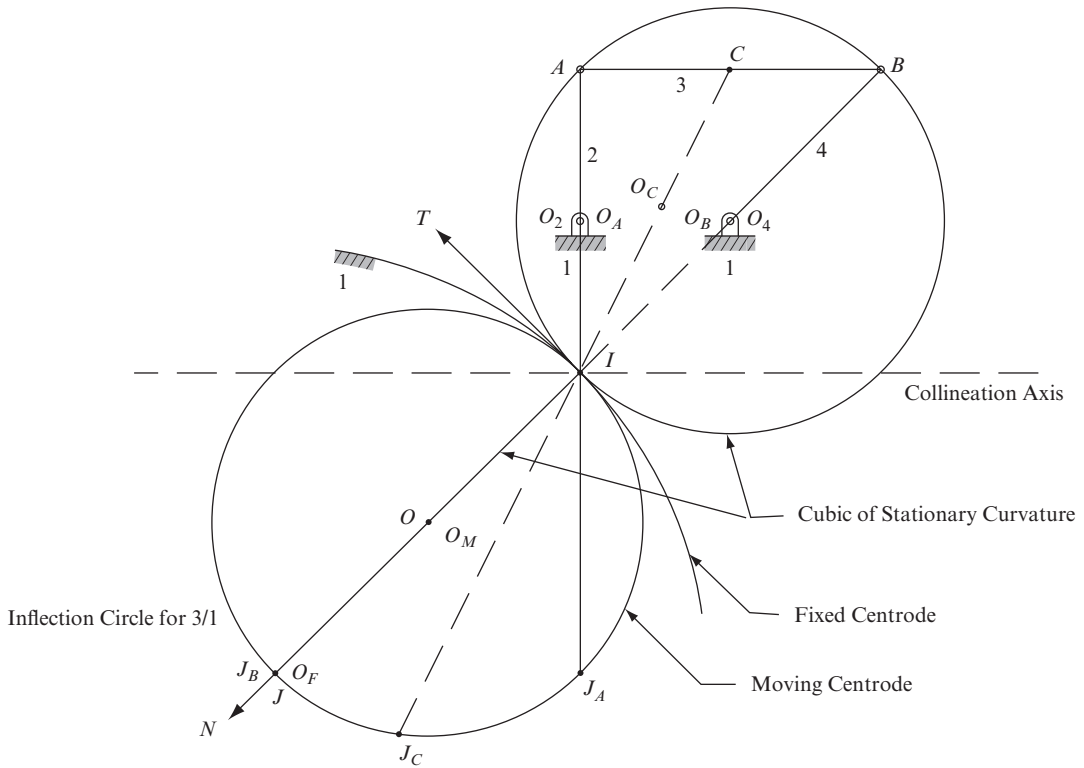


Figure 4.40 Cubic of stationary curvature and the two centrodes.

I , is not a point fixed in the coupler, it is not a solution. Therefore, Ball's point is coincident with the inflection pole, J .

- (a) Rearranging Eq. (4.56), the radius of curvature of the moving centrode can be written as

$$\frac{1}{R_{IO_M}} = \frac{1}{R_{JI}} - \frac{3}{M}. \quad (9)$$

Then, substituting Eq. (7) into this equation, gives

$$\frac{1}{R_{IO_M}} = \frac{1}{R_{JI}} - \frac{3}{R_{JI}} = -\frac{2}{R_{JI}}. \quad (10a)$$

Therefore, the radius of curvature of the moving centrode is

$$R_{IO_M} = -\frac{R_{JI}}{2} = \sqrt{2} \text{ in.} \quad \text{Ans. (10b)}$$

The negative signs in these last two equations indicate that the direction from O_M to I is opposite to the direction from I to J . The moving centrode is coincident with (or coalesces with) the inflection circle (Fig. 4.40).

- (b) The Euler-Savary equation can be written as

$$\frac{1}{R_{JI}} = \frac{1}{R_{IO_F}} - \frac{1}{R_{IO_M}}. \quad (11)$$

Substituting Eq. (10b) into this equation, and rearranging, the radius of curvature of the fixed centrode can be written as

$$R_{IO_F} = -R_{JI} = 2\sqrt{2} \text{ in.} \quad \text{Ans. (12)}$$

The negative sign indicates that the direction from O_F to I is opposite to the direction from I to J . The radius of the osculating circle for the fixed centrode is the same as the diameter of the inflection circle; that is, O_F is coincident with the inflection pole J (Fig. 4.40).

- (c) The position of coupler point C is shown in Fig. 4.40. Using complex polar notation, this can be written as

$$\mathbf{R}_{CI} = x_{CI}\hat{\mathbf{i}} + y_{CI}\hat{\mathbf{j}} = jR_{O_2I} + R_{AO_2}e^{j\theta_2} + R_{CA}e^{j\theta_3}, \quad (13)$$

which has real and imaginary components of

$$x_{CI} = R_{AO_2} \cos \theta_2 + R_{CA} \cos \theta_3 \quad \text{and} \quad y_{CI} = R_{O_2I} + R_{AO_2} \sin \theta_2 + R_{CA} \sin \theta_3. \quad (14)$$

Substituting the known data $R_{O_2I} = 1 \text{ in}$, $R_{AO_2} = 1 \text{ in}$, and $R_{CA} = 1 \text{ in}$ into Eq. (14) gives

$$x_{CI} = (1 \text{ in}) \cos \theta_2 + (1 \text{ in}) \cos \theta_3 \quad \text{and} \quad y_{CI} = (1 \text{ in}) + (1 \text{ in}) \sin \theta_2 + (1 \text{ in}) \sin \theta_3. \quad (15)$$

For this posture, $\theta_2 = 90^\circ$ and $\theta_3 = 0$; therefore, $x_{CI} = 1 \text{ in}$ and $y_{CI} = 2 \text{ in}$. Taking the derivative of Eqs. (14) with respect to the input angle θ_2 gives

$$x'_{CI} = -R_{AO_2} \sin \theta_2 - \theta'_3 R_{CA} \sin \theta_3 \quad \text{and} \quad y'_{CI} = R_{AO_2} \cos \theta_2 + \theta'_3 R_{CA} \cos \theta_3. \quad (16)$$

Next, we use the angular velocity ratio theorem, Eq. (3.30), to find the first-order kinematic coefficient:

$$\theta'_3 = \frac{R_{I_{23}I_{12}}}{R_{I_{23}I_{13}}} = \frac{1 \text{ in}}{2 \text{ in}} = 0.5 \text{ rad/rad.} \quad (17)$$

Then, using the dimensions given previously and evaluating at the same posture, we get

$$x'_{CI} = -1 \text{ in/rad} \quad \text{and} \quad y'_{CI} = 0.5 \text{ in/rad,} \quad (18)$$

and, from Eq. (3.34b),

$$r'_{CI} = \sqrt{(x'_{CI})^2 + (y'_{CI})^2} = \sqrt{(-1 \text{ in/rad})^2 + (0.5 \text{ in/rad})^2} = 1.12 \text{ in/rad}. \quad (19)$$

Measuring from the centrode tangent and centrode normal axes, the polar coordinates of coupler point C are

$$R_{CI} = \sqrt{(2 \text{ in})^2 + (1 \text{ in})^2} = \sqrt{5} \text{ in} \quad \text{and} \quad \psi_C = -71.57^\circ. \quad (20)$$

The inflection point for point C , from Eq. (4.42), is

$$R_{JCI} = R_{JI} \sin \psi_C = -2.68 \text{ in}. \quad (21)$$

Then, from Eq. (4.43), the radius of curvature of the path of coupler point C is

$$\rho_C = R_{CC'} = \frac{R_{CI}^2}{R_{CI} - R_{JCI}} = \frac{(\sqrt{5} \text{ in})^2}{(\sqrt{5} \text{ in}) - (-2.68 \text{ in})} = 1.02 \text{ in}. \quad \text{Ans. (22)}$$

Finally, measuring from point I , Eqs. (4.54) give the center of curvature of the coupler curve as

$$x_{C'I} = x_{CI} - \rho_C \left(\frac{y'_C}{r'_C} \right) = (1 \text{ in}) - (1.02 \text{ in}) \left(\frac{0.5 \text{ in/rad}}{1.12 \text{ in/rad}} \right) = 0.54 \text{ in}, \quad \text{Ans. (23)}$$

$$y_{C'I} = y_{CI} - \rho_C \left(\frac{x'_C}{r'_C} \right) = (2 \text{ in}) - (1.02 \text{ in}) \left(\frac{1 \text{ in/rad}}{1.12 \text{ in/rad}} \right) = 1.09 \text{ in}. \quad \text{Ans. (24)}$$

These results are in good agreement with the measurements from Fig. 4.40.

4.18 REFERENCES

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- [8] Tao, D. C., 1964. *Applied Linkage Synthesis*, Reading, MA: Addison-Wesley.

PROBLEMS*

- †4.1** The position vector of a point is defined by the equation

$$\mathbf{R} = \left(4t - \frac{t^3}{3}\right)\hat{\mathbf{i}} + 10t\hat{\mathbf{j}},$$

where R is in inches and t is in seconds. Find the acceleration of the point at $t = 2$ s.

- †4.2** A point moves according to the equation

$$\mathbf{R} = \left(t^2 - \frac{t^3}{6}\right)\hat{\mathbf{i}} + \frac{t^3}{3}\hat{\mathbf{j}},$$

where R is in meters and t is in seconds. Find the acceleration at $t = 3$ s.

- 4.3** The path of a point is described by the equation

$$\mathbf{R} = (t^2 + 4)e^{-j\pi t/10},$$

where R is in millimeters and t is in seconds. Find the unit tangent vector to the path, the normal and tangential components of the absolute acceleration, and the radius of curvature of the path, at $t = 20$ s.

- 4.4** The motion of a point is described by the equations

$$x = 4t\cos\pi t^3 \text{ and } y = \frac{t^3 \sin 2\pi t}{6},$$

where x and y are in feet, and t is in seconds. Find the acceleration of the point at $t = 1.40$ s.

- †4.5** Link 2, in the posture shown, has an angular velocity $\omega_2 = 120$ rad/s ccw and an angular acceleration $\alpha_2 = 4\,800$ rad/s² ccw. Determine the absolute acceleration of point A.

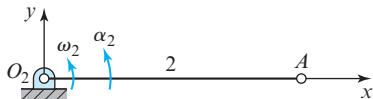


Figure P4.5 $R_{AO_2} = 500$ mm.

- 4.6** The accelerations of points A and B of link 2, which is rotating clockwise, are as given. Determine the angular velocity, the angular acceleration, and the acceleration of midpoint C of link 2.

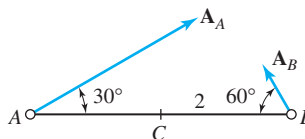


Figure P4.6 $A_A = 600$ ft/s² and $A_B = 150$ ft/s².

- 4.7** For the given kinematic data for link 2, find the velocity and acceleration of points B and C.

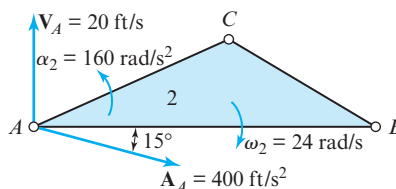


Figure P4.7 $R_{BA} = 16$ in, $R_{CA} = 10$ in, and $R_{CB} = 8$ in.

- †4.8** The angular velocity and angular acceleration of link 2 of the Scott-Russell linkage in the posture shown are $\omega_2 = 20$ rad/s cw and $\alpha_2 = 1\,493$ rad/s² cw, respectively. Determine the velocity and acceleration of point B and the angular acceleration of link 3.

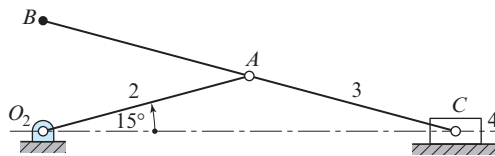


Figure P4.8 $R_{AO_2} = R_{CA} = R_{BA} = 100$ mm.

- †4.9** In the posture shown in Fig. P4.8, slider 4 is moving to the left with a constant velocity $V_C = 2$ m/s. Find the angular velocity and angular acceleration of link 2.

If the velocity of point B in the posture shown in Prob. 3.8 is constant, determine the acceleration of point A and the angular acceleration of link 3.

*When assigning problems, the instructor may wish to specify the method of solution to be used, because a variety of approaches are presented in the text.

- †4.11**  If the angular velocity of crank 2 in the posture shown in Prob. 3.9 is constant, determine the angular accelerations of links 3 and 4.
- †4.12**  If the angular velocity of crank 2 in the posture shown in Prob. 3.10 is constant, determine the acceleration of point *C* and the angular accelerations of links 3 and 4.
- †4.13**  If the angular velocity of crank 2 in the posture shown in Prob. 3.11 is constant, determine the acceleration of point *C* and the angular accelerations of links 3 and 4.
- †4.14** If the angular velocity of crank 2 in the posture shown in Prob. 3.13 is constant, determine the accelerations of points *C* and *D* and the angular acceleration of link 4.
- †4.15** If the angular velocity of crank 2 in the posture shown in Prob. 3.14 is constant, determine the acceleration of point *C* and the angular acceleration of link 4.
- †4.16** If the angular velocity of crank 2 in the posture shown in Prob. 3.16 is constant, determine the acceleration of point *C* and the angular acceleration of link 4.
- 4.17**  If the angular velocity of crank 2 in the posture shown in Prob. 3.17 is constant, determine the acceleration of point *B* and the angular accelerations of links 3 and 6.
- 4.18** For the four-bar linkage of Prob. 3.18 in the posture shown, determine the angular acceleration of crank 2 to ensure that the angular acceleration of link 4 is zero.
- 4.19** For the four-bar linkage of Prob. 3.19 in the posture shown, determine the angular acceleration of crank 2 to ensure that the angular acceleration of link 4 is 100 rad/s² cw.
- †4.20** If the angular velocity of crank 2 in the posture shown in Prob. 3.20 is constant, determine the acceleration of point *C* and the angular acceleration of link 3.
- †4.21** If the angular velocity of crank 2 in the posture shown in Prob. 3.21 is constant, determine the acceleration of point *C* and the angular acceleration of link 3.
- 4.22** If the angular velocity of crank 2 in the posture shown in Prob. 3.22 is constant, determine the accelerations of points *B* and *D*.
- 4.23** If the angular velocity of crank 2 in the posture shown in Prob. 3.23 is constant, determine the accelerations of points *B* and *D*.

†4.24 to 4.30 The nomenclature for the four-bar linkage is shown in Fig. P4.24; the dimensions and data are given in Table P4.24 to P4.30. Angular velocity ω_2 is constant for each problem (a negative sign indicates that the direction is clockwise). The dimensions of even-numbered problems are inches and odd-numbered problems are millimeters. For each problem, determine $\theta_3, \theta_4, \omega_3, \omega_4, \alpha_3$, and α_4 .

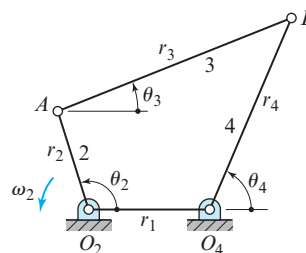


Figure P4.24

Table P4.24 to P4.30

Prob.	r_1	r_2	r_3	r_4	θ_2, deg	$\omega_2, \text{rad/s}$
P4.24	4	6	9	10	240	1
P4.25	100	150	250	250	-45	56
P4.26	14	4	14	10	0	10
P4.27	250	100	500	400	70	-6
P4.28	8	2	10	6	40	12
P4.29	400	125	300	300	210	-18
P4.30	16	5	12	12	315	-18

4.31 For the inverted slider-crank linkage in the posture shown, crank 2 has a constant angular velocity of 60 rev/min ccw. Find the velocity and acceleration of point *B*, and the angular velocity and acceleration of link 4.

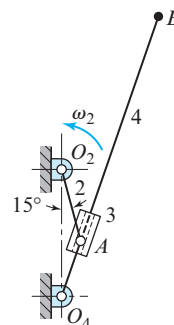


Figure P4.31 $R_{O_4O_2} = 12$ in, $R_{AO_2} = 7$ in, and $R_{BO_4} = 28$ in.

- 4.32** For the modified Scotch-yoke linkage in the posture shown in Prob. 3.26, determine the acceleration of link 4.
- 4.33** For the linkage in the posture shown in Prob. 3.27, determine the acceleration of point E .
- 4.34** For the inverted slider-crank linkage in the posture shown in Prob. 3.24, determine the acceleration of point B and the angular acceleration of link 4.
- 4.35** For the linkage in the posture shown in Prob. 3.25, determine the acceleration of point B and the angular acceleration of link 3.
- 4.36** For the linkage in the posture shown in Prob. 3.31, the input angular velocity is constant. Determine the accelerations of points A and B .
- 4.37** For the mechanism in the posture shown in Prob. 3.32, crank 2 has an angular acceleration of 2 rad/s^2 ccw. Determine the acceleration of point C_4 and the angular acceleration of link 3.
- 4.38** For the mechanism in the posture shown in Prob. 3.29, the input angular velocity is constant. Determine the angular accelerations of links 3 and 4.
- 4.39** For the mechanism in the posture shown in Prob. 3.30, the input angular velocity is constant. Determine the acceleration of point G and the angular accelerations of links 5 and 6.
- 4.40** Continue Prob. 3.40 and find the second-order kinematic coefficients of links 3 and 4. Assuming an input acceleration of $A_{A_2} = 5 \text{ m/s}^2$, find the angular accelerations of links 3 and 4.
- 4.41** Continue Prob. 3.49 and find the second-order kinematic coefficients of links 3, 4, and 5. Assuming constant angular velocity for link 2, find the angular accelerations of links 3, 4, and 5.
- 4.42** Continue Prob. 3.50 and find the second-order kinematic coefficients of links 3, 4, and 5. Assuming constant angular velocity for link 2, find the angular accelerations of links 3, 4, and 5.
- 4.43** Draw the inflection circle for the absolute motion of the coupler link of the double-slider linkage. Select several points on the centrode normal and find their conjugate points. Plot portions of the paths of these points to demonstrate for yourself that the conjugates are indeed the centers of curvature.
- 4.44** Draw the inflection circle for the absolute motion of the coupler of the four-bar linkage. Find the center of curvature of the coupler curve of point C and generate a portion of the path of C to verify your findings.

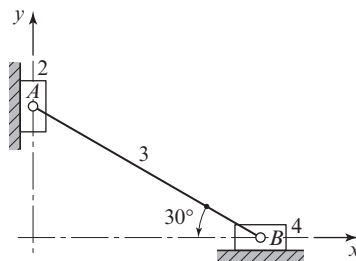


Figure P4.43 $R_{BA} = 125 \text{ mm}$.

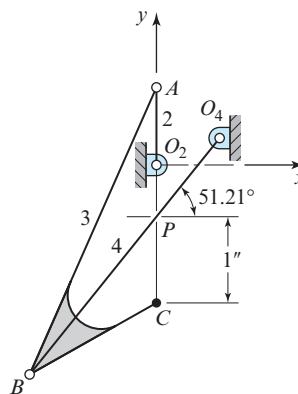


Figure P4.44 $R_{CA} = 2.5 \text{ in}$, $R_{AO_2} = 0.9 \text{ in}$, $R_{BO_4} = 3.5 \text{ in}$, and $R_{PO_4} = 1.17 \text{ in}$.

- 4.45** For the motion of the coupler relative to the frame, find the inflection circle, the centrode normal, the centrode tangent, and the centers of curvature of points C and D of the linkage of Prob. 3.13. Choose points on the coupler coincident with the instantaneous center of velocity and inflection pole, and plot nearby portions of their paths.
- 4.46** For the four-bar linkage in the posture shown, link 2 is 30° counterclockwise from the ground link, and the angular velocity and angular acceleration of the coupler link are $\omega_3 = 5 \text{ rad/s}$ ccw and $\alpha_3 = 20 \text{ rad/s}^2$ cw, respectively. For the instantaneous motion of the coupler link, show: (a) the velocity pole, I , the pole tangent, T , and the pole normal, N ; (b) the inflection circle and the Bresse circle; and (c) the instantaneous center of acceleration. Then determine: (d) the radius of curvature of the path of coupler point C ; (e) the velocity of C ; (f) the angular velocity of link 2; (g) the velocity of pole I ; (h) the acceleration of C ; and (i) the acceleration of the velocity pole.

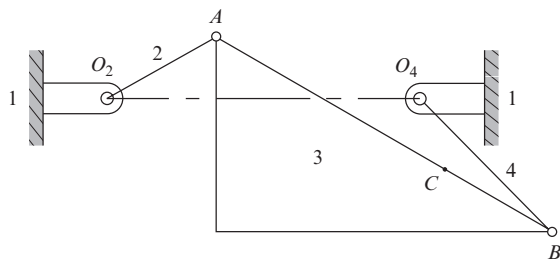


Figure P4.46 $R_{O_4O_2} = 2.5$ in, $R_{AO_2} = 1$ in, $R_{BA} = 3.15$ in, $R_{BO_4} = 1.5$ in., and $R_{CB} = 1$ in.

- 4.47** Consider the double-slider linkage in the posture given in Prob. 3.8. Point B moves with a constant velocity $V_B = 40$ m/s to the left, as shown in the figure. The angular velocity and angular acceleration of coupler link AB are $\omega_3 = 36.6$ rad/s ccw and $\alpha_3 = 1340$ rad/s² cw, respectively. For the absolute motion of coupler link AB in the specified posture, draw the inflection circle and the Bresse circle. Then determine: (a) the radius of curvature of the path of point C , which is a point of link 3 midway between points A and B ; and (b) the velocity of the velocity pole, I . Using the instantaneous center of acceleration determine: (c) the acceleration of the pole, I ; and (d) the accelerations of points A and C .
- 4.48** For the linkage of Prob. 3.17, link 2 is rotating with an angular velocity $\omega_2 = 15$ rad/s ccw and an angular acceleration $\alpha_2 = 320.93$ rad/s² cw. For the instantaneous motion of connecting rod 3, find: (a) the inflection circle and the Bresse circle; (b) the location of the instantaneous center of acceleration; (c) the center of curvature of the path traced by coupler point C ; (d) the accelerations of points A , B , and C ; and (e) the acceleration of inflection pole J .
- 4.49** Figure P3.32 illustrates an epicyclic gear train driven by the arm, link 2, with an angular velocity $\omega_2 = 3.33$ rad/s cw and an angular acceleration $\alpha_2 = 15$ rad/s² ccw. Define point E as a point on the circumference of planet gear 4 horizontal to the right of point B such that angle $\angle DBE = 90^\circ$. For the absolute motion of gear 4, draw the inflection circle and the Bresse circle on a scaled drawing of the epicyclic gear train. Then, determine: (a) the location of the instantaneous center of acceleration of the planet gear; (b) the radii of curvature of the paths of points B and E ; (c) the locations of the centers of curvature of the paths of points B and E ; and (d) the accelerations of points B and E and pole I .

- 4.50** On 18 in $\times$ 24 in paper, draw the four-bar linkage full size, placing A' 6 in up from the lower edge and 7 in left of the right edge. (Better utilization of the paper is obtained by tilting the frame through about 15° as indicated.) For the coupler link, draw the inflection circle and the cubic of stationary curvature. Choose a coupler point C coincident with the cubic, and plot a portion of its coupler curve in the vicinity of the cubic. Find the conjugate point C' . Draw a circle through C with center at C' , and compare this circle with the actual path of C . Find Ball's point. Locate a point D on the coupler at Ball's point, and plot a portion of its path. Compare the result with a straight line.

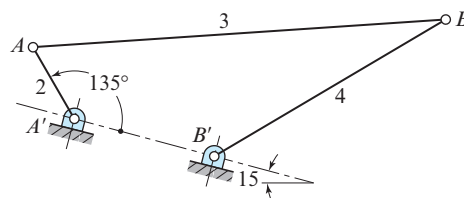


Figure P4.50 $R_{AA'} = 1$ in, $R_{BA} = 5$ in, $R_{B'A'} = 1.75$ in, and $R_{BB'} = 3.25$ in.

- 4.51** For the mechanism in the posture shown in Fig. P3.51, the first- and second-order kinematic coefficients are $\theta'_3 = -8.333$ rad/rad, $r'_2 = -100$ mm/rad, $\theta''_3 = -8.642$ rad/rad², and $r''_2 = -237.033$ mm/rad² (where θ_2 is the input and r_2 is the vector from ground pin O_2 to pin A). Wheel 3 is rolling without slipping on the ground link at point C and sliding in the slot that is cut in link 2. The radius of the ground link is $\rho_1 = 60$ mm, and the radius of the wheel is $\rho_3 = 15$ mm. Determine: (a) the unit normal vector to the path of point D ; (b) the radius of curvature of the path of this point; and (c) the x and y coordinates of the center of curvature of this path. If the angular velocity of link 2 is a constant $\omega_2 = 30$ rad/s ccw, then determine the acceleration of point D .
- 4.52** For the mechanism in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = 0.50$ rad/ft, $R'_4 = -1.00$ ft/ft, $\theta''_3 = 0.25$ rad/ft², and $R''_4 = -1.00$ ft/ft². Roller 4 is pinned to link 3 at B and is rolling without slipping on the vertical ground link at C . Determine: (a) the first- and second-order kinematic coefficients of point P ; (b) the unit tangent vector and the unit normal vector to the path traced by P ; (c) the radius

of curvature of this path; and (d) the x and y coordinates of the center of curvature of this path. If the constant velocity of the input is $\mathbf{V}_2 = -10\hat{\mathbf{i}}$ ft/s, determine the acceleration of P .

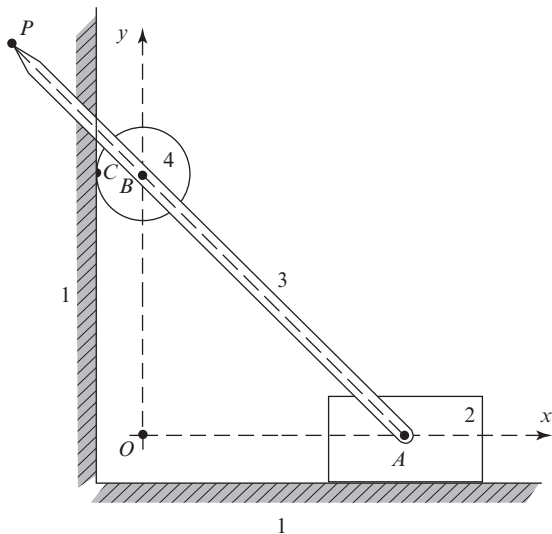


Figure P4.52 $\mathbf{R}_2 = \mathbf{R}_{AO} = 2\hat{\mathbf{i}}$ ft, $\mathbf{R}_4 = \mathbf{R}_{BO} = 2\hat{\mathbf{j}}$ ft, $R_{PA} = 4$ ft, and $\rho_4 = 0.375$ ft.

4.53 For the rack-and-pinion mechanism in the posture shown, the first- and second-order kinematic coefficients are $R'_2 = -400$ mm/rad, $R'_4 = -346.41$ mm/rad, $R''_2 = 1385.6$ mm/rad², and $R''_4 = 1400$ mm/rad² (where $\mathbf{R}_2$ is the vector from ground pin O_2 to the point of contact, C , of link 3). Determine: (a) the first- and second-order kinematic coefficients of point B ; (b) the unit tangent vector and the unit normal vector to the path traced by point B ; (c) the radius of curvature of this path; and (d) the x and y coordinates of the center of curvature of this path. If the constant angular velocity of the input link is $\omega_2 = -12\hat{\mathbf{k}}$ rad/s, determine the acceleration of point B .

4.54 For the gear train in Prob. 3.54, the angular velocity and acceleration of input gear 2 are $\omega_2 = 77$ rad/s ccw and $\alpha_2 = 5$ rad/s² cw, respectively. Determine: (a) the second-order kinematic coefficients of gear 3 and rack 5; (b) the angular accelerations of gear 3 and link 4; and (c) the acceleration of the rack.

4.55 For the gear train in Prob. 3.55, the angular velocity and acceleration of input arm 2 are $\omega_2 = 50$ rad/s cw and $\alpha_2 = 15$ rad/s² cw, respectively. Using the method of kinematic coefficients, determine the angular accelerations of gears 3, 4, and 5.

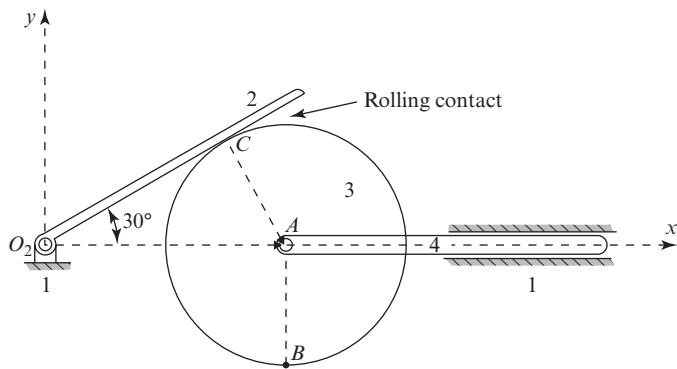


Figure P4.53 $\rho_3 = 100$ mm.

- 4.56** For the mechanism in the posture shown, link 4 is rolling without slipping on the ground at point C , and the first- and second-order kinematic coefficients are $\theta'_3 = -1.341$ rad/rad, $R'_4 = -3.097$ in/rad, $\theta''_3 = -2.475$ rad/rad², and $R''_4 = -7.372$ in/rad² (where R_4 is the vector from the origin to pin B , which connects links 3 and 4). Determine the radius of curvature of the path of point D , and the x and y coordinates of the center of curvature of this path. If the constant input angular velocity of link 2 is $\omega_2 = -9\hat{\mathbf{k}}$ rad/s, determine the acceleration of point D .

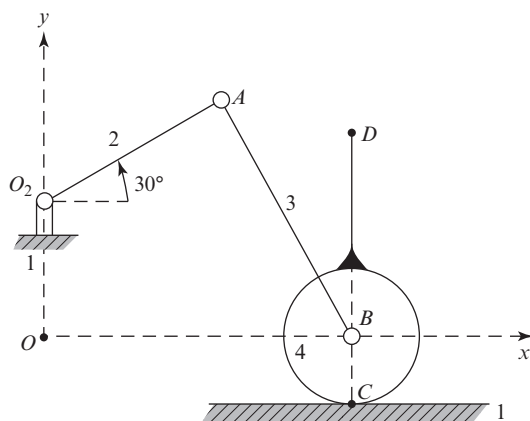


Figure P4.56 $R_{O_2} = 1.00$ in, $R_{AO_2} = 1.50$ in, $R_{BA} = 2.00$ in, $R_B = 2.267$ in, $R_{CB} = 0.50$ in, and $R_{DB} = 1.50$ in.

- 4.57** For the linkage in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = -3.0$ rad/rad, $R'_4 = 86.6$ mm/rad, $\theta''_3 = -13.856$ rad/rad², and $R''_4 = 150$ mm/rad² (where R_4 is the vector from ground pivot O_2 to pin B). Determine the radius of curvature of the path of point C , and the x and y coordinates of the center of curvature of this path. If the input angular velocity of link 2 is a constant $\omega_2 = 22$ rad/s ccw, determine the acceleration of point C .

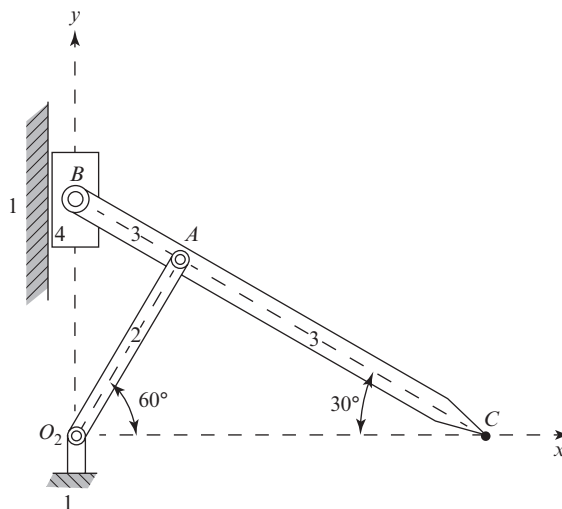


Figure P4.57 $R_{AO_2} = 43.3$ mm, $R_{BA} = 25$ mm, and $R_{CA} = 75$ mm.

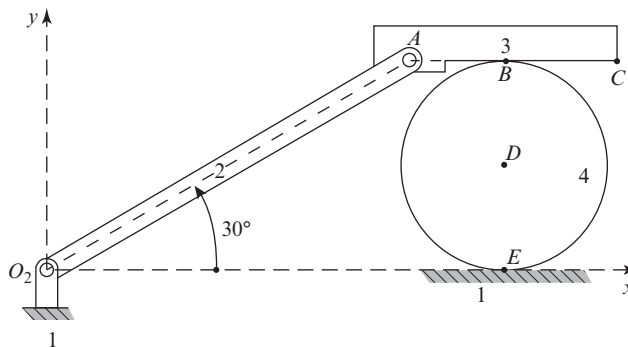


Figure P4.58 $R_{AO_2} = 4$ in, $R_{BA} = 1$ in, $R_{CA} = 2$ in, and $\rho_4 = 1$ in.

4.58 For the mechanism in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = -3.464$ rad/rad, $\theta'_4 = 1$ rad/rad, $\theta''_3 = 5.464$ rad/rad², and $\theta''_4 = 7.732$ rad/rad², respectively. Line AC in link 3 is parallel to the x axis. The circular wheel, link 4, is rolling on the ground link at point E and rolling on link 3 at point B . Determine: (a) the first- and second-order kinematic coefficients of point C ; (b) the unit tangent vector and the unit normal vector to the path traced by point C ; (c) the radius of curvature of this path; and (d) the x and y coordinates of the center of curvature of this path. If the input angular velocity of link 2 is a constant $\omega_2 = 15\hat{k}$ rad/s, determine the acceleration of point C .

4.59 For the linkage in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = \theta'_4 = 0.5$ rad/rad, $\theta''_3 = \theta''_4 = 0$, $R'_{34} = 0$, and $R''_{34} = -50$ mm/rad² (where R_{34} is the vector from point B fixed in link 3 to point C fixed in link 4).

Determine: (a) the radius of curvature of the path of point B and (b) the center of curvature of the path of this point. If the input angular velocity of link 2 is a constant $\omega_2 = 10$ rad/s cw, determine the acceleration of point B .

4.60 For the mechanism in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = -4.333$ rad/rad, $\theta'_4 = 0$, $R'_4 = 26$ in/rad, $\theta''_3 = 0$, $\theta''_4 = 0.813$ rad/rad², and $R''_4 = 4.875$ in/rad² (where the rotation of link 2 is the input and $\mathbf{R}_4$ is the vector from point O_4 to point C on link 3). The angle between line AB in link 3 and line O_2A is a right angle. Determine: (a) the first- and second-order kinematic coefficients of point B ; (b) the unit tangent vector and the unit normal vector to the path traced by this point; (c) the radius of curvature of the path traced by this point; and (d) the x and y coordinates of the center of curvature of the path traced by this point. If the input angular velocity of link 2 is a constant $\omega_2 = 15\hat{k}$ rad/s, determine the acceleration of point B .

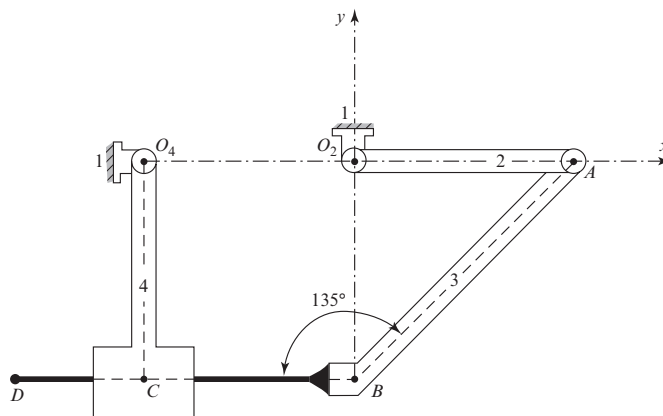


Figure P4.59 $R_{AO_2} = 100$ mm, $R_{BA} = 141.4$ mm, $R_{CB} = R_{O_2O_4} = 100$ mm, and $R_{CO_4} = 100$ mm.

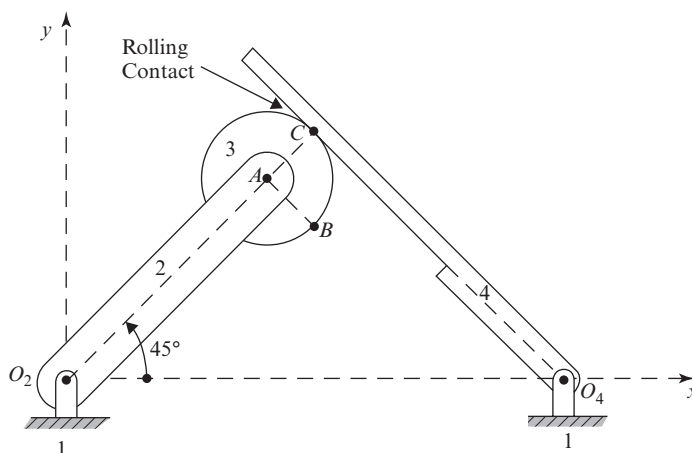


Figure P4.60 $R_{AO_2} = 26$ in and $R_{BA} = R_{CA} = \rho_3 = 6$ in.

4.61 For the linkage of Prob. 3.61 in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = \theta'_4 = 1$ rad/rad, $R'_{34} = -12$ in/rad, $\theta''_3 = \theta''_4 = +2.309$ rad/rad², and $R''_{34} = -20.785$ in/rad² (where R_{34} is the vector from O_4 to point B fixed in link 3). Determine the first- and second-order kinematic coefficients of coupler point C . Then, determine: (a) the unit tangent vector and the unit normal vector to the path traced by point C ; (b) the radius of curvature of this path; and (c) the x and y coordinates of the center of curvature of this path. If the angular velocity of input link 2 is a constant $\omega_2 = 15\mathbf{k}$ rad/s, determine the acceleration of point C .

4.62 For the mechanism in the posture shown, the first- and second-order kinematic coefficients are $\theta'_3 = 2.165$ rad/rad, $\theta'_4 = -7.143$ rad/rad, $\theta''_3 = -9.369$ rad/rad², and $\theta''_4 = 26.784$ rad/rad². Determine: (a) the first- and second-order kinematic coefficients for point C ; (b) the radius of curvature of the path of this point; and (c) the x and y coordinates of the center of curvature of this path. If the input angular velocity of link 2 is a constant $\omega_2 = 50$ rad/s ccw, determine the acceleration of point C .

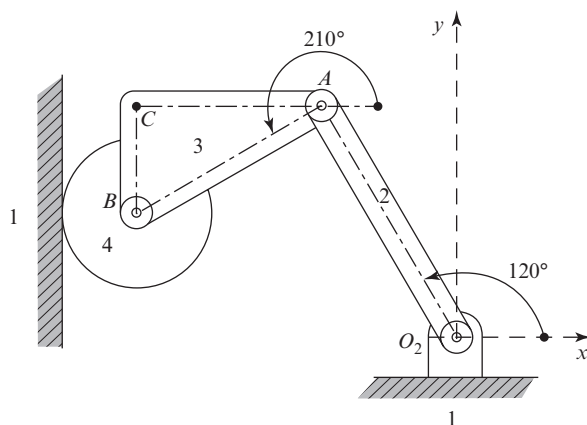
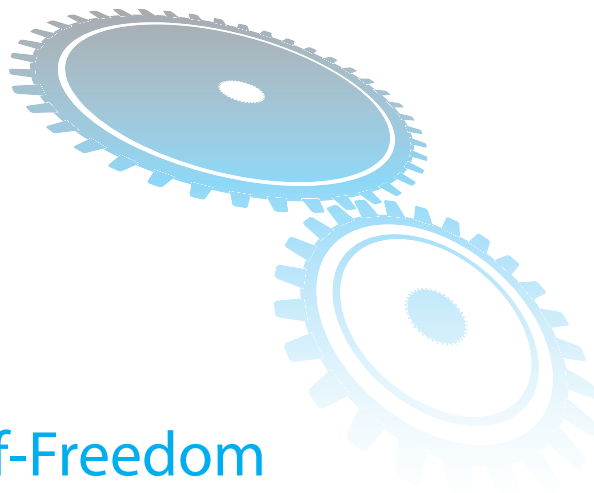


Figure P4.62 $R_{AO_2} = 500$ mm, $R_{BA} = 400$ mm, $R_{CA} = 346.4$ mm, and $\rho_4 = 140$ mm.

5

Multi-Degree-of-Freedom Mechanisms



5.1 INTRODUCTION

In Chaps. 2, 3, and 4, we concentrated totally on problems that exhibit a single degree of freedom and can be analyzed by specifying the motion of a single input variable. This was justifiable, since, by far, the vast majority of practical mechanisms are designed to have only one degree of freedom so that they can be driven by a single power source. However, there are mechanisms that have multiple degrees of freedom and can only be analyzed if more than one input motion is given. In this chapter, we will look at how our methods can be used to find the positions, velocities, and accelerations of these mechanisms.

Consider, for example, the planar five-bar linkage shown in Fig. 5.1. The Kutzbach criterion, Eq. (1.1), indicates that this linkage has a mobility of two and, therefore, requires two input motions to provide a unique output motion. This linkage is operated by rotating cranks 2 and 3 independently and can, therefore, produce a wide variety of motions for the two coupler links 4 and 5.

A practical application of the five-bar linkage is to position the end-effector of an industrial robotic manipulator (for example, the General Electric Model P80 robotic manipulator shown in Fig. 5.2).

Another common practical application of a five-bar linkage is the pantograph linkage shown in Fig. 1.26. A variation of the pantograph is the linkage shown in Fig. 5.3, where the path of point P is a magnified copy of the path of point C .

If the two input rotations of Fig. 5.1 are interconnected by gears that have rolling contact with each other, as shown in Fig. 5.4, then the resulting mechanism has only a

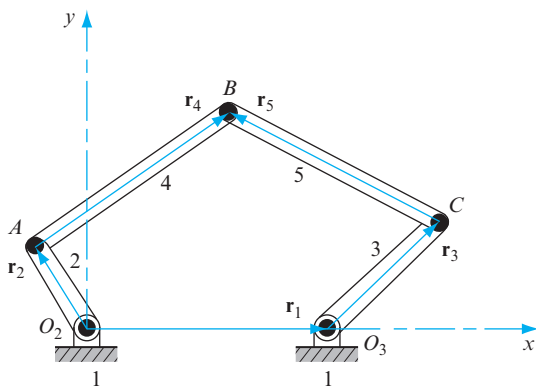


Figure 5.1 Vectors for the five-bar linkage.

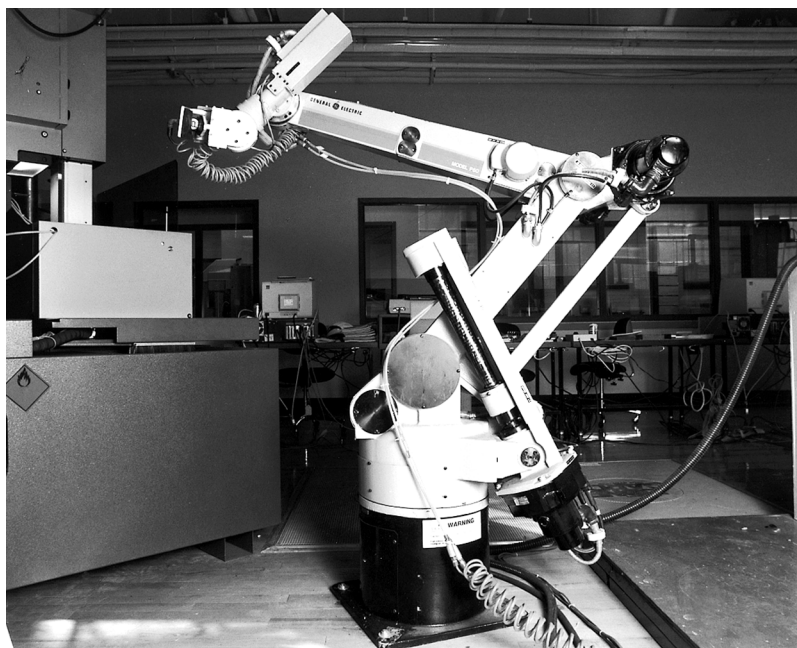


Figure 5.2 General Electric Model P80 robotic manipulator.

single degree of freedom and is commonly referred to as a *geared five-bar* mechanism. Such a mechanism is often found in machinery, since it can provide more complex motions than the well-known planar four-bar linkage that was investigated in previous chapters.

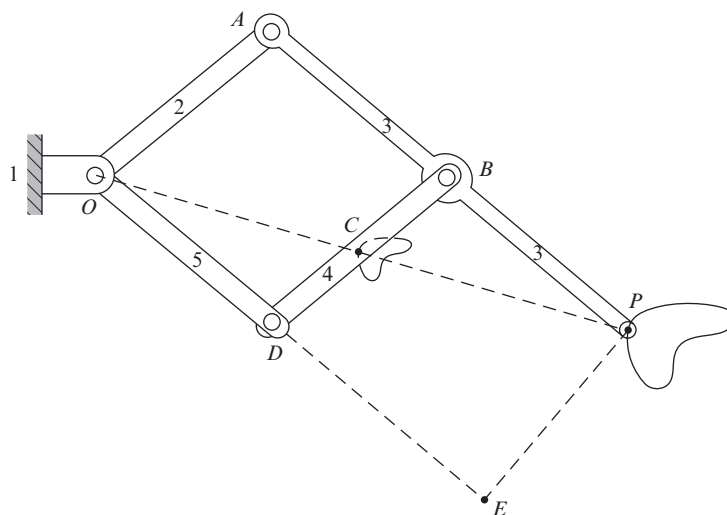


Figure 5.3 A pantograph.

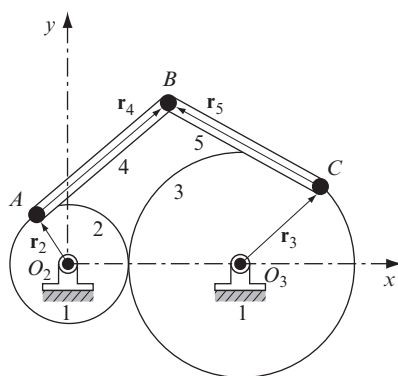


Figure 5.4 Geared five-bar mechanism.

EXAMPLE 5.1

For the five-bar linkage in the posture shown in Fig. 5.5, the postures of the two input link are $\theta_2 = 120^\circ$ and $\theta_3 = 45^\circ$. Determine the postures of the two coupler links, θ_4 and θ_5 .

SOLUTION

The vectors for the kinematic analysis of this linkage are shown in Fig. 5.5. The postures of links 4 and 5 are found from a scale drawing:

$$\theta_4 = 36.5^\circ \quad \text{and} \quad \theta_5 = 151.1^\circ.$$

Ans.

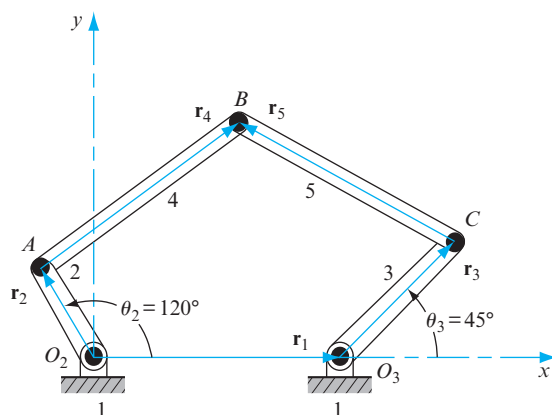


Figure 5.5 $r_1 = R_{O_3O_2} = 6$ in, $r_2 = R_{AO_2} = 2.5$ in, $r_3 = R_{CO_3} = 4$ in, $r_4 = R_{BA} = 6$ in, and $r_5 = R_{BC} = 6$ in.

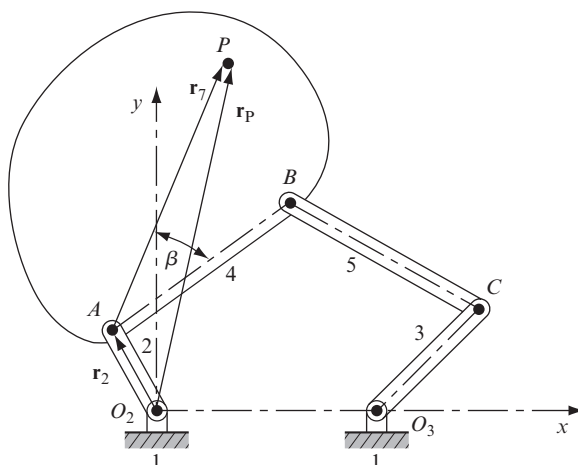


Figure 5.6 Vectors for coupler point P of link 4.

For a point fixed in coupler link 4 or coupler link 5, to follow a unique path, control of two independent input motions is required. Still, there is an extremely wide variety of curves that can be generated by a coupler point of the planar five-bar linkage. For purposes of illustration, consider the arbitrary point P of coupler link 4, as shown in Fig. 5.6.

EXAMPLE 5.2

Consider the continuation of Example 5.1 to include coupler point P shown in Fig. 5.6. The location of point P is given by $r_7 = R_{PA} = 8$ in and $\beta = 30^\circ$. For the given posture of the linkage, the problem is to determine the absolute position coordinates of point P .

SOLUTION

After Fig. 5.5 of Example 5.1 is completed to scale, vector $\mathbf{r}_7$ can be drawn at the orientation $\theta_7 = \theta_4 + \beta = 66.5^\circ$. The absolute coordinates of point P can then be measured, and the results are

$$x_P = 1.95 \text{ in} \quad \text{and} \quad y_P = 9.50 \text{ in.} \quad \text{Ans.}$$

5.2 POSTURE ANALYSIS; ALGEBRAIC SOLUTION

The analytic approaches for the solution of the posture analysis of multi-degree-of-freedom mechanisms are parallel to the methods presented in Chap. 2. When sufficient input data are given to represent the positions of all degrees of freedom, the loop-closure equations can be formulated. For planar mechanisms, these equations can be separated into real and imaginary parts (horizontal and vertical components), and these allow solutions for two unknowns per loop. We will demonstrate this by continuing with the planar five-bar linkage of Example 5.1 and Example 5.2.

EXAMPLE 5.3

For the planar five-bar linkage of Fig. 5.5, the loop-closure equation is

$$\overset{\sqrt{I}}{\mathbf{r}_2} + \overset{\sqrt{?}}{\mathbf{r}_4} - \overset{\sqrt{?}}{\mathbf{r}_5} - \overset{\sqrt{I}}{\mathbf{r}_3} - \overset{\sqrt{\sqrt{I}}}{\mathbf{r}_1} = \mathbf{0}, \quad (1)$$

where θ_2 and θ_3 are the input angles, marked as given above the appropriate terms in Eq. (1). For the given posture of the linkage, the problem is to determine: (a) unknown coupler angles θ_4 and θ_5 ; and (b) the absolute position coordinates of coupler point P .

SOLUTION

In complex polar notation, Eq. (1) becomes

$$r_2 e^{j\theta_2} + r_4 e^{j\theta_4} - r_5 e^{j\theta_5} - r_3 e^{j\theta_3} - r_1 = 0. \quad (2)$$

Separating the real and imaginary parts of this equation gives

$$r_2 \cos \theta_2 + r_4 \cos \theta_4 - r_5 \cos \theta_5 - r_3 \cos \theta_3 - r_1 = 0, \quad (3)$$

$$r_2 \sin \theta_2 + r_4 \sin \theta_4 - r_5 \sin \theta_5 - r_3 \sin \theta_3 = 0. \quad (4)$$

The solution to these equations can be determined either analytically or numerically. For example, using the Newton-Raphson iterative procedure (Sec. 2.9), the postures of the two coupler links are evaluated as

$$\theta_4 = 36.447^\circ \quad \text{and} \quad \theta_5 = 151.084^\circ. \quad \text{Ans.}$$

These verify the graphic results determined in Example 5.1 and have higher accuracy.

The vectors defining the location of coupler point P are shown in Fig. 5.6. The vector equation for this point can be written as

$$\mathbf{r}_P = \mathbf{r}_2 + \mathbf{r}_7, \quad (5)$$

where the constraint equation for the direction of vector $\mathbf{r}_7$ is

$$\theta_7 = \theta_4 + \beta. \quad (6)$$

From Eq. (5), the absolute position coordinates of point P are

$$x_P = r_2 \cos \theta_2 + r_7 \cos \theta_7, \quad (7)$$

$$y_P = r_2 \sin \theta_2 + r_7 \sin \theta_7. \quad (8)$$

Substituting the given data into these equations, the position of point P (for the given posture of the linkage) is

$$x_P = (2.5 \text{ in}) \cos 120^\circ + (8 \text{ in}) \cos 66.447^\circ = 1.947 \text{ in}, \quad \text{Ans.}$$

$$y_P = (2.5 \text{ in}) \sin 120^\circ + (8 \text{ in}) \sin 66.447^\circ = 9.499 \text{ in}. \quad \text{Ans.}$$

These verify the graphic results determined in Example 5.2 and have higher accuracy.

5.3 VELOCITY ANALYSIS; VELOCITY POLYGONS

Velocity analysis is performed quite easily by the velocity polygon methods of Secs. 3.3 through 3.8 once the velocities of each of the independent inputs are specified. No new methods are required for the utilization of velocity polygons when a mechanism has more than one degree of freedom. However, input velocities must be given for each degree of freedom. An example is presented here to demonstrate how the graphic methods can be extended.

EXAMPLE 5.4

For the five-bar linkage of Example 5.1, in the posture shown in Fig. 5.5, suppose that the constant angular velocities of links 2 and 3 are $\omega_2 = 10 \text{ rad/s}$ ccw and $\omega_3 = 5 \text{ rad/s}$ cw, respectively. Determine: (a) the angular velocities of coupler links 4 and 5; and (b) the velocity of coupler point P .

SOLUTION

First, the velocities of points A and C , respectively, are

$$V_A = V_{AO_2} = \omega_2 R_{AO_2} = (10 \text{ rad/s}) (2.5 \text{ in}) = 25.0 \text{ in/s},$$

$$V_C = V_{CO_3} = \omega_3 R_{CO_3} = (5 \text{ rad/s}) (4.0 \text{ in}) = 20.0 \text{ in/s}.$$

From these, we can write two vector equations for the velocity of point B :

$$\mathbf{V}_B = \overset{\sqrt{\vee}}{\mathbf{V}}_A + \overset{?\vee}{\mathbf{V}}_{BA} = \overset{\sqrt{\vee}}{\mathbf{V}}_C + \overset{?\vee}{\mathbf{V}}_{BC}.$$

Solving this equation with the velocity polygon shown in Fig. 5.7 locates velocity image point B .

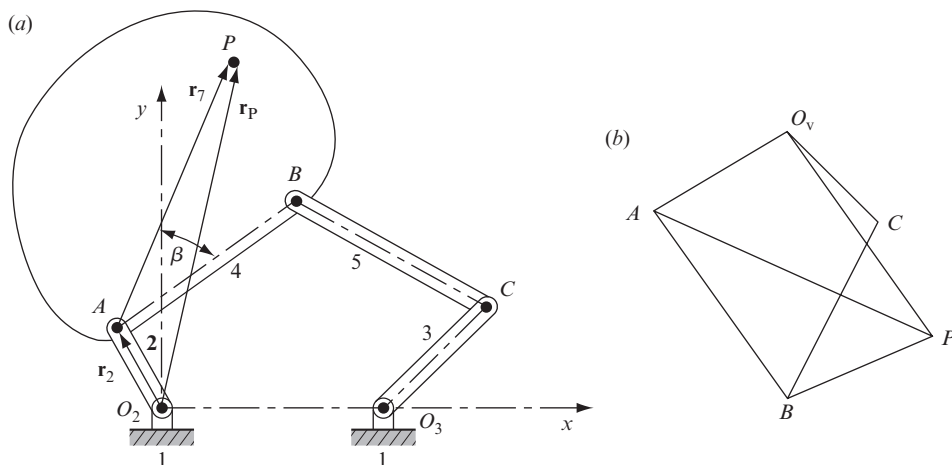


Figure 5.7 (a) Linkage posture; (b) velocity polygon.

Now we can measure the velocities V_{BA} and V_{BC} and obtain the angular velocities of coupler links 4 and 5; that is,

$$\omega_4 = \frac{V_{BA}}{R_{BA}} = \frac{35.34 \text{ in/s}}{6.0 \text{ in}} = 5.89 \text{ rad/s cw}, \quad \text{Ans.}$$

$$\omega_5 = \frac{V_{BC}}{R_{BC}} = \frac{30.60 \text{ in/s}}{6.0 \text{ in}} = 5.10 \text{ rad/s ccw}. \quad \text{Ans.}$$

Once the velocities of points A and B are known, we can construct the velocity image of coupler link 4 as explained in Sec. 3.4 and obtain the velocity of coupler point P ; that is,

$$\mathbf{V}_P = 38 \text{ in/s} \angle -55.5^\circ. \quad \text{Ans.}$$

5.4 INSTANTANEOUS CENTERS OF VELOCITY

If we attempt to use the method of instantaneous centers of velocity for planar mechanisms with multiple degrees of freedom, we encounter a complication. We find that the locations of the secondary instant centers cannot be determined by the methods of Secs. 3.13 through 3.15 alone. The locations of the secondary instant centers cannot, in fact, be determined using only the geometry of the mechanism; their locations depend on the ratios of the independent input velocities [1]. One method for finding the secondary instant centers is demonstrated in the following example.

EXAMPLE 5.5

For the five-bar linkage in the posture of Example 5.1, Fig. 5.5, suppose that links 2 and 3 are rotating with constant angular velocities $\omega_2 = 10 \text{ rad/s}$ ccw and $\omega_3 = 5 \text{ rad/s}$ cw, respectively. Using the method of instant centers, determine the angular velocities of coupler links 4 and 5, and the velocity of coupler point P .

SOLUTION

The five primary instant centers for the five-bar linkage, denoted $I_{12}, I_{24}, I_{45}, I_{35}$, and I_{13} , are the centers of the five pin joints, and are shown in Fig. 5.8.

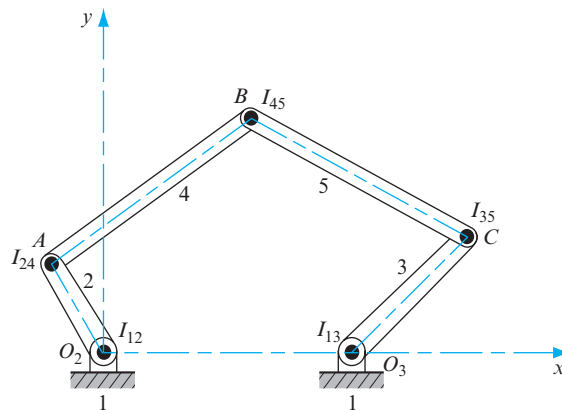


Figure 5.8 Primary instant centers.

From the Aronhold-Kennedy theorem (Sec. 3.14), we know that secondary instant center I_{23} , which relates the two known velocities, ω_2 and ω_3 , must lie on the line containing instant centers I_{12} and I_{13} .

Furthermore, from the angular velocity ratio theorem, Eq. (3.28), we know that the ratio of the angular velocity of link 2 to the angular velocity of link 3 can be written as

$$\frac{\omega_2}{\omega_3} = \frac{R_{I_{23}I_{13}}}{R_{I_{23}I_{12}}} = \frac{10.0 \text{ rad/s}}{-5.0 \text{ rad/s}} = -2.0. \quad (1)$$

From Sec. 3.17 we know that, if instant center I_{23} is located between the two absolute instant centers, I_{12} and I_{13} , then the angular velocity ratio is negative. Similarly, if instant center I_{23} is outside the two absolute instant centers, I_{12} and I_{13} , then the angular velocity ratio is positive. Note, however, that a sign convention is not necessary in Eq. (1), since a directed line is used for measuring the locations of the instant centers.

In this example, Eq. (1) gives

$$R_{I_{23}I_{13}} = -2.0R_{I_{23}I_{12}}. \quad (2)$$

But we also know from Fig. 5.8, using positive distances to the right, that

$$R_{I_{23}I_{13}} = R_{I_{23}I_{12}} + R_{I_{12}I_{13}} = R_{I_{23}I_{12}} - 6.0 \text{ in}, \quad (3)$$

and solving Eqs. (2) and (3) simultaneously gives

$$\begin{aligned} -2.0R_{I_{23}I_{12}} &= R_{I_{23}I_{12}} - 6.0 \text{ in}, \\ R_{I_{23}I_{12}} &= 2.0 \text{ in}. \end{aligned} \quad (4)$$

Therefore, instant center I_{23} is located 2.0 in to the right of instant center I_{12} , as shown on the scaled drawing of the linkage in Fig. 5.9.

The remaining secondary instant centers can be obtained directly from the Aronhold-Kennedy theorem. The Kennedy circle, used to help locate the secondary instant centers, is also shown in this figure. Secondary instant center I_{14} , for example, must lie on the line containing instant centers I_{12} and I_{24} . Similarly, secondary instant center I_{15} must lie on the line containing instant centers I_{13} and I_{35} . The locations of all secondary instant centers that lie within the limits of the page are shown in Fig. 5.9. The location of instant center I_{34} lies outside of the page, but it is indicated by two dashed lines. Note that it is not necessary to find the locations of all secondary instant centers in order to determine the unknown angular velocities ω_4 and ω_5 .

Choosing the upward direction along each line of centers as positive, the distances between instant centers I_{12} and I_{24} , and between instant centers I_{13} and I_{35} are known from the lengths of links 2 and 3, respectively; these are

$$R_{I_{24}I_{12}} = 2.50 \text{ in} \quad \text{and} \quad R_{I_{35}I_{13}} = 4.00 \text{ in}. \quad (5)$$

Distances between other instant centers are also measured from the scaled drawing as

$$R_{I_{24}I_{14}} = -4.24 \text{ in}, \quad R_{I_{25}I_{12}} = -13.41 \text{ in}, \quad R_{I_{25}I_{15}} = -26.29 \text{ in}, \quad \text{and} \quad R_{I_{35}I_{15}} = -3.92 \text{ in}.$$

Substituting these distances into Eq. (3.28), angular velocity ratios can be written as

$$\frac{\omega_4}{\omega_2} = \frac{R_{I_{24}I_{12}}}{R_{I_{24}I_{14}}} = \frac{2.50 \text{ in}}{-4.24 \text{ in}} = -0.590, \quad (6)$$

$$\frac{\omega_5}{\omega_2} = \frac{R_{I_{25}I_{12}}}{R_{I_{25}I_{15}}} = \frac{-13.41 \text{ in}}{-26.29 \text{ in}} = 0.510, \quad (7)$$

$$\frac{\omega_5}{\omega_3} = \frac{R_{I_{35}I_{13}}}{R_{I_{35}I_{15}}} = \frac{4.00 \text{ in}}{-3.92 \text{ in}} = -1.020. \quad (8)$$

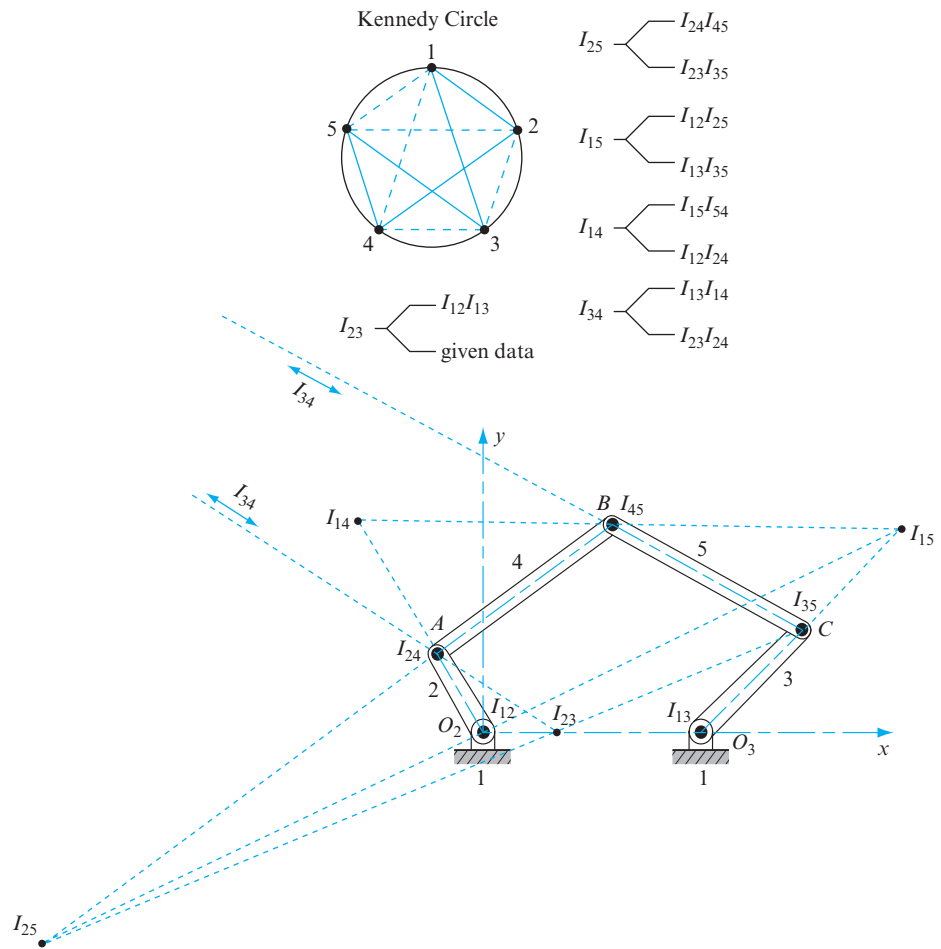


Figure 5.9 Secondary instant centers.

Then, substituting the angular velocity of link 2 into Eqs. (6) and (7), the angular velocities of links 4 and 5 are

$$\omega_4 = -0.590\omega_2 = -0.590(10 \text{ rad/s}) = -5.90 \text{ rad/s (cw)}, \quad \text{Ans.}$$

$$\omega_5 = 0.510\omega_2 = 0.510(10 \text{ rad/s}) = 5.10 \text{ rad/s (ccw)}. \quad \text{Ans.}$$

As a check, we can substitute the angular velocity of link 3 into Eq. (8) to verify that the angular velocity of link 5 is

$$\omega_5 = -1.020\omega_3 = -1.020(-5 \text{ rad/s}) = 5.10 \text{ rad/s (ccw)}. \quad \text{Ans.}$$

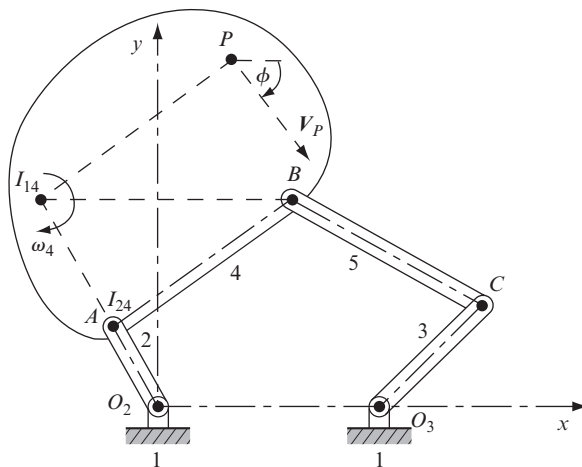


Figure 5.10 Absolute instant center for coupler link 4.

Using the method of instant centers, the magnitude of the velocity of point P can be written as

$$V_P = \omega_4 R_{PI_{14}}. \quad (9)$$

From the scaled drawing (Fig. 5.10), the distance from the absolute instant center of link 4 (I_{14}) to point P is measured as $R_{PI_{14}} = 6.46$ in. Substituting this value into Eq. (9), the magnitude of the velocity of point P is

$$V_P = (|-5.90 \text{ rad/s}|)(6.46 \text{ in}) = 38.11 \text{ in/s.} \quad \text{Ans.}$$

The direction of the velocity of point P is measured from Fig. 5.10 as

$$\phi = -55.5^\circ. \quad \text{Ans.}$$

Note that these results are in good agreement with those given in Example 5.4.

A special case of the five-bar linkage is when instant center I_{23} is stationary on the line of centers $I_{12}I_{13}$; then, links 2 and 3 can be replaced by two circular gears, as shown in Fig. 5.4. This mechanism, referred to as a *geared five-bar mechanism*, has a mobility of one (the input is the rotation of either link 2 or link 3). The two gears—that is, links 2 and 3—are in rolling contact at instant center I_{23} . The ratio of the angular velocity of gear 3 to the angular velocity of gear 2 (also referred to as the *gear ratio*) is the reciprocal of the ratio of Eq. (1).

5.5 FIRST-ORDER KINEMATIC COEFFICIENTS

If we require higher accuracy in solutions for velocities, we look for an analytic or numeric technique. The method of kinematic coefficients, which was presented in Sec. 3.12, provides such a technique.

When dealing with mechanisms having multiple degrees of freedom, however, the method of kinematic coefficients raises a new complication in notation. It requires an additional subscript to keep track of the multiple input motions. For example, the time rate of change of a dependent variable, such as θ_i , the angular velocity of link i , can be written as $\omega_i = d\theta_i/dt$. However, since there are multiple degrees of freedom, the dependent variable θ_i is a function of two or more independent variables, say θ_j and θ_k ; that is, $\theta_i = \theta_i(\theta_j, \theta_k)$. In this case, the time rate of change of the dependent variable θ_i can be written as

$$\omega_i = \frac{d\theta_i}{dt} = \frac{\partial \theta_i}{\partial \theta_j} \frac{d\theta_j}{dt} + \frac{\partial \theta_i}{\partial \theta_k} \frac{d\theta_k}{dt}. \quad (5.1)$$

Similarly, a point, P , on one of the moving links has a position that is a function of both independent variables, $\mathbf{r}_P = \mathbf{r}_P(\theta_j, \theta_k)$. Therefore, the velocity of point P can be written as

$$\mathbf{V}_P = \frac{d\mathbf{r}_P}{dt} = \frac{\partial \mathbf{r}_P}{\partial \theta_j} \frac{d\theta_j}{dt} + \frac{\partial \mathbf{r}_P}{\partial \theta_k} \frac{d\theta_k}{dt}. \quad (5.2)$$

Also, since $\mathbf{r}_P(\theta_j, \theta_k)$ has coordinates such as $\mathbf{r}_P = x_P \hat{\mathbf{i}} + y_P \hat{\mathbf{j}} + z_P \hat{\mathbf{k}}$, then the velocity of point P can be written as

$$\begin{aligned} \mathbf{V}_P &= \left(\frac{\partial x_P}{\partial \theta_j} \hat{\mathbf{i}} + \frac{\partial y_P}{\partial \theta_j} \hat{\mathbf{j}} + \frac{\partial z_P}{\partial \theta_j} \hat{\mathbf{k}} \right) \frac{d\theta_j}{dt} + \left(\frac{\partial x_P}{\partial \theta_k} \hat{\mathbf{i}} + \frac{\partial y_P}{\partial \theta_k} \hat{\mathbf{j}} + \frac{\partial z_P}{\partial \theta_k} \hat{\mathbf{k}} \right) \frac{d\theta_k}{dt} \\ &= \left(\frac{\partial x_P}{\partial \theta_j} \frac{d\theta_j}{dt} + \frac{\partial x_P}{\partial \theta_k} \frac{d\theta_k}{dt} \right) \hat{\mathbf{i}} + \left(\frac{\partial y_P}{\partial \theta_j} \frac{d\theta_j}{dt} + \frac{\partial y_P}{\partial \theta_k} \frac{d\theta_k}{dt} \right) \hat{\mathbf{j}} + \left(\frac{\partial z_P}{\partial \theta_j} \frac{d\theta_j}{dt} + \frac{\partial z_P}{\partial \theta_k} \frac{d\theta_k}{dt} \right) \hat{\mathbf{k}}. \end{aligned} \quad (5.3)$$

If we wish to continue to use the prime notation, where a first-order kinematic coefficient carries the symbol $\theta'_{ij} = \partial \theta_i / \partial \theta_j$ or $\mathbf{r}'_{Pj} = \partial \mathbf{r}_P / \partial \theta_j$, we see that a second subscript is required to indicate with respect to which independent variable the derivative is taken. With this additional subscript, Eq. (5.1) becomes

$$\omega_i = d\theta_i/dt = \theta'_{ij} \omega_j + \theta'_{ik} \omega_k. \quad (5.4)$$

Similarly, Eq. (5.2) becomes

$$\mathbf{V}_P = d\mathbf{r}_P/dt = \mathbf{r}'_{Pj} \omega_j + \mathbf{r}'_{Pk} \omega_k, \quad (5.5)$$

and Eq. (5.3) becomes

$$\mathbf{V}_P = \left(x'_{Pj} \hat{\mathbf{i}} + y'_{Pj} \hat{\mathbf{j}} + z'_{Pj} \hat{\mathbf{k}} \right) \omega_j + \left(x'_{Pk} \hat{\mathbf{i}} + y'_{Pk} \hat{\mathbf{j}} + z'_{Pk} \hat{\mathbf{k}} \right) \omega_k$$

or

$$\mathbf{V}_P = \left(x'_{Pj} \omega_j + x'_{Pk} \omega_k \right) \hat{\mathbf{i}} + \left(y'_{Pj} \omega_j + y'_{Pk} \omega_k \right) \hat{\mathbf{j}} + \left(z'_{Pj} \omega_j + z'_{Pk} \omega_k \right) \hat{\mathbf{k}}. \quad (5.6)$$

Although we find need for this additional subscript, first-order kinematic coefficients do provide insight into the velocity analysis of multi-degree-of-freedom mechanisms,

just as they did in Sec. 3.12 for single-degree-of-freedom mechanisms. This is illustrated in the following example.

EXAMPLE 5.6

Consider the five-bar linkage in the posture of Example 5.1 (Fig. 5.5), and with links 2 and 3 rotating with the same constant angular velocities as in Example 5.5. Using the method of kinematic coefficients, determine the angular velocities of coupler links 4 and 5, and the velocity of coupler point P .

SOLUTION

We begin the analysis by writing the loop-closure equation as we did in Example 5.3, and separating the real and imaginary parts as in Eqs. (3) and (4) of that example. As we demonstrated there, those equations can be solved for the position variables, θ_4 and θ_5 .

Next, we take the partial derivatives of Eqs. (3) and (4) of Example 5.3 with respect to independent variable θ_2 ; that is,

$$-r_2 \sin \theta_2 - r_4 \sin \theta_4 \theta'_{42} + r_5 \sin \theta_5 \theta'_{52} = 0, \quad (1a)$$

$$r_2 \cos \theta_2 + r_4 \cos \theta_4 \theta'_{42} - r_5 \cos \theta_5 \theta'_{52} = 0. \quad (1b)$$

Rearranging these two equations and expressing them in matrix form gives

$$\begin{bmatrix} -r_4 \sin \theta_4 & r_5 \sin \theta_5 \\ r_4 \cos \theta_4 & -r_5 \cos \theta_5 \end{bmatrix} \begin{bmatrix} \theta'_{42} \\ \theta'_{52} \end{bmatrix} = \begin{bmatrix} r_2 \sin \theta_2 \\ -r_2 \cos \theta_2 \end{bmatrix}. \quad (2)$$

Now we can solve this matrix equation for the first-order kinematic coefficients θ'_{42} and θ'_{52} . Using the data obtained so far, the results are $\theta'_{42} = -0.237$ and $\theta'_{52} = 0.456$, which are both dimensionless (rad/rad).

Similarly, taking the partial derivative of Eqs. (3) and (4) of Example 5.3 with respect to independent variable θ_3 , and rearranging the resulting equations into matrix form, gives

$$\begin{bmatrix} -r_4 \sin \theta_4 & r_5 \sin \theta_5 \\ r_4 \cos \theta_4 & -r_5 \cos \theta_5 \end{bmatrix} \begin{bmatrix} \theta'_{43} \\ \theta'_{53} \end{bmatrix} = \begin{bmatrix} -r_3 \sin \theta_3 \\ r_3 \cos \theta_3 \end{bmatrix}. \quad (3)$$

We can solve this matrix equation for the first-order kinematic coefficients θ'_{43} and θ'_{53} . The results are $\theta'_{43} = 0.705$ rad/rad and $\theta'_{53} = -0.109$ rad/rad.

Table 5.1 presents the angular variables, θ_4 and θ_5 , and the first-order kinematic coefficients of the coupler links, for input angles in the range $40^\circ \leq \theta_2 \leq 150^\circ$ and $85^\circ \geq \theta_3 \geq 30^\circ$.

As indicated in Sec. 4.12, it is not a coincidence that the (2×2) coefficient matrices in Eqs. (2) and (3) are identical. This is always true for all sets of derivative equations determined by differentiating loop-closure equations. This coefficient matrix is called the *Jacobian* of the system. In this example, the determinant of the Jacobian is

$$\Delta = r_4 r_5 \sin (\theta_4 - \theta_5). \quad (4)$$

Table 5.1 Input Angles, Coupler Angles, and First-Order Kinematic Coefficients

θ_2 deg	θ_3 deg	θ_4 deg	θ_5 deg	θ'_{42} —	θ'_{43} —	θ'_{52} —	θ'_{53} —
40	85	93.419	142.993	−0.533	0.743	−0.440	0.128
50	80	84.547	138.843	−0.513	0.703	−0.291	0.065
60	75	76.200	136.526	−0.466	0.674	−0.134	0.016
70	70	68.479	135.944	−0.412	0.659	0.012	−0.019
80	65	61.343	136.872	−0.360	0.654	0.138	−0.044
90	60	54.690	139.054	−0.316	0.658	0.242	−0.062
100	55	48.401	142.263	−0.281	0.667	0.327	−0.077
110	50	42.360	146.315	−0.254	0.683	0.397	−0.091
120	45	36.447	151.084	−0.237	0.705	0.456	−0.109
130	40	30.526	156.510	−0.230	0.737	0.508	−0.136
140	35	24.383	162.646	−0.241	0.793	0.564	−0.185
150	30	17.496	169.884	−0.306	0.927	0.663	−0.311

At the current posture, this determinant has a value of $\Delta = -32.72 \text{ in}^2$ and does not cause numeric difficulty. However, we note that this determinant becomes zero when $\theta_5 = \theta_4$ or when $\theta_5 = \theta_4 \pm 180^\circ$. This occurs when the coupler links are either fully extended or folded on top of each other—that is, when the five-bar linkage is in the posture of a quadrilateral with the two coupler links aligned. The kinematic coefficients become indeterminate in such a posture.

The angular velocities of links 4 and 5 can be determined from Eq. (5.4). At the posture shown in Fig. 5.1, the values are

$$\omega_4 = \theta'_{42}\omega_2 + \theta'_{43}\omega_3 = -0.237(10.0 \text{ rad/s}) + 0.705(-5.0 \text{ rad/s}) = -5.90 \text{ rad/s (cw)}, \quad \text{Ans.}$$

$$\omega_5 = \theta'_{52}\omega_2 + \theta'_{53}\omega_3 = 0.456(10.0 \text{ rad/s}) - 0.109(-5.0 \text{ rad/s}) = 5.10 \text{ rad/s (ccw)}. \quad \text{Ans.}$$

These confirm the results found in both Example 5.4 and Example 5.5.

Differentiating constraint Eq. (6) of Example 5.3 with respect to the two independent inputs, the first-order kinematic coefficients of the orientation of vector $\mathbf{r}_7$ are

$$\theta'_{72} = \theta'_{42} \text{ and } \theta'_{73} = \theta'_{43}. \quad (5)$$

Partially differentiating Eqs. (7) and (8) of Example 5.3 with respect to input variable θ_2 and using Eq. (5), the first-order kinematic coefficients of point P are

$$x'_{P2} = -r_2 \sin \theta_2 - r_7 \sin \theta_7 \theta'_{42}, \quad (6)$$

$$y'_{P2} = r_2 \cos \theta_2 + r_7 \cos \theta_7 \theta'_{42}. \quad (7)$$

Similarly, partially differentiating Eqs. (7) and (8) of Example 5.3 with respect to the input variable θ_3 and using Eq. (5), the first-order kinematic coefficients of point P are

$$x'_{P3} = -r_7 \sin \theta_7 \theta'_{43}, \quad (8)$$

$$y'_{P3} = r_7 \cos \theta_7 \theta'_{43}. \quad (9)$$

Substituting the specified data into Eqs. (6) through (9), the first-order kinematic coefficients of point P are

$$x'_{P2} = -0.4294 \text{ in/rad},$$

$$y'_{P2} = -2.0066 \text{ in/rad},$$

$$x'_{P3} = -5.1681 \text{ in/rad},$$

$$y'_{P3} = 2.2528 \text{ in/rad}.$$

and

Finally, substituting these values and the given angular velocities into Eq. (5.6), the velocity of point P is

$$\begin{aligned} \mathbf{V}_P = & [(-0.4294 \text{ in})(10 \text{ rad/s}) + (-5.1681 \text{ in})(-5 \text{ rad/s})]\hat{\mathbf{i}} \\ & + [(-2.0066 \text{ in})(10 \text{ rad/s}) + (2.2528 \text{ in})(-5 \text{ rad/s})]\hat{\mathbf{j}}. \end{aligned}$$

That is,

$$\mathbf{V}_P = 21.546 \text{ in/s } \hat{\mathbf{i}} - 31.330 \text{ in/s } \hat{\mathbf{j}} = 38.024 \text{ in/s } \angle -55.483^\circ. \quad \text{Ans.}$$

The velocity of point P is shown in Fig. 5.11. Note that these results are in good agreement with, and have higher accuracy than, those obtained in Example 5.4 and Example 5.5.

Table 5.2 indicates the first-order kinematic coefficients of point P and the magnitude and direction of the velocity of point P for the range of input angles used in Table 5.1.

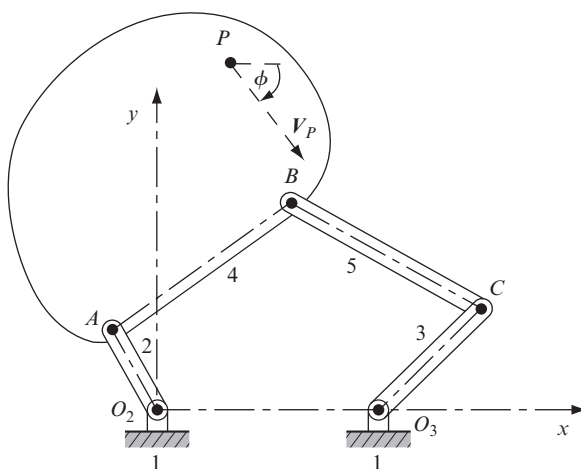


Figure 5.11 Velocity of coupler point P .

Table 5.2 First-order Kinematic Coefficients and Velocities of Coupler Point P

θ_2 deg	θ_3 deg	x'_{P2} in	y'_{P2} in	x'_{P3} in	y'_{P3} in	V_P in/s	ϕ deg
40	85	1.9543	4.2650	-4.9588	-3.2721	73.811	53.081
50	80	1.8180	3.3120	-5.1124	-2.3350	62.609	45.681
60	75	1.4177	2.2909	-5.1816	-1.5054	50.330	37.209
70	70	0.9102	1.3410	-5.2151	-0.7774	39.200	26.183
80	65	0.4201	0.5017	-5.2332	-0.1227	30.885	10.503
90	60	0.0191	-0.2342	-5.2391	0.4870	26.816	-10.260
100	55	-0.2610	-0.8859	-5.2303	1.0735	27.506	-31.146
110	50	-0.4107	-1.4715	-5.2054	1.6553	31.766	-46.367
120	45	-0.4294	-2.0066	-5.1681	2.2528	38.024	-55.483
130	40	-0.3144	-2.5116	-5.1347	2.9020	45.583	-60.380
140	35	-0.0397	-3.0379	-5.1568	3.6942	55.053	-62.539
150	30	0.5534	-3.8178	-5.4662	5.0096	71.258	-62.535

5.6 METHOD OF SUPERPOSITION

Equations (5.4), (5.5), and (5.6) suggest that, for mechanisms with more than one degree of freedom, another method of solution for velocities is to solve the problem multiple times, with all except one of the inputs considered inactive (frozen or locked) during each solution, and then to sum the results to find the total solution with all inputs active. This method is called the *method of superposition*, and it is valid for velocity analysis, since the velocity equations are linear—that is, since all dependent velocities are linear combinations of the input velocities. The procedure may be best understood through an example.

EXAMPLE 5.7

Let us again solve the velocity analysis of Example 5.3, this time using the method of superposition. There are two cases that must be considered: case (a), where link 3 is temporarily considered frozen whereas link 2 has an angular velocity of $\omega_2 = 10$ rad/s ccw, and case (b), where link 2 is temporarily considered frozen whereas link 3 has an angular velocity of $\omega_3 = 5$ rad/s cw.

SOLUTION

The angular velocity of link 4 can be written as a linear combination of case (a) and case (b). Using the angular velocity ratio theorem, this gives

$$\omega_4 = \left(\frac{R_{I_{24}^2 I_{12}^2}}{R_{I_{24}^2 I_{14}^2}} \right) \omega_2 + \left(\frac{R_{I_{34}^3 I_{13}^3}}{R_{I_{34}^3 I_{14}^3}} \right) \omega_3, \quad (1)$$

where the superscripts in the instant center labels denote the input variable that is moving whereas all others are temporarily considered frozen. Similarly, the angular velocity of link 5 can be written as

$$\omega_5 = \left(\frac{R_{I_{25}^2 I_{12}^2}}{R_{I_{25}^2 I_{15}^2}} \right) \omega_2 + \left(\frac{R_{I_{35}^3 I_{13}^3}}{R_{I_{35}^3 I_{15}^3}} \right) \omega_3. \quad (2)$$

Comparing Eqs. (1) and (2) with Eqs. (5.4), the first-order kinematic coefficients of links 4 and 5 can be written as

$$\theta'_{42} = \frac{R_{I_{24}^2 I_{12}^2}}{R_{I_{24}^2 I_{14}^2}}, \quad \theta'_{43} = \frac{R_{I_{34}^3 I_{13}^3}}{R_{I_{34}^3 I_{14}^3}}, \quad \theta'_{52} = \frac{R_{I_{25}^2 I_{12}^2}}{R_{I_{25}^2 I_{15}^2}}, \quad \text{and} \quad \theta'_{53} = \frac{R_{I_{35}^3 I_{13}^3}}{R_{I_{35}^3 I_{15}^3}}. \quad (3)$$

Note that a sign convention is not necessary if directed lines are used when measuring the relative locations of instant centers. However, if a sign convention is preferred, then each first-order kinematic coefficient is negative if the relative instant center I_{ij}^k is between the absolute instant centers I_{1i}^k and I_{1j}^k and positive if the relative instant center I_{ij}^k is outside the absolute instant centers I_{1i}^k and I_{1j}^k .

Case (a): Link 3 is temporarily considered frozen. The linkage is temporarily regarded as a four-bar linkage, as shown in Fig. 5.12, with input from the rotation of link 2 alone.

From the Aronhold-Kennedy theorem, the secondary instant center I_{14}^2 is the point of intersection of the line containing instant centers I_{15}^2 and I_{45}^2 , and the line containing instant centers I_{12}^2 and I_{24}^2 . Similarly, the secondary instant center I_{25}^2 is the point of intersection of the line containing instant centers I_{24}^2 and I_{45}^2 , and the line containing instant centers I_{12}^2 and I_{15}^2 .

Using the convention that lines proceed positive to the right, the distances between the instant centers, measured from the scaled drawing of Fig. 5.12, are

$$R_{I_{24}^2 I_{12}^2} = -2.50 \text{ in}, \quad R_{I_{24}^2 I_{14}^2} = 10.56 \text{ in}, \quad R_{I_{25}^2 I_{12}^2} = -7.76 \text{ in}, \quad \text{and} \quad R_{I_{25}^2 I_{15}^2} = -17.03 \text{ in}. \quad (4)$$

Case (b): Link 2 is temporarily considered frozen. The linkage is temporarily regarded as a four-bar linkage, as shown in Fig. 5.13, with input from the rotation of link 3 alone.

From the Aronhold-Kennedy theorem, the secondary instant center I_{34}^3 is the point of intersection of the line containing instant centers I_{13}^3 and I_{14}^3 , and the line containing instant centers I_{35}^3 and I_{45}^3 . Similarly, the secondary instant center I_{15}^3 is the point of intersection of the line containing instant centers I_{14}^3 and I_{45}^3 , and the line containing instant centers I_{13}^3 and I_{35}^3 .

Continuing with the convention that lines proceed positive to the right, the distances between the instant centers, measured from the scaled drawing of Fig. 5.13, are

$$R_{I_{34}^3 I_{13}^3} = 18.06 \text{ in}, \quad R_{I_{34}^3 I_{14}^3} = 25.62 \text{ in}, \quad R_{I_{35}^3 I_{13}^3} = 4.00 \text{ in}, \quad \text{and} \quad R_{I_{35}^3 I_{15}^3} = -36.67 \text{ in}. \quad (5)$$

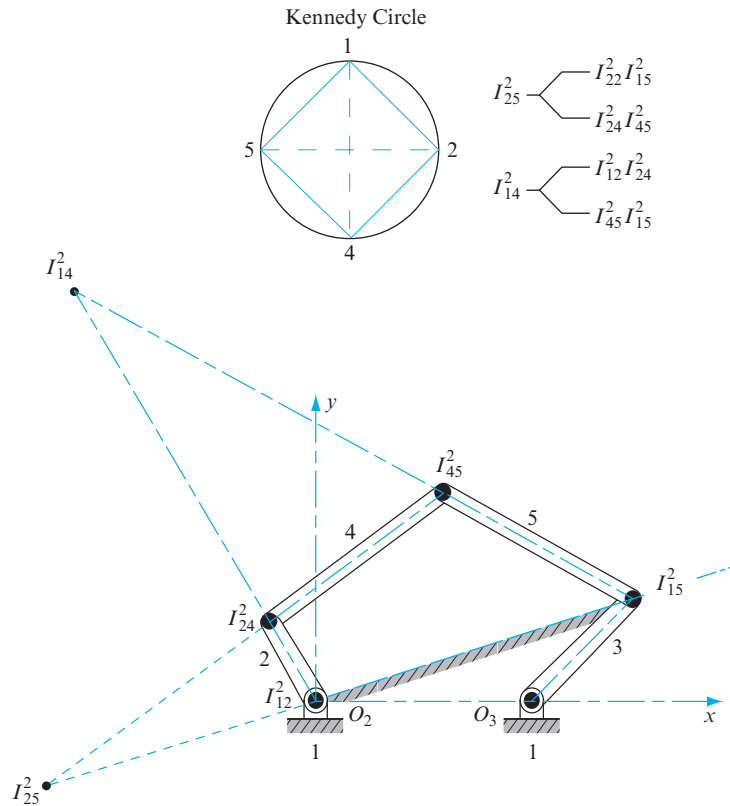


Figure 5.12 Case (a): Four-bar linkage with link 3 temporarily considered frozen.

Substituting Eqs. (4) and (5) into Eqs. (3), the first-order kinematic coefficients of links 4 and 5 are

$$\theta'_{42} = \frac{R_{I_{24}^2 I_{12}^2}}{R_{I_{24}^2 I_{14}^2}} = \frac{-2.50 \text{ in}}{10.56 \text{ in}} = -0.237, \quad \theta'_{43} = \frac{R_{I_{34}^3 I_{13}^3}}{R_{I_{34}^3 I_{14}^3}} = \frac{18.06 \text{ in}}{25.62 \text{ in}} = 0.705, \quad (6a)$$

$$\theta'_{52} = \frac{R_{I_{25}^2 I_{12}^2}}{R_{I_{25}^2 I_{15}^2}} = \frac{-7.76 \text{ in}}{-17.03 \text{ in}} = 0.455, \quad \theta'_{53} = \frac{R_{I_{35}^3 I_{13}^3}}{R_{I_{35}^3 I_{15}^3}} = \frac{4.00 \text{ in}}{-36.67 \text{ in}} = -0.109. \quad (6b)$$

Note that the values of the first-order kinematic coefficients of links 4 and 5, given by Eqs. (6), are all nondimensional (rad/rad) and are all in good agreement with the values found analytically in Example 5.6.

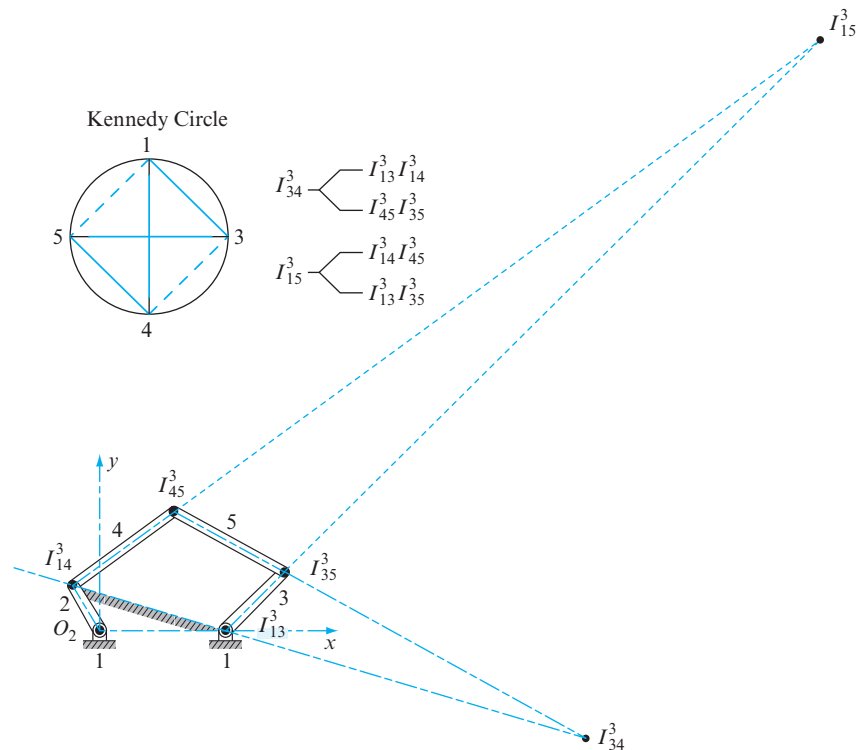


Figure 5.13 Case (b): Four-bar linkage with link 2 temporarily considered frozen.

Substituting Eqs. (6) into Eq. (5.4), the angular velocities of links 4 and 5, respectively, are

$$\omega_4 = -0.237(10 \text{ rad/s}) + 0.705(-5 \text{ rad/s}) = -5.90 \text{ rad/s (cw)}, \quad \text{Ans.}$$

$$\omega_5 = 0.455(10 \text{ rad/s}) - 0.109(-5 \text{ rad/s}) = 5.10 \text{ rad/s ccw}. \quad \text{Ans.}$$

Note that the answers for the angular velocities of coupler links 4 and 5 from the graphic methods (Example 5.4 and Example 5.5) are in good agreement with the answers obtained from the analytic methods (Example 5.6 and Example 5.7).

5.7 ACCELERATION ANALYSIS; ACCELERATION POLYGONS

If the accelerations of each of the independent inputs are known, then an acceleration analysis can be performed by the acceleration polygon method of Secs. 4.3 through 4.8. An example is presented here to demonstrate how the graphic method for single-degree-of-freedom mechanisms can be extended to multi-degree-of-freedom mechanisms.

EXAMPLE 5.8

For the five-bar linkage of Example 5.1 in the posture shown in Fig. 5.5, links 2 and 3 are rotating with constant angular velocities of $\omega_2 = 10 \text{ rad/s}$ ccw and $\omega_3 = 5 \text{ rad/s}$ cw, respectively. Determine the angular accelerations of coupler links 4 and 5.

SOLUTION

The velocities are determined, using the same input data, in the velocity polygon of Example 5.4, which is repeated here as Fig. 5.14*b*. Since the angular velocities of links 2 and 3 are constant, we have $\alpha_2 = \alpha_3 = 0$, and there remain only the normal components of acceleration for points *A* and *C*. Hence,

$$A_A = A_{AO_2}^n = \omega_2^2 R_{AO_2} = (10 \text{ rad/s})^2 (2.5 \text{ in}) = 250 \text{ in/s}^2,$$

$$A_C = A_{CO_3}^n = \omega_3^2 R_{CO_3} = (5 \text{ rad/s})^2 (4 \text{ in}) = 100 \text{ in/s}^2.$$

Next, two acceleration-difference equations are written to relate the acceleration of point *B* to those of points *A* and *C*; that is,

$$A_B = \overset{\vee}{A}_A + \overset{\vee}{A}_{BA}^n + \overset{?}{A}_{BA}^t = \overset{\vee}{A}_C + \overset{\vee}{A}_{BC}^n + \overset{?}{A}_{BC}^t. \quad (1)$$

Using results from the velocity polygon (Fig. 5.7*b*), the magnitudes of the two normal components in Eq. (1) are

$$A_{BA}^n = \frac{V_{BA}^2}{R_{BA}} = \frac{(35.34 \text{ in/s})^2}{6 \text{ in}} = 208.15 \text{ in/s}^2$$

and

$$A_{BC}^n = \frac{V_{BC}^2}{R_{BC}} = \frac{(30.60 \text{ in/s})^2}{6 \text{ in}} = 156.06 \text{ in/s}^2.$$

With these, the terms of Eq. (1) can be constructed graphically as shown in Fig. 5.14.

The angular accelerations of links 4 and 5 can now be determined from the two tangential components scaled from the acceleration polygon; namely,

$$\alpha_4 = \frac{A_{BA}^t}{R_{BA}} = \frac{1.2 \text{ in/s}^2}{6 \text{ in}} = 0.2 \text{ rad/s}^2 \text{ cw}, \quad \text{Ans.}$$

$$\alpha_5 = \frac{A_{BC}^t}{R_{BC}} = \frac{222.7 \text{ in/s}^2}{6 \text{ in}} = 37.1 \text{ rad/s}^2 \text{ ccw}. \quad \text{Ans.}$$

Once the accelerations of points *A* and *B* are known, we can construct the acceleration image of coupler link 4 as explained in Sec. 4.4, and obtain the acceleration of coupler point *P*; that is,

$$\mathbf{A}_P = 472 \text{ in/s}^2 \angle -88.2^\circ. \quad \text{Ans.}$$

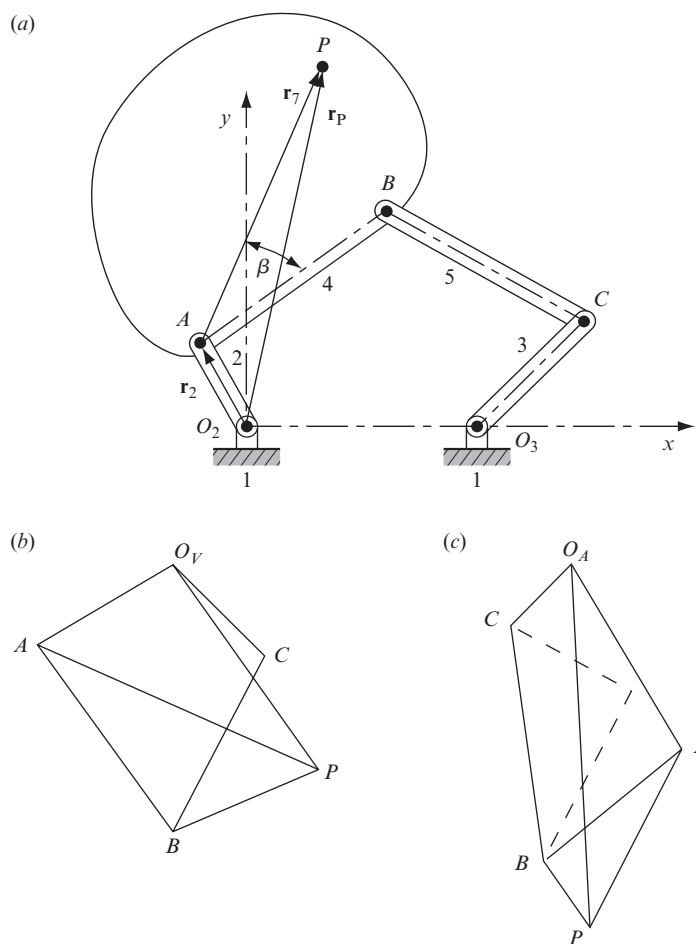


Figure 5.14 (a) Linkage posture; (b) velocity polygon; (c) acceleration polygon.

5.8 SECOND-ORDER KINEMATIC COEFFICIENTS

For higher accuracy in the solution of acceleration problems, we look for an analytic or numeric technique. The method of kinematic coefficients, which was presented in Sec. 4.12, provides such a technique.

As with first-order kinematic coefficients, we use subscripts to signify each independent variable when we have multiple degrees of freedom. For example, when we have the second-order kinematic coefficient of the dependent variable θ_i with respect to the two independent input variables θ_j and θ_k , we will use the notation

$$\theta''_{ijk} = \frac{\partial^2 \theta_i}{\partial \theta_j \partial \theta_k}. \quad (5.7)$$

Using this notation, the derivative of Eq. (5.4) with respect to time (that is, the angular acceleration of link i) can be written as

$$\alpha_i = \frac{d^2\theta_i}{dt^2} = \frac{d\omega_i}{dt} = \theta''_{ij}\omega_j^2 + 2\theta''_{ijk}\omega_j\omega_k + \theta''_{ikk}\omega_k^2 + \theta'_{ij}\alpha_j + \theta'_{ik}\alpha_k. \quad (5.8)$$

Differentiating Eq. (5.6) with respect to time, the acceleration of point P can be written as

$$\begin{aligned} \mathbf{A}_P = & \left(x''_{Pij}\omega_j^2 + 2x''_{Pjk}\omega_j\omega_k + x''_{Pkk}\omega_k^2 + x'_{Pj}\alpha_j + x'_{Pk}\alpha_k \right) \hat{\mathbf{i}} \\ & + \left(y''_{Pij}\omega_j^2 + 2y''_{Pjk}\omega_j\omega_k + y''_{Pkk}\omega_k^2 + y'_{Pj}\alpha_j + y'_{Pk}\alpha_k \right) \hat{\mathbf{j}}. \end{aligned} \quad (5.9)$$

The angle for the direction of the acceleration of point P can be written as

$$\psi = \tan^{-1} \left(\frac{A_P^y}{A_P^x} \right). \quad (5.10)$$

Substituting the components of Eq. (5.9) into this equation, the direction of $\mathbf{A}_P$ can be written as

$$\psi = \tan^{-1} \left(\frac{y''_{Pij}\omega_j^2 + 2y''_{Pjk}\omega_j\omega_k + y''_{Pkk}\omega_k^2 + y'_{Pj}\alpha_j + y'_{Pk}\alpha_k}{x''_{Pij}\omega_j^2 + 2x''_{Pjk}\omega_j\omega_k + x''_{Pkk}\omega_k^2 + x'_{Pj}\alpha_j + x'_{Pk}\alpha_k} \right). \quad (5.11)$$

For the special case where input velocities ω_j and ω_k are both constant—that is, where the input accelerations are $\alpha_j = 0$ and $\alpha_k = 0$ —Eq. (5.11) can be written as

$$\psi = \tan^{-1} \left(\frac{y''_{Pij}\omega_j^2 + 2y''_{Pjk}\omega_j\omega_k + y''_{Pkk}\omega_k^2}{x''_{Pij}\omega_j^2 + 2x''_{Pjk}\omega_j\omega_k + x''_{Pkk}\omega_k^2} \right). \quad (5.12)$$

Even with these additional subscripts, second-order kinematic coefficients can be used for acceleration analysis of multi-degree-of-freedom mechanisms, almost exactly as they were in Sec. 4.12 with a single degree of freedom. Let us demonstrate this by again solving Example 5.8 using second-order kinematic coefficients.

EXAMPLE 5.9

For the five-bar linkage of Example 5.1 in the posture shown in Fig. 5.5, links 2 and 3 are rotating with constant angular velocities of $\omega_2 = 10$ rad/s ccw and $\omega_3 = 5$ rad/s cw,

respectively. Determine the angular accelerations of coupler links 4 and 5, and the acceleration of coupler point P .

SOLUTION

Taking the partial derivative of Eqs. (1) of Example 5.6 with respect to input angle θ_2 , and writing this in matrix form, gives

$$\begin{bmatrix} -r_4 \sin \theta_4 & r_5 \sin \theta_5 \\ r_4 \cos \theta_4 & -r_5 \cos \theta_5 \end{bmatrix} \begin{bmatrix} \theta''_{422} \\ \theta''_{522} \end{bmatrix} = \begin{bmatrix} r_2 \cos \theta_2 + r_4 \cos \theta_4 \theta'_2 - r_5 \cos \theta_5 \theta'^2_{52} \\ r_2 \sin \theta_2 + r_4 \sin \theta_4 \theta'_2 - r_5 \sin \theta_5 \theta'^2_{52} \end{bmatrix}. \quad (1)$$

We can solve this matrix equation for the second-order kinematic coefficients θ''_{422} and θ''_{522} . Using the data obtained so far, the results are $\theta''_{422} = 0.139 \text{ rad/rad}^2$ and $\theta''_{522} = 0.208 \text{ rad/rad}^2$.

We also partially differentiate Eqs. (1) of Example 5.6 with respect to input angle θ_3 ; writing this in matrix form, gives

$$\begin{bmatrix} -r_4 \sin \theta_4 & r_5 \sin \theta_5 \\ r_4 \cos \theta_4 & -r_5 \cos \theta_5 \end{bmatrix} \begin{bmatrix} \theta''_{423} \\ \theta''_{523} \end{bmatrix} = \begin{bmatrix} r_4 \cos \theta_4 \theta'_{42} \theta'_{43} - r_5 \cos \theta_5 \theta'_{52} \theta'_{53} \\ r_4 \sin \theta_4 \theta'_{42} \theta'_{43} - r_5 \sin \theta_5 \theta'_{52} \theta'_{53} \end{bmatrix}. \quad (2)$$

Then, solving this matrix equation gives the second-order kinematic coefficients θ''_{423} and θ''_{523} . Note that taking the partial derivative of Eqs. (3) of Example 5.6 with respect to input angle θ_2 gives the same result. For the current geometry and posture of the linkage, the results are $\theta''_{423} = 0.131 \text{ rad/rad}^2$ and $\theta''_{523} = -0.206 \text{ rad/rad}^2$.

Next, we partially differentiate Eqs. (3) of Example 5.6 with respect to input angle θ_3 , which gives

$$\begin{bmatrix} -r_4 \sin \theta_4 & r_5 \sin \theta_5 \\ r_4 \cos \theta_4 & -r_5 \cos \theta_5 \end{bmatrix} \begin{bmatrix} \theta''_{433} \\ \theta''_{533} \end{bmatrix} = \begin{bmatrix} -r_3 \cos \theta_3 + r_4 \cos \theta_4 \theta'^2_{43} - r_5 \cos \theta_5 \theta'^2_{53} \\ -r_3 \sin \theta_3 + r_4 \sin \theta_4 \theta'^2_{43} - r_5 \sin \theta_5 \theta'^2_{53} \end{bmatrix}. \quad (3)$$

Then, we can solve this matrix equation for the second-order kinematic coefficients θ''_{433} and θ''_{533} . For the current geometry and posture of the linkage, the results are $\theta''_{433} = -0.038 \text{ rad/rad}^2$ and $\theta''_{533} = -0.173 \text{ rad/rad}^2$.

Table 5.3 presents the second-order kinematic coefficients of the two coupler links for the same range of input angles given in Table 5.1.

Substituting the numeric values of the first-order and second-order kinematic coefficients into Eqs. (5.8), the angular accelerations of the coupler links (for the posture shown in Fig. 5.5) are

$$\alpha_4 = -0.150 \text{ rad/s}^2 \text{ (cw)} \quad \text{and} \quad \alpha_5 = 37.075 \text{ rad/s}^2 \text{ (ccw)}. \quad \text{Ans.}$$

Differentiating Eq. (5) of Example 5.6 with respect to the two independent inputs, the second-order kinematic coefficients of vector $\mathbf{r}_7$ are

$$\theta''_{722} = \theta''_{422}, \quad \theta''_{723} = \theta''_{423}, \quad \text{and} \quad \theta''_{733} = \theta''_{433}. \quad (4)$$

Table 5.3 Second-Order Kinematic Coefficients of Coupler Links

θ_2 deg	θ_3 deg	θ''_{422} —	θ''_{423} —	θ''_{433} —	θ''_{522} —	θ''_{523} —	θ''_{533} —
40	85	−0.135	−0.263	−0.016	0.535	−0.472	−0.156
50	80	0.095	−0.236	−0.075	0.686	−0.430	−0.214
60	75	0.215	−0.177	−0.107	0.701	−0.361	−0.244
70	70	0.254	−0.112	−0.114	0.635	−0.294	−0.251
80	65	0.249	−0.055	−0.106	0.537	−0.242	−0.245
90	60	0.225	−0.005	−0.088	0.436	−0.208	−0.233
100	55	0.196	0.038	−0.068	0.346	−0.190	−0.217
110	50	0.167	0.081	−0.049	0.269	−0.188	−0.198
120	45	0.139	0.131	−0.038	0.208	−0.206	−0.173
130	40	0.104	0.208	−0.050	0.168	−0.259	−0.128
140	35	0.034	0.371	−0.144	0.174	−0.404	−0.002
150	30	−0.281	0.987	−0.752	0.434	−1.006	0.635

Differentiating Eqs. (6) and (7) of Example 5.6 with respect to input angle θ_2 , and using Eq. (5) of Example 5.6 and Eq. (4) of this example, the second-order kinematic coefficients of point P are

$$x''_{P22} = -r_2 \cos \theta_2 - r_7 \cos \theta_7 \theta_2'^2 - r_7 \sin \theta_7 \theta_2'', \quad (5)$$

$$y''_{P22} = -r_2 \sin \theta_2 - r_7 \sin \theta_7 \theta_2'^2 + r_7 \cos \theta_7 \theta_2''. \quad (6)$$

Similarly, differentiating Eqs. (6) and (7) of Example 5.6 with respect to input angle θ_3 , and using Eq. (5) of Example 5.6 and Eq. (4) of this example, the second-order kinematic coefficients of point P are

$$x''_{P23} = -r_7 \cos \theta_7 \theta_2' \theta_3' - r_7 \sin \theta_7 \theta_2'' \theta_3', \quad (7)$$

$$y''_{P23} = -r_7 \sin \theta_7 \theta_2' \theta_3' + r_7 \cos \theta_7 \theta_2'' \theta_3', \quad (8)$$

$$x''_{P33} = -r_7 \cos \theta_7 \theta_3'^2 - r_7 \sin \theta_7 \theta_3'', \quad (9)$$

$$y''_{P33} = -r_7 \sin \theta_7 \theta_3'^2 + r_7 \cos \theta_7 \theta_3''. \quad (10)$$

Substituting the results of Eqs. (1) and (3) of Example 5.6 and Eqs. (1) to (3) into Eqs. (5) to (10), the second-order kinematic coefficients of point P are

$$x''_{P22} = -2.5 \cos 120^\circ - 8 \cos 66.447^\circ (-0.237)^2 - 8 \sin 66.447^\circ (+0.139) = +0.054 \text{ in/rad}^2,$$

$$y''_{P22} = -2.5 \sin 120^\circ - 8 \sin 66.447^\circ (-0.237)^2 + 8 \cos 66.447^\circ (+0.139) = -2.133 \text{ in/rad}^2,$$

$$x''_{P23} = -8 \cos 66.447^\circ (-0.237) (+0.705) - 8 \sin 66.447^\circ (0.131) = -0.429 \text{ in/rad}^2,$$

$$y''_{P23} = -8 \sin 66.447^\circ (-0.237) (+0.705) + 8 \cos 66.447^\circ (0.131) = +1.642 \text{ in/rad}^2,$$

$$x''_{P33} = -8 \cos 66.447^\circ (+0.705)^2 - 8 \sin 66.447^\circ (-0.038) = -1.311 \text{ in/rad}^2,$$

$$y''_{P33} = -8 \sin 66.447^\circ (+0.705)^2 + 8 \cos 66.447^\circ (-0.038) = -3.762 \text{ in/rad}^2.$$

Then, substituting these results and the specified input angular velocities and angular accelerations into Eq. (5.9), the acceleration of point P is

$$\mathbf{A}_P = 15.51\hat{\mathbf{i}} - 471.57\hat{\mathbf{j}} \text{ in/s}^2.$$

Therefore, the magnitude and the direction of the acceleration of point P , respectively, are

$$A_P = \sqrt{(15.51 \text{ in/s}^2)^2 + (-471.57 \text{ in/s}^2)^2} = 471.82 \text{ in/s}^2$$

and

$$\psi = \tan^{-1} \left(\frac{-471.57 \text{ in/s}^2}{15.51 \text{ in/s}^2} \right) = -88.117^\circ.$$

The direction of the acceleration of point P is shown in Fig. 5.15.

Table 5.4 presents the second-order kinematic coefficients of point P and the magnitude and direction of the acceleration of point P for the same range of input angles given in Table 5.1.

Note that the answers here are in good agreement with, and have higher accuracy than, the answers obtained from the acceleration polygon (Example 5.8).

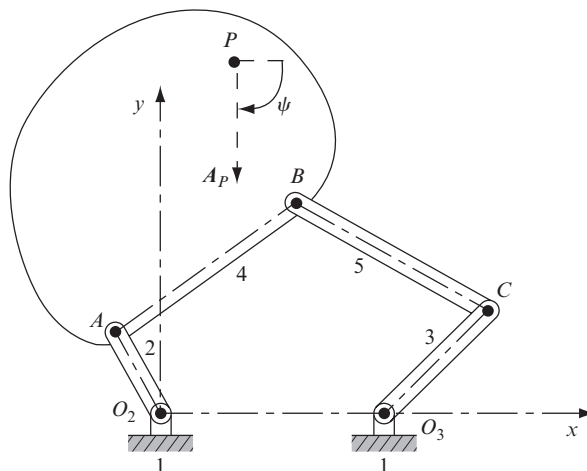


Figure 5.15 Acceleration of coupler point P .

Table 5.4 Second-Order Kinematic Coefficients and Acceleration of Coupler Point P

θ_2 deg	θ_3 deg	x''_{P22} in.	y''_{P22} in.	x''_{P23} in.	y''_{P23} in.	x''_{P33} in.	y''_{P33} in.	A_P in./s ²	ψ deg
40	85	0.236	-2.914	0.013	3.805	2.537	-3.612	766.98	-83.583
50	80	-1.425	-4.147	0.517	3.406	2.188	-3.342	850.35	-99.446
60	75	-2.417	-4.316	0.656	2.811	1.837	-3.256	836.00	-108.217
70	70	-2.666	-3.992	0.569	2.281	1.417	-3.302	766.03	-112.089
80	65	-2.402	-3.547	0.393	1.896	0.926	-3.404	679.65	-112.157
90	60	-1.870	-3.130	0.197	1.653	0.384	-3.511	599.38	-109.199
100	55	-1.232	-2.764	0.005	1.530	-0.184	-3.600	535.02	-103.873
110	50	-0.578	-2.436	-0.193	1.519	-0.758	-3.673	490.69	-96.725
120	45	0.054	-2.133	-0.429	1.642	-1.311	-3.762	471.82	-88.117
130	40	0.677	-1.875	-0.783	1.999	-1.794	-3.981	497.37	-78.265
140	35	1.423	-1.826	-1.520	2.969	-1.991	-4.761	646.57	-67.780
150	30	3.318	-3.321	-4.290	7.007	-0.210	-9.128	1470.05	-59.072

The following example shows how the method of kinematic coefficients can be applied to the kinematic analysis of a two-degrees-of-freedom mechanism with rolling contact.

EXAMPLE 5.10

For the mechanism in the posture shown in Fig. 5.16, the planet gear, j , is in rolling contact with the ring gear, h , and the sun gear, k . Link i (the arm) has constant angular velocity $\omega_i = 10$ rad/s ccw, and gear h has angular velocity $\omega_h = 5$ rad/s cw and angular acceleration $\alpha_h = 15$ rad/s² ccw. Determine: (a) the first- and second-order kinematic coefficients of gears j and k ; and (b) the angular velocity and acceleration of gears j and k .

SOLUTION

For the rolling contact between planet gear j and ring gear h , the constraint equation can be written in the form

$$\rho_j (\Delta\theta_j - \Delta\theta_i) = +\rho_h (\Delta\theta_h - \Delta\theta_i). \quad (1)$$

Similarly, for the rolling contact between planet gear j and sun gear k , the constraint equation can be written as

$$\rho_j (\Delta\theta_j - \Delta\theta_i) = -\rho_k (\Delta\theta_k - \Delta\theta_i). \quad (2)$$

These two equations can be put into matrix form, with the independent variables on the right, as follows

$$\begin{bmatrix} \rho_j & 0 \\ \rho_j & \rho_k \end{bmatrix} \begin{bmatrix} \Delta\theta_j \\ \Delta\theta_k \end{bmatrix} = \begin{bmatrix} \rho_h & \rho_h - \rho_j \\ 0 & \rho_j + \rho_k \end{bmatrix} \begin{bmatrix} \Delta\theta_h \\ \Delta\theta_i \end{bmatrix}. \quad (3)$$

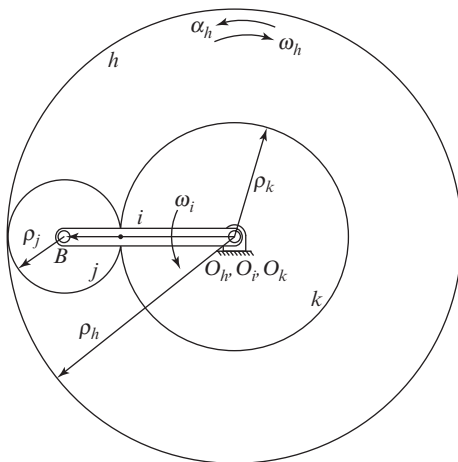


Figure 5.16 $\rho_h = 8$ in, $\rho_j = 2$ in, and $\rho_k = 4$ in.

- (a) Taking partial derivatives of Eqs. (3) with respect to the first independent variable, we find equations for the following first-order kinematic coefficients:

$$\begin{bmatrix} \rho_j & 0 \\ \rho_j & \rho_k \end{bmatrix} \begin{bmatrix} \theta'_{jh} \\ \theta'_{kh} \end{bmatrix} = \begin{bmatrix} \rho_h & \rho_h - \rho_j \\ 0 & \rho_j + \rho_k \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \rho_h \\ 0 \end{bmatrix}. \quad (4)$$

The determinant of the Jacobian matrix is

$$\Delta = \rho_j \rho_k = 8 \text{ in}^2,$$

and Cramer's rule gives the solution to Eqs. (4) as

$$\theta'_{jh} = \rho_h / \rho_j = 4.0 \text{ rad/rad} \quad \text{and} \quad \theta'_{kh} = -\rho_h / \rho_k = -2.0 \text{ rad/rad}. \quad \text{Ans. (5)}$$

Taking partial derivatives of Eqs. (3) with respect to the other independent variable, we find equations for the remaining first-order kinematic coefficients:

$$\begin{bmatrix} \rho_j & 0 \\ \rho_j & \rho_k \end{bmatrix} \begin{bmatrix} \theta'_{ji} \\ \theta'_{ki} \end{bmatrix} = \begin{bmatrix} \rho_h & \rho_h - \rho_j \\ 0 & \rho_j + \rho_k \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \rho_h - \rho_j \\ \rho_j + \rho_k \end{bmatrix}. \quad (6)$$

Cramer's rule gives the solutions to Eqs. (6) as

$$\theta'_{ji} = (\rho_h - \rho_j) / \rho_j = 3.0 \text{ rad/rad} \quad \text{and} \quad \theta'_{ki} = (-\rho_h + 2\rho_j + \rho_k) / \rho_k = 0.0. \quad \text{Ans. (7)}$$

The second-order kinematic coefficients can be found by taking partial derivatives of Eqs. (5) and (7) with respect to each of the independent variables. Since the right-hand sides of these equations are constants, we find the second-order kinematic coefficients to be

$$\theta''_{jhh} = \theta''_{khh} = \theta''_{jhi} = \theta''_{khi} = \theta''_{jii} = \theta''_{kii} = 0. \quad \text{Ans.}$$

(b) The angular velocities of links j and k are

$$\begin{aligned}\omega_j &= \theta'_{jh}\omega_h + \theta'_{ji}\omega_i \\ &= (4.0 \text{ rad/rad})(-5.0 \text{ rad/s}) + (3.0 \text{ rad/rad})(10.0 \text{ rad/s}) = 10.0 \text{ rad/s (ccw)},\end{aligned}$$

Ans.

$$\begin{aligned}\omega_k &= \theta'_{kh}\omega_h + \theta'_{ki}\omega_i \\ &= (-2.0 \text{ rad/rad})(-5.0 \text{ rad/s}) + (0.0 \text{ rad/rad})(10.0 \text{ rad/s}) = 10.0 \text{ rad/s (ccw)}.\end{aligned}$$

Ans.

The angular accelerations of links j and k are

$$\begin{aligned}\alpha_j &= \theta''_{jhh}\omega_h^2 + 2\theta''_{jhi}\omega_h\omega_i + \theta''_{jii}\omega_i^2 + \theta'_{jh}\alpha_h + \theta'_{ji}\alpha_i \\ &= 0 + 0 + 0 + (4.0 \text{ rad/rad})(15 \text{ rad/s}^2) + (3.0 \text{ rad/rad})(0) = 60.0 \text{ rad/s}^2 \text{ (ccw)},\end{aligned}$$

Ans.

$$\begin{aligned}\alpha_k &= \theta''_{khh}\omega_h^2 + 2\theta''_{khi}\omega_h\omega_i + \theta''_{kii}\omega_i^2 + \theta'_{kh}\alpha_h + \theta'_{ki}\alpha_i \\ &= 0 + 0 + 0 + (-2.0 \text{ rad/rad})(15 \text{ rad/s}^2) + 0 = -30.0 \text{ rad/s}^2 \text{ (cw)}.\end{aligned}$$

Ans.

5.9 PATH CURVATURE OF A COUPLER POINT TRAJECTORY

After the first- and second-order kinematic coefficients of a multi-degree-of-freedom linkage are known, a study of the curvature of the path traced by a coupler point, which was discussed in Sec. 4.17, can be undertaken. The radius of curvature of the path traced by coupler point P was given by Eq. (4.53). However, a modified version of this equation is required for multi-degree-of-freedom linkages.

The position of point P can be written as

$$\mathbf{r}_P = x_P \hat{\mathbf{i}} + y_P \hat{\mathbf{j}}, \quad (5.13)$$

and the velocity $\mathbf{V}_P$ of point P can be determined by differentiating this equation with respect to time. Recall that the velocity of point P can be written as in Eq. (5.6).

The unit vector tangent to the path of point P is given by Eq. (3.37); that is,

$$\hat{\mathbf{u}}_P^t = \frac{\mathbf{V}_P}{V_P}. \quad (5.14)$$

For the case where x_P and y_P are functions of two independent input variables, θ_j and θ_k , by substituting Eq. (5.6) into this equation, the unit tangent vector can be written as

$$\hat{\mathbf{u}}_P^t = \frac{\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)\hat{\mathbf{i}} + \left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)\hat{\mathbf{j}}}{\sqrt{\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)^2 + \left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)^2}}. \quad (5.15)$$

Then, the unit normal vector is 90° ccw from the unit tangent vector; that is,

$$\hat{\mathbf{u}}_P^n = \hat{\mathbf{k}} \times \hat{\mathbf{u}}_P^t = \frac{-\left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)\hat{\mathbf{i}} + \left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)\hat{\mathbf{j}}}{\sqrt{\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)^2 + \left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)^2}}. \quad (5.16)$$

The normal acceleration of point P is

$$A_P^n = \mathbf{A}_P \cdot \hat{\mathbf{u}}_P^n = A_P^x \hat{u}_P^{nx} + A_P^y \hat{u}_P^{ny}. \quad (5.17)$$

Substituting Eqs. (5.9) and (5.16) into Eq. (5.17), and performing the dot product operation, the normal acceleration of point P can be written as

$$A_P^n = \frac{\left(y''_{Pjj}\omega_j^2 + 2y''_{Pjk}\omega_j\omega_k + y''_{Pkk}\omega_k^2 + y'_{Pj}\alpha_j + y'_{Pk}\alpha_k\right)\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right) - \left(x''_{Pjj}\omega_j^2 + 2x''_{Pjk}\omega_j\omega_k + x''_{Pkk}\omega_k^2 + x'_{Pj}\alpha_j + x'_{Pk}\alpha_k\right)\left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)}{\sqrt{\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)^2 + \left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)^2}}. \quad (5.18)$$

For the special case where links j and k have constant angular velocities, then Eq. (5.18) can be written as

$$A_P^n = \frac{\left(y''_{Pjj}\omega_j^2 + 2y''_{Pjk}\omega_j\omega_k + y''_{Pkk}\omega_k^2\right)\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right) - \left(x''_{Pjj}\omega_j^2 + 2x''_{Pjk}\omega_j\omega_k + x''_{Pkk}\omega_k^2\right)\left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)}{\sqrt{\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)^2 + \left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)^2}}. \quad (5.19)$$

Substituting Eqs. (5.6) and (5.19) into Eq. (4.53) and simplifying, the radius of curvature of the path traced by coupler point P can be written as

$$\rho_P = \frac{\left[\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right)^2 + \left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)^2\right]^{3/2}}{\left(y''_{Pjj}\omega_j^2 + 2y''_{Pjk}\omega_j\omega_k + y''_{Pkk}\omega_k^2\right)\left(x'_{Pj}\omega_j + x'_{Pk}\omega_k\right) - \left(x''_{Pjj}\omega_j^2 + 2x''_{Pjk}\omega_j\omega_k + x''_{Pkk}\omega_k^2\right)\left(y'_{Pj}\omega_j + y'_{Pk}\omega_k\right)}. \quad (5.20)$$

If we define the input angular velocity ratio $\eta = \omega_k/\omega_j$, then Eq. (5.20) can be written as

$$\rho_P = \frac{\left[\left(x'_{Pj} + x'_{Pk}\eta\right)^2 + \left(y'_{Pj} + y'_{Pk}\eta\right)^2\right]^{3/2}}{\left(y''_{Pjj} + 2y''_{Pjk}\eta + y''_{Pkk}\eta^2\right)\left(x'_{Pj} + x'_{Pk}\eta\right) - \left(x''_{Pjj} + 2x''_{Pjk}\eta + x''_{Pkk}\eta^2\right)\left(y'_{Pj} + y'_{Pk}\eta\right)}. \quad (5.21)$$

The sign here has the same meaning that it has in earlier chapters. A negative value for the radius of curvature value indicates that the unit normal vector is pointing away from the center of curvature of the path traced by coupler point P .

The coordinates of the center of curvature of the path traced by point P , as given by Eq. (4.58), are written here as

$$x_C = x_P + \rho_P u_P^{nx} \quad \text{and} \quad y_C = y_P + \rho_P u_P^{ny}, \quad (5.22)$$

where x_P and y_P are found as indicated in Example 5.3, and u_P^{nx} and u_P^{ny} are given by the $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ components of Eq. (5.16).

EXAMPLE 5.11

Continuing the five-bar linkage of Example 5.1 through Example 5.9 in the posture shown in Fig. 5.5, links 2 and 3 are rotating with constant angular velocities of $\omega_2 = 10$ rad/s ccw and $\omega_3 = 5$ rad/s cw, respectively. Find the unit tangent vector and unit normal vector to the path of point P , the radius of curvature, and the Cartesian coordinates of the center of curvature of the path of point P .

SOLUTION

The velocity of coupler point P , obtained from the first-order kinematic coefficients with the same input data as in Example 5.6, is

$$\mathbf{V}_P = 21.546\hat{\mathbf{i}} - 31.330\hat{\mathbf{j}} \text{ in/s} = 38.024 \text{ in/s} \angle -55.483^\circ. \quad (1)$$

Substituting the velocity components of Eq. (1) into Eq. (5.14), the unit tangent vector is

$$\hat{\mathbf{u}}^t = \frac{21.546 \text{ in/s} \hat{\mathbf{i}} - 31.330 \text{ in/s} \hat{\mathbf{j}}}{38.02 \text{ in/s}} = 0.567\hat{\mathbf{i}} - 0.824\hat{\mathbf{j}}. \quad \text{Ans.}$$

Then, from Eq. (5.16), the unit normal vector is

$$\hat{\mathbf{u}}^n = 0.824\hat{\mathbf{i}} + 0.567\hat{\mathbf{j}}. \quad \text{Ans.}$$

The directions of the unit tangent vector and unit normal vector are shown in Fig. 5.17.

Substituting data from Example 5.6 and Example 5.9 for the first- and second-order kinematic coefficients and the specified input angular velocities into Eq. (5.20), the radius of curvature of the path of point P is

$$\rho_P = \frac{1 \text{ 445.81 in}^3/\text{s}^3}{-254.44 \text{ in}^2/\text{s}^3} = -5.682 \text{ in}. \quad \text{Ans.}$$

The negative sign indicates that the center of curvature is located in the direction of the negative unit normal vector $\hat{\mathbf{u}}^n$ from point P (Fig. 5.18).

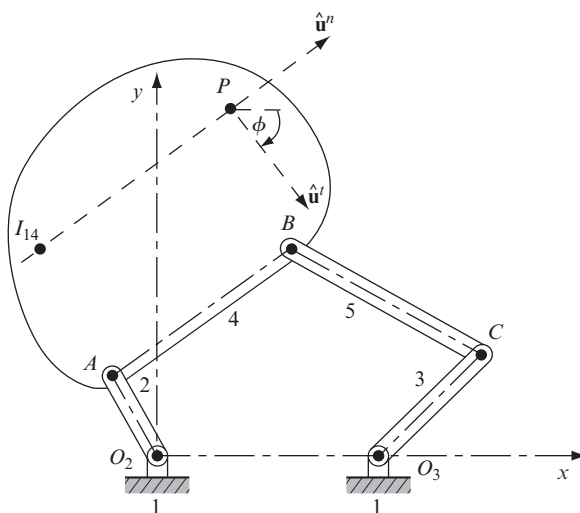


Figure 5.17 Unit tangent and unit normal vectors.

Finally, substituting the answers from Example 5.3 and the previous results into Eqs. (5.22), the coordinates of the center of curvature of the path of point P are

$$x_C = 1.947 \text{ in} + (-5.682 \text{ in})(0.824) = -2.735 \text{ in},$$

$$y_C = 9.499 \text{ in} + (-5.682 \text{ in})(0.567) = 6.277 \text{ in}.$$

The center of curvature of the path of point P is shown in Fig. 5.18. This figure also shows the unit tangent vector $\hat{\mathbf{u}}^t$ and the unit normal vector $\hat{\mathbf{u}}^n$ to the path traced by point P .

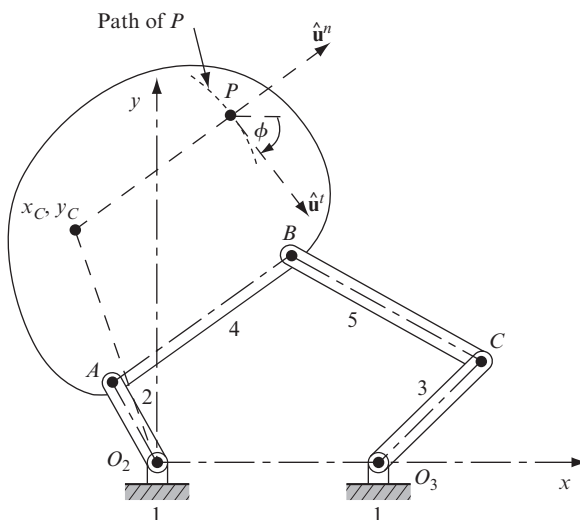


Figure 5.18 Center of curvature of the path traced by point P .

Table 5.5 presents the radius of curvature of the path traced by point P and the Cartesian coordinates of the center of curvature of this path for the range of input angles given in Table 5.1.

Table 5.5 Radius of Curvature and Coordinates of the Center of Curvature

θ_2 deg	θ_3 deg	ρ in	x_c in	y_c in
40	85	−10.350	5.784	2.067
50	80	−8.062	4.052	3.559
60	75	−5.340	2.247	5.595
70	70	−3.014	1.005	7.557
80	65	−1.667	0.551	8.821
90	60	−1.214	0.524	9.271
100	55	−1.481	0.408	9.031
110	50	−2.671	−0.364	8.130
120	45	−5.682	−2.735	6.279
130	40	−13.603	−9.496	2.157
140	35	−51.321	−42.794	−15.556
150	30	57.190	53.984	33.525

5.10 FINITE DIFFERENCE METHOD

The finite difference method can be used with the examples of this chapter to check: (a) the first-order kinematic coefficients of links 4 and 5, and (b) the second-order kinematic coefficients of links 4 and 5.

In general, the first-order kinematic coefficient of link i with respect to link j can be written as

$$\theta'_{ij} = \frac{\partial \theta_i}{\partial \theta_j} \approx \frac{\Delta \theta_i}{\Delta \theta_j}.$$

Keep in mind that the definition of a *partial* derivative implies that the symbol $\Delta \theta_j$ in the denominator corresponds to the change of an independent input θ_j while all other independent inputs are held frozen or locked during the change of $\Delta \theta_j$. However, data such as those in Tables 5.1 and 5.2, for example, do not correspond to this restriction, since they were derived with both $\Delta \theta_2$ and $\Delta \theta_3$ changing simultaneously. Therefore, the first-order kinematic coefficient of link i with respect to input j *cannot* be directly approximated by the following formula

$$\theta'_{ij} \neq \frac{\Delta \theta_i}{\Delta \theta_j} = \frac{(\theta_i)_A - (\theta_i)_B}{(\theta_j)_A - (\theta_j)_B},$$

where subscript A denotes the value after and B denotes the value before the value that is to be checked by the finite difference method. In situations such as Tables 5.1 and 5.2,

however, where the total change $\Delta\theta_i$ is the result of separate changes of more than one independent variable, say $\Delta\theta_j$ and $\Delta\theta_k$, the proper procedure is to write the finite change in the angular displacement of link i as

$$\Delta\theta_i \approx \theta'_{ij}\Delta\theta_j + \theta'_{ik}\Delta\theta_k$$

or as

$$[(\theta_i)_A - (\theta_i)_B] \approx \theta'_{ij}[(\theta_j)_A - (\theta_j)_B] + \theta'_{ik}[(\theta_k)_A - (\theta_k)_B]. \quad (5.23)$$

The following example will illustrate the use of these equations to check the first-order kinematic coefficients of Example 5.6.

EXAMPLE 5.12

Use finite differences to check the first-order kinematic coefficients of links 4 and 5 of the five-bar linkage of Example 5.6.

SOLUTION

By Eq. (5.23), the finite change in the angular displacement of link 4 can be written as

$$\Delta\theta_4 = [(\theta_4)_A - (\theta_4)_B] \approx \theta'_{42}[(\theta_2)_A - (\theta_2)_B] + \theta'_{43}[(\theta_3)_A - (\theta_3)_B], \quad (1)$$

and the finite change in the angular displacement of link 5 can be written as

$$\Delta\theta_5 = [(\theta_5)_A - (\theta_5)_B] \approx \theta'_{52}[(\theta_2)_A - (\theta_2)_B] + \theta'_{53}[(\theta_3)_A - (\theta_3)_B]. \quad (2)$$

For the input angles $\theta_2 = 120^\circ$ and $\theta_3 = 45^\circ$ (row nine of Table 5.1), the first-order kinematic coefficients are

$$\theta'_{42} = -0.237, \quad \theta'_{43} = 0.705, \quad \theta'_{52} = 0.455, \quad \text{and} \quad \theta'_{53} = -0.109.$$

Substituting these values into Eqs. (1) and (2), the finite change in the angular displacements of links 4 and 5, respectively, are

$$\Delta\theta_4 = [30.526^\circ - 42.360^\circ] \approx -0.237[130^\circ - 110^\circ] + 0.705[40^\circ - 50^\circ], \quad (3)$$

$$\Delta\theta_5 = [156.510^\circ - 146.315^\circ] \approx 0.455[130^\circ - 110^\circ] - 0.109[40^\circ - 50^\circ]. \quad (4)$$

Note that the answers given by the two separate calculations in Eqs. (3) and (4) are in reasonable agreement with each other. If the increments of the two input displacements are decreased, then even closer agreement between the two sets of calculations is obtained. The general conclusion, therefore, is that the first-order kinematic coefficients of the five-bar linkage as indicated by Table 5.1 are correct.

From Eqs. (5.7), the second-order kinematic coefficients of link i with respect to the input angles θ_j and θ_k can be written as

$$\theta''_{ijk} = \frac{\partial^2 \theta_i}{\partial \theta_j \partial \theta_k} = \frac{\partial \theta'_{ij}}{\partial \theta_k}.$$

Remembering the directional nature of finite differences, the finite change in the first-order kinematic coefficients of link i with respect to link j can be written as

$$\Delta\theta'_{ij} = [(\theta'_{ij})_A - (\theta'_{ij})_B] \approx \theta''_{ijj} [(\theta_j)_A - (\theta_j)_B] + \theta''_{ijk} [(\theta_k)_A - (\theta_k)_B],$$

where subscript A denotes the value after and B denotes the value before the value that is to be checked by the finite difference method.

The following example illustrates the value of these equations in checking the second-order kinematic coefficients.

EXAMPLE 5.13

Use finite differences to check the second-order kinematic coefficients of links 4 and 5 of the five-bar linkage of Example 5.9.

SOLUTION

The second-order kinematic coefficients can be checked by writing the finite change in the first-order kinematic coefficients of links 4 and 5 with respect to the input rotations of links 2 and 3 [Eq. (5.24)] as

$$\Delta\theta'_{42} = [(\theta'_{42})_A - (\theta'_{42})_B] \approx \theta''_{422} [(\theta_2)_A - (\theta_2)_B] + \theta''_{423} [(\theta_3)_A - (\theta_3)_B],$$

$$\Delta\theta'_{43} = [(\theta'_{43})_A - (\theta'_{43})_B] \approx \theta''_{432} [(\theta_2)_A - (\theta_2)_B] + \theta''_{433} [(\theta_3)_A - (\theta_3)_B],$$

$$\Delta\theta'_{52} = [(\theta'_{52})_A - (\theta'_{52})_B] \approx \theta''_{522} [(\theta_2)_A - (\theta_2)_B] + \theta''_{523} [(\theta_3)_A - (\theta_3)_B],$$

and

$$\Delta\theta'_{53} = [(\theta'_{53})_A - (\theta'_{53})_B] \approx \theta''_{532} [(\theta_2)_A - (\theta_2)_B] + \theta''_{533} [(\theta_3)_A - (\theta_3)_B],$$

where $\theta''_{432} = \theta''_{423}$ and $\theta''_{532} = \theta''_{523}$. From row nine of Tables 5.1 and 5.3, the values in these equations are

$$\Delta\theta'_{42} = [(-0.230) - (-0.254)] \approx 0.139 \left[\frac{130^\circ - 110^\circ}{57.296^\circ/\text{rad}} \right] + 0.131 \left[\frac{40^\circ - 50^\circ}{57.296^\circ/\text{rad}} \right],$$

$$\Delta\theta'_{43} = [0.737 - 0.683] \approx 0.131 \left[\frac{130^\circ - 110^\circ}{57.296^\circ/\text{rad}} \right] + (-0.038) \left[\frac{40^\circ - 50^\circ}{57.296^\circ/\text{rad}} \right],$$

$$\Delta\theta'_{52} = [0.508 - 0.397] \approx 0.208 \left[\frac{130^\circ - 110^\circ}{57.296^\circ/\text{rad}} \right] + (-0.206) \left[\frac{40^\circ - 50^\circ}{57.296^\circ/\text{rad}} \right],$$

$$\Delta\theta'_{53} = [(-0.136) - (-0.091)] \approx (-0.206) \left[\frac{130^\circ - 110^\circ}{57.296^\circ/\text{rad}} \right] + (-0.173) \left[\frac{40^\circ - 50^\circ}{57.296^\circ/\text{rad}} \right].$$

Note that the answers given by the two separate calculations in each of these equations are in reasonable agreement with each other. The general conclusion, therefore, is that the second-order kinematic coefficients of links 4 and 5 of the five-bar linkage as given by Table 5.3 are correct.

5.11 REFERENCE

- [1] Pennock, G. R., 2008. Curvature theory for a two-degree-of-freedom planar linkage, *Mech. Mach. Theory* 43:525–48.

PROBLEMS*

- 5.1** Slotted links 2 and 3 are driven independently at constant speeds of $\omega_2 = 30$ rad/s cw and $\omega_3 = 20$ rad/s cw, respectively. Find the absolute velocity and acceleration of the center of pin P carried in the two slots.

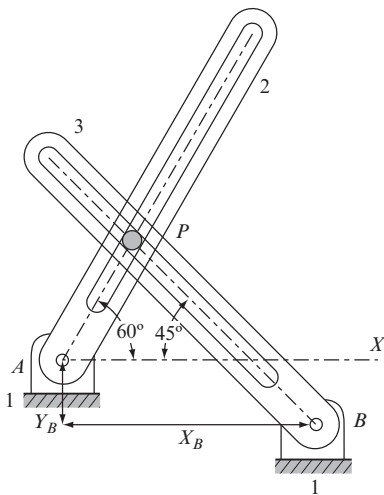


Figure P5.1 $x_B = 100$ mm and $y_B = 25$ mm.

- 5.2** For the five-bar linkage in the posture shown, the angular velocity of link 2 is 15 rad/s cw and the angular velocity of link 5 is 15 rad/s cw. Determine the angular velocity of link 3 and the apparent velocity $V_{B_4/5}$.
- 5.3** For the five-bar linkage in the posture shown in Fig. P5.2, the angular velocity of link 2 is $\omega_2 = 25$ rad/s ccw and the apparent velocity $V_{B_4/5}$ is 5 m/s upward along link 5. Determine the angular velocities of links 3 and 5.
- 5.4** For Prob. 5.2, assuming that the two given input velocities are constant, determine the angular acceleration of link 3 at the instant indicated.

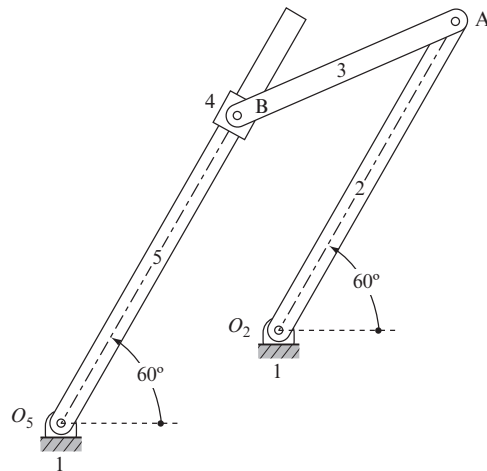


Figure P5.2 $R_{O_2O_5} = 200$ mm $\angle 23.1^\circ$, $R_{AO_2} = 300$ mm, and $R_{BA} = 200$ mm.

- 5.5** Link 2 rotates at a constant angular velocity of 10 rad/s ccw, while sliding block 3 slides toward point A on link 2 at a constant rate of 5 in/s. Find the absolute velocity and absolute acceleration of point P of block 3.

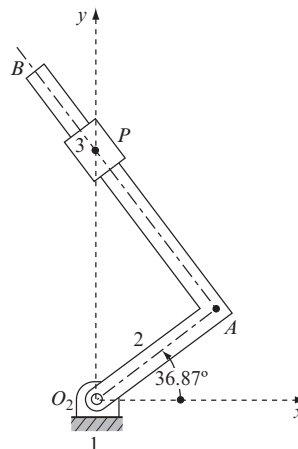


Figure P5.5 $R_{AO_2} = 3.0$ in, $R_{BA} = 6.0$ in, and $R_{PA} = 4.0$ in.

*When assigning problems, the instructor may wish to specify the method of solution to be used, since a variety of approaches are presented in the text.

5.6 For Prob. 5.5, determine the value of the sliding velocity, $V_{P_3/2}$, that minimizes the absolute velocity of point P of block 3. In addition, find the value of $V_{P_3/2}$ that minimizes the absolute acceleration of point P of block 3.

5.7 The left two-link planar robot is attempting to transfer a small object labeled P to the similar right robot. At the posture indicated, $\theta_2 = 45^\circ$ and $\theta_{3/2} = -15^\circ$. (Note that $\theta_{3/2} = \theta_3 - \theta_2$ is given, since that is the angle controlled by the motor in joint A .) Determine θ_4 and $\theta_{5/4}$ to allow the right robot to take possession of object P .

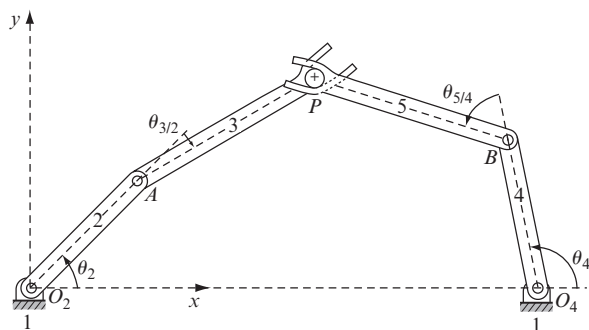


Figure P5.7 $R_{O_4O_2} = 1$ m, $R_{AO_2} = R_{BO_4} = 0.3$ m, and $R_{PA} = R_{PB} = 0.4$ m.

5.8 For the transfer of the object described in Prob. 5.7, it is necessary that the velocities of point P of the two robots match. If the two input velocities of the first robot are $\omega_2 = 10$ rad/s cw and $\omega_{3/2} = 15$ rad/s ccw, what angular velocities must be used for ω_4 and $\omega_{5/4}$?

5.9 For the transfer of the object described in Prob. 5.7, it is necessary that the velocities of point P of the two robots match. If the two input velocities of the first robot are $\omega_2 = 10$ rad/s cw and $\omega_{3/2} = 10$ rad/s ccw, what angular velocities must be used for ω_4 and $\omega_{5/4}$?

5.10 For the transfer of the object described in Prob. 5.7, it is necessary that the velocities of point P of the two robots match. If the two input velocities of the first robot are $\omega_2 = 10$ rad/s cw and $\omega_{3/2} = 0$, what angular velocities must be used for ω_4 and $\omega_{5/4}$?

5.11 To successfully transfer an object between two robots, as described in Probs. 5.7 and 5.8, it is helpful if the accelerations are also matched at point P . Assuming that the two input accelerations are $\alpha_2 = \alpha_3 = 0$ at this instant for the robot on the left, what angular accelerations must be given to the

two input joints of the robot on the right to achieve this?

5.12 The circular cam is driven by link 2 at a constant angular velocity $\omega_2 = 15$ rad/s ccw. Link 3 is rotating at a constant angular velocity $\omega_3 = 5$ rad/s cw, causing slipping at point C . Determine: (a) the first- and second-order kinematic coefficients of the mechanism; (b) the angular velocity and acceleration of link 4; and (c) the velocity of slipping at point C .

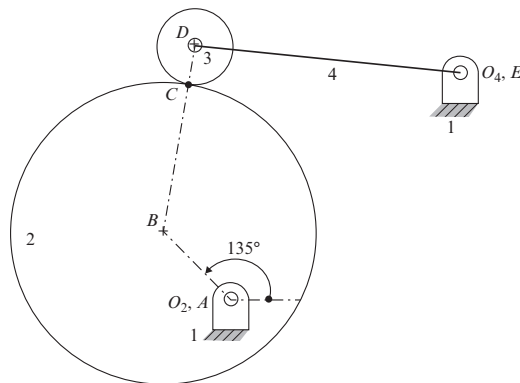


Figure P5.12 $R_{KO_2} = R_{O_4K} = 3$ in, $R_{BO_2} = 1.25$ in, $R_{CB} = \rho_2 = 2$ in, $R_{CD} = \rho_3 = 0.5$ in, and $R_{DO_4} = 3.5$ in.

5.13 Tracing point C of the pantograph linkage is required to follow a prescribed curve; that is, its two independent input variables are x_C and y_C . Then, point P , which carries a pen, traces a similar curve; that is, the outputs of the linkage are the x_P and y_P components of the motion of the pen. Determine the first-order kinematic coefficients of point P .

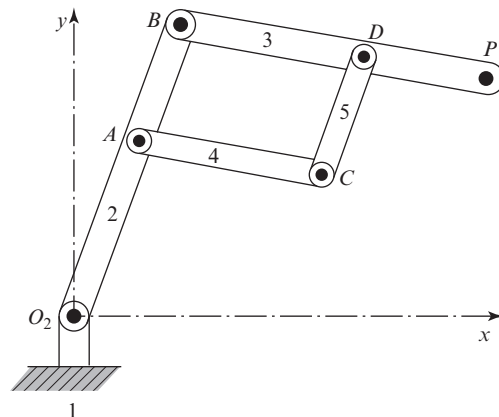


Figure P5.13 $R_{AO_2} = R_{CA} = R_{DB} = 300$ mm and $R_{BA} = R_{DC} = R_{PD} = 200$ mm.

- 5.14** The mechanism has rolling contact at point A , but there can be slipping at point B . Link 2 has a constant angular velocity of $\omega_2 = 20 \text{ rad/s}$ ccw, and link 3 has an angular velocity $\omega_3 = 5 \text{ rad/s}$ cw and an angular acceleration of $\alpha_3 = 2 \text{ rad/s}^2$ ccw. Determine: (a) the first- and second-order kinematic coefficients of link 5; and (b) the angular velocity and angular acceleration of link 5.

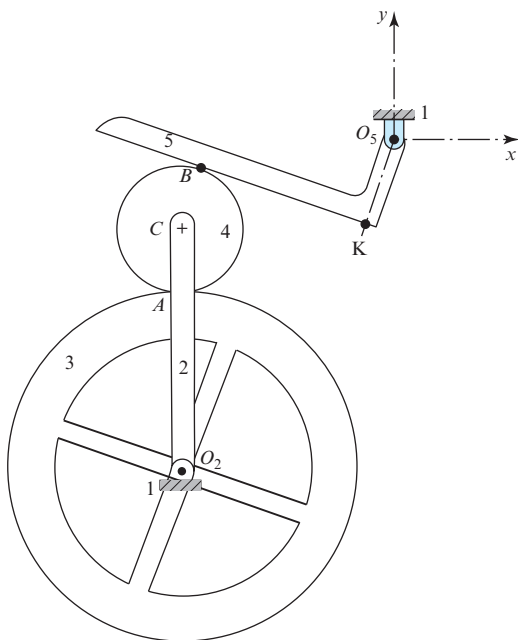


Figure P5.14 $\mathbf{R}_{O_5 O_2} = 2.50\mathbf{i} + 3.75\mathbf{j} \text{ in}$, $R_{CO_2} = 2.75 \text{ in}$, $R_{KO_5} = 1 \text{ in}$, $\rho_3 = 2 \text{ in}$, and $\rho_4 = 0.75 \text{ in}$.

- 5.15** The mechanism has rolling contact at point A . For the current posture $\theta_2 = 60^\circ$. Link 2 has constant angular velocity $\omega_2 = 50 \text{ rad/s}$ ccw, and link 3 has angular velocity $\omega_5 = 25 \text{ rad/s}$ cw and angular acceleration $\alpha_5 = 20 \text{ rad/s}^2$ ccw. Determine: (a) the first- and second-order kinematic coefficients for links 3 and 4; and (b) the angular velocities and angular accelerations of links 3 and 4.
- 5.16** The mechanism has rolling contact between rack 3 and gear 4 at point F , between gears 4 and 5 at point C , and between gears 5 and 6 at point E . Link 2 has constant angular velocity $\omega_2 = 50 \text{ rad/s}$ ccw, and

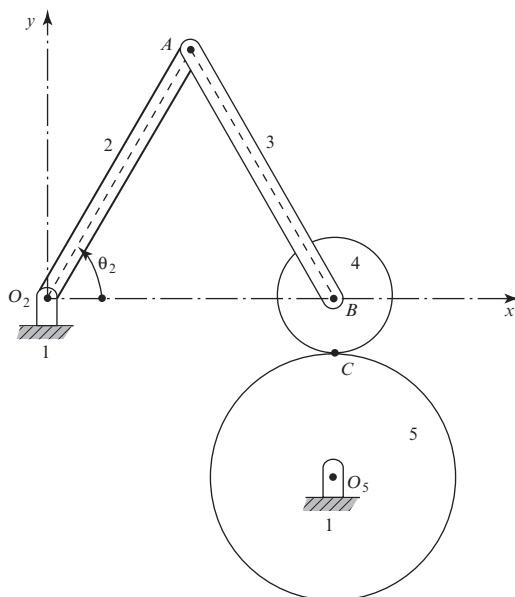


Figure P5.15 $\mathbf{R}_{O_5 O_2} = 300\mathbf{i} - 190\mathbf{j} \text{ mm}$, $R_{AO_2} = 300 \text{ mm}$, $R_{CB} = 300 \text{ mm}$, $\rho_4 = 60 \text{ mm}$, and $\rho_5 = 130 \text{ mm}$.

link 6 has angular velocity $\omega_6 = 25 \text{ rad/s}$ cw and angular acceleration $\alpha_6 = 20 \text{ rad/s}^2$ ccw. Determine: (a) the first- and second-order kinematic coefficients for gear 5 and the rack; and (b) the velocity and acceleration of the rack and the angular velocities and accelerations of links 4 and 5.

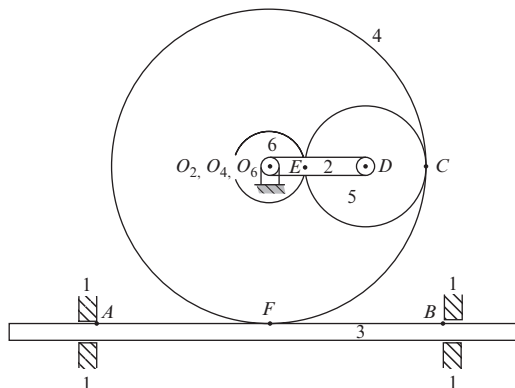
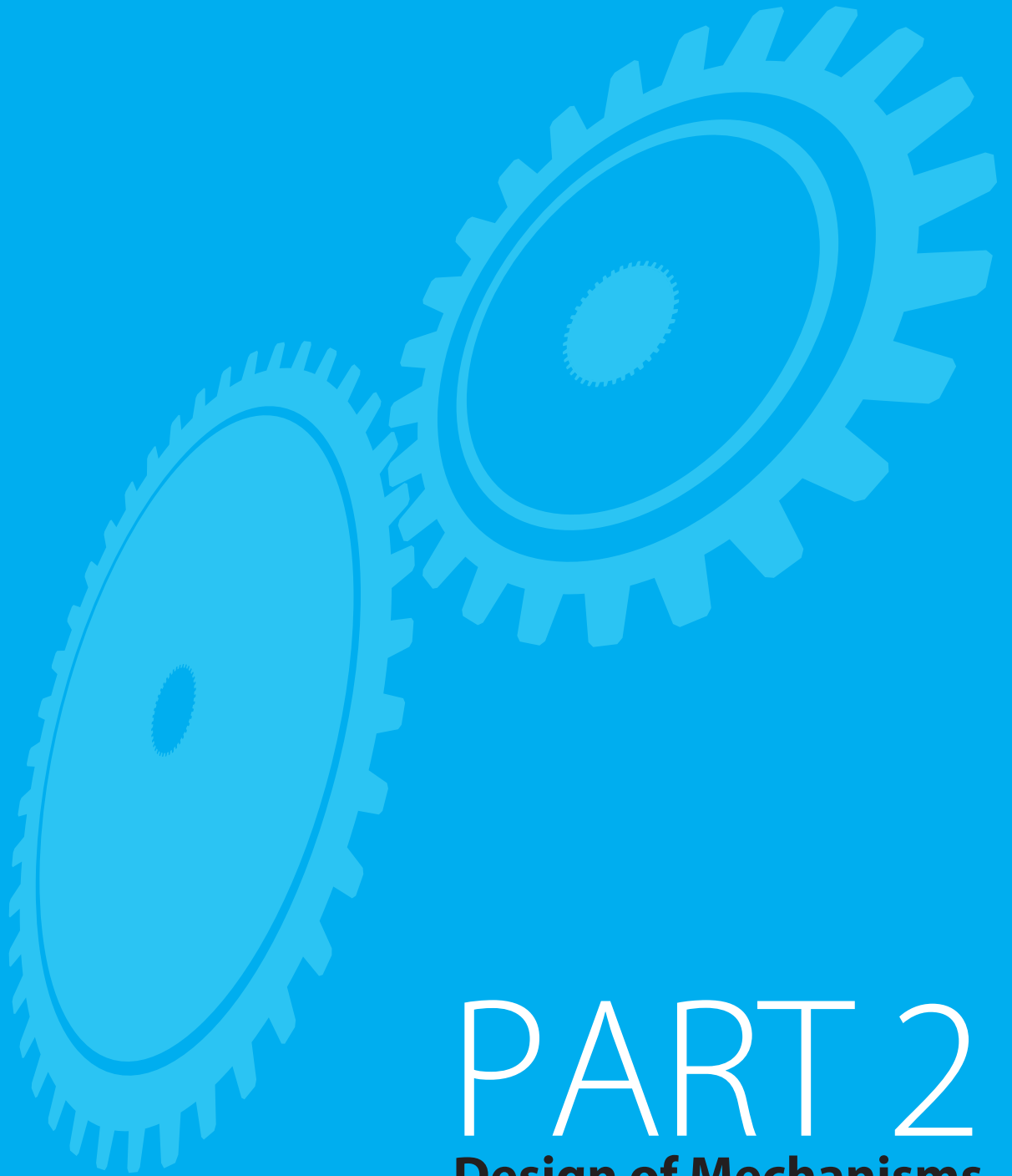


Figure P5.16 $R_{FA} = R_{BF} = 500 \text{ mm}$, $\rho_4 = R_{CO_4} = 450 \text{ mm}$, $\rho_5 = R_{CD} = 175 \text{ mm}$, $\rho_6 = R_{EO_2} = 100 \text{ mm}$, and $R_{DO_2} = 275 \text{ mm}$.

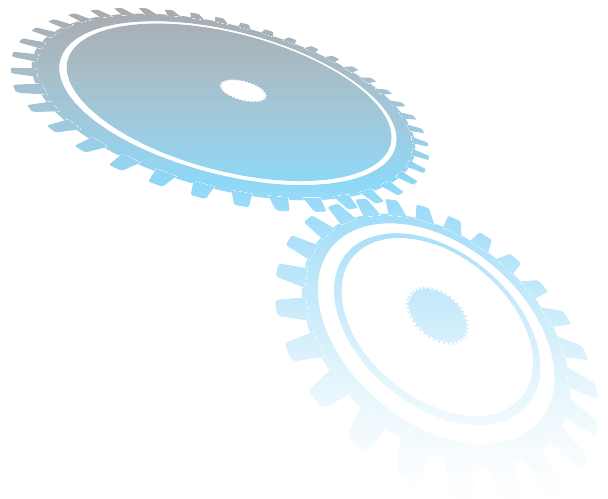


PART 2

Design of Mechanisms

6

Cam Design



6.1 INTRODUCTION

In previous chapters, we have learned how to analyze the kinematic characteristics of a given mechanism. We were given the design of a mechanism, and we studied ways to determine its mobility, its posture, its velocity, and its acceleration, and we even discussed its suitability for given types of tasks. However, we have said little about how the mechanism is designed—that is, how the sizes and shapes of the links are chosen by the designer.

The next several chapters introduce this *design* point of view as it relates to mechanisms. We will find ourselves looking more at individual types of machine components and learning when and why such components are chosen and how they are sized. In Chap. 6, which is devoted to the design of cams, for example, we assume that we know the task to be accomplished. However, we do not know but we seek techniques to help discover the size and shape of the cam to perform this task.

Of course, there is the creative step of deciding whether a cam should be used in the first place, as opposed to a gear train, a linkage, or some other mechanical device. This question often cannot be answered on the basis of scientific principles alone; it requires experience and imagination, and involves such factors as economics, marketability, reliability, maintenance, esthetics, ergonomics, ability to manufacture, and suitability for the task. These aspects are not well studied by a general scientific approach; they require human judgment of factors that are often not easily reduced to numbers or formulae. There is usually not a single “right” answer, and generally these questions cannot be answered by this or any other text or reference book.

On the other hand, this is not to say that there is no place for a general science-based approach in design situations. Most mechanical design is based on repetitive analysis.

Therefore, in Chap. 6 and in several of the upcoming chapters, we will use the principles of analysis presented in Part 1 of the book. Also, we will use the governing analysis equations to help in our choices of part sizes and shapes, and to help us assess the quality of our design as we proceed. It is important to point out that the forthcoming chapters are still based on the laws of mechanics. The primary shift for Part 2 is that the component dimensions are often the unknowns of the problem, whereas the input and output speeds, for example, may be given information. In Chap. 6, we will discover how to determine a cam contour, or profile, that delivers a specified motion characteristic.

6.2 CLASSIFICATION OF CAMS AND FOLLOWERS

A *cam* is a machine element used to drive another element, called a *follower*, through a specified motion by direct contact. Cam-and-follower mechanisms are simple and inexpensive, have few moving parts, and occupy very little space. Furthermore, follower motions having almost any desired characteristics are not difficult to design. For these reasons, cam mechanisms are used extensively in modern machinery.

The versatility and flexibility in the design of cam systems are among their more attractive features, yet this also leads to a wide variety of shapes and forms, and the need for terminology to distinguish them.

Sometimes, cams are classified according to their basic shapes. Figure 6.1 shows four different types of cams:

- (a) A *plate cam*, also called a *disk cam* or a *radial cam*.
- (b) A *wedge cam*.
- (c) A *cylindric cam* or *barrel cam*.
- (d) An *end cam* or *face cam*.

The least common of these in practical applications is the wedge cam, because of its need for a reciprocating motion rather than a continuous input motion. By far the most common is the plate cam. For this reason, most of the remainder of Chap. 6 specifically addresses plate cams, although the concepts presented pertain universally.

Cam systems can also be classified according to the basic shape of the follower. Figure 6.2 shows plate cams actuating four different types of followers:

- (a) A *knife-edge* follower.
- (b) A *flat-face* follower.
- (c) A *roller* follower.
- (d) A *spheric-face* or *curved-shoe* follower.

Note that the follower face is usually chosen to have a simple geometric shape, and the motion is achieved by careful design of the shape of the cam to mate with it. This is not always the case, and examples of *inverse cams*, where the follower is machined to a complex shape, can be found.

Another method of classifying cams is according to the characteristic output motion produced between the follower and the frame. Thus, some cams have *reciprocating* (translating) followers, as in Figs. 6.1a, b, and d, and Figs. 6.2a and b, whereas others

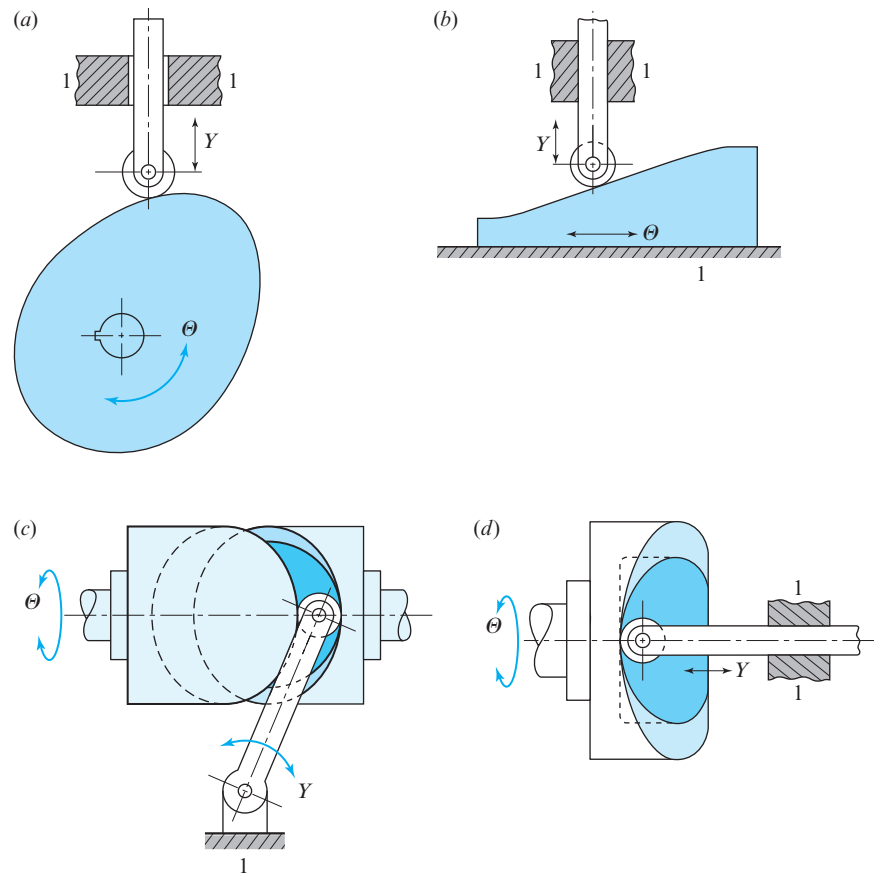


Figure 6.1 (a) Plate cam; (b) wedge cam; (c) barrel cam; (d) face cam.

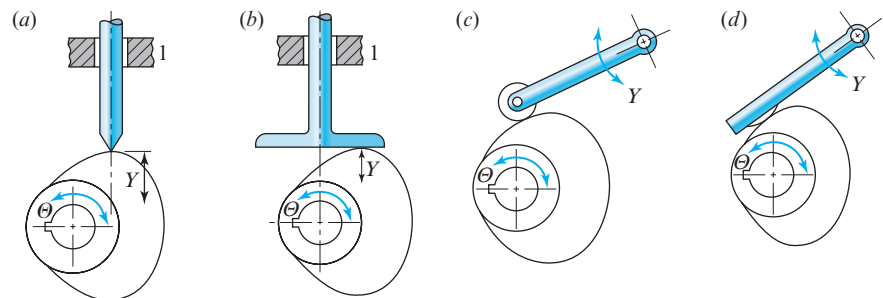
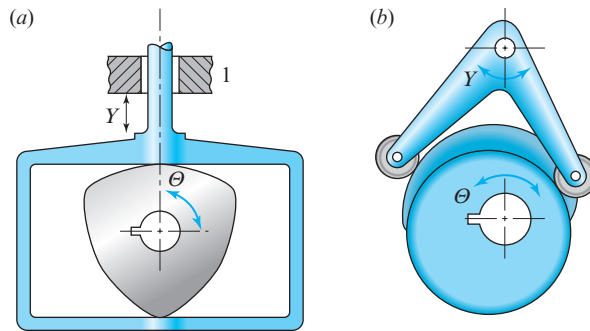


Figure 6.2 Plate cams with (a) an offset reciprocating knife-edge follower; (b) a reciprocating flat-face follower; (c) an oscillating roller follower; (d) an oscillating curved-shoe follower.

**Figure 6.3**

(a) Constant-breadth cam with a reciprocating flat-face follower; (b) conjugate cams with an oscillating roller follower.

have *oscillating* (rotating) followers, as in Figs. 6.1c, 6.2c, and 6.2d. Further classification of reciprocating followers distinguishes whether the centerline of the follower stem relative to the center of the cam is *offset*, as in Fig. 6.2a, or *radial*, as in Fig. 6.2b.

In all cam systems, the designer must ensure that the follower maintains contact with the cam at all times. This can be accomplished by depending on gravity, by the inclusion of a suitable spring, or by a mechanical constraint. In Fig. 6.1c, the follower is constrained by a groove. Figure 6.3a shows an example of a *constant-breadth* cam, where two contact points between the cam and the follower provide the constraint. Mechanical constraint can also be introduced by employing *dual* or *conjugate* cams in an arrangement such as that shown in Fig. 6.3b. Here each cam has its own roller, but the rollers are mounted on a common follower.

Throughout this chapter we use the symbol Θ to represent the total motion of the cam and the symbol Y to represent the total displacement of the follower. To investigate the design of cams in general, we denote the known input variable by $\Theta(t)$ and the output variable by Y . A review of Figs. 6.1, 6.2, and 6.3 will demonstrate the definitions of Θ and Y for various types of cams. These figures show that the input, Θ , is an angle for most cams, but it can be a distance, as in Fig. 6.1b. Also, the output, Y , is a translation distance for a reciprocating follower, but it is an angle for an oscillating follower.

6.3 DISPLACEMENT DIAGRAMS

Despite the wide variety of cam types used and their differences in form, they also have certain features in common that allow a systematic approach to their design. Usually a cam system is a single-degree-of-freedom device. It is driven by a known input motion, usually a shaft that rotates at constant speed, and it is intended to produce a certain desired periodic output motion for the follower.

During the rotation of a cam through one cycle, the follower executes a series of events as demonstrated in graphic form in the *displacement diagram* of Fig. 6.4. In such a diagram, the abscissa represents one cycle of the input, Θ (usually, one revolution of the cam), and is drawn to any convenient scale. The ordinate represents the follower travel, Y , and, for a reciprocating follower, is usually drawn at full scale to help in the layout of the cam. On a displacement diagram, it is possible to identify a portion of the graph called the *rise*, where the displacement of the follower is away from the cam center. The total rise is called the

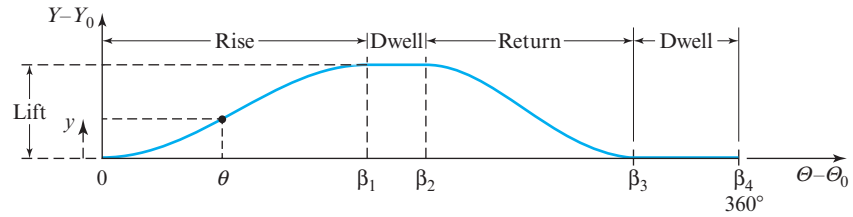


Figure 6.4 Displacement diagram for a cam.

lift. The *return* is the portion in which the displacement of the follower is toward the cam center. Portions of the cycle during which the follower is at rest are referred to as *dwells*.

For a particular motion segment (say, segment number k) of the total cam, the rotation lies in the range starting with Θ_k and extending by θ within the segment to

$$\Theta_{k+1} = \Theta_k + \beta_k, \quad (a)$$

where β_k is the total cam angle for segment k .

Similarly, the displacement of the follower is in the range starting from $Y_k + y(0)$ and extending by $y(\theta)$ within the segment to

$$Y_{k+1} = Y_k + y(\beta_k). \quad (b)$$

Thus, within a segment, we can write

$$y = y(\theta). \quad (6.1)$$

Many of the essential features of a displacement diagram, such as the total lift and the placement and duration of dwells, are usually dictated by the requirements of the application. There are, however, many possible choices of follower motions that might be used for the rise and return segments, and some are preferable over others, depending on the situation. One of the key steps in the design of a cam is the choice of suitable forms for these motions. Once the motions have been chosen—that is, once the exact relationship is set between the input Θ and the output Y —the displacement diagram can be constructed precisely and is a graphic representation of the functional relationship

$$Y = Y(\Theta).$$

This equation has stored within it the exact nature of the shape of the final cam: the necessary information for its layout and manufacture, and also the important characteristics that determine the quality of its dynamic performance. Before looking further at these topics, however, we will describe graphic methods of constructing the displacement diagrams for the following rise motions and the similar return motions: uniform motion, parabolic motion, simple harmonic motion, and cycloidal motion.

The displacement diagram for *uniform motion* is a straight line with constant slope. Thus, for constant input speed, the velocity of the follower is also constant. This motion

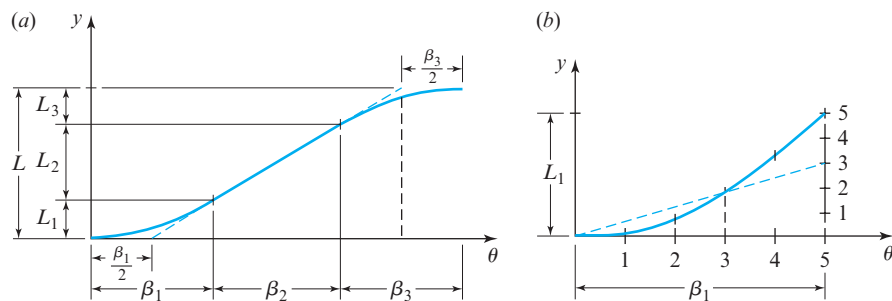


Figure 6.5 (a) Parabolic motion interfaces with uniform motion; (b) graphic construction.

is not useful for the full lift because of the sharp corners produced at the junctions with neighboring segments of the displacement diagram. It is often used, however, between other curve segments that eliminate the corners (called *modified uniform motion*).

Parabolic motion is one possible example of modified uniform motion, and the displacement diagram for this motion is shown in Fig. 6.5a. The central portion of the diagram, bounded by the cam angle β_2 and with lift L_2 , is uniform motion. The two ends, with cam angles β_1 and β_3 , and with corresponding lifts L_1 and L_3 , are shaped to deliver *parabolic motion* to the follower. Soon we shall learn that these produce constant acceleration for the follower. Figure 6.5a shows a graphic method for matching the slopes of parabolic motions with that of uniform motion. With the cam angles β_1 , β_2 , β_3 , and the total lift, L , known, individual lifts L_1 , L_2 , and L_3 are determined by locating the midpoints of the β_1 and β_3 segments and constructing a straight line as indicated. Figure 6.5b shows a graphic construction for a parabola to fit within a given rectangular boundary defined, first by L_1 and β_1 , and then by L_3 and β_3 . The abscissa and ordinate are divided into a convenient but equal number of divisions and numbered as indicated. The construction of each point of the parabola then follows that indicated by dashed lines for point 3.

In the layout of an actual cam, if this might be done graphically, a great many divisions are usually used to obtain good accuracy. At the same time, the drawing is made to a large scale, perhaps ten times full size, and then reduced to actual size by a pantographic method. However, for clarity in reading, the figures in this chapter are shown with fewer divisions to define the curves and illustrate the graphic techniques.

The displacement diagram for *simple harmonic motion* is shown in Fig. 6.6. The graphic construction makes use of a semicircle having a diameter equal to lift L of the segment. The semicircle and abscissa are divided into an equal number of divisions, and the construction then follows that indicated by dashed lines for division number 2.

Cycloidal motion obtains its name from the geometric curve called a *cycloid*. As shown on the left-hand side of Fig. 6.7, a circle of radius $L/2\pi$, where L is the lift of the segment, makes exactly one revolution by rolling along the ordinate from the origin, $y = 0$, to full lift, $y = L$. Point P of the circle, originally located at the origin, traces a cycloid, as shown. As the circle rolls without slip, the graph of the vertical displacement, y , of the point versus the rotation angle, θ , gives the displacement diagram shown at the right of Fig. 6.7. We find

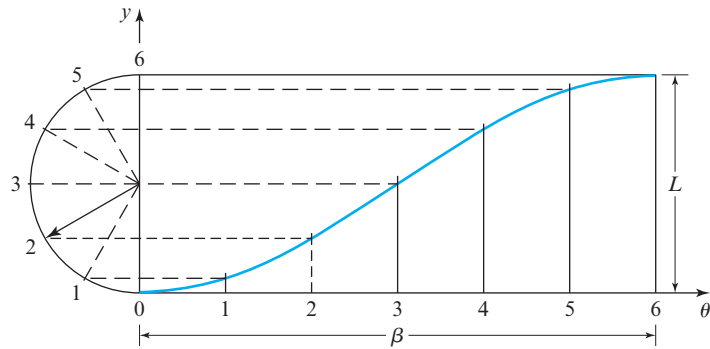


Figure 6.6 Simple harmonic motion displacement diagram; graphic construction.

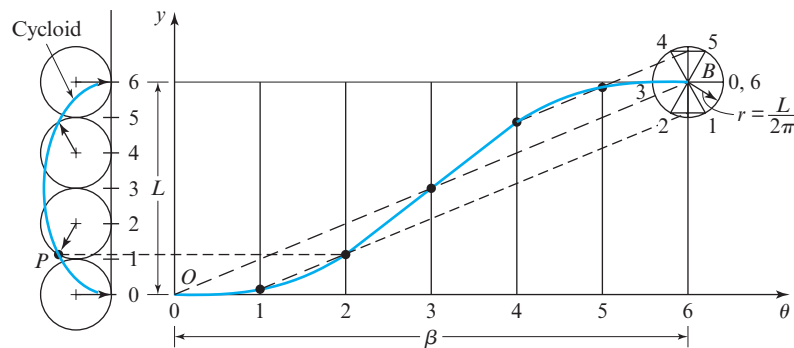


Figure 6.7 Cycloidal motion displacement diagram; graphic construction.

it much more convenient for graphic purposes to draw the circle only once, using either the origin, O , or point B as its center. After dividing the circle and the abscissa into an equal number of divisions and numbering them as indicated, we project each point of the circle horizontally until it intersects the ordinate; next, from the ordinate, we project parallel to the diagonal OB to obtain the corresponding point on the displacement diagram.

6.4 GRAPHIC LAYOUT OF CAM PROFILES

Let us now study the problem of determining the exact shape of the cam profile required to deliver a specified follower motion. We assume here that the required motion has been completely defined—graphically, analytically, or numerically—as discussed in later sections. Thus, a complete displacement diagram can be drawn to scale for the entire cam rotation. The requirement now is to lay out the proper cam profile to achieve the follower motion represented by this displacement diagram.

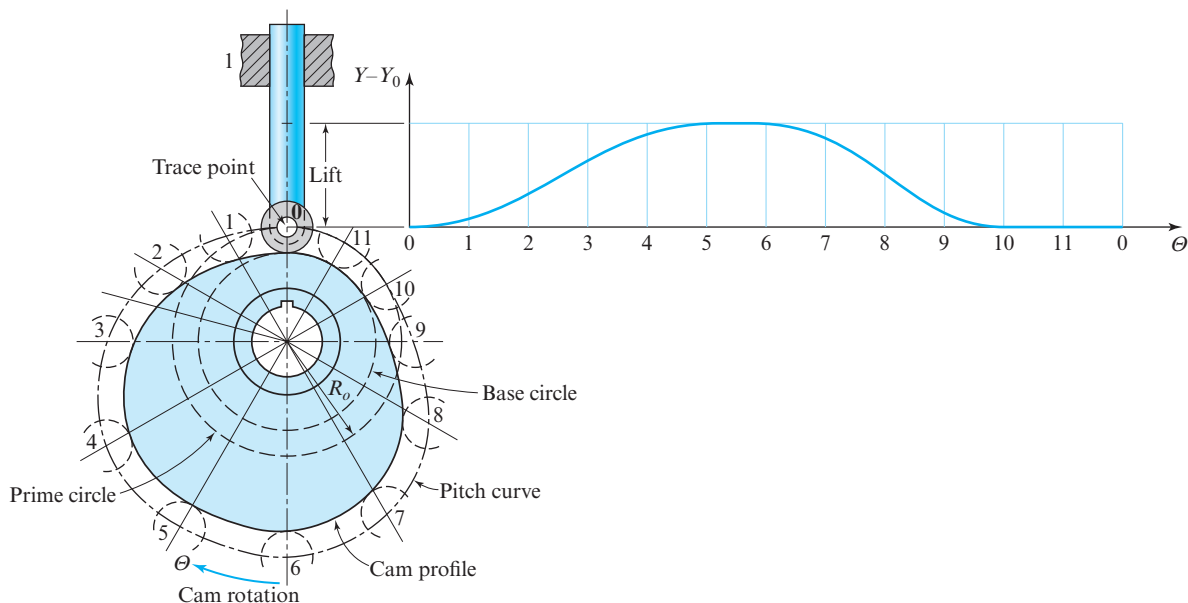


Figure 6.8 Cam nomenclature. The cam surface is developed by holding the cam stationary and rotating the follower from station 0 through stations 1, 2, 3, and so on.

We explain the procedure using the case of a plate cam with a radial roller follower, as shown in Fig. 6.8. Let us first note some additional nomenclature, illustrated in Fig. 6.8.

The *trace point* is a theoretic point of the follower, useful primarily in graphic constructions; it corresponds to the tip of a fictitious knife-edge follower. It is located at the center of a roller follower or along the surface of a flat-face follower.

The *pitch curve* is the locus generated by the trace point as the follower moves with respect to the cam. For a knife-edge follower, the pitch curve and cam profile are identical. For a roller follower they are separated by the radius of the roller.

The *prime circle* is the smallest circle that can be drawn tangent to the pitch curve with a center at the cam rotation axis. The radius of this circle is denoted R_0 .

The *base circle* is the smallest circle tangent to the cam profile centered on the cam rotation axis. For a roller follower, it is smaller than the prime circle by the radius of the roller; for a knife-edge or flat-face follower, it is identical to the prime circle.

In constructing the cam profile, we employ the principle of kinematic inversion. We imagine the sheet of paper on which we are working to be fixed to the cam, and we note that, as the cam rotates, the follower appears to rotate (with respect to the paper) *opposite to the direction of cam rotation*. As shown in Fig. 6.8, we divide the prime circle into a number of divisions and assign station numbers to these divisions. Dividing the displacement-diagram abscissa into corresponding divisions, we transfer $Y - Y_0$ distances from the displacement diagram directly onto the cam layout, measured along radial lines from the base circle to the trace point. The smooth curve through these points is the *pitch*

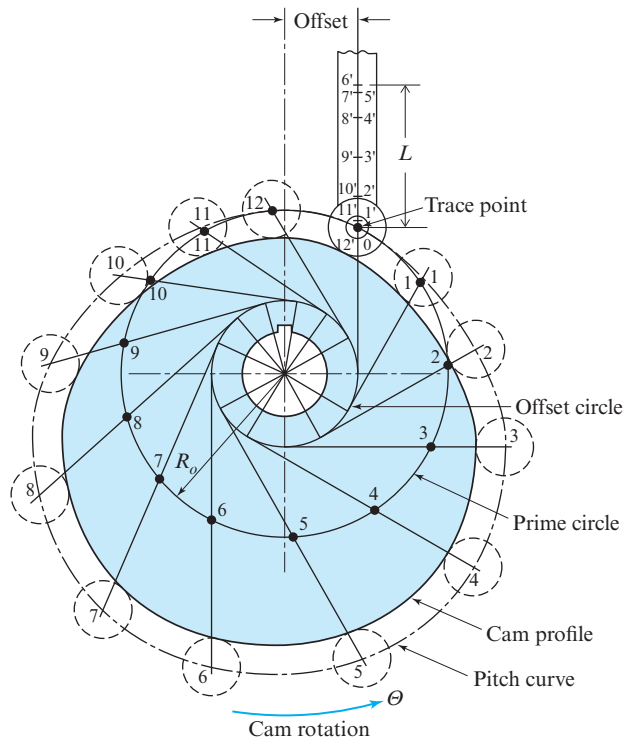


Figure 6.9 Graphic layout of a plate-cam profile with an offset reciprocating roller follower.

curve. For the case of a roller follower, as in this example, we simply draw the roller in its proper location at each station and then construct the cam profile as a smooth curve tangent to all of these roller locations.

Figure 6.9 shows how the method of construction is modified for an offset roller follower. We begin by constructing an *offset circle*, using a radius equal to the offset distance. After identifying station numbers around the prime circle, the centerline of the follower is constructed for each station, making it tangent to the offset circle. The roller centers for each station are established by transferring $Y - Y_0$ distances from the displacement diagram directly to these follower centerlines, always measuring positive outward from the prime circle. An alternative procedure is to identify points $0'$, $1'$, $2'$, and so on, on a single follower centerline and then to rotate them about the cam center to the corresponding follower centerline locations. In either case, the roller circle locations are drawn next and a smooth curve tangent to all roller locations is the required cam profile.

Figure 6.10 shows the construction for a plate cam with a reciprocating flat-face follower. The pitch curve is constructed using a method similar to that used for the roller follower in Fig. 6.8. Instead of roller locations, however, a line representing the flat face of the follower is constructed at each station. The cam profile is a smooth curve drawn tangent to all the follower location lines. It may be helpful to extend each straight line representing

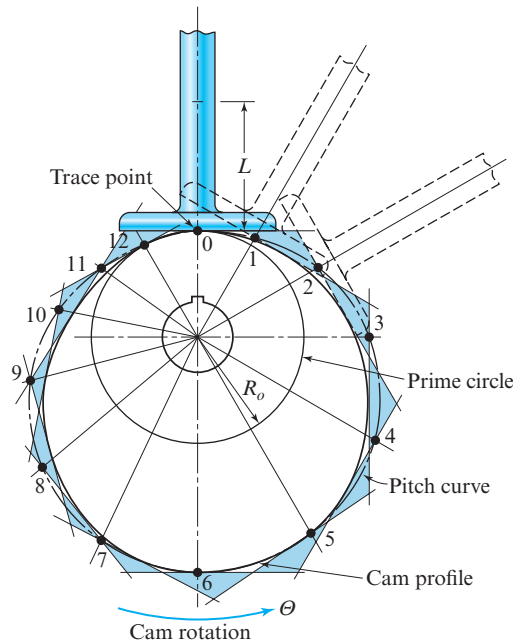


Figure 6.10 Graphic layout of a plate-cam profile with a reciprocating flat-face follower.

a location of the follower face to form a series of triangles. If these triangles are lightly shaded, as suggested in the illustration, it may be easier to draw the cam profile inside all the shaded triangles and tangent to the inner sides of the triangles. Note that the cam profile need not pass through points 0, 1, 2, 3, and so on, constructed from the displacement diagram.

Figure 6.11 shows the layout of the profile of a plate cam with an oscillating roller follower. In this case, to develop the cam profile, we must rotate the fixed pivot center of the follower opposite to the direction of cam rotation. To perform this inversion, first, a circle is drawn about the cam-shaft center through the fixed pivot of the follower. This circle is then divided and given station numbers to correspond to divisions on the displacement diagram. Next, arcs are drawn about each of these centers, all with equal radii corresponding to the length of the follower.

In the case of an oscillating follower, the ordinate values $Y - Y_0$ of the displacement diagram represent angular movements of the follower. If the ordinate scale of the displacement diagram is properly chosen initially, however, and if the total lift of the follower is a reasonably small angle, then ordinate distances from the displacement diagram at each station can be transferred directly to the corresponding arc traveled by the roller center using dividers and measuring positive outward along the arc from the prime circle to locate the trace point for each station. Finally, a circle representing the roller location is drawn with its center at the trace point for each station, and the cam profile is constructed as a smooth curve tangent to each of these roller locations.

From the examples presented in this section, it should be clear that each different type of cam-and-follower system requires its own graphic method of construction to

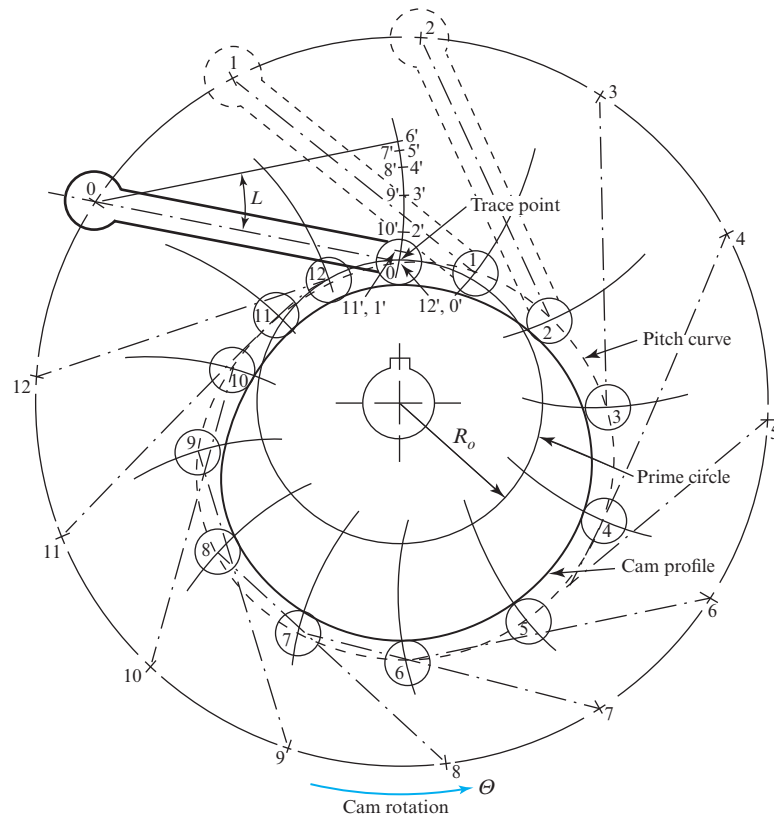


Figure 6.11 Graphic layout of a plate-cam profile with an oscillating roller follower.

determine the cam profile from the displacement diagram. The examples presented here are not intended to be exhaustive of those possible, but they illustrate the general approach. They also illustrate and reinforce the discussion of the previous section; it is now clear that much of the detailed shape of the cam itself results directly from the displacement diagram. Although different types of cams and followers have different shapes for the same displacement diagram, once a few parameters are chosen, such as the prime-circle radius, which determines the size of a cam, the remainder of its shape results directly from the motion requirements specified in the displacement diagram.

6.5 KINEMATIC COEFFICIENTS OF FOLLOWER

We have seen that, regardless of the type of cam or the type of follower, the displacement diagram is plotted with the cam input angle, Θ , as the abscissa and the follower output displacement, $Y - Y_0$, as the ordinate. This diagram is made up of a number of segments. In each segment, the abscissa is designated as θ and the ordinate as y . The displacement diagram is, therefore, a graph representing some mathematical function relating the input,

Θ , and output, Y , motions of the cam system. In general terms, one segment of this relationship is given by Eq. (6.1):

$$y = y(\theta).$$

Additional graphs can be plotted representing the derivatives of y with respect to the input θ —that is, the kinematic coefficients of the follower. The first-order kinematic coefficient of a segment is denoted

$$y'(\theta) = \frac{dy}{d\theta} \quad (6.2)$$

and represents the slope of the displacement diagram at the input position, θ , in the segment. The combined graph of first-order kinematic coefficients of all segments, although it may seem to be of little practical value, is a measure of the “steepness” of the displacement diagram throughout the cycle. We will find later that this is closely related to the mechanical advantage of the cam system and manifests itself in such things as the pressure angle (Sec. 6.10). If we consider a wedge cam (Fig. 6.1*b*) with a knife-edge follower (Fig. 6.2*a*), the displacement diagram itself is of the same shape as the corresponding cam. In such a case, we can visualize that difficulties will occur if any segment of the cam is too steep—that is, if the first-order kinematic coefficient y' of any segment has too high a value.

The second-order kinematic coefficient (that is, the second derivative of y with respect to input variable θ) of each segment is also significant. The second-order kinematic coefficient within a segment is denoted

$$y''(\theta) = \frac{d^2y}{d\theta^2}. \quad (6.3)$$

Although it is not as easy to visualize the reason, the second-order kinematic coefficient is very closely related to the curvature of the cam at locations along its profile. Recall that curvature is the reciprocal of the radius of curvature, [Sec. 4.1, Eq. (e)]. Therefore, if y'' becomes large, the radius of curvature becomes small. In particular, if the second-order kinematic coefficient becomes infinite, then the radius of curvature becomes zero; that is, the cam profile at such a position becomes pointed. This would be a highly unsatisfactory condition from the point of view of contact stress between the cam and the follower, and would very quickly cause surface damage.

The third-order kinematic coefficient of a segment, denoted

$$y'''(\theta) = \frac{d^3y}{d\theta^3}, \quad (6.4)$$

can also be plotted if desired. Although it is not easy to describe geometrically, this demonstrates the rate of change of y'' with respect to input variable θ . We will see that the third-order kinematic coefficient can also be controlled when choosing the detailed shape of the displacement diagram.

EXAMPLE 6.1

Derive equations to describe segments of a displacement diagram of a plate cam that rises with parabolic motion from a dwell to another dwell such that the total lift is L and the total cam angle is β . Plot these segments of the displacement diagram and the first-, second-, and third-order kinematic coefficients with respect to input variable θ . The abscissa of each segment graph should be normalized so that the ratio θ/β ranges from $\theta/\beta = 0$ at the left boundary to $\theta/\beta = 1$ at the right boundary of the segment.

SOLUTION

As shown in Fig. 6.5a, two parabolic segments are required, meeting at a common tangent taken here at midrange. For the first segment of the motion, we choose the general equation of a parabola; that is,

$$y = A\theta^2 + B\theta + C. \quad (1)$$

The first three derivatives of Eq. (1), with respect to input variable θ , are

$$y' = 2A\theta + B, \quad (2)$$

$$y'' = 2A, \quad (3)$$

$$y''' = 0. \quad (4)$$

To properly match the position and slope with those of the preceding dwell, at $\theta = 0$ we must have $y(0) = y'(0) = 0$. Thus, Eqs. (1) and (2) require that $B = C = 0$. Looking next at the inflection point, at $\theta = \beta/2$, we require $y = L/2$. Substituting these conditions into Eq. (1) and rearranging gives

$$A = \frac{2L}{\beta^2}.$$

Therefore, the displacement equation for the first segment of the *parabolic motion* becomes

$$y = 2L \left(\frac{\theta}{\beta} \right)^2. \quad (6.5a)$$

Differentiating this equation with respect to input variable θ , the first-, second-, and third-order kinematic coefficients, respectively, are

$$y' = \frac{4L}{\beta} \left(\frac{\theta}{\beta} \right), \quad (6.5b)$$

$$y'' = \frac{4L}{\beta^2}, \quad (6.5c)$$

$$y''' = 0. \quad (6.5d)$$

The maximum value for the first-order kinematic coefficient (that is, the maximum slope of y) occurs at the midpoint, where $\theta = \beta/2$. Substituting this value into Eq. (6.5b), the

maximum value for the first-order kinematic coefficient is

$$y'_{\max} = \frac{2L}{\beta}. \quad (5)$$

For the second segment of the parabolic motion, we return to the general equations (1) through (4) for a parabola. Substituting the conditions that $y = L$ and $y' = 0$ at $\theta = \beta$ into Eqs. (1) and (2) gives

$$L = A\beta^2 + B\beta + C, \quad (6)$$

$$0 = 2A\beta + B. \quad (7)$$

Since the slope must match that of the first parabola at $\theta = \beta/2$, then from Eqs. (2) and (5) we have

$$\frac{2L}{\beta} = 2A\frac{\beta}{2} + B. \quad (8)$$

Solving Eqs. (6), (7), and (8) simultaneously gives

$$A = -\frac{2L}{\beta^2}, \quad B = \frac{4L}{\beta}, \quad C = -L. \quad (9)$$

Substituting these constants into Eq. (1), the displacement equation for the second segment of the parabolic motion can be written as

$$y = L \left[1 - 2 \left(1 - \frac{\theta}{\beta} \right)^2 \right]. \quad (6.6a)$$

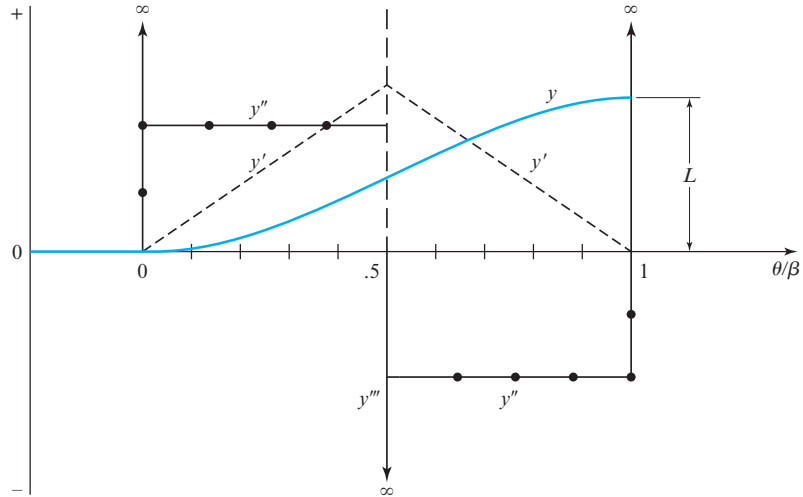


Figure 6.12 Displacement diagram and derivatives for full-rise parabolic motion, Eqs. (6.5) and (6.6).

Also, substituting Eqs. (9) into Eqs. (2), (3), and (4), the first-, second-, and third-order kinematic coefficients for the second segment of the parabolic motion, respectively, are

$$y' = \frac{4L}{\beta} \left(1 - \frac{\theta}{\beta} \right), \quad (6.6b)$$

$$y'' = -\frac{4L}{\beta^2}, \quad (6.6c)$$

$$y''' = 0. \quad (6.6d)$$

The displacement diagram and the first-, second-, and third-order kinematic coefficients for this example of *full-rise parabolic motion* are shown in Fig. 6.12.

The previous discussion relates to the kinematic coefficients of the follower motion. These coefficients are derivatives with respect to the input variable, θ , and relate to the geometry of the cam system. Let us now consider the derivatives of the follower motion with respect to time. First, we assume that the time variation of the input motion, $\Theta(t)$, and, therefore, $\theta(t)$, is known. The angular velocity, $\omega = d\theta/dt$, the angular acceleration, $\alpha = d^2\theta/dt^2$, and the next derivative (often called *angular jerk* or *second angular acceleration*), $\dot{\alpha} = d^3\theta/dt^3$, are all assumed to be known. Usually, a plate cam is driven by a constant-speed input shaft. In this case, ω is a known constant, $\theta = \omega t$, and $\alpha = \dot{\alpha} = 0$. During start-up of the cam system, however, this is not the case, and we will consider the more general situation first.

From the general equation of the displacement diagram, for chosen segment number k , we can write from Eqs. (a) and (b) in Sec. 6.3

$$y = Y - Y_k = y(\theta) \quad \text{and} \quad \theta = \Theta - \Theta_k = \theta(t). \quad (6.7)$$

Therefore, we can differentiate to find the time derivatives of the follower motion. The velocity of the follower, for example, is given by

$$\dot{y} = \frac{dy}{dt} = \left(\frac{dy}{d\theta} \right) \left(\frac{d\theta}{dt} \right),$$

which, using the first-order kinematic coefficient, can be written as

$$\dot{y} = y' \omega. \quad (6.8)$$

Similarly, the acceleration and the jerk (the third time derivative) of the follower can be written, respectively, as

$$\ddot{y} = \frac{d^2y}{dt^2} = y'' \omega^2 + y' \alpha \quad (6.9)$$

and

$$\ddot{\ddot{y}} = \frac{d^3y}{dt^3} = y''' \omega^3 + 3y'' \omega \alpha + y' \dot{\alpha}. \quad (6.10)$$

When the camshaft speed is constant, then $\alpha = 0$, and Eqs. (6.8) through (6.10) reduce to

$$\dot{y} = y'\omega, \quad \ddot{y} = y''\omega^2, \quad \dddot{y} = y'''\omega^3. \quad (6.11)$$

For this reason, it has become somewhat common to refer to the graphs of the kinematic coefficients y' , y'' , y''' , such as those shown in Fig. 6.12, as the “velocity,” “acceleration,” and “jerk” curves for a given segment of the motion. These are only appropriate names for a constant-speed cam, and then only when scaled by ω , ω^2 , and ω^3 , respectively.* However, it is helpful to use these names for the kinematic coefficients when considering the physical implications of a certain choice of displacement diagram. For example, considering the parabolic motion of Fig. 6.12, it is intuitively meaningful to say that the “velocity” of the follower rises linearly to a maximum at the midpoint $\theta = \beta/2$ and then decreases linearly to zero. The “acceleration” of the follower is zero during the initial dwell and changes abruptly (that is, a step change) to a constant positive value upon beginning the rise. There are two more step changes in the “acceleration” of the follower—namely, at the midpoint and at the end of the rise. At each of these three step changes in the “acceleration” of the follower, the “jerk” of the follower becomes infinite.

6.6 HIGH-SPEED CAMS

Continuing with our discussion of parabolic motion, let us consider briefly the implications of the “acceleration” curve segments of Fig. 6.12 on the dynamic performance of the cam system. Any real follower, of course, has at least some mass and, when this is multiplied by acceleration, exerts an inertia force (Chap. 12). Therefore, the “acceleration” curve of Fig. 6.12 can also be thought of as indicating the inertia force of the follower, which, in turn, is felt at the follower bearings and at the contact point on the cam surface. An “acceleration” curve with abrupt changes (that is, where the “jerk” becomes infinite), such as those demonstrated for parabolic motion, exert abruptly changing contact stresses at the bearings and on the cam surface, and will lead to noise, surface wear, and early failure. Thus, it is very important in choosing and joining segments of a displacement diagram to ensure that the first- and second-order kinematic coefficients (that is, the “velocity” and “acceleration” curves) are continuous—meaning, that they contain no step changes.

Sometimes in low-speed cam applications compromises are made with the “velocity” and “acceleration” relationships. It is sometimes simpler to employ a reverse procedure and design the cam shape first, obtaining the displacement diagram as a second step. Such cams are sometimes composed of a combination of curves, such as straight lines and circular arcs, which are readily produced by machine tools. Two examples are the *circle-arc cam* and the *tangent cam* shown in Fig. 6.13. The design approach is by iteration. A trial cam is designed and its kinematic characteristics are found. The process is then repeated until a cam with acceptable characteristics is obtained. Points *A*, *B*, *C*, and *D* of the circle-arc cam

* Accepting the word “velocity” literally, for example, leads to consternation when it is discovered that, for a plate cam with a reciprocating follower, the units of “velocity” y' are length per radian. Multiplying these units by radians per second, the units of ω , gives units of length per second for $\dot{y}$, however.

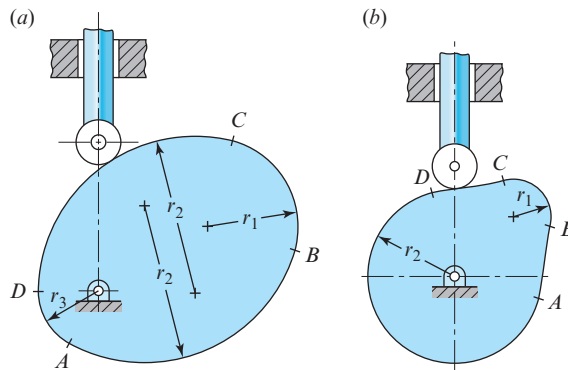


Figure 6.13 (a) Circle-arc cam; (b) tangent cam.

and the tangent cam are points of tangency or blending points. It is worth noting, as with the previous parabolic-motion example, that the acceleration changes abruptly at each of the blending points because of the instantaneous change in the radius of curvature of the cam profile.

Although cams with discontinuous acceleration characteristics have sometimes been accepted to save cost in low-speed applications, such cams have invariably exhibited major difficulties at some later time when the input speed of the machine was raised to increase the productivity of the application. For any high-speed cam application, it is extremely important that not only the displacement and “velocity” curves, but also the “acceleration” curve be made continuous for the entire motion cycle. No discontinuities should be allowed within or at the junctions of different segments of a cam.

As confirmed by Eq. (6.11), the importance of continuous derivatives becomes more serious as the cam-shaft speed is increased. The higher the speed, the greater the need for smooth curves. At very high speeds, it might also be desirable to require that jerk, which is related to rate of change of force, and perhaps even higher derivatives, be made continuous as well. In many applications, however, this is not necessary.

There is no simple answer as to how high a speed one must have before considering the application to require high-speed design techniques. The answer depends not only on the mass of the follower, but also on the stiffness of the return spring, the materials used, the flexibility of the follower, and many other factors [9]. Further analysis techniques on cam dynamics are presented in Secs. 6.11 to 6.16. Still, with the methods presented here, it is not difficult to achieve continuous displacement diagrams with continuous derivatives. Therefore, it is recommended that this be undertaken as standard practice. Cycloidal-motion cams, for example, are no more difficult to manufacture than parabolic-motion cams, and there is no good reason for use of the latter. The circle-arc cam and the tangent cam may be easy to produce, but, with modern machining methods, cutting more complex cam shapes is not expensive and is recommended.

6.7 STANDARD CAM MOTIONS

Example 6.1 in Sec. 6.5 gave a detailed derivation of the equations for parabolic motion and its first three derivatives [Eqs. (6.5) and (6.6)]. Then, in Sec. 6.6, reasons were provided

for avoiding the use of parabolic motion in high-speed cam systems. The purpose of this section is to present equations for a number of standard types of displacement curve segments that can be used to address most high-speed cam-motion requirements. The derivations parallel those of Example 6.1 and are not presented.

The displacement equation and the first-, second-, and third-order kinematic coefficients for a *full-rise simple harmonic motion* segment are

$$y = \frac{L}{2} \left(1 - \cos \frac{\pi \theta}{\beta} \right), \quad (6.12a)$$

$$y' = \frac{\pi L}{2\beta} \sin \frac{\pi \theta}{\beta}, \quad (6.12b)$$

$$y'' = \frac{\pi^2 L}{2\beta^2} \cos \frac{\pi \theta}{\beta}, \quad (6.12c)$$

$$y''' = -\frac{\pi^3 L}{2\beta^3} \sin \frac{\pi \theta}{\beta}. \quad (6.12d)$$

The displacement diagram and the first-, second-, and third-order kinematic coefficients for a full-rise simple harmonic motion segment are shown in Fig. 6.14. Unlike parabolic motion, simple harmonic motion exhibits no discontinuity at the inflection point, but it does contain nonzero “accelerations” at its two boundaries.

The displacement equation and the first-, second-, and third-order kinematic coefficients for a *full-rise cycloidal motion* segment are

$$y = L \left(\frac{\theta}{\beta} - \frac{1}{2\pi} \sin \frac{2\pi \theta}{\beta} \right), \quad (6.13a)$$

$$y' = \frac{L}{\beta} \left(1 - \cos \frac{2\pi \theta}{\beta} \right), \quad (6.13b)$$

$$y'' = \frac{2\pi L}{\beta^2} \sin \frac{2\pi \theta}{\beta}, \quad (6.13c)$$

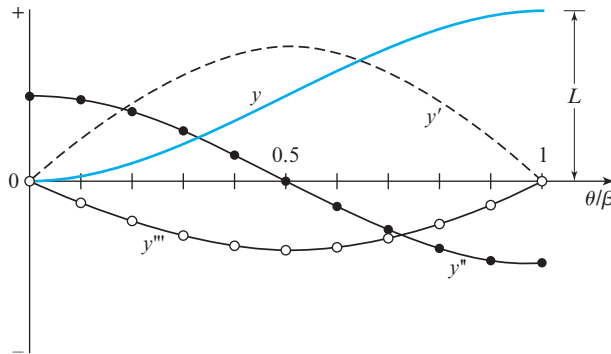


Figure 6.14 Displacement diagram and derivatives for a full-rise simple harmonic motion segment, Eqs. (6.12).

$$y''' = \frac{4\pi^2 L}{\beta^3} \cos \frac{2\pi\theta}{\beta}. \quad (6.13d)$$

The displacement diagram and the first-, second-, and third-order kinematic coefficients for a full-rise cycloidal motion segment are shown in Fig. 6.15. Note that all derivatives at the boundaries of this segment have zero values. Note also that this is the only standard motion with zeroes for all derivatives at the boundaries. However, the peak “velocity,” “acceleration,” and “jerk” values are higher than those for a simple harmonic motion segment.

The displacement equation and the first-, second-, and third-order kinematic coefficients for a *full-rise eighth-order polynomial motion* segment are

$$y = L \left[6.097\,55 \left(\frac{\theta}{\beta} \right)^3 - 20.780\,40 \left(\frac{\theta}{\beta} \right)^5 + 26.731\,55 \left(\frac{\theta}{\beta} \right)^6 - 13.609\,65 \left(\frac{\theta}{\beta} \right)^7 + 2.560\,95 \left(\frac{\theta}{\beta} \right)^8 \right], \quad (6.14a)$$

$$y' = \frac{L}{\beta} \left[18.292\,65 \left(\frac{\theta}{\beta} \right)^2 - 103.902\,00 \left(\frac{\theta}{\beta} \right)^4 + 160.389\,30 \left(\frac{\theta}{\beta} \right)^5 - 95.267\,55 \left(\frac{\theta}{\beta} \right)^6 + 20.487\,60 \left(\frac{\theta}{\beta} \right)^7 \right], \quad (6.14b)$$

$$y'' = \frac{L}{\beta^2} \left[36.585\,30 \left(\frac{\theta}{\beta} \right) - 415.608\,00 \left(\frac{\theta}{\beta} \right)^3 + 801.946\,50 \left(\frac{\theta}{\beta} \right)^4 - 571.605\,30 \left(\frac{\theta}{\beta} \right)^5 + 143.413\,20 \left(\frac{\theta}{\beta} \right)^6 \right], \quad (6.14c)$$

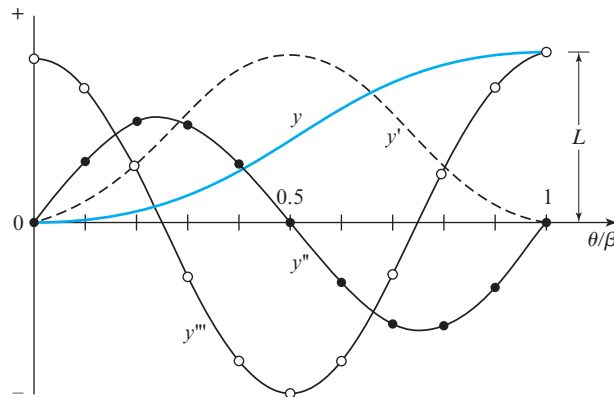


Figure 6.15 Displacement diagram and derivatives for a full-rise cycloidal motion segment, Eqs. (6.13).

$$y''' = \frac{L}{\beta^3} \left[36.585\,30 - 1\,246.824\,00 \left(\frac{\theta}{\beta} \right)^2 + 3\,207.786\,00 \left(\frac{\theta}{\beta} \right)^3 - 2\,858.026\,50 \left(\frac{\theta}{\beta} \right)^4 + 860.479\,20 \left(\frac{\theta}{\beta} \right)^5 \right]. \quad (6.14d)$$

The displacement diagram and the first-, second-, and third-order kinematic coefficients for the full-rise motion segment formed from an eighth-order polynomial are shown in Fig. 6.16. Equations (6.14) have seemingly awkward coefficients, since they have been specially derived to have many “nice” properties [6]. Among these, Fig. 6.16 shows not only that several of the kinematic coefficients are zero at both ends of the segment, but also that the “acceleration” characteristics are nonsymmetric. Also, the peak values of “acceleration” are kept as small as possible (that is, the magnitudes of the positive and negative peak “accelerations” are equal).

The displacement diagrams of simple harmonic, cycloidal, and eighth-order polynomial motions segments look quite similar at first glance. Each rises through a lift of L in a cam rotation angle of β , and each begins and ends with zero slope. For these reasons, they are all referred to as *full-rise* motion segments. However, their “acceleration” curves are quite different. A simple harmonic motion segment has nonzero “acceleration” at the boundaries, a cycloidal motion segment has zero “acceleration” at both boundaries, and an eighth-order polynomial motion segment has one zero and one nonzero “acceleration” at its two boundaries. This variety provides the selections necessary when matching these curves with neighboring curves of different types.

Full-return motion segments of the same three types are shown in Figs. 6.17 through 6.19.

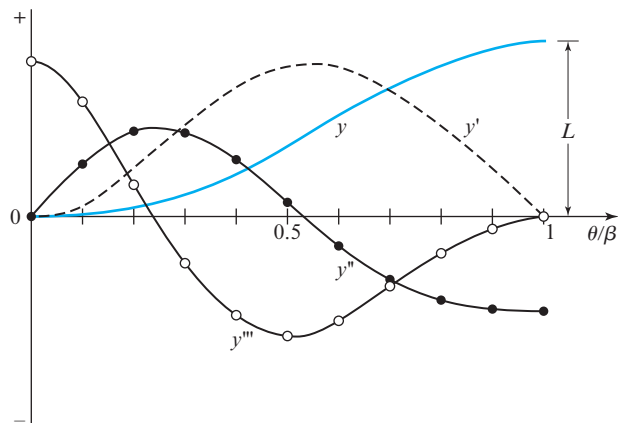


Figure 6.16 Displacement diagram and derivatives for a full-rise eighth-order polynomial motion segment, Eqs. (6.14).

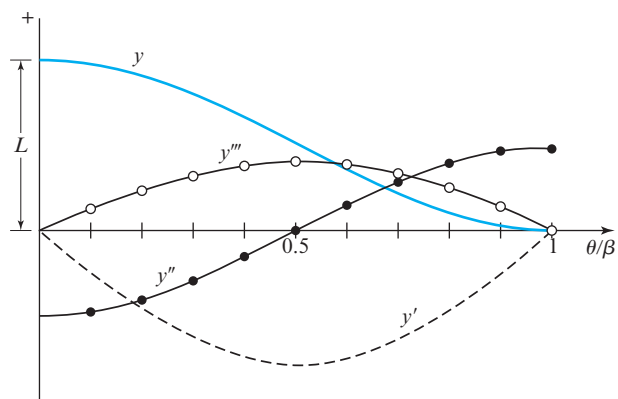


Figure 6.17 Displacement diagram and derivatives for a full-return simple harmonic motion segment, Eqs. (6.15).

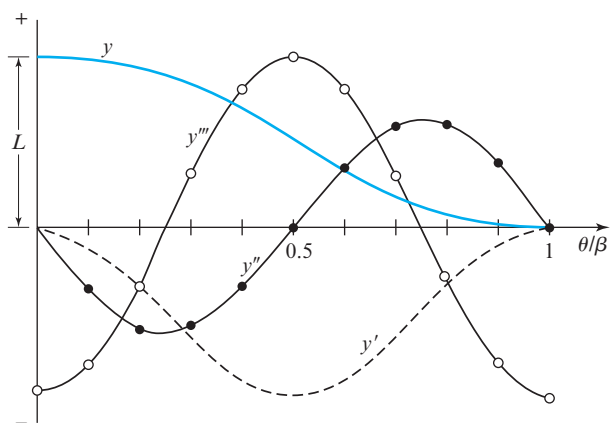


Figure 6.18 Displacement diagram and derivatives for a full-return cycloidal motion segment, Eqs. (6.16).

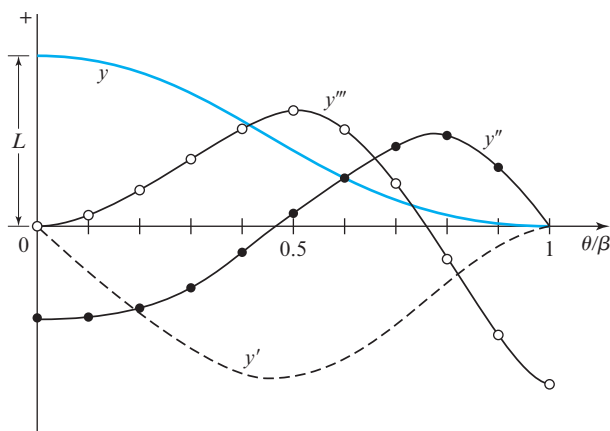


Figure 6.19 Displacement diagram and derivatives for a full-return eighth-order polynomial motion segment, Eqs. (6.17).

The displacement equation and the first-, second-, and third-order kinematic coefficients for a *full-return simple harmonic motion* segment are

$$y = \frac{L}{2} \left(1 + \cos \frac{\pi \theta}{\beta} \right), \quad (6.15a)$$

$$y' = -\frac{\pi L}{2\beta} \sin \frac{\pi \theta}{\beta}, \quad (6.15b)$$

$$y'' = -\frac{\pi^2 L}{2\beta^2} \cos \frac{\pi \theta}{\beta}, \quad (6.15c)$$

$$y''' = \frac{\pi^3 L}{2\beta^3} \sin \frac{\pi \theta}{\beta}. \quad (6.15d)$$

For a *full-return cycloidal motion* segment, the displacement equation and the first-, second-, and third-order kinematic coefficients are

$$y = L \left(1 - \frac{\theta}{\beta} + \frac{1}{2\pi} \sin \frac{2\pi \theta}{\beta} \right), \quad (6.16a)$$

$$y' = -\frac{L}{\beta} \left(1 - \cos \frac{2\pi \theta}{\beta} \right), \quad (6.16b)$$

$$y'' = -\frac{2\pi L}{\beta^2} \sin \frac{2\pi \theta}{\beta}, \quad (6.16c)$$

$$y''' = -\frac{4\pi^2 L}{\beta^3} \cos \frac{2\pi \theta}{\beta}. \quad (6.16d)$$

For a *full-return eighth-order polynomial motion* segment, the displacement equation and the first-, second-, and third-order kinematic coefficients are

$$y = L \left[1.000 \ 00 - 2.634 \ 15 \left(\frac{\theta}{\beta} \right)^2 + 2.780 \ 55 \left(\frac{\theta}{\beta} \right)^5 + 3.170 \ 60 \left(\frac{\theta}{\beta} \right)^6 - 6.877 \ 95 \left(\frac{\theta}{\beta} \right)^7 + 2.560 \ 95 \left(\frac{\theta}{\beta} \right)^8 \right], \quad (6.17a)$$

$$y' = -\frac{L}{\beta} \left[5.268 \ 30 \frac{\theta}{\beta} - 13.902 \ 75 \left(\frac{\theta}{\beta} \right)^4 - 19.023 \ 60 \left(\frac{\theta}{\beta} \right)^5 + 48.145 \ 65 \left(\frac{\theta}{\beta} \right)^6 - 20.487 \ 60 \left(\frac{\theta}{\beta} \right)^7 \right], \quad (6.17b)$$

$$y'' = -\frac{L}{\beta^2} \left[5.268\,30 - 55.611\,00 \left(\frac{\theta}{\beta} \right)^3 - 95.118\,00 \left(\frac{\theta}{\beta} \right)^4 + 288.873\,90 \left(\frac{\theta}{\beta} \right)^5 - 143.413\,20 \left(\frac{\theta}{\beta} \right)^6 \right], \quad (6.17c)$$

$$y''' = \frac{L}{\beta^3} \left[166.833\,00 \left(\frac{\theta}{\beta} \right)^2 + 380.472\,00 \left(\frac{\theta}{\beta} \right)^3 - 1\,444.369\,50 \left(\frac{\theta}{\beta} \right)^4 + 860.479\,20 \left(\frac{\theta}{\beta} \right)^5 \right]. \quad (6.17d)$$

Polynomial displacement equations of much higher order, and meeting many more conditions than those presented here are also in common use. Automated procedures for determining the coefficients have been developed by Stoddart [8], who also indicates how the choice of coefficients can be made to compensate for elastic deformation of the follower system under dynamic conditions. Such cams are referred to as *polydyne cams*.

In addition to the full-rise and full-return motions presented earlier, it is also useful to have a selection of standard *half-rise* and *half-return* motion segments available. These are curves for which one segment boundary has a nonzero slope and can be used to blend with uniform motion. The displacement diagrams and the first-, second-, and third-order kinematic coefficients for *half-rise simple harmonic motion* segments, sometimes called *half-harmonic rise motion* segments, are shown in Fig. 6.20. The equations corresponding to Fig. 6.20a are

$$y = L \left(1 - \cos \frac{\pi \theta}{2\beta} \right), \quad (6.18a)$$

$$y' = \frac{\pi L}{2\beta} \sin \frac{\pi \theta}{2\beta}, \quad (6.18b)$$

$$y'' = \frac{\pi^2 L}{4\beta^2} \cos \frac{\pi \theta}{2\beta}, \quad (6.18c)$$

$$y''' = -\frac{\pi^3 L}{8\beta^3} \sin \frac{\pi \theta}{2\beta}. \quad (6.18d)$$

The displacement equation and the first-, second-, and third-order kinematic coefficients corresponding to the half-rise simple harmonic motion segments of Fig. 6.20b are

$$y = L \sin \frac{\pi \theta}{2\beta}, \quad (6.19a)$$

$$y' = \frac{\pi L}{2\beta} \cos \frac{\pi \theta}{2\beta}, \quad (6.19b)$$

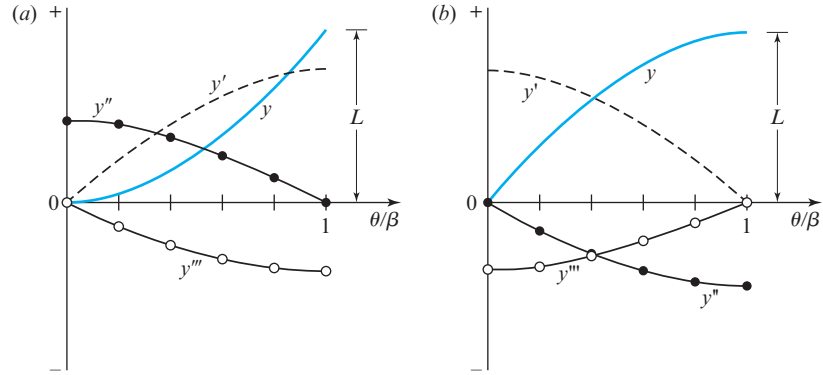


Figure 6.20 Displacement diagram and derivatives for half-rise simple harmonic motion segments: (a) Eqs. (6.18); (b) Eqs. (6.19).

$$y'' = -\frac{\pi^2 L}{4\beta^2} \sin \frac{\pi\theta}{2\beta}, \quad (6.19c)$$

$$y''' = -\frac{\pi^3 L}{8\beta^3} \cos \frac{\pi\theta}{2\beta}. \quad (6.19d)$$

The curves for *half-return simple harmonic motion* segments are shown in Fig. 6.21. The equations corresponding to Fig. 6.21a are

$$y = L \cos \frac{\pi\theta}{2\beta}, \quad (6.20a)$$

$$y' = -\frac{\pi L}{2\beta} \sin \frac{\pi\theta}{2\beta}, \quad (6.20b)$$

$$y'' = -\frac{\pi^2 L}{4\beta^2} \cos \frac{\pi\theta}{2\beta}, \quad (6.20c)$$

$$y''' = \frac{\pi^3 L}{8\beta^3} \sin \frac{\pi\theta}{2\beta}. \quad (6.20d)$$

The displacement equation and the first-, second-, and third-order kinematic coefficients corresponding to the half-return simple harmonic motion segments of Fig. 6.21b are

$$y = L \left(1 - \sin \frac{\pi\theta}{2\beta} \right), \quad (6.21a)$$

$$y' = -\frac{\pi L}{2\beta} \cos \frac{\pi\theta}{2\beta}, \quad (6.21b)$$

$$y'' = \frac{\pi^2 L}{4\beta^2} \sin \frac{\pi\theta}{2\beta}, \quad (6.21c)$$

$$y''' = \frac{\pi^3 L}{8\beta^3} \cos \frac{\pi\theta}{2\beta}. \quad (6.21d)$$

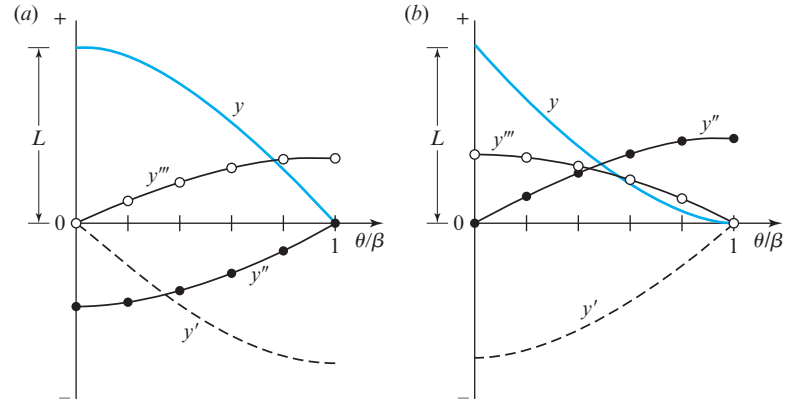


Figure 6.21 Displacement diagram and derivatives for half-return simple harmonic motion segments: (a) Eqs. (6.20); (b) Eqs. (6.21).

In addition to the half-harmonics, half-cycloidal motion segments are also useful, since their “accelerations” are zero at both segment boundaries. The displacement diagrams and first-, second-, and third-order kinematic coefficients for *half-rise cycloidal motion* segments are shown in Fig. 6.22. The equations corresponding to Fig. 6.22a are

$$y = L \left(\frac{\theta}{\beta} - \frac{1}{\pi} \sin \frac{\pi \theta}{\beta} \right), \quad (6.22a)$$

$$y' = \frac{L}{\beta} \left(1 - \cos \frac{\pi \theta}{\beta} \right), \quad (6.22b)$$

$$y'' = \frac{\pi L}{\beta^2} \sin \frac{\pi \theta}{\beta}, \quad (6.22c)$$

$$y''' = \frac{\pi^2 L}{\beta^3} \cos \frac{\pi \theta}{\beta}. \quad (6.22d)$$

The displacement equation and the first-, second-, and third-order kinematic coefficients corresponding to the *half-rise cycloidal motion* segments of Fig. 6.22b are

$$y = L \left(\frac{\theta}{\beta} + \frac{1}{\pi} \sin \frac{\pi \theta}{\beta} \right), \quad (6.23a)$$

$$y' = \frac{L}{\beta} \left(1 + \cos \frac{\pi \theta}{\beta} \right), \quad (6.23b)$$

$$y'' = -\frac{\pi L}{\beta^2} \sin \frac{\pi \theta}{\beta}, \quad (6.23c)$$

$$y''' = -\frac{\pi^2 L}{\beta^3} \cos \frac{\pi \theta}{\beta}. \quad (6.23d)$$

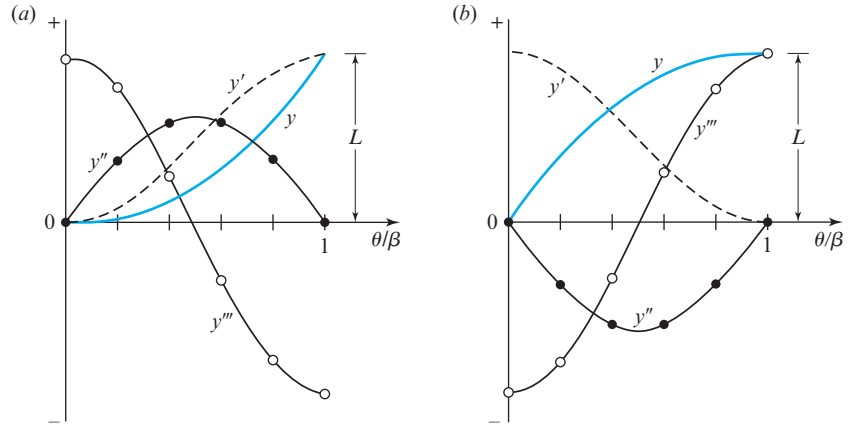


Figure 6.22 Displacement diagram and derivatives for half-rise cycloidal motion segments: (a) Eqs. (6.22); (b) Eqs. (6.23).

The curves for *half-return cycloidal motion* segments are shown in Fig. 6.23. The equations corresponding to Fig. 6.23a are

$$y = L \left(1 - \frac{\theta}{\beta} + \frac{1}{\pi} \sin \frac{\pi \theta}{\beta} \right), \quad (6.24a)$$

$$y' = -\frac{L}{\beta} \left(1 - \cos \frac{\pi \theta}{\beta} \right), \quad (6.24b)$$

$$y'' = -\frac{\pi L}{\beta^2} \sin \frac{\pi \theta}{\beta}, \quad (6.24c)$$

$$y''' = -\frac{\pi^2 L}{\beta^3} \cos \frac{\pi \theta}{\beta}. \quad (6.24d)$$

The displacement equation and the first-, second-, and third-order kinematic coefficients corresponding to the half-return cycloidal motion segments of Fig. 6.23b are

$$y = L \left(1 - \frac{\theta}{\beta} - \frac{1}{\pi} \sin \frac{\pi \theta}{\beta} \right), \quad (6.25a)$$

$$y' = -\frac{L}{\beta} \left(1 + \cos \frac{\pi \theta}{\beta} \right), \quad (6.25b)$$

$$y'' = \frac{\pi L}{\beta^2} \sin \frac{\pi \theta}{\beta}, \quad (6.25c)$$

$$y''' = \frac{\pi^2 L}{\beta^3} \cos \frac{\pi \theta}{\beta}. \quad (6.25d)$$

We will see shortly how the “standard” segment graphs and equations presented in this section can greatly reduce the analytic effort involved in designing the full displacement

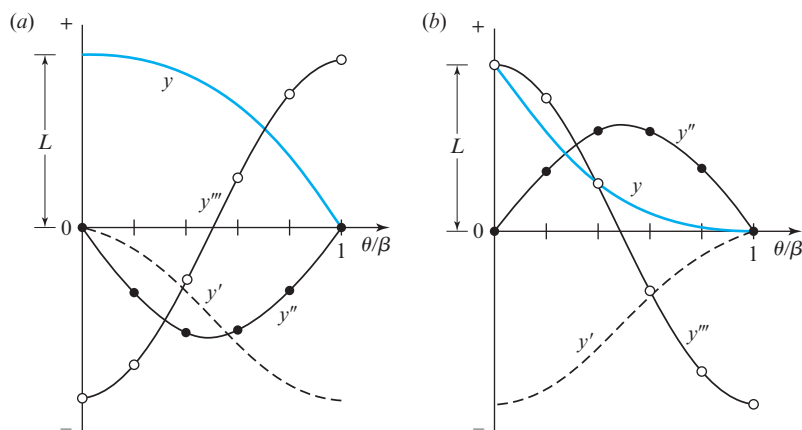


Figure 6.23 Displacement diagram and derivatives for half-return cycloidal motion segments: (a) Eqs. (6.24); (b) Eqs. (6.25).

diagram for a high-speed cam. First, however, we will summarize several important features of the segment graphs of Fig. 6.14 through Fig. 6.23:

1. Each graph shows one segment of a full displacement diagram.
2. The total lift for that segment is labeled L for each, and the total cam travel is labeled β .
3. The abscissa of each segment graph is normalized so that the ratio θ/β ranges from $\theta/\beta = 0$ at one boundary to $\theta/\beta = 1$ at the other boundary of the segment.
4. The scales used in plotting the graphs are not depicted but are consistent for all full-rise and full-return curves, and for all half-rise and half-return curves. Thus, in judging the suitability of one curve compared with another, the magnitudes of the “accelerations,” for example, can be compared. For this reason, when other factors are equivalent, simple harmonic motion should be used where possible and, in order to keep “accelerations” small, cycloidal motion should be avoided except where necessary.

Finally, it should be noted that the standard cam motion segments presented in this section do not form an exhaustive set. The set presented here is sufficient for most practical applications. However, cams with good dynamic characteristics can also be formed from a wide variety of other possible motion segment curves. A much more extensive set can be found, for example, in the text by F. Y. Chen [2].

6.8 MATCHING DERIVATIVES OF DISPLACEMENT DIAGRAMS

In the previous section, a great many equations were presented that might be used to represent different segments of the displacement diagram of a cam. In this section, we will study how these can be joined together to form the motion specification for a complete

cam. The procedure is one of solving for proper values of L and β for each segment so that:

1. The motion requirements of the particular application are met.
2. The displacement diagram, as well as the diagrams of the first- and second-order kinematic coefficients, are continuous across the boundaries of the merged segments. The diagram of the third-order kinematic coefficient may be allowed discontinuities if necessary, but must not become infinite; that is, the “acceleration” curve may contain corners but not discontinuities (jumps).
3. The maximum magnitudes of the “velocity” and “acceleration” peaks are kept as low as possible consistent with the first two conditions.

The procedure may best be understood through an example.

EXAMPLE 6.2

A plate cam with a reciprocating follower is to be driven by a constant-speed motor at 150 rpm. The follower is to start from a dwell, accelerate to a uniform velocity of 25 in/s, maintain this velocity for 1.25 in of rise, decelerate to the top of the lift, return, and then dwell for 0.10 s. The total lift is to be 3.00 in. Determine the complete specifications of the displacement diagram.

SOLUTION

The speed of the input shaft is

$$\omega = 150 \text{ rev/min} = 15.707 \text{ 96 rad/s.} \quad (1)$$

Using Eq. (6.8), the first-order kinematic coefficient (that is, the slope of the uniform “velocity” segment) is

$$y' = \frac{\dot{y}}{\omega} = \frac{25 \text{ in/s}}{15.707 \text{ 96 rad/s}} = 1.591 \text{ 55 in/rad.} \quad (2)$$

Since this “velocity” is held constant for 1.25 in of rise, the total cam rotation in this segment is

$$\beta_2 = \frac{L_2}{y'} = \frac{1.25 \text{ in}}{1.591 \text{ 55 in/rad}} = 0.785 \text{ 40 rad} = 45.000^\circ. \quad (3)$$

Similarly, from Eq. (1), the total cam rotation during the final dwell is

$$\beta_5 = 0.10 \text{ s} (15.707 \text{ 96 rad/s}) = 1.570 \text{ 796 rad} = 90.000^\circ. \quad (4)$$

Note that several digits of accuracy higher than usual are utilized here and are recommended as standard practice when matching cam motion derivatives. Any inaccuracies in the L and β values result in discontinuities in the smoothness of derivatives at the boundaries of the segments and discontinuities in force, as explained in Sec. 6.6.

From these results and the given information, we can sketch the beginnings of the displacement diagram, not necessarily working to scale, but to visualize the motion requirements. This gives the general shapes shown by the heavy segments of the curves of Fig. 6.24a. The lighter segments of the displacement curve are not yet accurately known, but can be sketched by lightly outlining a smooth curve for visualization. Working from this approximate curve, we can also sketch the general nature of the derivative curves. From the changing slope of the displacement diagram we sketch the “velocity” curve (Fig. 6.24b), and from the changing slope of this curve we sketch the “acceleration” curve (Fig. 6.24c). At this stage, no attempt is made to produce accurate curves drawn to scale but only to provide an idea of the desired curve shapes.

Next, using the sketches of Fig. 6.24, we compare these desired motion curves with the various standard curve segments of Figs. 6.14 through 6.23 to choose an appropriate set of equations for each segment of the cam. In segment AB , for example, we find that Fig. 6.22a is the only standard-motion curve segment available with half-rise characteristics,

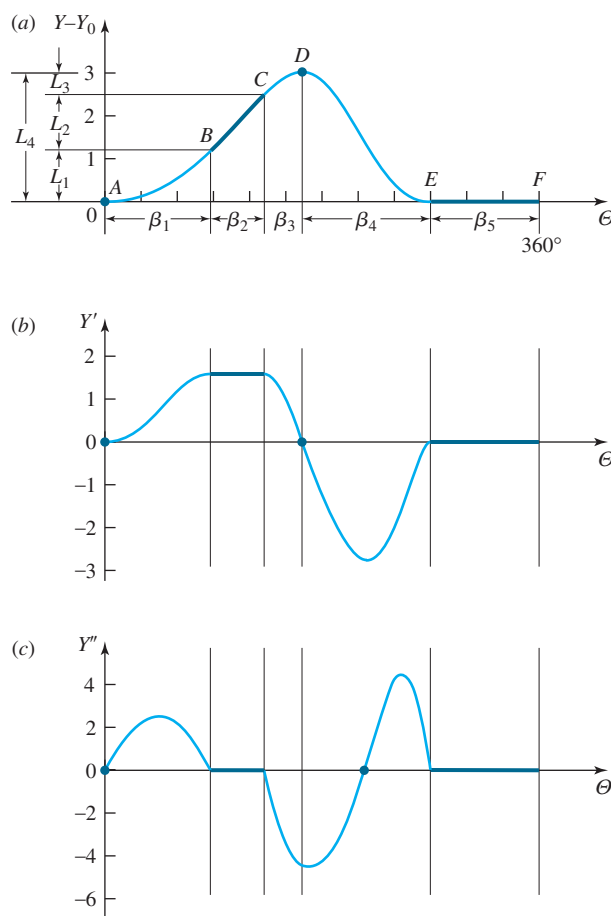


Figure 6.24 (a) Displacement diagram, in; (b) “velocity” diagram, in/rad; (c) “acceleration” diagram, in/rad².

an appropriate slope curve, and the necessary zero “acceleration” at both boundaries of the segment. Thus, we choose the half-rise cycloidal motion of Eq. (6.22) for this segment of the cam. There are two sets of choices possible for segments CD and DE . One set might be the choice of Fig. 6.22*b* matched with Fig. 6.18. However, to keep the peak “accelerations” low and to keep the “jerk” curves as smooth as possible, we choose Fig. 6.20*b* matched with Fig. 6.19. Thus, for segment CD , we will use the half-rise harmonic motion of Eq. (6.19), and for segment DE , we choose the eighth-order polynomial return motion of Eq. (6.17).

Choosing the curve types, however, is not sufficient to fully specify the segment characteristics. We must also find values for the unknown parameters of the segment equations; these are L_1 , L_3 , β_1 , β_3 , and β_4 . We do this by equating the kinematic coefficients at each nonzero segment boundary. For example, to match the “velocities” at point B we must equate the first-order kinematic coefficient from Eq. (6.22*b*) at the AB segment right boundary (that is, at $\theta_1/\beta_1 = 1$) with the first-order kinematic coefficient of the BC segment; that is,

$$y'_B = \frac{2L_1}{\beta_1} = \frac{L_2}{\beta_2} = \frac{1.25 \text{ in}}{0.78540 \text{ rad}} = 1.59155 \text{ in/rad}$$

or

$$L_1 = (0.79577 \text{ in/rad}) \beta_1. \quad (5)$$

Similarly, to match the “velocities” at point C , we equate the first-order kinematic coefficient of segment BC with the first-order kinematic coefficient of Eq. (6.19*b*) at the CD segment left boundary (that is, at $\theta_3/\beta_3 = 0$). This gives

$$y'_C = \frac{L_2}{\beta_2} = \frac{\pi L_3}{2\beta_3} = 1.59155 \text{ in/rad}$$

or

$$L_3 = (1.01321 \text{ in/rad}) \beta_3. \quad (6)$$

To match the “accelerations” (that is, the curvatures) at point D , we equate the second-order kinematic coefficient of Eq. (6.19*c*) at the CD segment right boundary (that is, at $\theta_3/\beta_3 = 1$) with the second-order kinematic coefficient of Eq. (6.17*c*) at the DE segment left boundary (that is, at $\theta_4/\beta_4 = 0$). This gives

$$y''_D = -\frac{\pi^2 L_3}{4\beta_3^2} = -5.26830 \frac{L_4}{\beta_4^2},$$

where the total lift is $L_4 = 3$ in. Substituting Eq. (6) and the total lift into this result and rearranging gives

$$\beta_3 = 0.15818 \beta_4^2. \quad (7)$$

Finally, for geometric compatibility, we have the constraints

$$L_1 + L_3 = L_4 - L_2 = 1.750 \text{ in}, \quad (8)$$

and, considering Eqs. (3) and (4),

$$\beta_1 + \beta_3 + \beta_4 = 2\pi - \beta_2 - \beta_5 = 3.926\ 99\ \text{rad}. \quad (9)$$

Solving the five equations—that is, Eqs. (5) through (9)—simultaneously for the five unknowns, L_1 , L_3 , β_1 , β_3 , and β_4 , provides the proper values for the remaining parameters. In summary, the segment parameters are

$$\begin{aligned} L_1 &= 1.183\ 1\ \text{in}, & \beta_1 &= 1.486\ 74\ \text{rad} = 85.184^\circ, \\ L_2 &= 1.250\ 0\ \text{in}, & \beta_2 &= 0.785\ 40\ \text{rad} = 45.000^\circ, \\ L_3 &= 0.566\ 9\ \text{in}, & \beta_3 &= 0.559\ 51\ \text{rad} = 32.058^\circ, \\ L_4 &= 3.000\ 0\ \text{in}, & \beta_4 &= 1.880\ 74\ \text{rad} = 107.758^\circ, \\ L_5 &= 0.000\ 0\ \text{in}, & \beta_5 &= 1.570\ 80\ \text{rad} = 90.000^\circ. \end{aligned} \quad \text{Ans.}$$

At this time, an accurate layout of the displacement diagram and, if desired, the kinematic coefficients can be made to replace the sketches. The curves of Fig. 6.24 have been drawn to scale using these values.

6.9 PLATE CAM WITH RECIPROCATING FLAT-FACE FOLLOWER

Once the displacement diagram of a cam system has been completely determined, as described in Sec. 6.8, the layout of the actual cam shape can be attempted, as demonstrated in Sec. 6.4. In laying out the cam, however, we find the need for a few more parameters, depending on the type of cam and follower—for example, the prime-circle radius, any offset distance, the roller radius, and so on. Also, as we will see, each different type of cam-and-follower system can be subject to certain further difficulties unless these remaining parameters are properly chosen.

In this section, we study the troubles that may be encountered in the design of a plate cam with a reciprocating flat-face follower. The geometric parameters of such a system that must yet be chosen are the prime-circle radius, R_0 , the offset (eccentricity), ε , of the follower stem, and the width of the follower face.

Figure 6.25 shows the layout of a plate cam with a radial reciprocating flat-face follower. In this illustration, the displacement chosen was a full-rise cycloidal motion segment with $L_1 = 100\ \text{mm}$ during $\beta_1 = 90^\circ$ of cam rotation, followed by a full-return cycloidal motion segment during the remaining $\beta_2 = 270^\circ$ of cam rotation. The layout procedure of Fig. 6.10 was followed to develop the cam shape, and the radius chosen for the prime circle was $R_0 = 25\ \text{mm}$. Obviously, there is a problem, since the resulting cam profile intersects itself. During machining, part of the cam shape is lost, and, when in operation, the intended cycloidal motion is not fully achieved. Such a cam is said to be *undercut*.

Why did undercutting occur in this example and how can it be avoided? It resulted from attempting to achieve too great a lift in too little cam rotation with too small a cam. One possible cure for this trouble is to decrease the desired lift, L_1 , or to increase the cam rotation angle, β_1 . However, this is not possible while still achieving the original design specifications. Another cure is to continue with the same displacement characteristics but

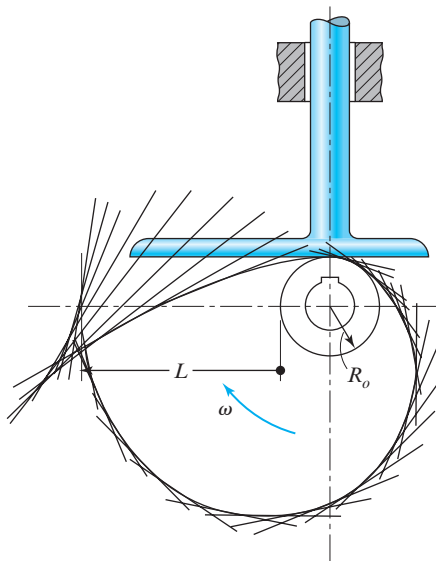


Figure 6.25 Undercut plate-cam profile layout with reciprocating flat-face follower.

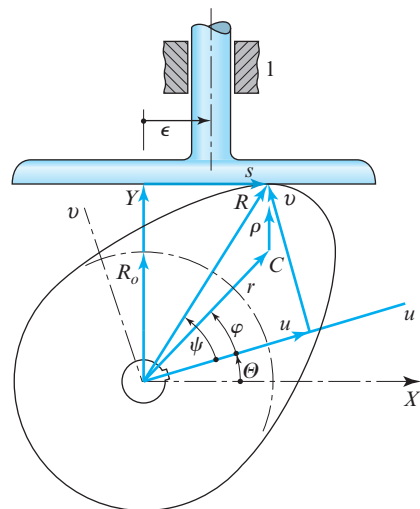


Figure 6.26 Vectors for plate-cam profile with reciprocating flat-face follower.

to increase the prime-circle radius, R_0 , to avoid undercutting. This does produce a larger cam, but with sufficient increase, it does overcome the undercutting difficulty.

The minimum value of R_0 that avoids undercutting can be found by developing an equation for the radius of curvature of the cam profile. We start by writing the loop-closure equation using the vectors shown in Fig. 6.26. Using complex polar notation, the loop-closure equation for a general plate cam with a reciprocating flat-face follower is

$$\mathbf{R} = re^{j(\Theta + \varphi)} + j\rho = j(R_0 + Y) + s. \quad (a)$$

Recall that the symbol Θ represents the total rotation of a general plate cam.

We have carefully chosen the vectors so that point C is located at the instantaneous center of curvature, and ρ is the radius of curvature corresponding to the current contact point. The line along the u axis that separates the angles Θ and φ is fixed on the cam and is horizontal for the cam posture $\Theta = 0$. The value of Y_0 is zero. The angle Θ specifies the rotation of the cam, and the u and v axes rotate with the cam.

Separating Eq. (a) into real and imaginary parts, respectively, gives

$$r \cos(\Theta + \varphi) = s, \quad (b)$$

$$r \sin(\Theta + \varphi) + \rho = R_0 + Y. \quad (c)$$

Since point C is the center of curvature, the magnitudes of r , φ , and ρ do not change for a small increment in cam rotation;* that is,

$$\frac{dr}{d\Theta} = \frac{d\varphi}{d\Theta} = \frac{d\rho}{d\Theta} = 0.$$

Therefore, differentiating Eq. (a) with respect to the cam rotation angle, Θ , gives

$$jre^{j(\Theta+\varphi)} = jY' + s', \quad (d)$$

where $Y' = dY/d\Theta = dy/d\theta = y'$ and $s' = ds/d\Theta = ds/d\theta$. Separating Eq. (d) into real and imaginary parts, the first-order kinematic coefficients are

$$-r \sin(\Theta + \varphi) = s', \quad (e)$$

$$r \cos(\Theta + \varphi) = y'. \quad (f)$$

Equating Eqs. (b) and (f), we find the location of the trace point along the surface of the follower as

$$s = y'. \quad (6.26)$$

Differentiating this equation with respect to the cam rotation angle, Θ , we find

$$s' = y''. \quad (g)$$

Substituting Eq. (g) into Eq. (e) and then substituting the result into Eq. (c), the radius of curvature of the cam profile can be written as

$$\rho = R_0 + Y + y''. \quad (6.27)$$

We should carefully note the importance of Eq. (6.27); it states that the radius of curvature of the cam profile can be obtained for each cam rotation angle, Θ , directly from the displacement equations, *before laying out the cam profile*. All that is needed is the choice of the prime-circle radius, R_0 , and values for the displacement, Y , and the second-order kinematic coefficient, y'' .

We can use Eq. (6.27) to select a value for R_0 that will avoid undercutting. When undercutting occurs, the radius of curvature of the cam profile switches sign from positive to negative. On the verge of undercutting, the cam comes to a point, and the radius of curvature becomes zero for some value of the cam rotation angle, Θ . However, we can choose R_0 large enough that this is never the case. In fact, to avoid high contact stresses, we may wish to ensure that ρ is everywhere larger than some specified value, $\rho_{\min}$. To do this, from Eq. (6.27), we require that

$$\rho = R_0 + Y + y'' > \rho_{\min}.$$

* The values of r , φ , and ρ are not truly constant but are currently at stationary values; their higher derivatives are nonzero.

Since R_0 and Y are always positive, the critical situation occurs at or near the posture where the second-order kinematic coefficient, y'' , has its largest negative value. Denoting this minimum value of y'' as $y''_{\min}$ and remembering that Y corresponds to the same posture, defined by cam angle Θ , we have the condition

$$R_0 > \rho_{\min} - Y - y''_{\min}, \quad (6.28)$$

which must be satisfied. This can easily be checked once the displacement equations have been established, and an appropriate value of R_0 can be chosen before the cam layout is attempted.

Returning now to Fig. 6.26, we see that Eq. (6.26) can also be of value. This equation states that the length of travel of the point of contact on either side of the cam rotation center corresponds precisely to the plot of the first-order kinematic coefficient. Thus, the minimum face width for a flat-face follower must extend at least $y'_{\max}$ to the right and $-y'_{\min}$ to the left of the camshaft center to maintain contact; that is,

$$\text{Face width} > y'_{\max} - y'_{\min}. \quad (6.29)$$

EXAMPLE 6.3

Assuming that the displacement characteristics in Example 6.2 are to be achieved by a plate cam with a reciprocating flat-face follower, determine the minimum face width and the minimum prime-circle radius to ensure that the radius of curvature of the cam is everywhere greater than $\rho_{\min} = 0.25$ in.

SOLUTION

From Fig. 6.24*b*, the maximum “velocity” (that is, the maximum value of the first-order kinematic coefficient) occurs in segment *BC* and is

$$y'_{\max} = \frac{L_2}{\beta_2} = \frac{1.250 \text{ in}}{0.785 \text{ 40 rad}} = 1.592 \text{ in/rad}. \quad (1)$$

The minimum “velocity” occurs in segment *DE* at approximately $\theta/\beta_4 = 0.5$. From Eq. (6.17*b*), the minimum value of the first-order kinematic coefficient is approximately

$$y'_{\min} \approx y'(\theta/\beta_4 = 0.5) = -2.812 \text{ in/rad}. \quad (2)$$

Substituting Eqs. (1) and (2) into Eq. (6.29), the minimum face width is

$$\text{Face width} > (1.592 \text{ in}) - (-2.812 \text{ in}) = 4.404 \text{ in}. \quad \text{Ans.}$$

Therefore, the follower would be positioned 1.592 in to the right and 2.812 in to the left of the cam rotation axis, and some appropriate additional allowance may be added on each side.

The largest negative “acceleration” (the minimum value of the second-order kinematic coefficient) occurs at D and can be obtained from Eq. (6.19c) at $\theta/\beta_3 = 1$; that is,

$$y''_{\min} = -\frac{\pi^2 L_3}{4\beta_3^2} = -\frac{\pi^2 (0.5669 \text{ in})}{4(0.55951 \text{ rad})^2} = -4.46818 \text{ in/rad}^2.$$

Substituting this result and the known parameters into Eq. (6.28), the minimum prime-circle radius is

$$R_0 > 0.250 \text{ in} - (-4.468 \text{ in}) - 3.000 \text{ in} = 1.718 \text{ in.} \quad \text{Ans.}$$

From this calculation, we would choose the actual prime-circle radius as, say, $R_0 = 1.75 \text{ in}$.

We see that the eccentricity of the flat-face follower stem does not affect the geometry of the cam. This eccentricity is usually chosen to avoid high bending stress in the follower. Also, there may be a higher load in the follower during the working stroke, say the lift stroke, than during the return motion. In such a case, the eccentricity may be chosen to locate the follower stem more centrally over the contact point during the lift portion of the motion cycle.

Looking again at Fig. 6.26, we can write another loop-closure equation; that is,

$$ue^{j\Theta} + ve^{j(\Theta+\pi/2)} = j(R_0 + Y) + s,$$

where we recall that u and v denote the coordinates of the contact point in a coordinate system attached to the cam. Dividing this equation by $e^{j\Theta}$ gives

$$u + jv = j(R_0 + Y)e^{-j\Theta} + se^{-j\Theta}.$$

Using Eq. (6.26), the real and imaginary parts of this equation can be written as

$$u = (R_0 + Y)\sin\Theta + y'\cos\Theta, \quad (6.30a)$$

$$v = (R_0 + Y)\cos\Theta - y'\sin\Theta. \quad (6.30b)$$

These two equations give the coordinates of the cam profile and provide an alternative to the graphic layout procedure of Fig. 6.10. They can be used to generate a table of numeric rectangular coordinate data from which the cam can be machined. Polar coordinate equations for this same curve are

$$R = \sqrt{(R_0 + Y)^2 + (y')^2} \quad (6.31a)$$

and

$$\psi = \frac{\pi}{2} - \Theta - \tan^{-1} \frac{y'}{R_0 + Y}. \quad (6.31b)$$

6.10 PLATE CAM WITH RECIPROCATING ROLLER FOLLOWER

Figure 6.27 shows a plate cam with a reciprocating roller follower. We see that three geometric parameters remain to be chosen after the displacement diagram is completed and before the cam layout is finalized. These three parameters are the radius of the prime circle, R_0 , the eccentricity, ϵ , and the radius of the roller, R_f . There are also two potential problems to be considered when choosing these parameters. One difficulty is undercutting and the other is an excessive pressure angle.

Pressure angle is the angle between the axis of motion of the follower stem and the line of action of the force exerted by the cam onto the roller follower—that is, the normal to the pitch curve through the trace point. The pressure angle is labeled ϕ in Fig. 6.27. Only the component of force along the line of motion of the follower is useful in overcoming the output load; the perpendicular component and, therefore, the angle ϕ , should be kept low to reduce sliding friction between the follower and its guideway and to ease bending of the follower stem. Too high a pressure angle increases the deleterious effect of friction and may cause the translating follower to chatter or perhaps even to jam. Cam pressure angles of up to about 30° or 35° are about the largest that can be used without difficulties.

In Fig. 6.27, we see that the normal to the pitch curve intersects the horizontal axis at point I_{24} —that is, at the instantaneous center of velocity between cam 2 and follower stem 4. Since the follower stem is translating, all points of the follower stem have velocities equal to that of the instant center I_{24} . This velocity must also be equal to the velocity of the coincident point of link 2; that is,

$$V_{I_{24}} = \dot{y} = \omega R_{I_{24}O_2}.$$

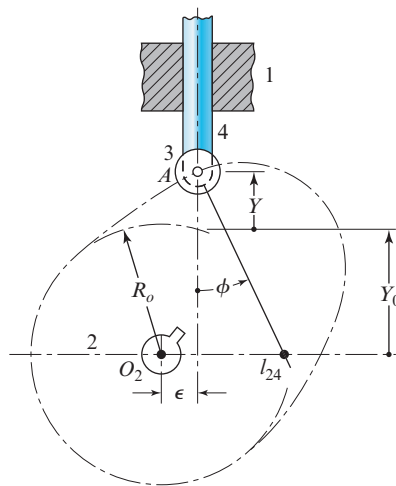


Figure 6.27 Vectors for plate cam with reciprocating roller follower.

Dividing this equation by the angular velocity of the cam, ω [Eq. (6.11)], the first-order kinematic coefficient is

$$y' = \frac{\dot{y}}{\omega} = R_{I_{24}O_2}.$$

This first-order kinematic coefficient can also be expressed in terms of the eccentricity of the follower stem and the pressure angle of the cam as

$$y' = \varepsilon + (Y_0 + Y) \tan \phi, \quad (a)$$

where, as shown in Fig. 6.27, the vertical distance from the cam axis to the prime circle is

$$Y_0 = \sqrt{R_0^2 - \varepsilon^2}. \quad (b)$$

Substituting Eq. (b) into Eq. (a) and rearranging, the pressure angle of the cam can be written as

$$\phi = \tan^{-1} \left(\frac{y' - \varepsilon}{Y + \sqrt{R_0^2 - \varepsilon^2}} \right). \quad (6.32)$$

From this equation, we observe that, once the displacement equations and the first-order kinematic coefficient have been determined, the two parameters, R_0 and ε , can be adjusted to seek a suitable pressure angle. We also note that the pressure angle is continuously changing as the cam rotates, and therefore we are particularly interested in studying its extreme values.

Let us first consider the effect of eccentricity. From Eq. (6.32), we observe that increasing ε either increases or decreases the magnitude of the numerator, depending on the sign of the first-order kinematic coefficient y' . Thus, a small eccentricity, ε , can be used to reduce the pressure angle, ϕ , during the rise motion when y' is positive, but only at the expense of an increased pressure angle during the return motion when y' is negative. Still, since the magnitudes of the forces are usually greater during rise, it is common practice to offset the follower to take advantage of this reduction in pressure angle.

A much more significant effect can be made in reducing the pressure angle by increasing the prime-circle radius, R_0 . To study this effect, let us take the conservative approach and assume a radial follower—that is, where there is no eccentricity. Substituting $\varepsilon = 0$ into Eq. (6.32), the equation for the pressure angle reduces to

$$\phi = \tan^{-1} \left(\frac{y'}{Y + R_0} \right). \quad (6.33)$$

To find the extremum values of the pressure angle, it is possible to differentiate this equation with respect to the cam rotation angle and equate it to zero, thus finding the values of the rotation angle, Θ , that yield the maximum and the minimum pressure angles. This is a tedious process, however, and can be avoided by using the nomogram of Fig. 6.28. This nomogram was produced by searching out on a digital computer the maximum value

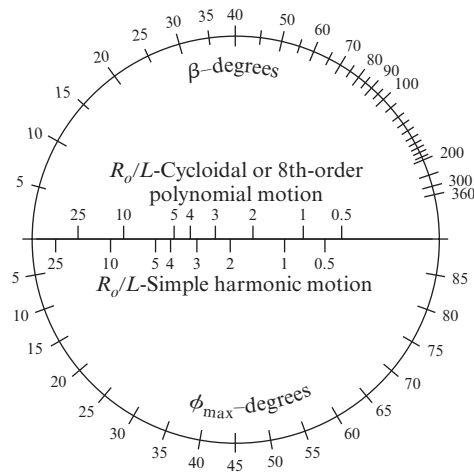


Figure 6.28 Nomogram relating the maximum pressure angle, $\phi_{\max}$, to the prime-circle radius, R_0 , lift, L , and segment angle, β , for radial reciprocating roller-follower cams with full-rise or full-return simple harmonic, cycloidal, or eighth-order polynomial motion.

of ϕ from Eq. (6.33) for each of the standard full-rise and full-return motion curves of Sec. 6.7. With the nomogram, it is possible to use the known values of L and β for each segment of the displacement diagram and to read directly the maximum pressure angle occurring in that motion segment for a particular choice of R_0 . Alternatively, a desired maximum pressure angle can be chosen, and a corresponding minimum value of R_0 can be determined. The process is best illustrated by an example.

EXAMPLE 6.4

Assuming that the displacements determined in Example 6.2 are to be achieved by a plate cam with a reciprocating radial roller follower, determine the minimum prime-circle radius that ensures that the pressure angle is everywhere less than 30° .

SOLUTION

Each segment of the displacement diagram can be checked in succession using the nomogram of Fig. 6.28.

For segment AB of Fig. 6.24, we have half-rise cycloidal motion with $L_1 = 1.183$ in and $\beta_1 = 85.184^\circ$. Since this is a half-rise curve, whereas the nomogram of Fig. 6.28 was developed only for full-rise curves, it is necessary to double both L_1 and β_1 , thus imagining that the curve is full rise. This gives $L_1^* = 2.366$ in and $\beta_1^* \approx 170^\circ$. Next, connecting a straight line from $\beta^* = 170^\circ$ to $\phi_{\max} = 30^\circ$, we read from the upper scale on the central axis of the nomogram a value of $R_0/L_1^* \approx 0.75$, from which

$$R_0 \geq 0.75(2.366 \text{ in}) = 1.775 \text{ in.} \quad (1)$$

The segment BC need not be checked, since the maximum pressure angle for this segment occurs at boundary B and cannot be greater than that for segment AB .

Segment CD has half-rise harmonic motion with $L_3 = 0.567$ in and $\beta_3 = 32.058^\circ$. Again, since this is a half-rise curve, these values are doubled, and $L_3^* = 1.134$ in and

$\beta_3^* \approx 64^\circ$ are used instead. Then, from the nomogram, we find $R_0^*/L_3^* \approx 2.15$, from which

$$R_0^* \geq 2.15(1.134 \text{ in}) = 2.438 \text{ in.}$$

However, here we must be careful. This value is the radius of a fictitious prime circle for which the horizontal axis of our fictional doubled “full-rise” harmonic curve would have $y = 0$. This is not the R_0 we seek, since our imagined full-harmonic curve has a nonzero Y^* value at its base. Referring to Fig. 6.24, we find

$$Y_3^* = Y_D - 2L_3 = 3.000 - 1.134 = 1.866 \text{ in.}$$

The appropriate value of R_0 for this segment is

$$R_0 \geq 2.438 - 1.866 = 0.572 \text{ in.} \quad (2)$$

Next we check segment DE , which has eighth-order polynomial motion with $L_4 = 3.000$ in and $\beta_4 = 107.758^\circ$. Since this is a full-return motion curve with $y = 0$ at its base, no adjustments are necessary for use of the nomogram. We find $R_0/L_4 \approx 1.3$ and

$$R_0 \geq 1.3(3.000 \text{ in}) = 3.900 \text{ in.} \quad (3)$$

To ensure that the pressure angle does not exceed 30° throughout all segments of the cam motion, we must choose the prime-circle radius to be at least as large as the maximum of these discovered values, Eqs. (1), (2), and (3). Remembering the inability to read the nomogram with great precision, we might choose a yet larger value, such as

$$R_0 = 4.000 \text{ in.} \quad \text{Ans.}$$

Once a final value has been chosen, we can use Fig. 6.28 again to find the actual maximum pressure angle in each segment of the motion:

$$\begin{aligned} AB: \quad \frac{R_0}{L_1^*} &= \frac{4.000}{2.366} = 1.691 & \beta_1^* &= 170^\circ & \phi_{\max} &= 18^\circ, \\ CD: \quad \frac{R_0^*}{L_3^*} &= \frac{5.866}{1.134} = 5.173 & \beta_3^* &= 64^\circ & \phi_{\max} &= 14^\circ, \\ DE: \quad \frac{R_0}{L_4} &= \frac{4.000}{3.000} = 1.333 & \beta_4 &= 108^\circ & \phi_{\max} &= 29^\circ. \end{aligned}$$

Although the prime circle has been sized to give a satisfactory pressure angle, the follower may still not complete the desired motion. It is still possible that the curvature of the pitch curve is too sharp and that the cam profile may be undercut. Figure 6.29a shows a portion of a cam pitch curve and two cam profiles generated by two different-size rollers. It is clear from Fig. 6.29a that a small roller moving on the given pitch curve generates a satisfactory cam profile. The cam profile generated by the larger roller, however, intersects itself and is said to be *undercut*. The result, after machining, is a pointed cam that does not produce the desired motion. Still, if the prime circle and thus the cam size is increased enough, even the larger roller generates a cam profile that operates satisfactorily.

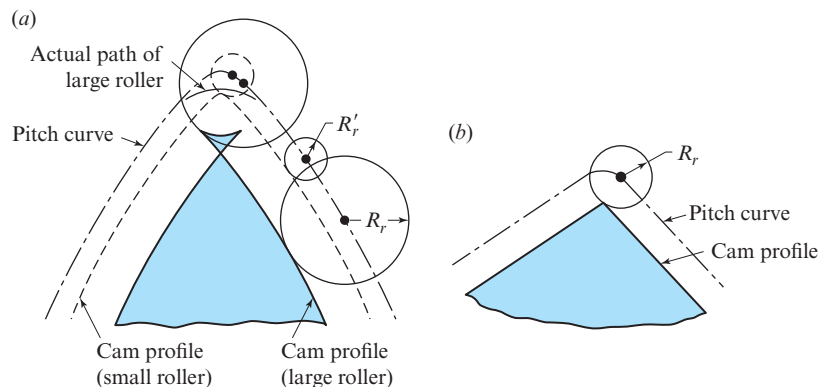


Figure 6.29 (a) Undercut; (b) pointed cam profile.

In Fig. 6.29b, we see that the cam profile becomes pointed when the roller radius, R_r , is equal to the magnitude* of the radius of curvature of the pitch curve. Therefore, to achieve some chosen minimum size, ρ^* , for the radius of curvature of the cam profile, the radius of curvature of the pitch curve must be of greater magnitude than this value by the radius of the roller; that is,

$$|\rho| \geq \rho^* + R_r. \quad (c)$$

Recall Sec. 4.17, where the radius of curvature of a point trajectory was given by Eq. (4.56).

In concept, it is possible to search out the minimum value $|\rho|_{\min}$ for a particular choice of the displacement equation, Y , and a particular prime-circle radius, R_0 . However, since it would be burdensome to perform such a search for each new cam design, the minimum-size radius of curvature (normalized with respect to R_0) has been sought out by a digital computer program for each of the standard cam motions of Sec. 6.7. The results are presented graphically in Fig. 6.30 through Fig. 6.34. Each of these figures shows a graph of $(|\rho|_{\min} + R_r)/R_0$ versus β for one type of standard-motion curve with various ratios of R_0/L . Since we have already chosen the displacement equations and have found a suitable value of R_0 , each segment of the cam can now be checked to find its minimum radius of curvature.

Saving even more effort, it is not necessary to check those segments of the cam where the second-order kinematic coefficient y'' remains positive throughout the segment, such as the half-rise motions of Eqs. (6.18) and (6.22) or the half-return motions of Eqs. (6.21) and (6.25). Assuming that the “acceleration” curve is continuous, the minimum radius of curvature of the cam cannot occur in these segments. For each of these segments,

$$|\rho|_{\min} = R_0 - R_r. \quad (6.34)$$

* Remember that radius of curvature can have either a positive or a negative value (Sec. 4.17). However, here we are only concerned with its size—that is, its absolute value.

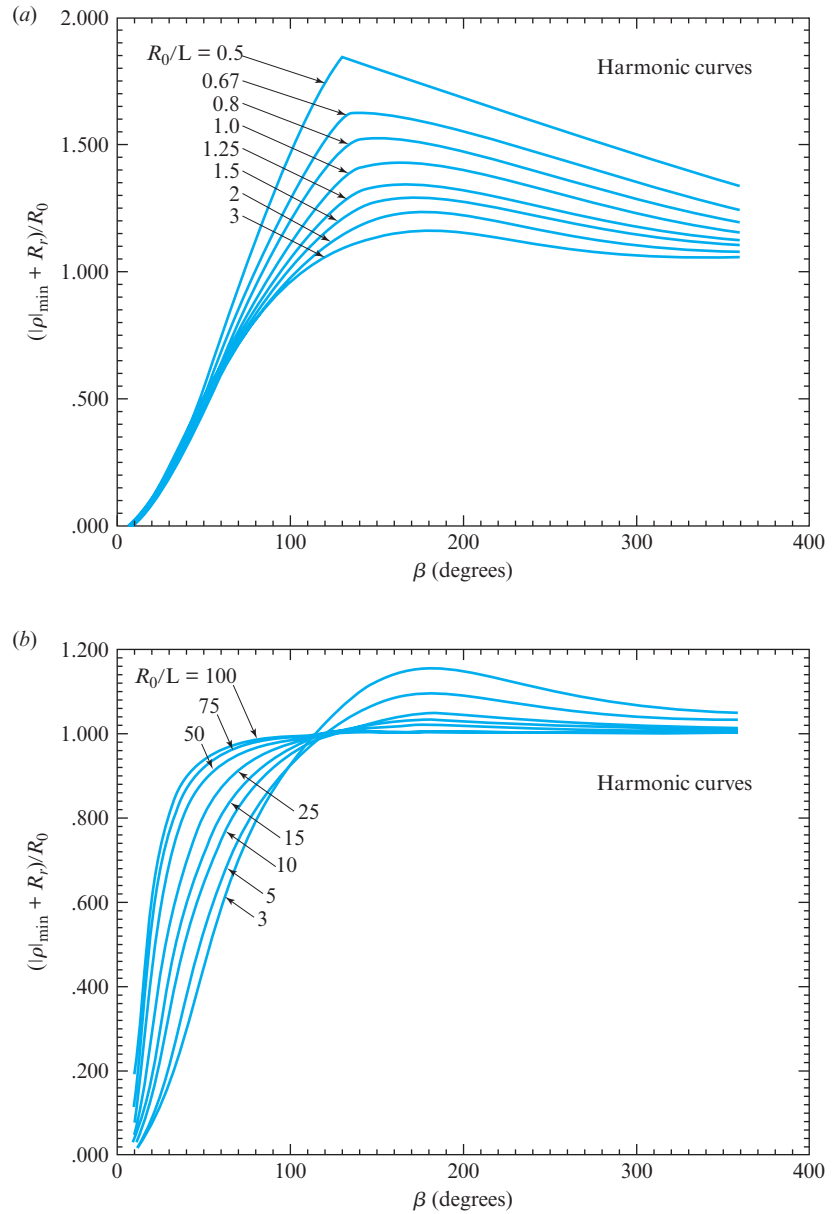


Figure 6.30 Minimum radius of curvature of radial reciprocating roller follower cams with full-rise or full-return simple harmonic motion, Eqs. (6.12) or (6.15). (From [3].)

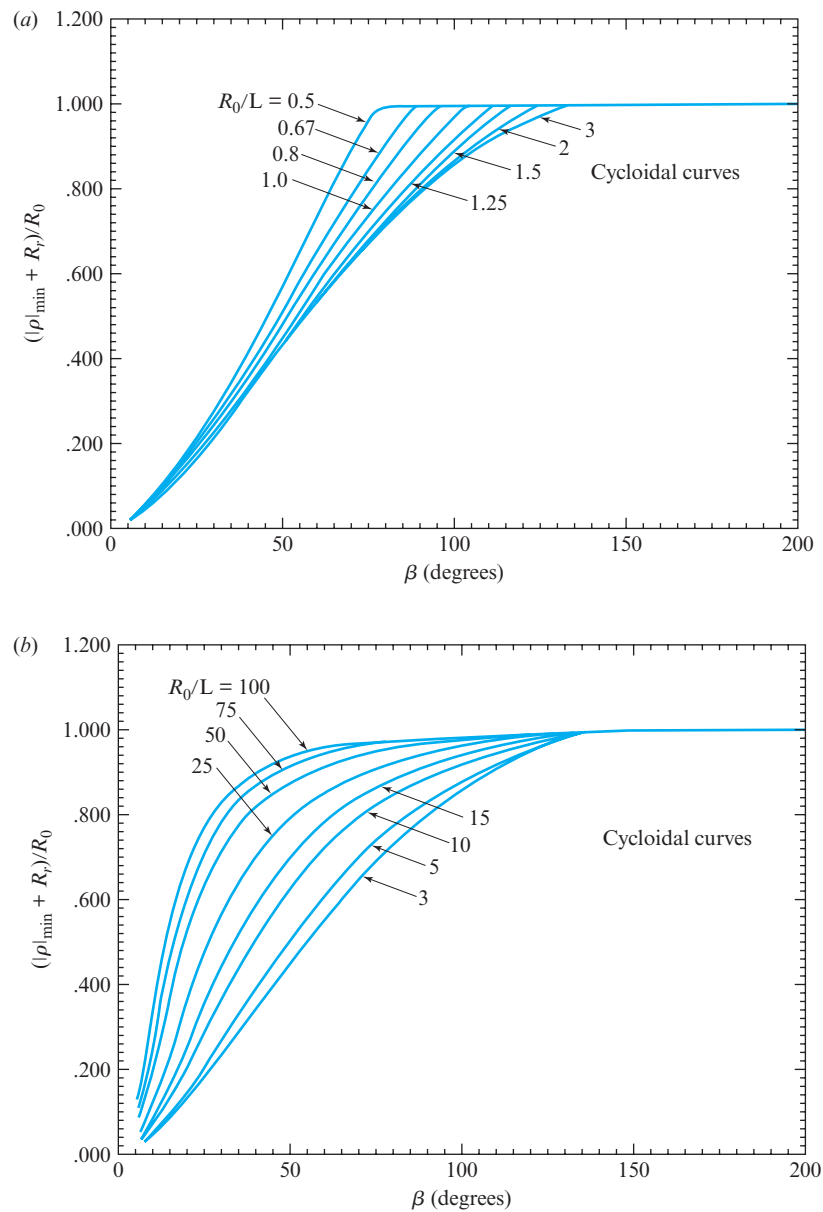


Figure 6.31 Minimum radius of curvature of radial reciprocating roller follower cams with full-rise or full-return cycloidal motion, Eqs. (6.13) or (6.16). (From [3].)

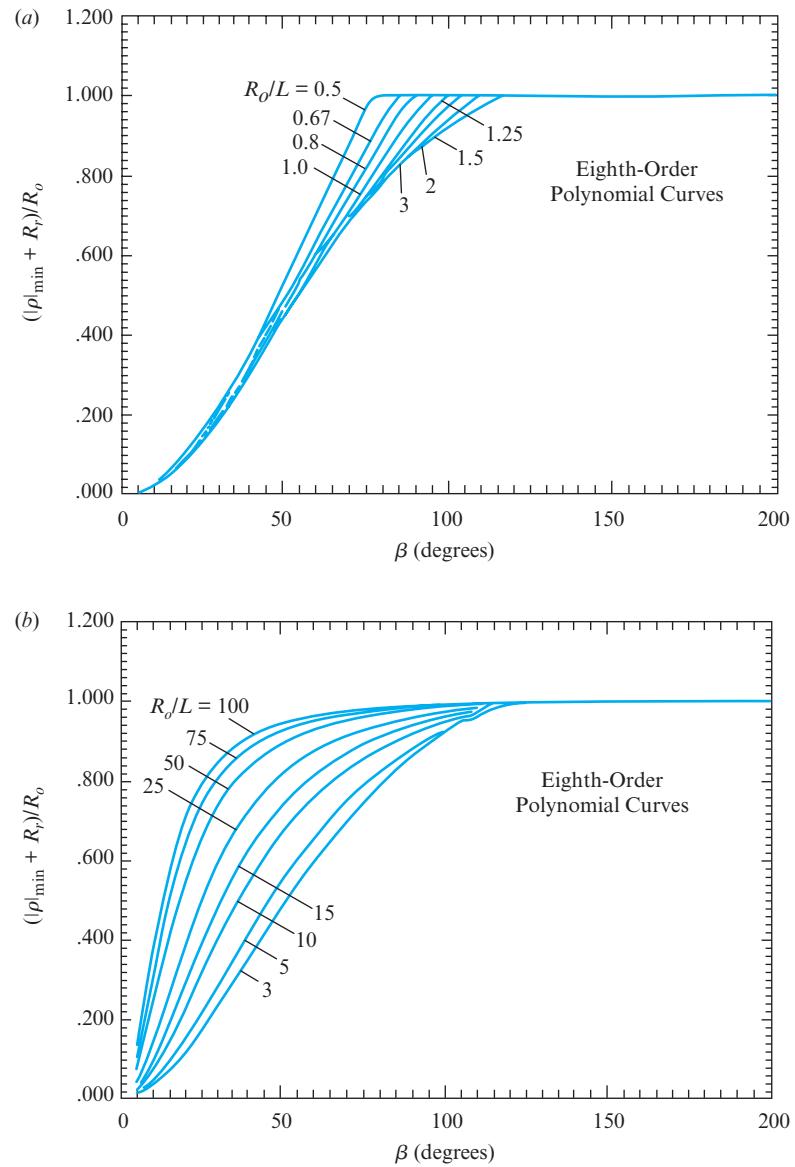


Figure 6.32 Minimum radius of curvature of radial reciprocating roller follower cams with full-rise or full-return eighth-order polynomial motion, Eqs. (6.14) or (6.17). (From [3].)

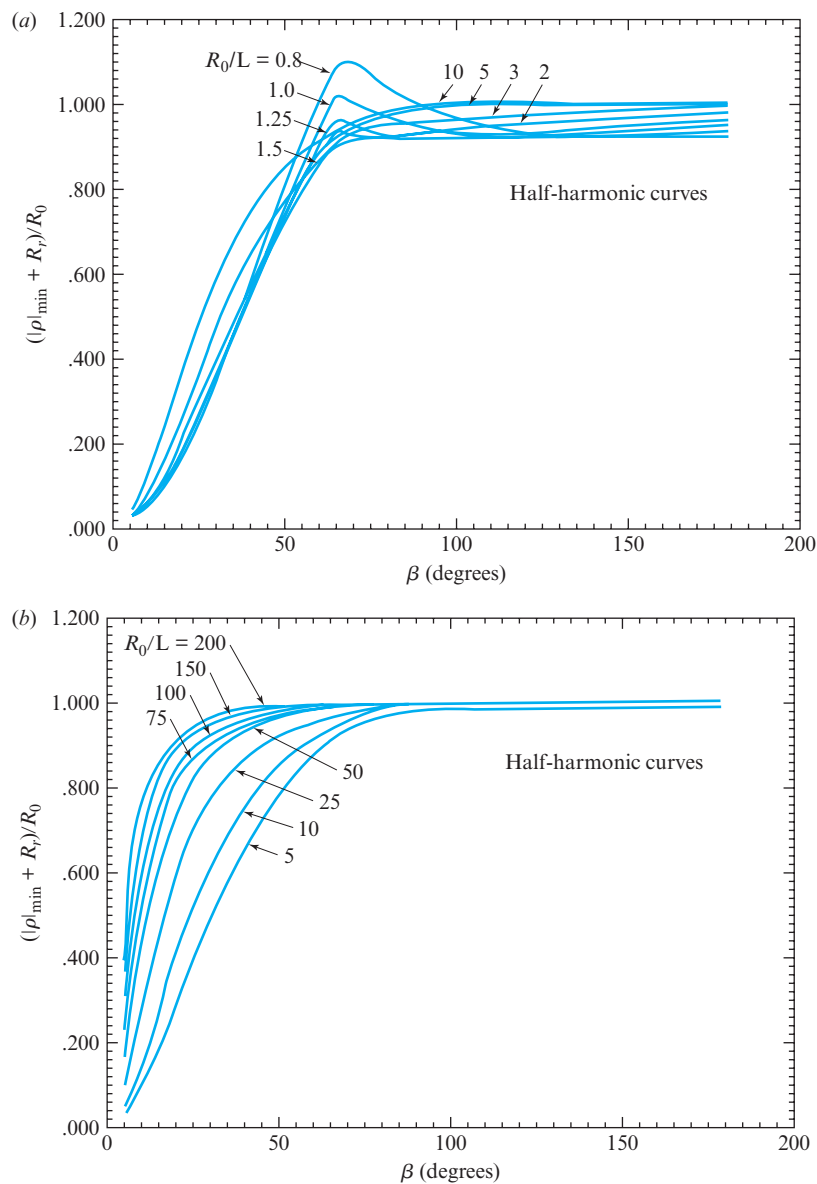


Figure 6.33 Minimum radius of curvature of radial reciprocating roller follower cams with half-harmonic motion, Eqs. (6.19) or (6.20). (From [3].)

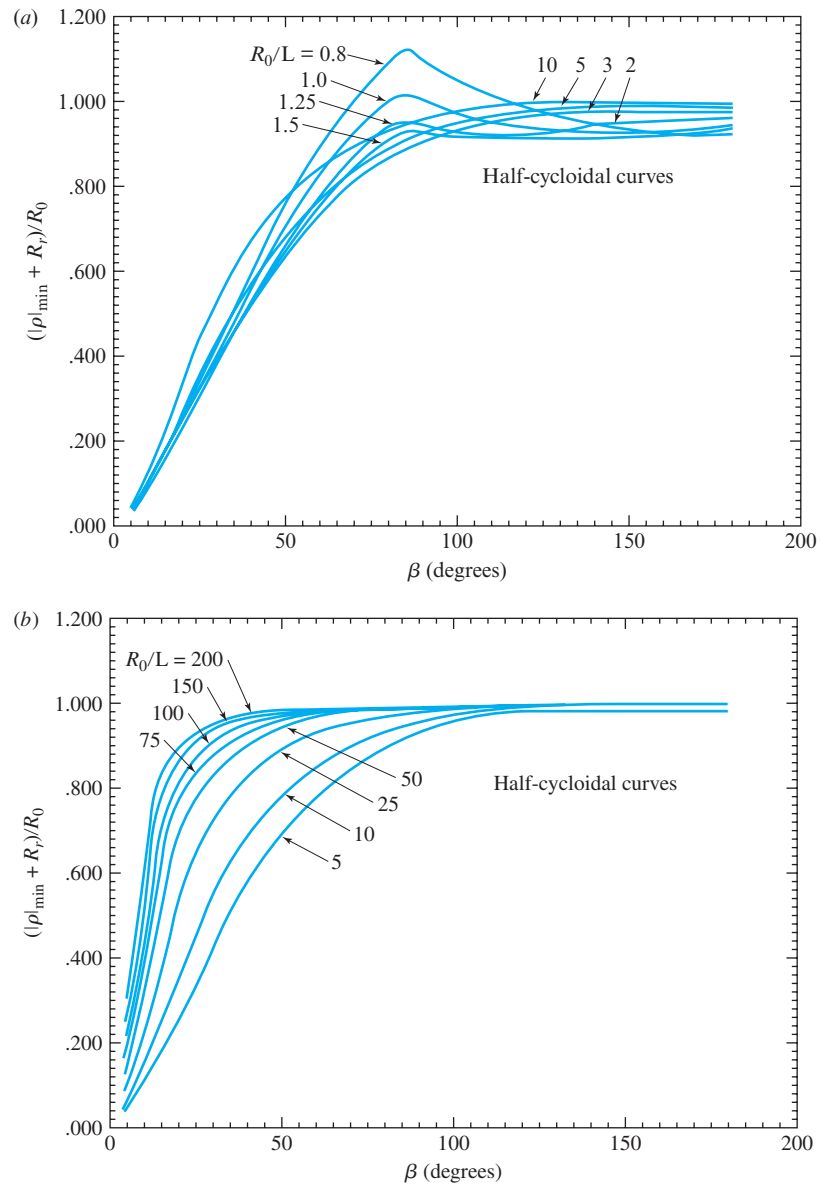


Figure 6.34 Minimum radius of curvature of radial reciprocating roller follower cams with half-cycloidal motion, Eqs. (6.23) or (6.24). (From [3].)

EXAMPLE 6.5

Assuming that the displacement characteristics of Example 6.2 are to be achieved by a plate cam with a reciprocating roller follower, determine the minimum radius of curvature of the cam profile if a prime-circle radius of $R_0 = 4.000$ in (from Example 6.4) and a roller radius of $R_r = 0.500$ in are used.

SOLUTION

For the segment AB of Fig. 6.24, we find from Eq. (6.34) that

$$|\rho|_{\min} = R_0 - R_r = 4.000 \text{ in} - 0.500 \text{ in} = 3.500 \text{ in.} \quad (1)$$

For the segment CD , we have half-harmonic rise motion with $L_3 = 0.5669$ in and $\beta_3 = 32.058^\circ$, from which we find

$$\frac{R_0^*}{L_3} = \frac{4.000 \text{ in} + (L_1 + L_2)}{L_3} = \frac{6.433 \text{ in}}{0.567 \text{ in}} = 11.346,$$

where R_0^* was adjusted by $Y_3 = (L_1 + L_2)$, since the curves of Fig. 6.33 were plotted for the condition where $Y = 0$ at the base of each motion segment. Now, using Fig. 6.33*b*, we find $(|\rho|_{\min} + R_r)/R_0^* \approx 0.66$. Therefore,

$$|\rho|_{\min} \approx 0.66R_0^* - R_r = 0.66(6.433 \text{ in}) - 0.500 \text{ in} = 3.746 \text{ in,} \quad (2)$$

where, again, the adjusted value of R_0^* is used.

For the segment DE , we have eighth-order polynomial motion with $L_4 = 3.0000$ in and $\beta_4 = 107.758^\circ$, from which $R_0/L_4 = 1.33$. Using Fig. 6.32*a*, we find $(|\rho|_{\min} + R_r)/R_0 \approx 0.80$, and

$$|\rho|_{\min} \approx 0.80R_0 - R_r = 0.80(4.000 \text{ in}) - 0.500 \text{ in} = 2.700 \text{ in.} \quad (3)$$

Choosing the smallest value among Eqs. (1), (2), and (3), the minimum size of the radius of curvature of the entire cam profile is

$$|\rho|_{\min} = 2.700 \text{ in.} \quad \text{Ans.}$$

To machine the cam to the proper shape, we need the coordinates of the cam surface in the rotating uv coordinate system attached to the cam. The rectangular coordinates of the follower center (the pitch curve) of a plate cam with a reciprocating roller follower in the rotating uv coordinate system are

$$u = \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right) \sin \Theta + \varepsilon \cos \Theta, \quad (d)$$

$$v = \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right) \cos \Theta - \varepsilon \sin \Theta. \quad (e)$$

Differentiating these with respect to the rotation of the cam, the first-order kinematic coefficients of the follower center are

$$u' = y' \sin \Theta + \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right) \cos \Theta - \varepsilon \sin \Theta, \quad (f)$$

$$v' = y' \cos \Theta - \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right) \sin \Theta - \varepsilon \cos \Theta, \quad (g)$$

and, differentiating again, the second-order kinematic coefficients of the follower center are

$$u'' = y'' \sin \Theta + 2y' \cos \Theta - \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right) \sin \Theta - \varepsilon \cos \Theta, \quad (h)$$

$$v'' = y'' \cos \Theta - 2y' \sin \Theta - \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right) \cos \Theta + \varepsilon \sin \Theta. \quad (i)$$

We note that positive (counterclockwise) rotation of the cam causes the center point of the follower to increment in the clockwise direction around the pitch curve of the cam in the rotating uv coordinate system. This defines the positive sense of the unit tangent vector $\hat{\mathbf{u}}^t$ for the pitch curve, and the unit normal vector is then given by $\hat{\mathbf{u}}^n = \hat{\mathbf{k}} \times \hat{\mathbf{u}}^t$ in the rotating coordinate system. Since this unit normal points away from the center of curvature of the pitch curve, the radius of curvature of the pitch curve has a negative value.

To normalize Eqs. (f) and (g), we define

$$w' = +\sqrt{u'^2 + v'^2} \quad (6.35)$$

or

$$w' = + \left[(y' - \varepsilon)^2 + \left(\sqrt{R_0^2 - \varepsilon^2} + Y \right)^2 \right]^{1/2}, \quad (j)$$

and with this we find the unit tangent,

$$\hat{\mathbf{u}}^t = \left(\frac{u'}{w'} \right) \hat{\mathbf{i}} + \left(\frac{v'}{w'} \right) \hat{\mathbf{j}}, \quad (k)$$

and the unit normal,

$$\hat{\mathbf{u}}^n = \left(\frac{-v'}{w'} \right) \hat{\mathbf{i}} + \left(\frac{u'}{w'} \right) \hat{\mathbf{j}}, \quad (l)$$

in the rotating uv coordinate system attached to the cam.

The coordinates of the point of contact between the cam and the roller follower in the rotating coordinate system of the cam can now be written as

$$u_{\text{cam}} = u + R_r \left(\frac{v'}{w'} \right) \quad \text{and} \quad v_{\text{cam}} = v - R_r \left(\frac{u'}{w'} \right). \quad (6.36)$$

The radius of curvature of the pitch curve can be written from Eq. (4.57) as

$$\rho = \frac{w'^3}{u'v'' - v'u''}. \quad (6.37)$$

Remembering that the radius of curvature of the pitch curve is expected to have a negative value, as explained earlier, the radius of curvature of the cam profile is

$$\rho_{\text{cam}} = \rho + R_r. \quad (6.38)$$

Note that Eq. (6.38) still yields a negative value, although smaller than Eq. (6.37), since the unit normal vector still points away from the center of curvature of the cam.

The unit vector in the direction of motion of the follower (the Y direction), when expressed in the rotating uv coordinate system, is

$$\hat{\mathbf{u}}^Y = \sin \Theta \hat{\mathbf{i}} + \cos \Theta \hat{\mathbf{j}}. \quad (m)$$

The pressure angle can be written from Fig. 6.27 with the aid of Eqs. (l) and (m) as

$$\cos \phi = \hat{\mathbf{u}}^n \cdot \hat{\mathbf{u}}^Y = -\left(\frac{v'}{w'}\right) \sin \Theta + \left(\frac{u'}{w'}\right) \cos \Theta. \quad (6.39)$$

EXAMPLE 6.6

A plate cam with a reciprocating radial roller follower is to be designed such that the displacement of the follower is

$$Y = 15(1 - \cos 2\Theta) \text{ mm}.$$

The prime-circle radius is to be $R_0 = 40$ mm, and the roller is to have a radius of $R_r = 12$ mm. The cam is to rotate counterclockwise. For the cam rotation angle $\Theta = 30^\circ$, determine the following:

- The coordinates of the point of contact between the cam and the follower in the rotating coordinate system.
- The radius of curvature of the cam profile.
- The pressure angle of the cam.

SOLUTION

At the cam rotation angle $\Theta = 30^\circ$, the lift of the follower from the specified displacement equation is

$$Y = 15(1 - \cos 2\Theta) \text{ mm} = 7.500 \text{ mm}. \quad (1)$$

By differentiating the displacement equation with respect to the cam rotation angle, we find

$$y' = 30 \sin 2\Theta \text{ mm/rad} = 25.981 \text{ mm/rad}, \quad (2a)$$

$$y'' = 60 \cos 2\Theta \text{ mm/rad}^2 = 30.000 \text{ mm/rad}^2. \quad (2b)$$

From Eqs. (d) and (e), with offset $\varepsilon = 0$, the coordinates of the roller center in the rotating coordinate system attached to the cam are

$$u = (R_0 + Y) \sin \Theta = (55 - 15 \cos 2\Theta) \sin \Theta \text{ mm} = 23.750 \text{ mm}, \quad (3a)$$

$$v = (R_0 + Y) \cos \Theta = (55 - 15 \cos 2\Theta) \cos \Theta \text{ mm} = 41.136 \text{ mm}. \quad (3b)$$

Then, from Eqs. (f) and (g), the first-order kinematic coefficients of the trace point are

$$\begin{aligned} u' &= y' \sin \Theta + (R_0 + Y) \cos \Theta \\ &= (25.981 \text{ mm/rad}) \sin \Theta + (47.500 \text{ mm}) \cos \Theta = 54.127 \text{ mm/rad}, \end{aligned} \quad (4a)$$

$$\begin{aligned} v' &= y' \cos \Theta - (R_0 + Y) \sin \Theta \\ &= (25.981 \text{ mm/rad}) \cos \Theta - (47.500 \text{ mm}) \sin \Theta = -1.250 \text{ mm/rad}, \end{aligned} \quad (4b)$$

and, from Eqs. (h) and (i), the second-order kinematic coefficients of the trace point are

$$\begin{aligned} u'' &= y'' \sin \Theta + 2y' \cos \Theta - (R_0 + Y) \sin \Theta \\ &= (30.000 \text{ mm/rad}^2) \sin \Theta + (51.962 \text{ mm/rad}) \cos \Theta - (47.500 \text{ mm}) \sin \Theta \\ &= 36.250 \text{ mm/rad}^2, \end{aligned} \quad (5a)$$

$$\begin{aligned} v'' &= y'' \cos \Theta - 2y' \sin \Theta - (R_0 + Y) \cos \Theta \\ &= (30.000 \text{ mm/rad}^2) \cos \Theta - (51.962 \text{ mm/rad}) \sin \Theta - (47.500 \text{ mm}) \cos \Theta \\ &= -41.136 \text{ mm/rad}^2, \end{aligned} \quad (5b)$$

From Eq. (6.35) we have

$$\begin{aligned} w' &= +\sqrt{u'^2 + v'^2} \\ &= +\sqrt{(54.127 \text{ mm/rad})^2 + (-1.250 \text{ mm/rad})^2} = +54.141 \text{ mm/rad}, \end{aligned} \quad (6)$$

- (a) Substituting Eqs. (3), (4), and (6), and the given dimensions into Eqs. (6.36), the coordinates of the point of contact between the cam and the roller follower, in the rotating coordinate system, are

$$u_{\text{cam}} = 23.750 \text{ mm} + (12 \text{ mm}) \left(\frac{-1.250 \text{ mm/rad}}{54.141 \text{ mm/rad}} \right) = 23.413 \text{ mm}, \quad \text{Ans.}$$

$$v_{\text{cam}} = 41.136 \text{ mm} + (12 \text{ mm}) \left(\frac{-54.127 \text{ mm/rad}}{54.141 \text{ mm/rad}} \right) = 29.139 \text{ mm}. \quad \text{Ans.}$$

- (b) From Eq. (6.37), using Eqs. (4) through (6), the radius of curvature of the pitch curve is

$$\begin{aligned} \rho &= \frac{(54.141 \text{ mm/rad})^3}{(54.127 \text{ mm/rad})(-41.136 \text{ mm/rad}^2) - (-1.250 \text{ mm/rad})(36.250 \text{ mm/rad}^2)} \\ &= -72.775 \text{ mm}, \end{aligned}$$

and we note that this value is negative, as explained earlier. With this, Eq. (6.38) gives the radius of curvature of the cam profile as

$$\rho_{\text{cam}} = \rho + R_r = -72.775 \text{ mm} + 12.0 \text{ mm} = -60.775 \text{ mm}, \quad \text{Ans.}$$

which we note is still negative. This confirms that the center of curvature is still further inward and that there is no undercutting at this location on the cam profile.

- (c) From Eq. (6.39), we find the pressure angle for this posture,

$$\cos \phi = - \left(\frac{-1.250 \text{ mm/rad}}{54.141 \text{ mm/rad}} \right) \sin 30^\circ + \left(\frac{54.127 \text{ mm/rad}}{54.141 \text{ mm/rad}} \right) \cos 30^\circ; \quad \phi = 28.68^\circ. \quad \text{Ans.}$$

In this and the previous section, we have considered troubles that result from poor choice of the prime-circle radius for a plate cam with a reciprocating follower. Although the equations are different for oscillating followers or other types of cams, a similar approach can be used to guard against undercutting [5] and severe pressure angle [4]. Similar equations can also be developed for cam profile data [7]. An extensive survey of the cam design literature has been compiled by Chen [1]. We present one more example here to illustrate a general approach for the design of cams that requires vector analysis in addition to the previous equations.

EXAMPLE 6.7

A plate cam with an oscillating roller follower, shown in Fig. 6.35a, is to be designed such that the cam rotates counterclockwise, and the displacement of the follower follows the equation

$$Y = 0.5(1 - \cos 2\theta) \text{ rad.}$$

The prime-circle radius is $R_0 = 1.500$ in, the roller radius is $R_r = 0.500$ in, the distance between the center of the camshaft and the follower pivot is $R_{O_4O_2} = r_1 = 3.000$ in, and

the length of the follower arm is $R_{CO_4} = r_4 = 2.598$ in. For the cam rotation angle of $\theta_2 = \Theta = 45^\circ$, determine the following:

- The coordinates of the point of contact between the cam and the follower.
- The radius of curvature of the cam profile.
- The pressure angle of the cam.

SOLUTION

Let us designate the rotation of the oscillating follower by the angle θ_4 , as shown in Fig. 6.35*b*. With the specified dimensions, the angle of the follower with zero displacement must be $\theta_4 = 150^\circ = 5\pi/6$ rad. Combining this with the specified displacement gives the complete equation for the rotation of the follower:

$$\theta_4 = 5\pi/6 - y = 5\pi/6 - 0.5(1 - \cos 2\Theta) \text{ rad.} \quad (1)$$

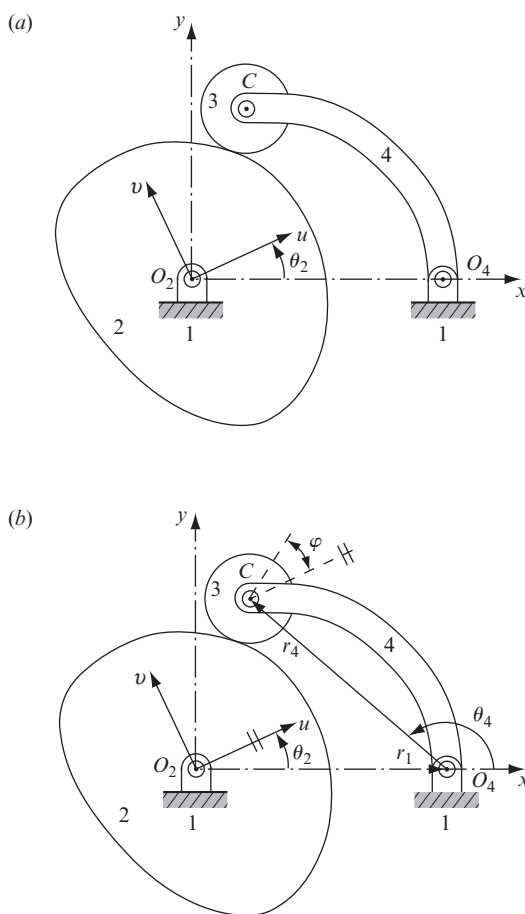


Figure 6.35 Plate cam with oscillating roller follower.

Note that a negative sign is used, since the follower displacement, as shown by Fig. 6.35*b*, is in the clockwise direction.

From Fig. 6.35*b*, the coordinates of the trace point, *C*, in the fixed coordinate system are

$$x = r_1 + r_4 \cos \theta_4, \quad (2a)$$

$$y = r_4 \sin \theta_4. \quad (2b)$$

These coordinates can be transformed to the moving coordinate system of the cam by the relations

$$u = x \cos \Theta + y \sin \Theta, \quad (3a)$$

$$v = -x \sin \Theta + y \cos \Theta. \quad (3b)$$

Therefore, substituting Eqs. (2) into Eqs. (3), the coordinates of the trace point, *C*, in the moving coordinate system are

$$u = (r_4 \sin \theta_4) \sin \Theta + (r_1 + r_4 \cos \theta_4) \cos \Theta,$$

$$v = (r_4 \sin \theta_4) \cos \Theta - (r_1 + r_4 \cos \theta_4) \sin \Theta,$$

which can be written as

$$u = r_1 \cos \Theta + r_4 \cos(\theta_4 - \Theta), \quad (4a)$$

$$v = -r_1 \sin \Theta + r_4 \sin(\theta_4 - \Theta). \quad (4b)$$

Differentiating Eqs. (4) with respect to the rotation angle, Θ , of the cam, the first- and second-order kinematic coefficients of the trace point, *C*, are

$$u' = -r_1 \sin \Theta - r_4 (\theta_4' - 1) \sin(\theta_4 - \Theta), \quad (5a)$$

$$v' = -r_1 \cos \Theta + r_4 (\theta_4' - 1) \cos(\theta_4 - \Theta), \quad (5b)$$

$$u'' = -r_1 \cos \Theta - r_4 (\theta_4' - 1)^2 \cos(\theta_4 - \Theta) - r_4 \theta_4'' \sin(\theta_4 - \Theta), \quad (6a)$$

$$v'' = +r_1 \sin \Theta - r_4 (\theta_4' - 1)^2 \sin(\theta_4 - \Theta) + r_4 \theta_4'' \cos(\theta_4 - \Theta), \quad (6b)$$

where the first- and second-order kinematic coefficients of the follower arm, from Eq. (1), are

$$\theta_4' = -\sin 2\Theta \text{ and } \theta_4'' = -2\cos 2\Theta. \quad (7)$$

The coordinates of the point of contact between the cam and the follower, in the rotating coordinate system, are determined in the same way as for Eqs. (6.35) and (6.36).

At the cam rotation angle $\Theta = 45^\circ$, from Eqs. (1) and (7), we have

$$\theta_4 = 2.118 \text{ rad} = 121.35^\circ, \quad \theta_4' = -1.000 \text{ rad/rad}, \quad \text{and} \quad \theta_4'' = 0.000. \quad (8)$$

Substituting Eqs. (8) into Eqs. (4), the coordinates of the trace point, C , are

$$u = (3.000 \text{ in}) \cos 45^\circ + (2.598 \text{ in}) \cos(121.35^\circ - 45^\circ) = 2.734 \text{ in}, \quad (9a)$$

$$v = -(3.000 \text{ in}) \sin 45^\circ + (2.598 \text{ in}) \sin(121.35^\circ - 45^\circ) = 0.403 \text{ in}. \quad (9b)$$

Substituting Eqs. (8) and the given geometry into Eqs. (5), the first-order kinematic coefficients of the trace point are

$$u' = -(3.000 \text{ in}) \sin 45^\circ - (2.598 \text{ in}) (-2.000) \sin(121.35^\circ - 45^\circ) = 2.928 \text{ in/rad}, \quad (10a)$$

$$v' = -(3.000 \text{ in}) \cos 45^\circ + (2.598 \text{ in}) (-2.000) \cos(121.35^\circ - 45^\circ) = -3.347 \text{ in/rad}, \quad (10b)$$

and from Eq. (6.35), we have

$$w' = +\sqrt{(2.928 \text{ in/rad})^2 + (-3.347 \text{ in/rad})^2} = 4.447 \text{ in/rad}. \quad (11)$$

- (a) Then, substituting Eqs. (9) through (11) into Eqs. (6.36), the coordinates of the contact point between the cam and the follower in the rotating coordinate system are

$$u_{\text{cam}} = (2.734 \text{ in}) + (0.5 \text{ in}) \left(\frac{-3.347 \text{ in/rad}}{4.447 \text{ in/rad}} \right) = 2.358 \text{ in}, \quad \text{Ans.}$$

$$v_{\text{cam}} = (0.403 \text{ in}) - (0.5 \text{ in}) \left(\frac{2.928 \text{ in/rad}}{4.447 \text{ in/rad}} \right) = 0.074 \text{ in}. \quad \text{Ans.}$$

- (b) In the rotating coordinate system of the cam, the direction of motion of the roller center, C , is defined by angle ϕ (Fig. 6.35b), which can be written as

$$\phi = \theta_4 - 90^\circ - \theta = 121.35^\circ - 90^\circ - 45^\circ = -13.65^\circ. \quad (12)$$

Therefore, from Eq. (6.39), the pressure angle of the cam can be written as

$$\cos \phi = \left(-\frac{v'}{w'} \right) \cos \phi + \left(\frac{u'}{w'} \right) \sin \phi.$$

Substituting Eqs. (10), (11), and (12) into this equation, the pressure angle (at the cam rotation angle $\theta = 45^\circ$) is

$$\phi = 54.83^\circ. \quad \text{Ans.}$$

- (c) Substituting Eqs. (8) and the given geometry into Eqs. (6), the second-order kinematic coefficients of the follower center are

$$\begin{aligned} u'' &= -(3.000 \text{ in}) \cos 45^\circ - (2.598 \text{ in}) (-2.000)^2 \cos(121.35^\circ - 45^\circ) \\ &= -4.573 \text{ in/rad}^2, \end{aligned} \quad (13a)$$

$$\begin{aligned} v'' &= (3.000 \text{ in}) \sin 45^\circ - (2.598 \text{ in}) (-2.000)^2 \sin(121.35^\circ - 45^\circ) \\ &= -7.977 \text{ in/rad}^2. \end{aligned} \quad (13b)$$

Then, substituting Eqs. (10), (11), and (13) into Eq. (6.37), the radius of curvature of the pitch curve at this location is

$$\begin{aligned}\rho &= \frac{(4.447 \text{ in/rad})^3}{(2.928 \text{ in/rad})(-7.977 \text{ in/rad}^2) - (-3.347 \text{ in/rad})(-4.573 \text{ in/rad}^2)} \\ &= -2.275 \text{ in.}\end{aligned}$$

The negative sign indicates that the unit normal vector to the pitch curve, through the follower center, points away from the center of curvature, as explained earlier. Therefore, from Eq. (6.38), the radius of curvature of the cam profile is

$$\rho_{\text{cam}} = \rho + R_r = -2.275 \text{ in} + 0.500 \text{ in} = -1.775 \text{ in.} \quad \text{Ans.}$$

6.11 RIGID AND ELASTIC CAM SYSTEMS

Figure 6.36a is a cross-sectional view illustrating the overhead valve arrangement in an automobile engine. In analyzing the dynamics of this or any other cam system, we expect to determine the contact force at the cam surface, the spring force, and the cam-shaft torque, all for a complete rotation of the cam. In one method of analysis, all parts of the cam-follower train, consisting of the push rod, the rocker arm, and the valve stem, together with the cam shaft, are assumed to be rigid. If this is an accurate assumption, and if the speed of the cam-follower train is moderate, then such an analysis usually produces quite satisfactory results. In any event, such a rigid-body analysis should always be attempted as the first step.

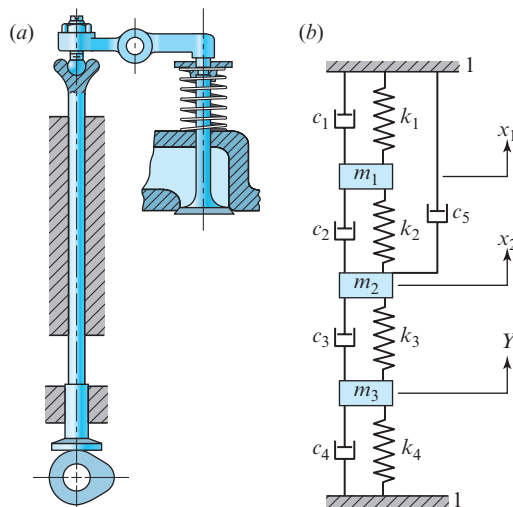


Figure 6.36 An overhead valve arrangement for an automotive engine.

Sometimes the speeds are so high or the members are so elastic (perhaps because of extreme lengths or slenderness) that an elastic-body analysis must be used. This fact is usually discovered when troubles are encountered with the system. Such troubles are usually evidenced by noise, chatter, unusual wear, poor product performance, or perhaps fatigue failure of some of the parts. In other cases, laboratory investigation of the operation of a prototype system may reveal substantial differences between the theoretical and the observed performance.

Figure 6.36b is a mathematical model of an elastic-body cam system. Here, m_3 represents the mass of the cam and a portion of the cam shaft. The motion machined into the cam profile is the coordinate $Y(\Theta)$, a function of the cam-shaft angle, Θ . Masses m_1 and m_2 , and stiffnesses k_2 and k_3 are lumped characteristics of the follower train. The stiffness of the follower retaining spring is k_1 , and the bending stiffness of the cam shaft is k_4 . The dashpots, c_i ($i = 1, 2, 3, 4$, and 5), are inserted to represent the effects of friction, which, in the analysis, may indicate either viscous damping or sliding friction or any combination of the two. The system of Fig. 6.36b is a rather sophisticated one requiring the solution of three simultaneous differential equations. This is not presented here; instead, we focus our attention on simpler systems.

6.12 DYNAMICS OF AN ECCENTRIC CAM

An eccentric is the name given to a circular plate cam with the cam shaft mounted off-center. The distance, e , between the center of the disk and the center of the shaft is the eccentricity. Figure 6.37a shows a simple reciprocating follower eccentric cam system. It consists of an eccentric plate cam, a flat-face follower mass, and a retaining spring of stiffness k . The coordinate Y designates the motion of the follower, as long as the cam remains in contact with the follower. We arbitrarily select $Y = 0$ at the bottom of the stroke. Then, the kinematic quantities of interest are

$$Y = e - e \cos \omega t, \quad \dot{Y} = \omega e \sin \omega t, \quad \text{and} \quad \ddot{Y} = \omega^2 e \cos \omega t, \quad (6.40)$$

where ωt is the cam angle Θ .

To make a rigid-body analysis, we assume no friction and construct a free-body diagram of the follower. In Fig. 6.37b, F_{23} is the cam contact force and F_S is the spring force. In general, F_{23} and F_S do not have the same line of action, so a pair of frame forces, $F_{13,A}$ and $F_{13,B}$, is exerted at bearings A and B .

Before writing the equation of motion, let us investigate the spring force in more detail. Recall that the *spring stiffness*, k , also called the *spring rate*, is defined as the amount of force necessary to deform the spring a unit length. Thus, the units of k are usually expressed in Newtons per meter or pounds per inch. The purpose of the spring is to keep or retain the follower in contact with the cam. Thus, the spring should exert some force even at the bottom of the stroke, where it is extended the most. This force, called the *preload*, P , is the force exerted by the spring when $Y = 0$. Thus, $P = k\delta$, where δ is the deformation through which the spring must be compressed to assemble it.

Summing forces on the follower mass in the Y direction gives

$$\sum F^Y = F_{23} - k(Y + \delta) = m\ddot{Y}. \quad (a)$$

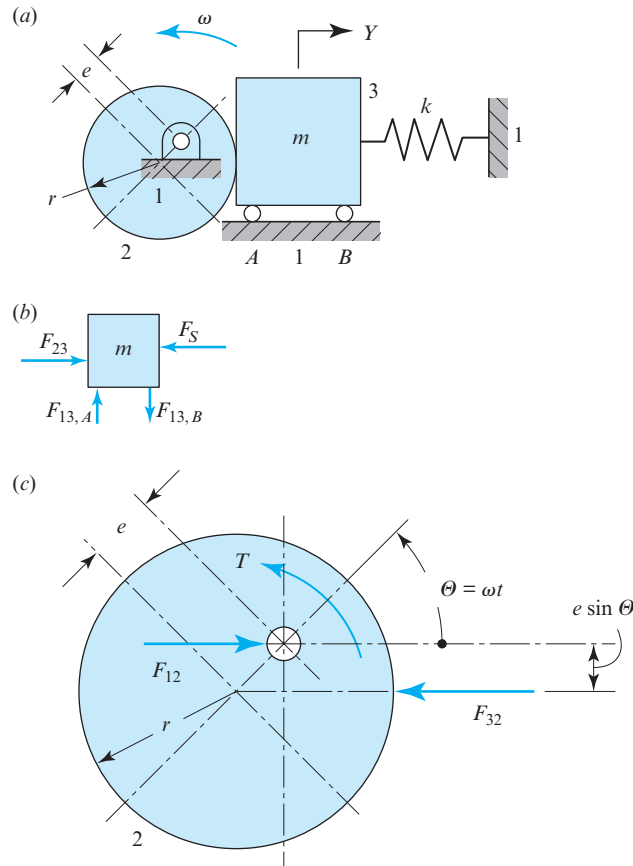


Figure 6.37 (a) Eccentric plate cam with flat-face follower; (b) free-body diagram of the follower; (c) free-body diagram of the cam.

Note that the contact force F_{23} can have only a positive value. Rearranging this equation gives

$$m\ddot{Y} + kY = F_{23} - k\delta \quad \text{or} \quad m\ddot{Y} + kY = F_{23} - P. \quad (b)$$

Then, substituting the first and third equations of Eqs. (6.40) into this equation, and rearranging, the contact force can be written as

$$F_{23} = (m\omega^2 - k)e \cos \omega t + (ke + P). \quad (6.41)$$

This equation and Fig. 6.38a demonstrate that the contact force, F_{23} , consists of a constant term $ke + P$ with a cosine wave superimposed on it. The maximum occurs at $\Theta = 0^\circ$ and the minimum occurs at $\Theta = 180^\circ$. The cosine, or variable, component has an amplitude that depends upon the square of the cam-shaft speed. Thus, as the speed increases, this

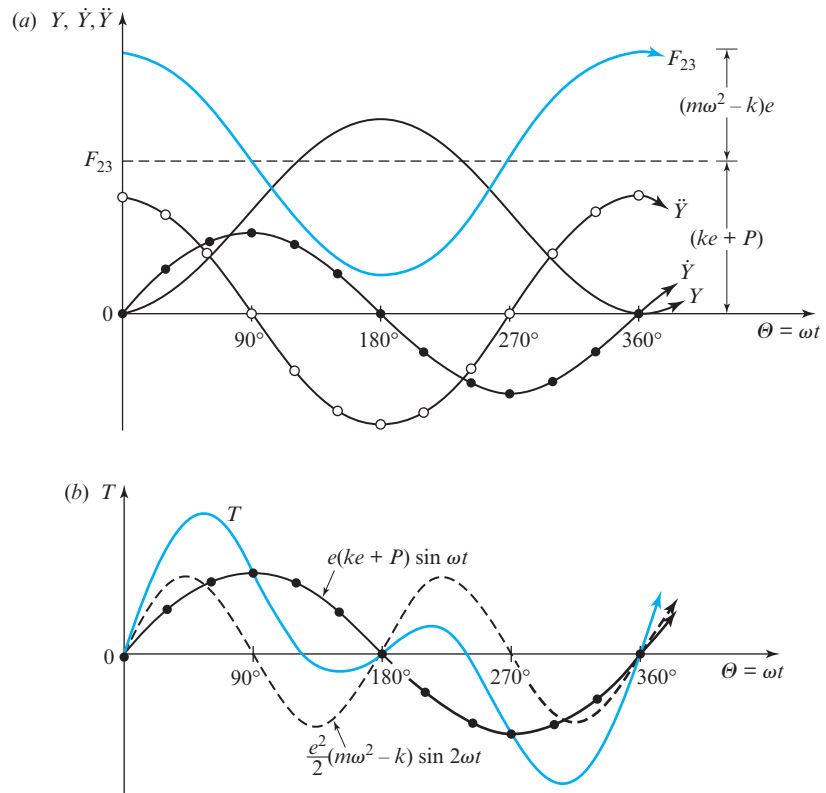


Figure 6.38 (a) Plot of displacement, velocity, acceleration, and contact force for an eccentric cam system; (b) graph of torque components and total cam-shaft torque.

term increases at a greater rate. At some speed, the contact force could become zero at or near $\Theta = 180^\circ$. When this happens, there is usually some impact between the cam and the follower, resulting in clicking, rattling, or very noisy operation. In effect, the sluggishness, or inertia, of the follower prevents it from remaining in contact with the cam. The result is often called *jump* or *float*. The noise occurs when contact is reestablished. Of course, the purpose of the retaining spring is to prevent this. Since the contact force consists of a cosine wave superimposed on a constant term, all we must do to prevent jump is to move, or elevate, the cosine wave away from the zero position. To do this we can increase the constant term, $ke + P$, by increasing the preload, P , or the spring rate, k , or both.

Having learned that jump begins at $\cos \omega t = -1$ with zero contact force (that is, $F_{23} = 0$), we can solve Eq. (6.41) for the jump speed. The result is

$$\omega = \sqrt{\frac{2ke + P}{me}}. \quad (6.42)$$

Using the same procedure, we find that jump will not occur for the range of preload

$$P > e(m\omega^2 - 2k). \quad (6.43)$$

Figure 6.37c is a free-body diagram of the cam. The torque, T , applied by the shaft onto the cam, is

$$T = F_{23}e \sin \omega t.$$

Then, substituting Eq. (6.41) into this equation gives

$$T = \left[(m\omega^2 - k)e \cos \omega t + (ke + P) \right] e \sin \omega t,$$

which, using a trigonometric identity, can be written as

$$T = e(ke + P) \sin \omega t + 1/2e^2(m\omega^2 - k) \sin 2\omega t. \quad (6.44)$$

A plot of this equation is presented in Fig. 6.38b. Note that the torque consists of a double-frequency component, whose amplitude is a function of the cam velocity squared, superimposed on a single-frequency component, whose amplitude is independent of velocity. In this example, the area of the torque-displacement diagram in the positive T direction is the same as in the negative T direction. This means that the energy required to drive the follower in the forward direction is recovered when the follower returns. A flywheel, or inertia, on the cam shaft can be used to store and release this fluctuating energy requirement. Of course, if an external load is connected in some manner to the follower system, the energy required to drive this load will raise the torque curve in the positive direction and increase the area in the positive loop of the T curve.

EXAMPLE 6.8

A cam-and-follower mechanism similar to Fig. 6.37a has the cam machined so that it moves the follower to the right through a distance of 40 mm with parabolic motion in 120° of cam rotation, dwells for 30° , then returns with parabolic motion to the starting posture in the remaining cam angle. The spring rate is 5 kN/m, and the mechanism is assembled with a 35-N preload. The follower mass is 18 kg. Assume no friction. (a) Without computing numeric values, sketch approximate graphs of the displacement, the acceleration, and the cam contact force, all versus the cam angle for the full cycle of events from $\Theta = 0^\circ$ to $\Theta = 360^\circ$ of cam rotation. On this graph, indicate where jump or liftoff is most likely to begin. (b) Using the given data, at what speed would jump begin?

SOLUTION

(a) The cam contact force can be written from Eq. (a) as

$$F = kY + P + m\ddot{Y}, \quad (1)$$

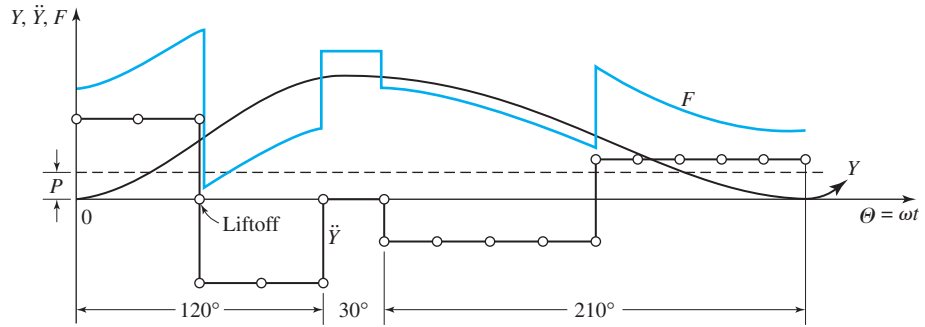


Figure 6.39 Plots of displacement, acceleration, and contact force.

which is composed of the term $m\ddot{Y}$, which varies with the acceleration, the term kY , which varies with the displacement, and the constant term, P . Figure 6.39 shows the displacement diagram of the cam motion described, along with the acceleration, $\ddot{Y}$, and the contact force, F . Note that jump will first occur at $\Theta = \omega t = 60^\circ$, since this is the first posture where F approaches zero.

- (b) Liftoff will occur at the midpoint of the rise where (for $\beta_1 = 120^\circ$) the cam angle is $\Theta = \beta_1/2 = 60^\circ$ when the acceleration becomes negative. The acceleration at this posture [Eq. (6.6c)] is

$$\ddot{Y} = -\frac{4L_1\omega^2}{\beta_1^2} = -\frac{4(0.040\text{m})\omega^2}{[120^\circ(\pi \text{ rad}/180^\circ)]^2} = (-0.0365 \text{ m/rad}^2)\omega^2.$$

Substituting this value, $P = 35 \text{ N}$, and $kY = (5\,000 \text{ N/m})(0.020 \text{ m}) = 100 \text{ N}$ into Eq. (1) with $F = 0$ gives

$$0 = 100 \text{ N} + 35 \text{ N} + (18 \text{ kg})(-0.0365 \text{ m/rad}^2)\omega^2.$$

Then, rearranging this equation gives

$$\omega = \sqrt{\frac{100 \text{ N} + 35 \text{ N}}{(18 \text{ kg})(0.0365 \text{ m/rad}^2)}} = 14.3 \text{ rad/s} = 137 \text{ rev/min.} \quad \text{Ans.}$$

6.13 EFFECT OF SLIDING FRICTION

Let F_μ be the force of sliding (Coulomb) friction as defined by Eq. (11.10). Since the friction force is opposite in sense to the velocity, let us define a sign function as follows:

$$\text{sgn}Z = \begin{cases} +1 & \text{for } Z \geq 0 \\ -1 & \text{for } Z < 0 \end{cases}. \quad (6.45)$$

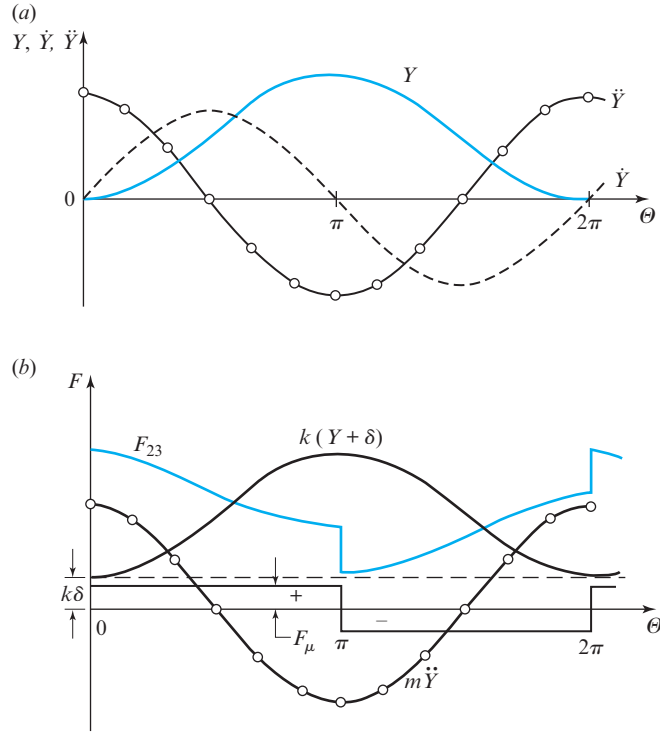


Figure 6.40 Effect of sliding friction on a cam system with harmonic motion: (a) graph of displacement, velocity, and acceleration for one motion cycle; (b) graph of force components F_{μ} , $k\delta$, $k(Y + \delta)$, and the resultant contact force F_{23} .

With this notation, Eq. (a) of Sec. 6.12 can be written as

$$\sum F^Y = F_{23} - F_{\mu} \operatorname{sgn} \dot{Y} - k(Y + \delta) - m\ddot{Y} = 0$$

or

$$F_{23} = F_{\mu} \operatorname{sgn} \dot{Y} + k(Y + \delta) + m\ddot{Y}. \quad (6.46)$$

This equation is plotted for simple harmonic motion with no dwells in Fig. 6.40. By studying both parts of this diagram, we note that F_{μ} is positive when $\dot{Y}$ is positive, and we see how F_{23} is obtained by graphically summing the four component curves.

6.14 DYNAMICS OF DISK CAM WITH RECIPROCATING ROLLER FOLLOWER

In Chap. 11, we will analyze a cam system incorporating a reciprocating roller follower. In this section, we present an analytic approach to a similar difficulty in which sliding friction is also included. The geometry of such a system is shown in Fig. 6.41a. In the analysis to follow here, the effect of follower weight on bearings B and C is neglected.

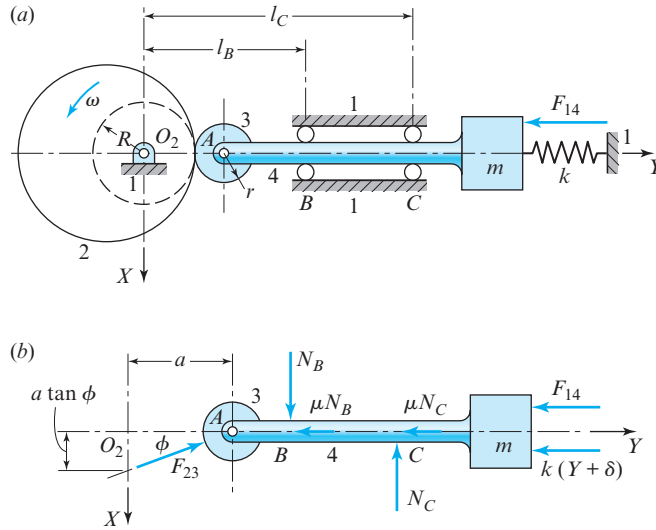


Figure 6.41 (a) Plate cam driving a reciprocating roller-follower system; (b) free-body diagram of the follower system.

Figure 6.41b is a free-body diagram of the follower and roller. If $Y(\Theta)$ is any motion machined into the cam, and $\Theta = \omega t$ is the cam angle, at $Y = 0$ the follower is at the bottom of its stroke and so $Y_0 = R_{AO_2} = R + r$. Therefore,

$$a = Y_0 + Y = R + r + Y. \quad (6.47)$$

In Fig. 6.41b, the roller contact force forms an angle, ϕ , the pressure angle, with the Y axis. Since the direction of the force $\mathbf{F}_{23}$ is the same as the normal to the contacting surfaces, the intersection of this line with the X axis is the common instant center of the cam and follower. This means that the velocity of this point is the same whether it is considered as a point on the follower or a point on the cam. Therefore,

$$\dot{Y} = a\omega \tan \phi,$$

and so

$$\tan \phi = \frac{\dot{Y}}{a\omega} = \frac{Y'}{a}, \quad (6.48)$$

where Y' is the first-order kinematic coefficient of the cam motion. Note that this agrees identically with Eq. (6.33).

In the analysis to follow, the two bearing reactions are N_B and N_C , the coefficient of sliding friction is μ , and δ is the precompression of the retaining spring. Summing forces in the X and Y directions gives

$$\sum F_{13,4}^X = -F_{23}^X + N_B - N_C = 0 \quad (a)$$

and

$$\sum F_{13,4}^Y = F_{23}^Y - \mu \operatorname{sgn} \dot{Y} (N_B + N_C) - F_{14} - k(Y + \delta) - m\ddot{Y} = 0. \quad (b)$$

A third equation is obtained by taking moments about A:

$$\sum M_A = -N_B(l_B - a) + N_C(l_C - a) = 0. \quad (c)$$

With the help of Eq. (6.48), these three equations can be solved for the unknowns F_{23} , N_B , and N_C .

First, we solve Eq. (c) for N_C . This gives

$$N_C = N_B \frac{l_B - a}{l_C - a}. \quad (d)$$

Now we substitute Eq. (d) into Eq. (a) and solve for the bearing reaction N_B . The result is

$$N_B = \frac{l_C - a}{l_C - l_B} F_{23}^X. \quad (e)$$

Substituting $F_{23}^X = F_{23}^Y \tan \phi$ into this equation, and using Eq. (6.48), gives

$$N_B = \frac{F_{23}^Y (l_C - a) \tan \phi}{l_C - l_B} = \frac{(l_C - a) Y'}{(l_C - l_B) a} F_{23}^Y. \quad (6.49)$$

Next, we substitute Eqs. (d) and (6.49) into the friction term of Eq. (b):

$$\mu \operatorname{sgn} \dot{Y} (N_B + N_C) = Y' \operatorname{sgn} \dot{Y} \left[\frac{l_C + l_B - 2a}{(l_C - l_B) a} \right] F_{23}^Y. \quad (f)$$

Substituting this result back into Eq. (b) and solving for F_{23}^Y gives

$$F_{23}^Y = \frac{F_{14} + k(Y + \delta) + m\ddot{Y}}{1 - Y' \operatorname{sgn} \dot{Y} \left[\frac{l_C + l_B - 2a}{(l_C - l_B) a} \right]}. \quad (6.50)$$

For a computer or calculator solution, a simple computation for the sgn function is

$$\operatorname{sgn} Z = \frac{Z}{|Z|}. \quad (6.51)$$

Finally, the cam-shaft torque is

$$T = -a \tan \phi F_{23}^Y = -Y' F_{23}^Y. \quad (6.52)$$

The equations of this section require the kinematic expressions for the appropriate rise and return motions, developed in Sec. 6.7.

6.15 DYNAMICS OF ELASTIC CAM SYSTEMS

Figure 6.42 shows the effect of follower elasticity upon the actual measured displacement and velocity of a follower system driven by a cycloidal cam. To understand what has happened, we must compare these diagrams with the theoretic ones earlier in the chapter. Although the effect of elasticity is more pronounced for velocity, it is usually the variation of the displacement, especially at the top of rise, that causes the most trouble in practical situations. These troubles are usually evidenced by poor or unreliable product quality when elastic systems are used in manufacturing or assembly lines, and they result in noise, unusual wear, and fatigue failure.

A complete analysis of elastic cam systems requires a good background in vibration analysis. To avoid the necessity for this background while still developing a basic understanding, we use an extremely simplified cam system with a linear motion cam. It must be observed, however, that such a cam system would never be used for an actual high-speed application.

In Fig. 6.43a, k_1 is the stiffness of the retaining spring, m is the lumped mass of the follower, and k_2 represents the stiffness of the follower. Since the follower is usually a rod or a lever, k_2 is many times greater than k_1 .

Spring k_1 is assembled with a preload. The coordinate X of the follower motion is chosen at the equilibrium position of the mass after spring k_1 is assembled. Thus, k_1 and k_2 exert equal and opposite preload forces on the mass of the follower. Assuming no friction, the free-body diagram of the mass is as shown in Fig. 6.43b. To determine the directions of the forces, the coordinate X , representing the actual motion of the follower, has been assumed to be larger than the coordinate Y , representing the theoretic follower motion

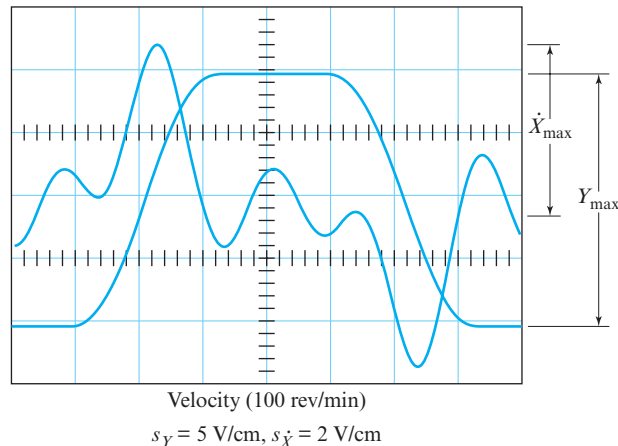


Figure 6.42 Photograph of the oscilloscope traces of the measured displacement and velocity of a dwell-rise-dwell-return cam-and-follower system machined for cycloidal motion. The zero axis of the displacement diagram has been translated downward to obtain a larger diagram in the space available.

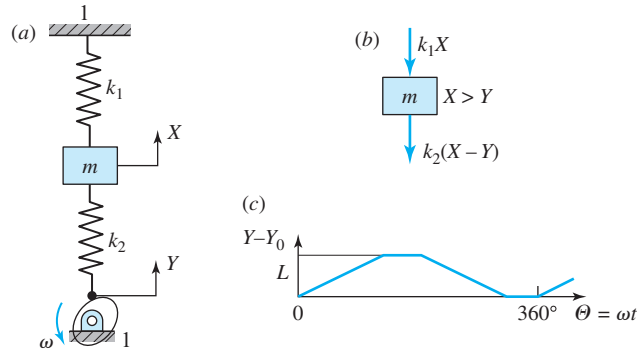


Figure 6.43 (a) Undamped model of a cam-and-follower system; (b) free-body diagram of the follower mass; (c) displacement diagram.

machined into the cam. However, the same result is obtained if Y is assumed to be larger than X .

Using Fig. 6.43b, we find the equation of motion to be

$$\sum F = -k_1 X - k_2 (X - Y) - m\ddot{X} = 0 \quad (a)$$

or

$$\ddot{X} + \frac{k_1 + k_2}{m} X = \frac{k_2}{m} Y. \quad (6.53)$$

This is the differential equation for the motion of the follower. It can be solved as shown in Chap. 13 when function Y is specified. This equation can be solved piecewise for each cam segment; that is, the ending conditions for one segment of motion must be used as the beginning or starting conditions for the next segment.

Let us analyze the first segment of motion using uniform motion, as shown in Fig. 6.43c. First we use the notation

$$\omega_n = \sqrt{\frac{k_1 + k_2}{m}}. \quad (6.54)$$

We should not confuse ω_n with the angular velocity of the cam, ω . The quantity ω_n is called the *undamped natural frequency*. The units of ω_n and ω are both radians per second.

Equation (6.53) can now be written as

$$\ddot{X} + \omega_n^2 X = \frac{k_2 Y}{m}. \quad (6.55)$$

The solution to this equation is

$$X = A \cos \omega_n t + B \sin \omega_n t + \frac{k_2 Y}{m \omega_n^2}, \quad (b)$$

where, for the linear rise segment of duration β_1 , the cam motion is

$$Y = \frac{L}{\beta_1} \Theta = \frac{L\omega t}{\beta_1}. \quad (6.56)$$

Of course Eq. (6.56) is valid only during the rise segment. We can verify Eq. (b) as the solution by substituting it and its second derivative into Eq. (6.55).

The first derivative of Eq. (b) is

$$\dot{X} = -A\omega_n \sin \omega_n t + B\omega_n \cos \omega_n t + \frac{k_2 \dot{Y}}{m\omega_n^2}. \quad (c)$$

For $t = 0$ at the beginning of the rise with $X = \dot{X} = 0$, we find from Eqs. (b) and (c) that

$$A = 0 \quad \text{and} \quad B = -\frac{k_2 \dot{Y}}{m\omega_n^3}.$$

Thus, Eq. (b) becomes

$$X = \frac{k_2}{m\omega_n^2} \left(Y - \frac{\dot{Y}}{\omega_n} \sin \omega_n t \right). \quad (6.57)$$

This equation is plotted in Fig. 6.44. Note that the motion consists of a negative sine term superimposed upon a ramp representing the uniform rise. Because of the additional compression of spring k_2 during the rise, the ramp term $k_2 Y / m\omega_n^2$, called the *follower command* in Fig. 6.44, becomes less than the intended cam rise motion Y .

After the end of the rise, Eqs. (6.55) through (6.57) are no longer valid, and a new segment of motion, a dwell, begins. The follower response for this era is shown in Fig. 6.44, but we will not solve for it here.

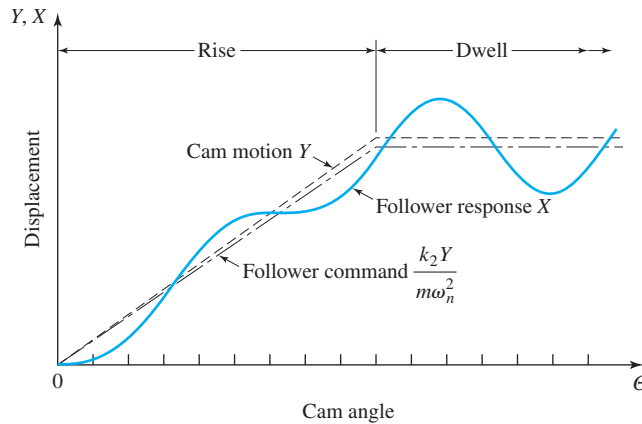


Figure 6.44 Displacement diagram of a uniform-motion cam mechanism showing the follower response.

Equation (6.57) shows that the vibration amplitude $\dot{Y}/\omega_n$ can be reduced by making ω_n large, and Eq. (6.54) demonstrates that this can be done by increasing k_2 , which means that a very rigid follower should be used.

6.16 UNBALANCE, SPRING SURGE, AND WINDUP

Unbalance. As shown in Fig. 6.45a, a disk cam produces unbalance, since its mass is not symmetric about the axis of rotation. This means that two sets of vibratory forces exist, one caused by the eccentric cam mass and the other caused by the reaction of the follower against the cam. By keeping these effects in mind during design, the engineer can do much to guard against difficulties during operation.

Figures 6.45b and 6.45c show that face and cylindric cams have good balance characteristics. For this reason, these are good choices when high-speed operation is involved.

Spring Surge Texts on spring design demonstrate that helical springs may themselves vibrate because of the mass of the spring coils. When serious vibrations exist, a clear wave motion can be seen traveling up and down the spring. This vibration within the retaining spring, called *spring surge*, has been photographed with high-speed cameras, and the results have been exhibited in slow motion. For example, poorly designed automotive valve springs operating near their critical frequency permit the valve to open for short intervals during the period the valve is supposed to be closed. Such conditions result in very poor operation of the engine and rapid fatigue failure of the springs themselves.

Windup Figure 6.38b is a plot of cam-shaft torque, illustrating that the shaft exerts torque on the cam during a portion of the cycle and that the cam exerts torque on the shaft during another portion of the cycle. This varying torque requirement may cause the shaft to twist, or wind up, as the torque increases during follower rise. Also, during this period, the angular cam velocity is slowed and so is the follower velocity. Near the end of rise, the energy stored in the shaft by the windup is released, causing both the follower velocity and acceleration to rise above normal values. The resulting kick may produce follower jump or impact. This effect is most pronounced when heavy loads are being moved by the follower, when the follower moves at high speed, and when the shaft is flexible.

In most cases, a flywheel must be employed in a cam system to provide for varying torque requirements. Cam-shaft windup can be prevented to a large extent by mounting the flywheel as close as possible to the cam. Mounting it a long distance from the cam may actually make matters worse.

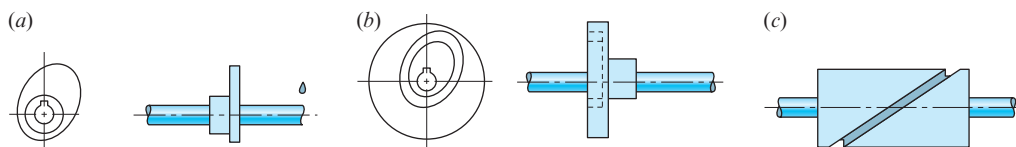


Figure 6.45 (a) A disk cam is inherently unbalanced; (b) a face cam is usually well balanced; (c) a cylindric cam has good balance.

6.17 REFERENCES

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PROBLEMS

- 6.1** The reciprocating radial roller follower of a plate cam is to rise 2 in with simple harmonic motion in 180° of cam rotation and return with simple harmonic motion in the remaining 180° . If the roller radius is 0.375 in and the prime-circle radius is 2 in, construct the displacement diagram, the pitch curve, and the cam profile for clockwise cam rotation.
- 6.2** A plate cam with a reciprocating flat-face follower has the same motion as in Prob. 6.1. The prime-circle radius is 2 in, and the cam rotates counterclockwise. Construct the displacement diagram and the cam profile, offsetting the follower stem by 0.75 in in the direction that reduces the bending of the follower during rise.
- 6.3** Construct the displacement diagram and the cam profile for a plate cam with an oscillating radial flat-face follower that rises through 30° with cycloidal motion in 150° of counterclockwise cam rotation, then dwells for 30° , returns with cycloidal motion in 120° , and dwells for 60° . Determine the necessary length for the follower face, allowing 5 mm clearance at the free end. The prime-circle radius is 30 mm and the follower pivot is 120 mm to the right.
- 6.4** A plate cam with an oscillating roller follower is to produce the same motion as in Prob. 6.3. The prime-circle radius is 60 mm, the roller radius is 10 mm, the length of the follower is 100 mm, and it is pivoted at 125 mm to the right of the cam rotation axis. The cam rotation is clockwise. Determine the maximum pressure angle.
- 6.5** For full-rise simple harmonic motion, write the equations for the velocity and the jerk at the midpoint of the motion. Also, determine the acceleration at the beginning and the end of the motion.
- 6.6** For full-rise cycloidal motion, determine the values of θ for which the acceleration is maximum and minimum. What are the formulae for the accelerations at these positions? Find the equations for the velocity and the jerk at the midpoint of the motion.
- 6.7** A plate cam with a reciprocating follower is to rotate clockwise at 400 rev/min. The follower is to dwell for 60° of cam rotation, after which it is to rise to a lift of 2.5 in. During 1 in of the return motion, it must have a constant velocity of -40 in/s. Recommend standard cam motions from Sec. 6.7 to be used for high-speed operation, and determine the corresponding lifts and cam rotation angles for each segment of the cam.
- 6.8** Repeat Prob. 6.7 except with a dwell for 20° of cam rotation.
- 6.9** If the cam of Prob. 6.7 is driven at constant speed, determine the time of the dwell and the maximum and minimum velocity and acceleration of the follower for the cam cycle.

- 6.10** A plate cam with an oscillating follower is to rise through 20° in 60° of cam rotation, dwell for 45° , then rise through an additional 20° , return, and dwell for 60° of cam rotation. Assuming high-speed operation, recommend standard cam motions from Sec. 6.7 to be used, and determine the lifts and cam rotation angles for each segment of the cam.
- 6.11** Determine the maximum velocity and acceleration of the follower for Prob. 6.10, assuming that the cam is driven at a constant speed of 600 rev/min.
- 6.12** The boundary conditions for a polynomial cam motion are as follows: for $\theta = 0$, $y = 0$, and $y' = 0$, whereas for $\theta = \beta$, $y = L$, and $y' = 0$. Determine the appropriate displacement equation and the first three derivatives of this equation with respect to cam rotation. Sketch the corresponding diagrams.
- 6.13** Determine the minimum face width using 0.1-in allowances at each end and determine the minimum radius of curvature for the cam of Prob. 6.2.
- 6.14** Determine the maximum pressure angle and the minimum radius of curvature for the cam of Prob. 6.1.
- 6.15** A radial reciprocating flat-face follower is to have the motion described in Prob. 6.7. Determine the minimum prime-circle radius if the radius of curvature of the cam is not to be less than 0.5 in. Using this prime-circle radius, what is the minimum length of the follower face using allowances of 0.15 in on each side?
- 6.16** Graphically construct the cam profile of Prob. 6.15 for clockwise cam rotation.
- 6.17** A radial reciprocating roller follower is to have the motion described in Prob. 6.7. Using a prime-circle radius of 20 in, determine the maximum pressure angle and the maximum roller radius that can be used without undercutting.
- 6.18** Graphically construct the cam profile of Prob. 6.17 using a roller radius of 0.75 in. Cam rotation is to be clockwise.
- 6.19** A plate cam rotates at 300 rev/min and drives a reciprocating radial roller follower through a full rise of 75 mm in 180° of cam rotation. Find the minimum radius of the prime circle if simple harmonic motion is used and the pressure angle is not to exceed 25° . Find the maximum acceleration of the follower.

- 6.20** Repeat Prob. 6.19 except that the motion is cycloidal.
- 6.21** Repeat Prob. 6.19 except that the motion is eighth-order polynomial.
- 6.22** Using a roller diameter of 20 mm, determine whether the cam of Prob. 6.19 is undercut.
- 6.23** Equations (6.30) and (6.31) describe the profile of a plate cam with a reciprocating flat-face follower. If such a cam is to be cut on a milling machine with cutter radius R_c , determine similar equations for the center of the cutter.
- 6.24** Write computer programs for each of the displacement equations of Sec. 6.7.
- 6.25** Write a computer program to plot the cam profile for Prob. 6.2.
- 6.26** A plate cam with an offset reciprocating roller follower has a dwell of 60° and then rises in 90° to another dwell of 120° , after which it returns in 90° of cam rotation. The radius of the base circle is 40 mm, the radius of the roller follower is 15 mm, and the follower offset is 20 mm. For the rise motion $60^\circ \leq \theta \leq 150^\circ$, the equation of the displacement (the lift) is

$$y = 40 \left[\frac{\theta}{\pi} + \sin \theta \right],$$

where y is in millimeters and θ is the cam rotation angle in radians. (a) Find equations for the first- and second-order kinematic coefficients of the lift, y , for this rise motion. (b) Sketch the displacement diagram and the first- and second-order kinematic coefficients for the follower motion described. Comment on the suitability of this rise motion in the context of the other displacements specified. At the cam rotation angle $\theta = 120^\circ$, determine the following: (c) the location of the point of contact between the cam and follower, expressed in the moving Cartesian coordinate system attached to the cam; (d) the radius of curvature of the pitch curve and the radius of curvature of the cam surface; and (e) the pressure angle of the cam. Is this pressure angle acceptable?

- 6.27** A plate cam with an offset reciprocating roller follower is to be designed using the input, the rise and fall, and the output motion shown in Table P6.27. The radius of the base circle is 30 mm, the radius of the roller follower is 12.5 mm, and the follower offset (eccentricity) is 15 mm.

Table P6.27 Displacement information for plate cam with reciprocating roller follower.

Cam Angle (θ , deg)	Rise or Fall (L , mm)	Follower Motion
0°–20°	0	Dwell
20°–110°	+25	Full-rise simple harmonic motion
110°–120°	0	Dwell
120°–200°	+5	Full-rise cycloidal motion
200°–270°	0	Dwell
270°–360°	–30	Full-return cycloidal motion

Comment on the suitability of the motions specified. At the cam rotation angle $\theta = 50^\circ$, determine the following: (a) the first-, second-, and third-order kinematic coefficients of the lift curve; (b) the coordinates of the point of contact between the roller follower and the cam surface, expressed in the Cartesian coordinate system rotating with the cam; (c) the radius of curvature of the pitch curve; (d) the unit tangent and the unit normal vectors to the pitch curve; and (e) the pressure angle of the cam.

- 6.28** A plate cam with a radial reciprocating roller follower is to be designed using the input, the rise and fall, and the output motion shown in Table P6.28. The base circle diameter is 3 in and the diameter of the roller is 1 in. Displacements are specified as follows:

Table P6.28 Displacement information for plate cam with reciprocating roller follower.

Input θ (deg)	Lift L (in)	Output y
0°–90°	3.0	Cycloidal rise
90°–105°	0	Dwell
105°–195°	–3.0	Cycloidal fall
195°–210°	0	Dwell
210°–270°	2.0	Simple harmonic rise
270°–285°	0	Dwell
285°–345°	–2.0	Simple harmonic fall
345°–360°	0	Dwell

Plot the lift curve (displacement diagram) and the profile of the cam. (a) Comment on the lift curves at appropriate positions of the cam (for example, when the cam rotation angle is $\theta = 0^\circ$, $\theta = 45^\circ$,

$\theta = 180^\circ$, $\theta = 210^\circ$, $\theta = 225^\circ$, and $\theta = 300^\circ$). (b) Identify on your cam profile the location(s) and the value(s) of the largest pressure angle. Would this pressure angle cause difficulties for a practical cam-and-follower system? (c) Identify on your cam profile the location(s) of any discontinuities in position, velocity, acceleration, and/or jerk. Are these discontinuities acceptable (why or why not)? (d) Identify on your cam profile any regions of positive radius of curvature of the cam profile. Are these regions acceptable (why or why not)? (e) For the values given in Table P6.28, what design changes would you suggest to improve this cam design?

- 6.29** Continue using the same displacement information and the same design parameters as in Prob. 6.28. Use a spreadsheet to determine and plot the following for a complete rotation of the cam: (a) the first-order kinematic coefficients of the follower center; (b) the second-order kinematic coefficients of the follower center; (c) the third-order kinematic coefficients of the follower center; (d) the lift curve (displacement diagram); (e) the radius of curvature of the cam surface; and (f) the pressure angle of the cam-and-follower system. Is the pressure angle suitable for a practical cam-and-follower system?

- 6.30** The cam rotation angle, the rise and fall, and the output motion of a disk cam with a reciprocating roller follower are given in Table P6.30. The diameter of the base circle of the cam is 90 mm, the diameter of the roller follower is 30 mm, and the follower eccentricity is 20 mm.

Table P6.30 Displacement information for plate cam with reciprocating roller follower.

Cam angle θ (degrees)	Lift L (mm)	Output y
0°–45°	0	Dwell
45°–120°	35	Full-rise simple harmonic motion
120°–130°	0	Dwell
130°–180°	15	Full-rise cycloidal motion
180°–210°	0	Dwell
210°–290°	20	Full-return simple harmonic motion
290°–310°	0	Dwell
310°–360°	30	Full-return cycloidal motion

Sketch the displacement diagram and its first two derivatives. At the cam rotation angle $\Theta = 230^\circ$, determine: (a) the first and second-order kinematic coefficients of the displacement diagram; (b) the coordinates of the point of contact between the cam and the roller follower, expressed in the moving Cartesian coordinate system attached to the cam; (c) the radius of the curvature of the cam profile; and (d) the pressure angle of the cam. Is this pressure angle acceptable for this cam-and-follower system?

6.31 The cam angle, the rise and fall, and the output motion of a disk cam with a reciprocating roller follower are as given in Table P6.31. The diameter of the base circle of the cam is 9.60 in, the diameter of the roller follower is 2.40 in, and the eccentricity (offset) of the roller follower is 2.80 in.

Table P6.31 Displacement information for plate cam with reciprocating roller follower.

Cam angle Θ (degrees)	Lift L (in)	Output y
$0^\circ - 5^\circ$	0	Dwell
$5^\circ - 115^\circ$	2.00	Full-rise cycloidal motion
$115^\circ - 120^\circ$	0	Dwell
$120^\circ - 180^\circ$	3.40	Full-rise simple harmonic motion
$180^\circ - 210^\circ$	0	Dwell
$210^\circ - 310^\circ$	4.80	Full-return eighth-order polynomial motion
$310^\circ - 325^\circ$	0	Dwell
$325^\circ - 360^\circ$	0.60	Full-return simple harmonic motion

Sketch the displacement diagram and its first two derivatives. At the cam rotation angle $\Theta = 235^\circ$, determine: (a) the first- and second-order kinematic coefficients of the displacement diagram; (b) the coordinates of the point of contact between the cam and the roller follower, expressed in the rotating Cartesian coordinate system attached to the cam; (c) the radius of the curvature of the cam profile; and (d) the pressure angle of the cam.

6.32 The cam angle, the rise and fall, and the output motion of a disk cam with a reciprocating roller follower are as given in Table P6.32. The diameter of the base circle of the cam is 180 mm, the diameter of the roller follower is 80 mm, and the eccentricity of the roller follower is 40 mm.

Table P6.32 Displacement information for disk cam and reciprocating roller follower.

Cam angle Θ (degrees)	Lift L (mm)	Output y
$0^\circ - 40^\circ$	0	Dwell
$40^\circ - 100^\circ$	60	Half-rise simple harmonic motion
$100^\circ - 180^\circ$	20π	Half-rise cycloidal motion
$180^\circ - 260^\circ$	0	Dwell
$260^\circ - 360^\circ$	$60 + 20\pi$	Full-return cycloidal motion

Part I: Sketch the displacement diagram and its first two derivatives. At the cam rotation angle $\Theta = 85^\circ$, determine: (a) the first-, second-, and third-order kinematic coefficients of the displacement diagram; (b) the radius of curvature of the cam surface; (c) the unit tangent and normal vectors to the cam at the point of contact with the follower; (d) the coordinates of the point of contact between the cam and the follower. Express your answers in the moving Cartesian coordinate system attached to the cam; and (e) the pressure angle of the cam. Part II: Repeat the problem for the cam angle $\Theta = 120^\circ$.

6.33 The cam angle, the rise and fall, and the output motion of a disk cam with a reciprocating roller follower are as given in Table P6.33. The diameter of the base circle of the cam is 2.80 in, the diameter of the roller follower is 1.20 in, and the follower eccentricity is 0.40 in.

Table P6.33 Displacement information for disk cam and reciprocating roller follower.

Cam Angle Θ (degrees)	Lift L (in)	Output y
$0^\circ - 30^\circ$	0	Dwell
$30^\circ - 90^\circ$	1.20	Full-rise simple harmonic motion
$90^\circ - 120^\circ$	0	Dwell
$120^\circ - 180^\circ$	0.80	Full-rise cycloidal motion
$180^\circ - 210^\circ$	0	Dwell
$210^\circ - 270^\circ$	0.80	Full-return cycloidal motion
$270^\circ - 300^\circ$	0	Dwell
$300^\circ - 360^\circ$	1.20	Full-return simple harmonic motion

Sketch the displacement diagram and its first two derivatives. At the cam angle $\Theta = 150^\circ$, determine: (a) the first-, second-, and third-order kinematic coefficients of the displacement diagram; (b) the radius of the curvature of the cam surface; (c) the unit tangent and normal vectors to the cam at the point of contact with the follower; (d) the coordinates of the point of contact between the cam and the follower. Express your answers in the moving Cartesian coordinate system attached to the cam; and (e) the pressure angle of the cam.

- 6.34** The cam angle, the rise and fall, and the output motion of a disk cam with a reciprocating roller follower are given in Table P6.34. The diameter of the base circle of the cam is 75 mm, the diameter of the roller follower is 25 mm, and the follower eccentricity is 20 mm.

Table P6.34 Displacement information for disk cam and the reciprocating roller follower.

Cam Angle Θ (degrees)	Lift L (mm)	Output motion y
$0^\circ - 60^\circ$	0	Dwell
$60^\circ - 180^\circ$	90	Full-rise cycloidal motion
$180^\circ - 240^\circ$	0	Dwell
$240^\circ - 360^\circ$	90	Full-return cycloidal motion

Sketch the displacement diagram and its first two derivatives. At the cam angle $\Theta = 300^\circ$, determine: (a) the first- and second-order kinematic coefficients of the displacement diagram; (b) the coordinates of the point of contact between the cam and the roller follower. Express your answers in the moving Cartesian coordinate system attached to the cam; (c) the radius of the curvature of the cam surface; and (d) the pressure angle of the cam.

- 6.35** The cam rotation angle, the rise and fall, and the output motion of a disk cam with a reciprocating roller follower are as given in Table P6.35. The diameter of the base circle of the cam is 2.80 in, the diameter of the roller follower is 1.20 in, and the follower eccentricity is 0.40 in.

Sketch the displacement diagram and its first two derivatives. At the cam rotation angle $\Theta = 230^\circ$,

Table P6.35 Displacement information for disk cam and the reciprocating roller follower.

Cam Angle Θ (degrees)	Lift L (in)	Output motion y
$0^\circ - 30^\circ$	0	Dwell
$30^\circ - 90^\circ$	1.20	Full-rise simple harmonic motion
$90^\circ - 120^\circ$	0	Dwell
$120^\circ - 180^\circ$	0.80	Full-rise cycloidal motion
$180^\circ - 210^\circ$	0	Dwell
$210^\circ - 270^\circ$	0.80	Full-return cycloidal motion
$270^\circ - 300^\circ$	0	Dwell
$300^\circ - 360^\circ$	1.20	Full-return simple harmonic motion

determine: (a) the first-, second-, and third-order kinematic coefficients of the displacement diagram; (b) the radius of the curvature of the cam surface; (c) the unit tangent and normal vectors to the cam at the point of contact with the follower; and (d) the coordinates of the point of contact between the cam and the follower. Express your answers in the moving Cartesian coordinate system attached to the cam; and (e) the pressure angle of the cam.

- 6.36** The mass, m , is constrained to move only in the vertical direction. The circular cam has an eccentricity of 2 in, a speed of 20 rad/s, and a weight of 8 lb. Neglecting friction, find the angle $\Theta = \omega t$ at the instant the cam–follower jumps.

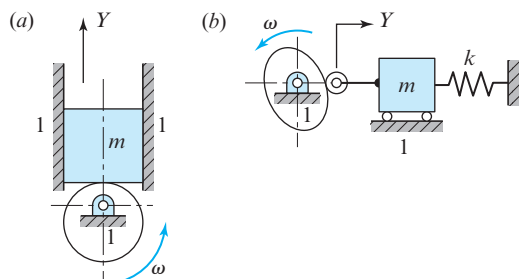


Figure P6.36

- 6.37** In Fig. P6.36a, the mass, m , is driven up and down by the eccentric cam and it has a weight of 10 lb. The cam eccentricity is 1 in. Assume no friction. (a) Derive the equation for the contact force. (b) Find the cam velocity, ω , corresponding to the beginning of the cam–follower jump.

- 6.38** In Fig. P6.36a, the slider has a mass of 2.5 kg. The cam is a simple eccentric and causes the slider to rise 25 mm with no friction. At what cam speed in revolutions per minute will the slider first lose contact with the cam? Sketch a graph of the contact force at this speed for 360° of cam rotation.
- 6.39** The cam-and-follower system in Fig. P6.36b has $k = 1 \text{ kN/m}$, $m = 0.90 \text{ kg}$, $Y = 15 - 15 \cos \omega t \text{ mm}$, and $\omega = 60 \text{ rad/s}$. The retaining spring is assembled with a preload of 2.5 N. (a) Compute the maximum and minimum values of the contact force. (b) If the follower is found to jump off the cam, compute the angle $\Theta = \omega t$ corresponding to the beginning of jump.
- 6.40** Figure P6.36b shows the model of a cam-and-follower system. The motion machined into the cam is to move the mass to the right through a distance of 2 in with parabolic motion in 150° of cam rotation, dwell for 30° , return to the starting position with simple harmonic motion, and dwell for the remaining 30° of cam rotation. There is no friction or damping. The spring rate is 40 lb/in, and the spring preload is 6 lb, corresponding to the $Y = 0$ position. The weight of the mass is 36 lb. (a) Sketch a displacement diagram showing the follower motion for the entire 360° of cam rotation. Without computing numeric values, superimpose graphs of the acceleration and cam contact force onto the same axes. Show where jump is most likely to begin. (b) At what speed in revolutions per minute would jump begin?
- 6.41** A cam-and-follower mechanism is shown in abstract form in Fig. P6.36b. The cam is cut so that it causes the mass to move to the right a distance of 25 mm with harmonic motion in 150° of cam rotation, dwell for 30° , then return to the starting position in the remaining 180° of cam rotation, also with harmonic motion. The spring is assembled with a 22-N preload and it has a rate of 4.4 kN/m. The follower mass is 17.5 kg. Compute the cam speed in revolutions per minute at which jump would begin.
- 6.42** Lever OAB is driven by a cam cut to give the roller a rise of 1 in with parabolic motion and parabolic return with no dwells. The lever and roller are to be assumed weightless and there is no friction. Calculate the jump speed if $l = 5 \text{ in}$ and mass B weighs 5 lb.

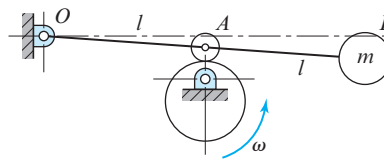
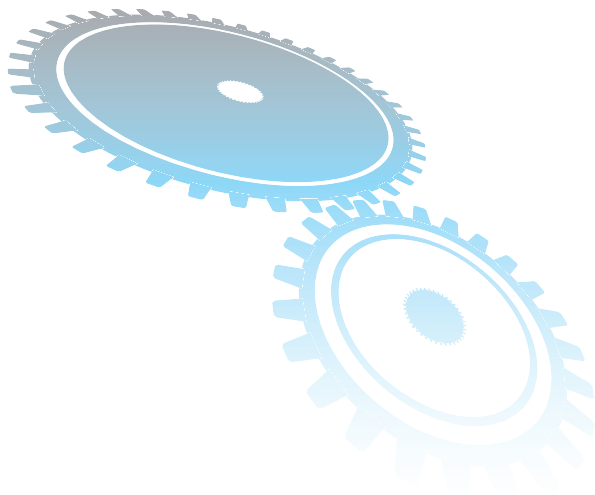


Figure P6.42

- 6.43** A cam-and-follower system similar to the one of Fig. 6.41 uses a plate cam driven at a speed of 600 rev/min and employs simple harmonic rise and parabolic return motions. The events are: rise in 150° , dwell for 30° , and return in 180° . The retaining spring has rate $k = 14 \text{ kN/m}$ with a pre-compression of 12.5 mm. The follower has a mass of 1.6 kg. The external load is related to follower motion Y by the equation $F_{14} = 0.325 - 10.75Y$, where Y is in meters and F_{14} is in kilonewtons. Dimensions corresponding to Fig. 6.41 are $R = 20 \text{ mm}$, $r = 5 \text{ mm}$, $l_B = 60 \text{ mm}$, and $l_C = 90 \text{ mm}$. Using a rise of $L = 20 \text{ mm}$ and assuming no friction, plot the displacement, cam-shaft torque, and radial component of the cam force for one complete revolution of the cam.
- 6.44** Repeat Prob. 6.43 with a speed of 900 rev/min and $F_{14} = 0.110 + 10.75Y \text{ kN}$, where Y is in meters and the coefficient of sliding friction is $\mu = 0.025$.
- 6.45** A plate cam drives a reciprocating roller follower through distance $L = 1.25 \text{ in}$ with parabolic motion in 120° of cam rotation, dwells for 30° , and returns with cycloidal motion in 120° , followed by a dwell for the remaining cam angle. The external load on the follower is $F_{14} = 36 \text{ lb}$ during the rise, and is zero during the dwells and the return. In the notation of Fig. 6.41, $R = 3 \text{ in}$, $r = 1 \text{ in}$, $l_B = 6 \text{ in}$, $l_C = 8 \text{ in}$, and $k = 150 \text{ lb/in}$. The spring is assembled with a preload of 37.5 lb when the follower is at the bottom of its stroke. The weight of the follower is 1.8 lb and the cam velocity is 140 rad/s. Assuming no friction, plot the displacement, the torque exerted on the cam by the shaft, and the radial component of the contact force exerted by the roller against the cam surface for one complete cycle of motion.
- 6.46** Repeat Prob. 6.45 if friction exists with $\mu = 0.04$ and the cycloidal return takes place in 180° .

7

Spur Gears



Gears are machine elements used to transmit rotary motion between two shafts, usually with a constant speed ratio. In this chapter, we will discuss the case where the axes of the two shafts are parallel, and the teeth are straight and parallel to the axes of rotation of the shafts; such gears are called *spur gears*.

7.1 TERMINOLOGY AND DEFINITIONS

A pair of spur gears in mesh is shown in Fig. 7.1. The *pinion* is a name given to the smaller of the two mating gears; the larger is often called the *gear* or the *wheel*. The pair of gears, chosen to work together, is often called a *gearset*.

The terminology of gear teeth is shown in Fig. 7.2, where most of the following definitions are shown.

The *pitch circle* is a theoretical circle on which all calculations are based. The pitch circles of a pair of mating gears are tangent to each other, and it is these pitch circles that were pictured in earlier chapters as rolling against each other without slip.

The *diametral pitch* P is the ratio of the number of teeth on the gear to its pitch diameter; that is,

$$P = \frac{N}{2R}, \quad (7.1)$$

where N is the number of teeth, and R is the pitch circle radius. Note that the diametral pitch cannot be directly measured on the gear itself. Also, note that, as the value of the diametral pitch becomes larger, the teeth become smaller; this is shown clearly in Fig. 7.3. The diametral pitch is used to indicate the tooth size in US customary units and usually has units of teeth per inch. A pair of mating gears has the same diametral pitch.

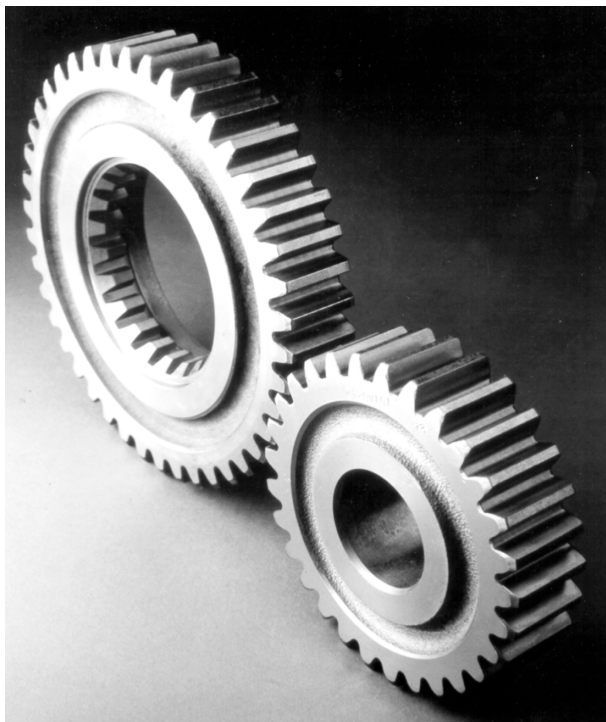


Figure 7.1 Pair of spur gears in mesh. (Courtesy of Gleason Works, Rochester, NY).

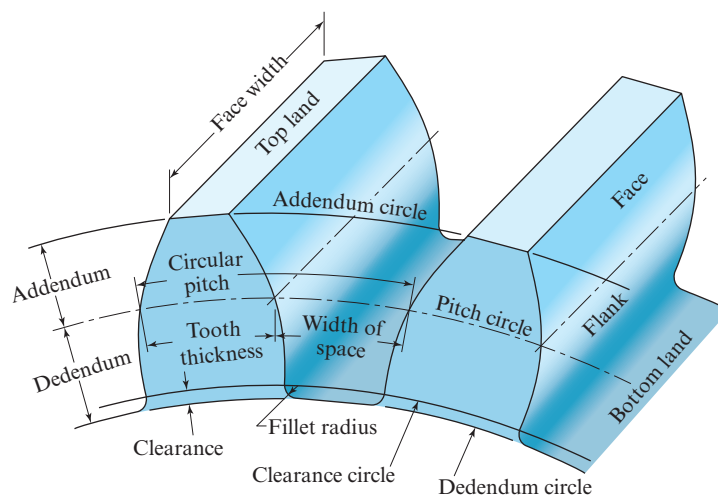


Figure 7.2 Gear tooth terminology.

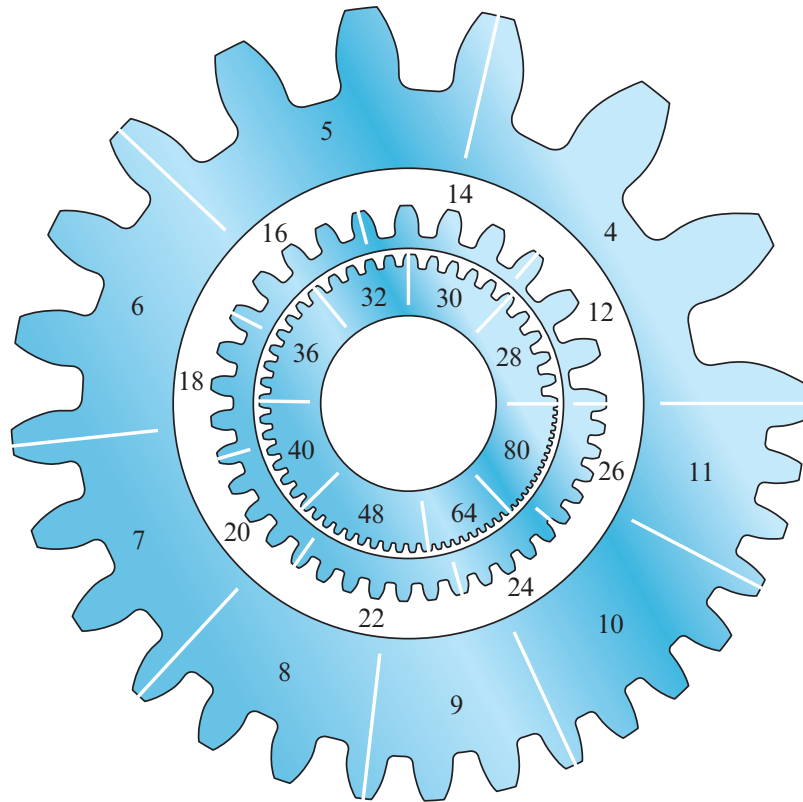


Figure 7.3 Tooth sizes in teeth per inch for various diametral pitches. (Courtesy of Gleason Cutting Tools Corp., Loves Park, IL.)

The *module* m is the ratio of the pitch diameter of the gear to its number of teeth; that is,

$$m = \frac{2R}{N}. \quad (7.2)$$

The module is the usual unit for indicating tooth size in International System (SI) units, and it customarily has units of millimeters per tooth. Note that the module is the reciprocal of the diametral pitch, and the relationship can be written as

$$m = \frac{25.4 \text{ (mm/in)}}{P \text{ (teeth/in)}} = \frac{25.4}{P} \text{ mm/tooth.}$$

Also note that metric gears should not be interchanged with US gears, since their standards for tooth sizes are not the same.

The *circular pitch*, p , is the distance from one tooth to the adjacent tooth, measured along the pitch circle. Therefore, it can be determined from

$$p = \frac{2\pi R}{N}. \quad (7.3)$$

Circular pitch is related to the previous definitions, depending on the units, by

$$p = \frac{\pi}{P} = \pi m. \quad (7.4)$$

The *addendum*, a , is the radial distance from the pitch circle to the top land of each tooth.

The *dedendum*, d , is the radial distance from the pitch circle to the bottom land of each tooth.

The *whole depth* is the sum of the addendum and dedendum.

The *clearance*, c , is the amount by which the dedendum of a gear exceeds the addendum of the mating gear.

The *backlash* is the amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured along the pitch circles.

7.2 FUNDAMENTAL LAW OF TOOTHED GEARING

Gear teeth mating with each other to produce rotary motion are similar to a cam and follower. When the tooth profiles (or cam and follower profiles) are shaped to produce a constant angular velocity ratio between the two shafts, then the two mating surfaces are said to be *conjugate*. It is possible to specify an arbitrary profile for one tooth and then to find a profile for the mating tooth so that the two surfaces are conjugate. One possible choice for such conjugate profiles is the *involute* profile, which, with few exceptions, is in universal use for gear teeth.

A single pair of mating gear teeth must be shaped such that, as they pass through their entire period of contact, the ratio of the angular velocity of the driven gear to that of the driving gear—that is, the first-order kinematic coefficient—*must remain constant*. This is the fundamental criterion that governs the choice of the tooth profiles. If this were not true in gearing, very serious vibration and impact problems would result, even at low speeds.

In Sec. 3.17 we learned the angular-velocity-ratio theorem, which states that the first-order kinematic coefficient of any mechanism is inversely proportional to the segments into which the common instant center cuts the line of centers. In Fig. 7.4, two profiles are in contact at point T ; let profile 2 represent the driver and profile 3 represent the driven element. The normal to the surfaces, CD , is called the *line of action*. The normal to the profiles at the point of contact, T , intersects the line of centers O_2O_3 at the instant center of velocity. In gearing, this instant center is referred to as the *pitch point* and usually carries the label P .

Designating the pitch circle radii of the two gear profiles as R_2 and R_3 , from the angular-velocity-ratio theorem, Eq. (3.28), we see that

$$\frac{\omega_2}{\omega_3} = \frac{R_3}{R_2}. \quad (7.5)$$

This equation is frequently used to define the fundamental law of gearing, which states that as gears go through their mesh, *the pitch point must remain stationary on the line of*

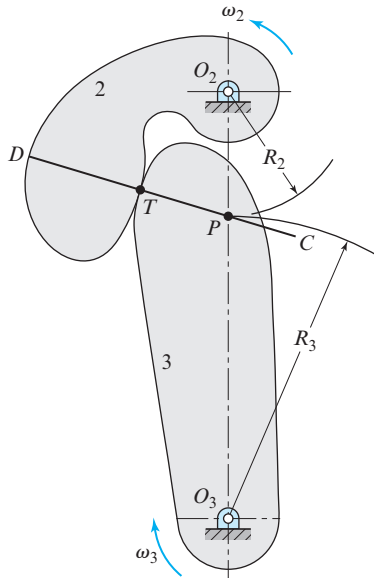


Figure 7.4

centers so that the speed ratio remains constant. This means that the line of action for every new instantaneous point of contact must always pass through the stationary pitch point, P . Thus, the problem of finding a conjugate profile for a given shape is to find a mating shape that satisfies the fundamental law of gearing.

It should not be assumed that any shape or profile is satisfactory just because a conjugate profile can be found. Although, theoretically, conjugate curves might be found, the practical problems of reproducing these curves from steel gear blanks or other materials while using existing machinery still exist. In addition, the sensitivity of the law of gearing to small dimensional changes of the shaft center distance caused either by misalignment or by large forces must also be considered. Finally, the tooth profile selected must be one that can be reproduced quickly and economically in very large quantities. A major portion of this chapter is devoted to illustrating how the involute curve profile fulfills these requirements.

7.3 INVOLUTE PROPERTIES

An *involute* curve is the path generated by a tracing point on a taut cord as the cord is unwrapped from a cylinder called the *base cylinder*. This is shown in Fig. 7.5, where T is the tracing point. Note that the cord, AT , is normal to the involute at T , and distance AT is the instantaneous value of the radius of curvature. As the involute is generated from its origin, T_0 , to T_1 , the radius of curvature varies continuously; it is zero at T_0 and increases continuously to T_1 . Thus, the cord is the generating line, and it is always normal to the involute.

If the two mating tooth profiles both have the shapes of involute curves, the condition that the pitch point, P , remain stationary is satisfied. This is shown in Fig. 7.6, where two gear blanks with fixed centers O_2 and O_3 are shown having base cylinders with respective

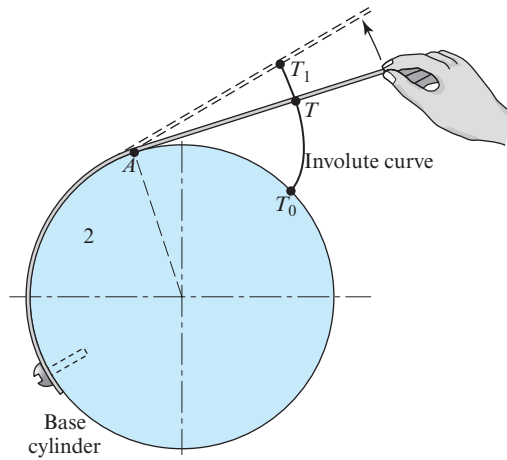


Figure 7.5 Involute curve.

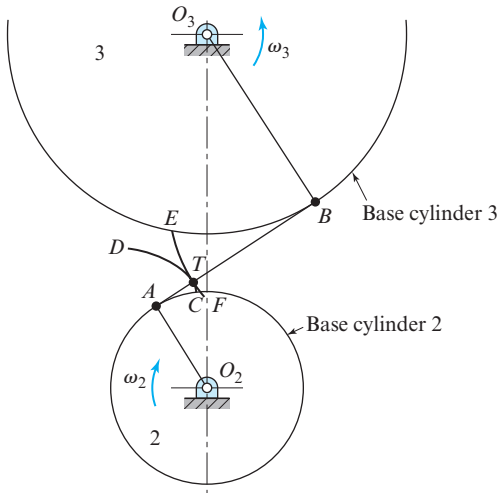


Figure 7.6 Conjugate involute curves.

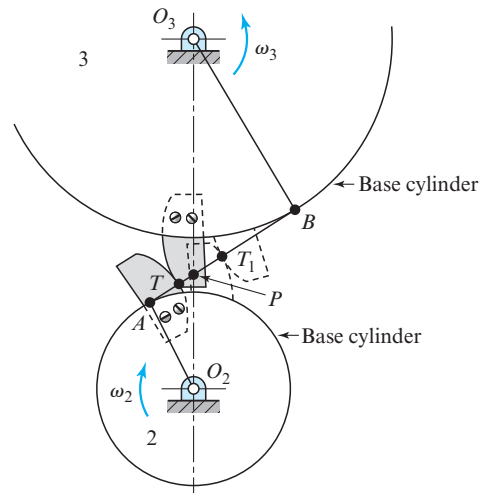


Figure 7.7 Involute action.

radii of O_2A and O_3B . We now imagine that a cord is wound clockwise about the base cylinder of gear 2, pulled taut between points A and B , and wound counterclockwise around the base cylinder of gear 3. If the two base cylinders are now rotated in opposite directions to keep the cord tight, a tracing point, T , traces out the involutes CD on gear 2 and EF on gear 3. The two involutes thus generated simultaneously by the single tracing point, T , are conjugate profiles.

Next imagine that the involutes of Fig. 7.6 are scribed on plates and that the plates are cut along the scribed curves and then bolted to the respective cylinders in the same postures. The result is shown in Fig. 7.7. The cord can now be removed, and, if gear 2 is moved clockwise, gear 3 is caused to move counterclockwise by the camlike action of the

two curved plates. The path of contact is line AB , formerly occupied by the cord. Since line AB is the generating line for each involute, it is normal to both profiles at all points of contact. Also, it always occupies the same position since it is always tangent to both base cylinders. Therefore point P is the pitch point. Point P does not move; therefore, the involute curves are conjugate curves and satisfy the fundamental law of gearing.

7.4 INTERCHANGEABLE GEARS; AGMA STANDARDS

A *tooth system* is the name given to a *standard** that specifies the relationships among addendum, dedendum, clearance, tooth thickness, and fillet radius to attain interchangeability of gears of different tooth numbers but of the same pressure angle and the same diametral pitch or module. We should be aware of the advantages and disadvantages of such a tooth system so that we can choose the best gears for a given design and have a basis for comparison if we depart from a standard tooth profile.

For a pair of spur gears to properly mesh, they must share the same pressure angle and the same tooth size, as specified by the diametral pitch or module. The numbers of teeth and the pitch diameters of the two gears in mesh need not match but are chosen to give the desired speed ratio, as demonstrated in Eq. (7.5).

The sizes of the teeth used are chosen by selecting the diametral pitch, P , or module, m . Standard cutters are generally available for the sizes listed in Table 7.1. Once the diametral pitch or module is chosen, the remaining dimensions of the tooth are set by the standards in Table 7.2. Tables 7.1 and 7.2 contain the standards for the spur gears most in use today, and they include values for both SI and US customary units.

Table 7.1 Standard Gear Tooth Sizes

Standard diametral pitches P , US customary, teeth/in	
Coarse	1, 1 $\frac{1}{4}$, 1 $\frac{1}{2}$, 1 $\frac{3}{4}$, 2, 2 $\frac{1}{2}$, 3, 4, 5, 6, 8, 10, 12, 14, 16, 18
Fine	20, 24, 32, 40, 48, 64, 72, 80, 96, 120, 150, 200
Standard modules m , SI, mm/tooth	
Preferred	1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25, 32, 40, 50
Next choice	1.125, 1.375, 1.75, 2.25, 2.75, 3.5, 4.5, 5.5, 7, 9, 11, 14, 18, 22, 28, 36, 45

* Standards are defined by the American Gear Manufacturers Association (AGMA) and the American National Standards Institute (ANSI). The AGMA standards may be quoted or extracted in their entirety, provided that an appropriate credit line is included—for example, “Extracted from AGMA Information Sheet—Strength of Spur, Helical, Herringbone, and Bevel Gear Teeth (AGMA 225.01) with permission of the publisher, the American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, VA 22314.” These standards have been used extensively in Chaps. 7 and 8.

Table 7.2 Standard Tooth Systems for Spur Gears

System	Pressure angle, ϕ (deg)	Addendum, a	Dedendum, d
Full depth	20°	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Full depth	$22\frac{1}{2}^\circ$	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Full depth	25°	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Stub teeth	20°	$0.8/P$ or $0.8m$	$1/P$ or $1m$

Let us illustrate the design choices by an example.

EXAMPLE 7.1

Two parallel shafts, separated by a distance (commonly referred to as the *center distance*) of 3.5 in, are to be connected by a gear pair so that the output shaft rotates at 40% of the speed of the input shaft. Design a gearset to fit this situation.

SOLUTION

The center distance can be written as $R_2 + R_3 = 3.5$ in, and substituting the given information into Eq. (7.5), we have $\omega_3/\omega_2 = R_2/R_3 = 0.40$. Then, substituting the second equation into the first equation and rearranging, we find that $R_2 = 1.0$ in and $R_3 = 2.5$ in. Next, we must choose the size of the teeth by picking a value for the diametral pitch or module. From Eq. (7.1), we find the number of teeth on the two gears to be $N_2 = 2PR_2 = 2P$ and $N_3 = 2PR_3 = 5P$. The choice of P or m for tooth size is often iterative. First, we might choose a value of $P = 6$ teeth/in; this gives the numbers of teeth as $N_2 = 12$ teeth and $N_3 = 30$ teeth; if we choose $P = 10$ teeth/in, then we get $N_2 = 20$ teeth and $N_3 = 50$ teeth. At this time, either choice appears acceptable, and we choose $P = 10$ teeth/in. However, this choice of P (or m) must later be checked for possible undercutting, as we will study in Sec. 7.7; for contact ratio, which we will study in Sec. 7.8; and for strength and wear of the teeth [4].

7.5 FUNDAMENTALS OF GEAR-TOOTH ACTION

To illustrate the fundamentals, we now proceed, step by step, through the actual graphic layout of a pair of spur gears. The dimensions used are those of Example 7.1, assuming standard 20° full-depth involute tooth form as specified in Table 7.2. The various steps, in the correct order, are shown in Figs. 7.8 and 7.9 and are as follows.

STEP 1 Calculate the two pitch circle radii, R_2 and R_3 , as in Example 7.1; identify O_2 and O_3 as the two shaft centers (Fig. 7.8); and draw the two pitch circles tangent to each other.

STEP 2 Draw the common tangent to the pitch circles perpendicular to the line of centers and through the pitch point P (Fig. 7.8). Draw the *line of action* at an angle equal to the pressure angle $\phi = 20^\circ$ from the common tangent. This line of action corresponds to the generating line discussed in Sec. 7.3; it is always normal to the involute curves and always passes through the pitch point. It is called the *line of action*, since the point of contact

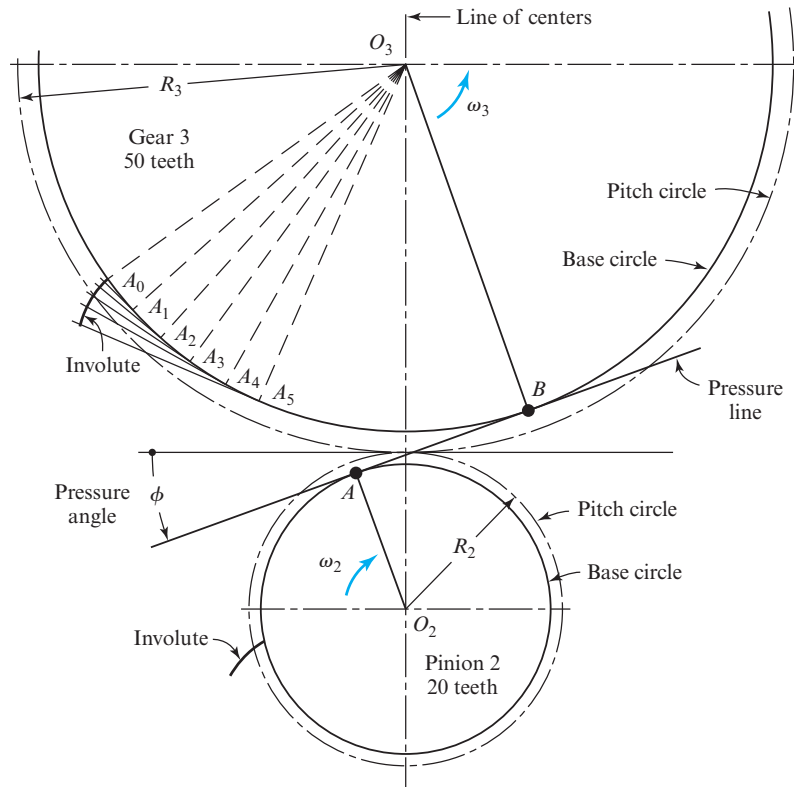


Figure 7.8 Start of layout of a pair of spur gears.

between the gear teeth always lies on this line, the pressure line, since, assuming no friction, the resultant tooth force acts along this line.

STEP 3 Through the centers of the two gears, draw the two perpendiculars, O_2A and O_3B , to the line of action (Fig. 7.8). Draw the two *base circles* with radii of $r_2 = O_2A$ and $r_3 = O_3B$; these correspond to the base cylinders of Sec. 7.3.

STEP 4 From Table 7.2, with $P = 10$ teeth/in, the addendum for both of the gears is found to be

$$a = \frac{1}{P} = \frac{1}{10 \text{ teeth/in}} = 0.10 \text{ in.}$$

Adding this to each of the pitch circle radii, draw the two addendum circles that define the top lands of the teeth on each gear. Carefully identify and label point C where the addendum circle of gear 3 intersects the line of action (Fig. 7.9). Similarly, identify and label point D where the addendum circle of gear 2 intersects the line of action.

Visualizing the rotation of the two gears in the directions given, we see that contact is not possible before point C , since the teeth of gear 3 are not of sufficient height; thus, C is the first point of contact between the teeth. Similarly, the teeth of gear 2 are too short to

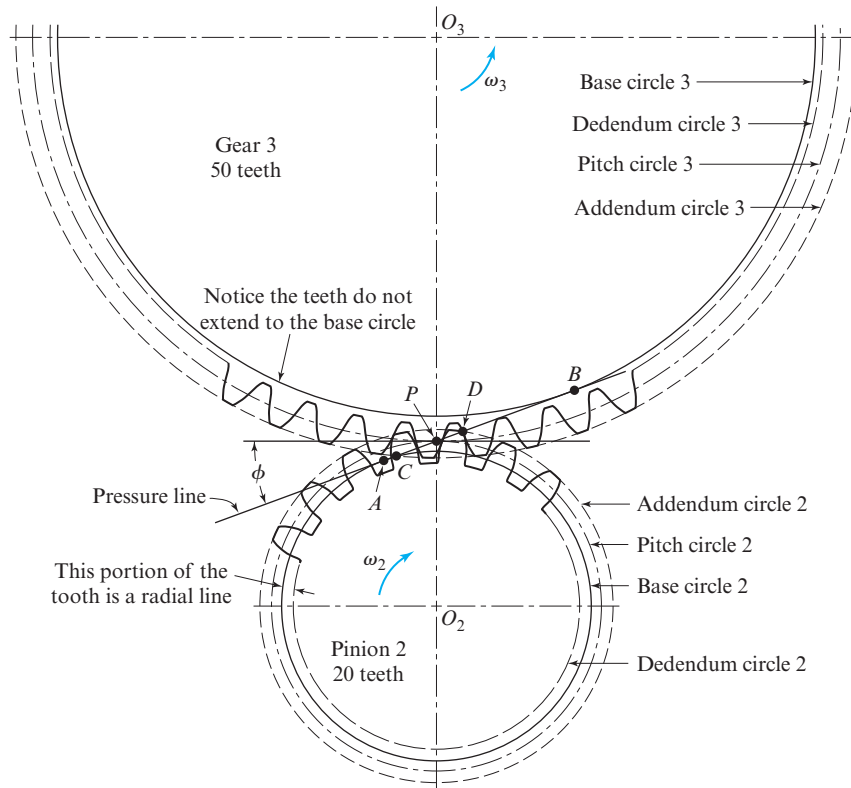


Figure 7.9 Continued layout of the pair of spur gears.

allow further contact after reaching point *D*; thus, contact between a pair of mating teeth begins at *C*, continues to *D*, and then ceases. When contact of a pair of teeth ceases at *D*, one or more pairs of trailing teeth must still be in contact in the span *CD* and take on the sharing of the transmitted load.

Steps 1 through 4 are critical for verifying the choice of any gear pair. We will continue with the diagram shown in Fig. 7.9 when we check for interference, undercutting, and contact ratio in later sections. However, to complete our visualization of gear tooth action, let us first proceed to the construction of the complete involute tooth shapes as shown in Fig. 7.8.

STEP 5 From Table 7.2, the dedendum for each gear is found to be

$$d = \frac{1.25}{P} = \frac{1.25}{10 \text{ teeth/in}} = 0.125 \text{ in.}$$

Subtracting this from each of the pitch circle radii, draw the two dedendum circles that define the bottom lands of the teeth on each gear (Fig. 7.9). Note that the dedendum circles often lie quite close to the base circles; however, they have distinctly different meanings. In

this example, the dedendum circle of gear 3 is larger than its base circle, and the dedendum circle of pinion 2 is smaller than its base circle. However, this is not always the case.

STEP 6 Generate an involute curve on each base circle as shown for gear 3 in Fig. 7.8. This is done by first dividing a portion of the base circle into a series of equal small parts, A_0, A_1, A_2 , and so on. Next, the radial lines O_3A_0, O_3A_1, O_3A_2 , and so on, are constructed, and tangents to the base circle are drawn perpendicular to each of these. The involute begins at A_0 . The second point is obtained by striking an arc, with center A_1 and radius A_0A_1 , up to the tangent line through A_1 . The next point is found by striking a similar arc with center at A_2 , and so on. This construction is continued until the involute curve is generated far enough to meet the addendum circle of gear 3. If the dedendum circle lies inside of the base circle, as is true for pinion 2 of this example, then, except for the fillet, the involute curve is extended inward to the dedendum circle by a radial line; this portion of the curve is not involute.

STEP 7 Using cardboard or preferably a sheet of clear plastic, cut a template for the involute curve and mark on it the center point of the corresponding gear. Note that two templates are needed, since the involute curves are different for gears 2 and 3.

STEP 8 Calculate the circular pitch using Eq. (7.4):

$$p = \frac{\pi}{P} = \frac{\pi}{10 \text{ teeth/in}} = 0.314 \text{ 16 in/tooth.}$$

This distance from one tooth to the next is now marked along the pitch circle, and the template is used to draw the involute portion of each tooth (Fig. 7.9). The width of a tooth and that of a tooth space are each equal to half of the circular pitch or $(0.314 \text{ 16 in/tooth})/2 = 0.157 \text{ 08 in/tooth}$. These distances are marked along the pitch circle, and the same template is turned over and used to draw the opposite sides of the teeth. The portion of the tooth space between the clearance and the dedendum may be used for a fillet radius. The top and bottom lands are now drawn as circular arcs along the addendum and dedendum circles to complete the tooth profiles. The same process is performed on the other gear using the other template.

Remember that steps 5 through 8 are not necessary for the proper design of a gearset. They are only included here to help us to visualize the relation between real tooth shapes and the theoretical properties of the involute curve.

Involute Rack We may imagine a *rack* as a spur gear having an infinitely large pitch diameter. Therefore, in theory, a rack is infinitely long and has an infinite number of teeth, and its base circle is located an infinite distance from the pitch point. For involute teeth, the curves on the sides of the teeth of a rack become straight lines making an angle with the line of centers equal to the pressure angle. The addendum and dedendum distances are the same as those given in Table 7.2. Figure 7.10 shows an involute rack in mesh with the pinion of the previous example.

Base Pitch Corresponding sides of involute teeth are parallel curves. The *base pitch* is the constant and fundamental distance between these curves—that is, the distance from one tooth to the next—measured along the common normal to the tooth profiles, which is the line of action (Fig. 7.10). The base pitch, p_b , and the circular pitch, p , are related as follows:

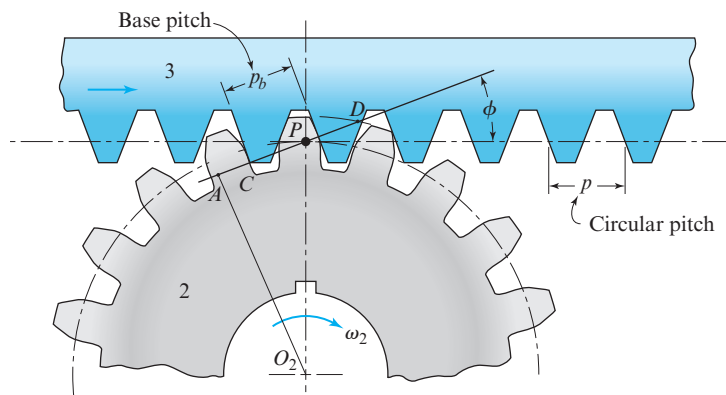


Figure 7.10 Involute pinion and rack.

$$p_b = p \cos \phi. \quad (7.6)$$

The base pitch is a much more fundamental measurement, as we will see later.

Internal Gear Figure 7.11 depicts the pinion of the preceding example in mesh with an *internal*, or *annular*, gear. With internal contact, both centers are on the same side of the pitch point. Thus, the locations of the addendum and dedendum circles of an internal gear are reversed with respect to the pitch circle; the addendum circle of the internal gear lies *inside* the pitch circle, whereas the dedendum circle lies *outside* the pitch circle. The base circle lies inside the pitch circle as with an external gear, but is now near the addendum circle. Otherwise, Fig. 7.11 is constructed in the same manner as was Fig. 7.9.

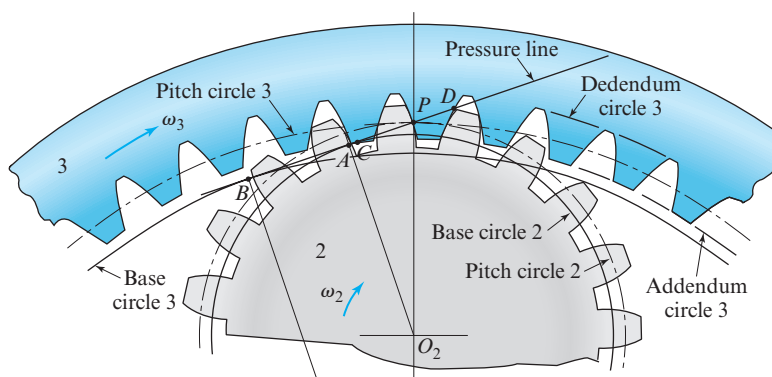


Figure 7.11 Involute pinion and internal gear.

7.6 MANUFACTURE OF GEAR TEETH

There are many ways of manufacturing the teeth of gears; for example, they can be made by sand casting, shell molding, investment casting, permanent-mold casting, die casting, or centrifugal casting. They can be formed by the powder-metallurgy process, or a single bar of aluminum can be formed by extrusion and then sliced into gears. Gears that carry large loads in comparison with their size are usually made of steel and are cut with either *form cutters* or *generating cutters*. In form cutting, the cutter is of the exact shape of the tooth space. With generating cutters, a tool having a shape different from the tooth space is moved through several cuts relative to the gear blank to obtain the proper shape for the teeth.

Probably the oldest method of cutting gear teeth is *milling*. A form milling cutter corresponding to the shape of the tooth space, such as that shown in Fig. 7.12a, is used to machine one tooth space at a time, as shown in Fig. 7.12b, after which the gear is indexed through one circular pitch to the next posture. Theoretically, with this method, a different cutter is required for each gear to be cut, since, for example, the shape of the tooth space in a 25-tooth gear is different from the shape of the tooth space in, say, a 24-tooth gear. Actually, the change in tooth space shape is not very large, and eight form cutters can be used to cut any gear in the range from 12 teeth to a rack with reasonable accuracy. Of course, a separate set of form cutters is required for each pitch.

Shaping is a highly favored method of *generating* gear teeth. The cutting tool may be either a rack cutter or a pinion cutter. The operation is explained by reference to Fig. 7.13. For shaping, the reciprocating cutter is first fed into the gear blank until the pitch circles are tangent. Then, after each cutting stroke, the gear blank and the cutter roll slightly on their pitch circles. When the blank and cutter have rolled by a distance equal to the circular pitch, one tooth has been generated and the cutting continues with the next tooth until all teeth have been cut. Shaping of an internal gear with a pinion cutter is shown in Fig. 7.14.

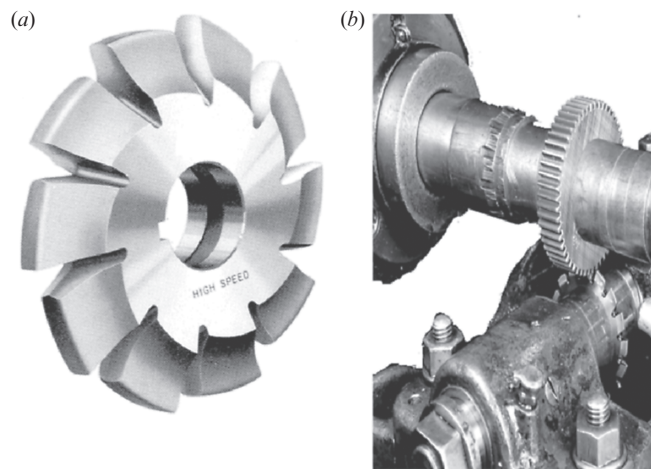
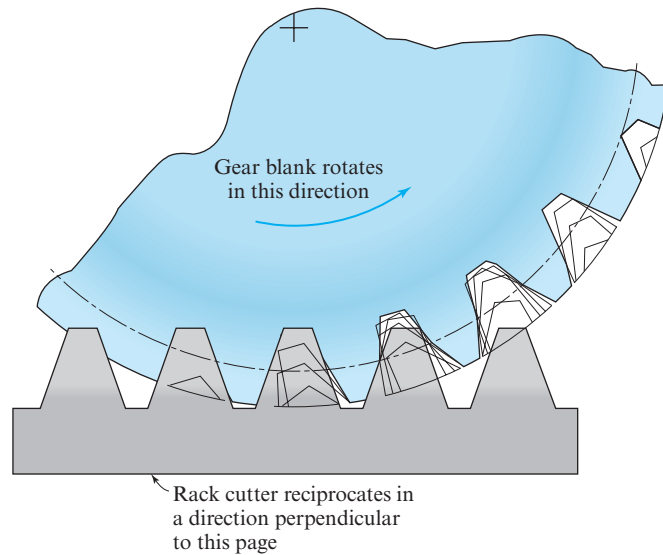


Figure 7.12 Manufacture of gear teeth by a form cutter: (a) a single-tooth involute hob; (b) machining of a single tooth space. (Courtesy of Gleason Works, Rochester, NY.)



 **Figure 7.13** Shaping of involute teeth with a rack cutter.

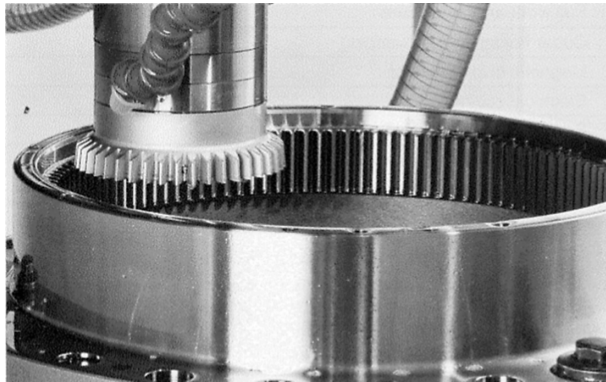


Figure 7.14 Shaping of an internal gear with a pinion cutter. (Courtesy of Gleason Works, Rochester, NY.)

Hobbing is another method of generating gear teeth, which is quite similar to shaping them with a rack cutter. However, hobbing is done with a special tool called a *hob*, a cylindric cutter with one or more helical threads quite like a screw-thread tap; the threads have straight sides like a rack. A number of different gear hobs are displayed in Fig. 7.15. A view of the hobbing of a gear is shown in Fig. 7.16. The hob and the gear blank are both



Figure 7.15 A variety of involute gear hobs. (Courtesy of Gleason Works, Rochester, NY.)



Figure 7.16 The hobbing of a gear. (Courtesy of Gleason Works, Rochester, NY.)

rotated continuously at the proper angular velocity ratio, and the hob is fed slowly across the face of the blank to cut the full thickness of the teeth.

Following the cutting process, grinding, lapping, shaving, and burnishing are often used as final finishing processes when tooth profiles of very high accuracy and surface finish are desired.

7.7 INTERFERENCE AND UNDERCUTTING

Figure 7.17 shows the pitch circles of the same gears used for discussion in Sec. 7.5. Let us assume that the pinion is the driver and that it is rotating clockwise.

We saw in Sec. 7.5 that for involute teeth, contact always takes place along the line of action, AB . Contact begins at point C , where the addendum circle of the driven gear crosses the line of action. Thus, initial contact is on the tip of the driven gear tooth and on the flank of the pinion tooth.

As the pinion tooth drives the gear tooth, contact approaches the pitch point, P . Near the pitch point, contact slides *up* the flank of the pinion tooth and *down* the face of the gear tooth. At the pitch point, contact is at the pitch circles; note that P is the instant center of velocity, and therefore the motion must be rolling with no slip at that point. Note also that this is the only location where the motion is true rolling.

As the teeth recede from the pitch point, the point of contact continues to travel in the same direction as before along the line of action. Contact continues to slide *up* the face of the pinion tooth and *down* the flank of the gear tooth. The last point of contact occurs at the tip of the pinion and the flank of the gear tooth, at the intersection, D , of the line of action and the addendum circle of the pinion.

The *approach* phase of the motion is the interval between the initial contact at point C and the pitch point, P . The *angles of approach* are the angles through which the two gears rotate as the point of contact progresses from C to P . However, reflecting on the unwrapping cord analogy of Fig. 7.6, we see that the distance CP is equal to a length

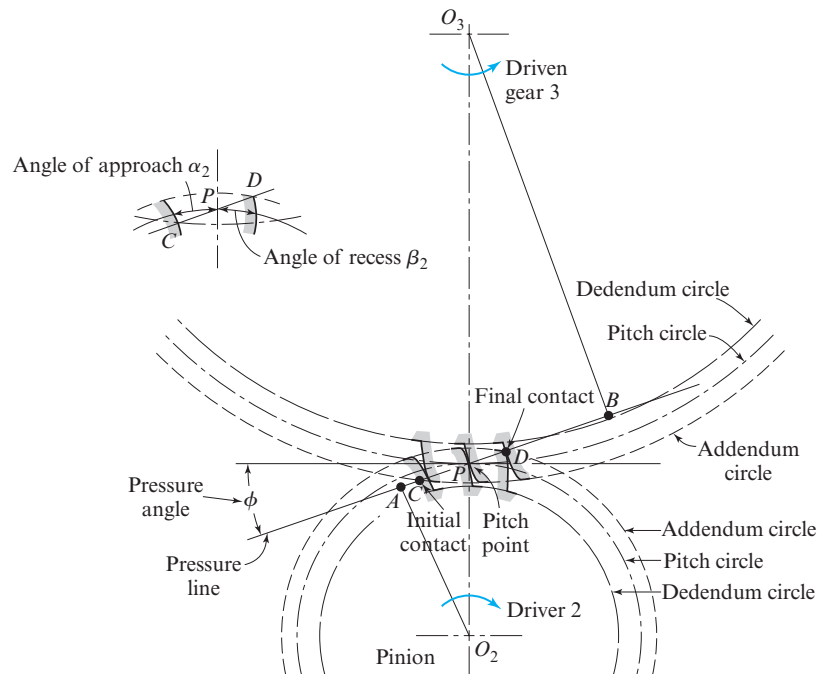


Figure 7.17 Approach and recess phases of gear tooth action.

of cord unwrapped from the base circle of the pinion during the approach phase of the motion. Similarly, an equal amount of cord has wrapped onto the driven gear during that same phase. Thus, the angles of approach for the pinion and the gear, in radians, are

$$\alpha_2 = \frac{CP}{r_2} \quad \text{and} \quad \alpha_3 = \frac{CP}{r_3}. \quad (7.7)$$

The *recess* phase of the motion is the interval during which contact progresses from the pitch point, P , to final contact at point D . The *angles of recess* are the angles through which the two gears rotate as the point of contact progresses from P to D . Again, from the unwrapping cord analogy, we find these angles, in radians, to be

$$\beta_2 = \frac{PD}{r_2} \quad \text{and} \quad \beta_3 = \frac{PD}{r_3}. \quad (7.8)$$

If the teeth come into contact such that they are not conjugate, this is called *interference*. Consider Fig. 7.18; shown here are two 16-tooth $14\frac{1}{2}^\circ$ pressure angle gears* with

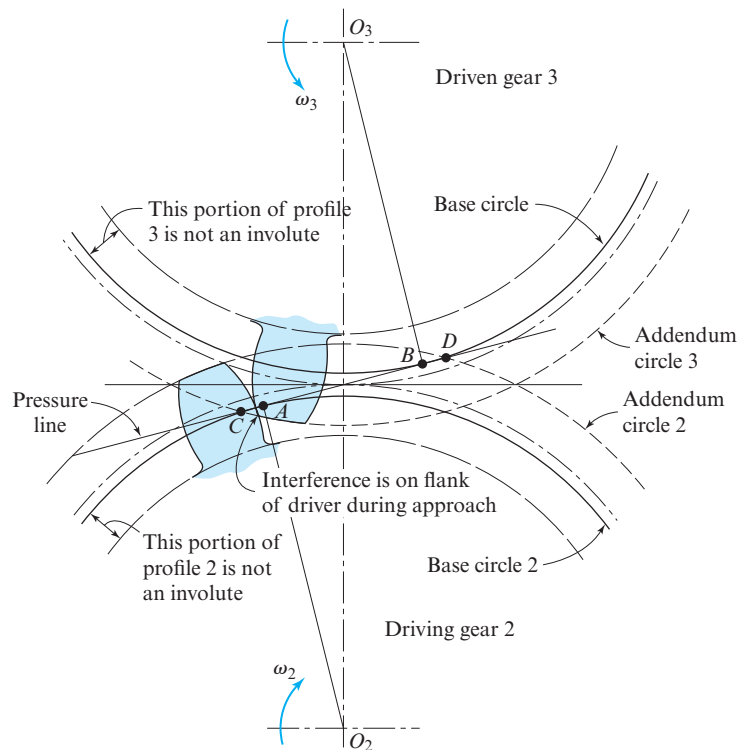


Figure 7.18 Interference in gear tooth action.

* Such gears were part of an older standard and are now obsolete. They are chosen here only to illustrate an example of interference.

full-depth involute teeth. The driver, gear 2, turns clockwise. As with previous figures, the points labeled *A* and *B* indicate the points of tangency of the line of action with the two base circles, whereas the points labeled *C* and *D* indicate the initial and final points of contact. Note that points *C* and *D* are now *outside* points *A* and *B*. This indicates interference.

The interference is explained as follows. Contact begins when the tip of the driven gear 3 tooth contacts the flank of the driving tooth. In this case, the flank of the driving tooth first tries to make contact with the driven tooth at point *C*, and this occurs *before* the involute portion of the driving tooth comes within range. In other words, contact occurs before the two teeth become tangent. The actual effect is that the non-tangent tip of the driven gear tooth interferes with and digs into the flank of the driving tooth.

In this example, a similar effect also occurs as the teeth leave contact. Contact should end at or before point *B*. Since, for this example, contact does not end until point *D*, the effect is for the non-tangent tip of the driving tooth to interfere with and dig into the flank of the driven tooth.

When gear teeth are produced by a generating process, interference is automatically eliminated, since the cutting tool removes the interfering portion of the tooth flank. This effect is called *undercutting*. If undercutting is at all pronounced, the undercut tooth is considerably weakened. Thus, the effect of eliminating interference by a generation process is merely to substitute another problem for the original.

The importance of the problem of teeth that have been weakened by undercutting cannot be emphasized too strongly. Of course, interference can be eliminated by using more teeth on the gears. However, if the gears are to transmit a given amount of power, more teeth can be used only by increasing the pitch diameter. This makes the gears larger, which is seldom desirable. It also increases the pitch-line velocity, which makes the gears noisier and somewhat reduces the power transmitted, although not in direct proportion. In general, however, the use of more teeth to eliminate interference or undercutting is seldom an acceptable solution.

Another method of reducing interference and the resulting undercutting is to employ a larger pressure angle. The larger pressure angle creates smaller base circles, so that a greater portion of the tooth profile has an involute shape. In effect, this means that fewer teeth can be used; as a result, gears with a larger pressure angle are often smaller.

Of course, the use of standard gears is far less expensive than manufacturing specially made nonstandard gears. However, as indicated in Table 7.2, gears with larger pressure angles can be made without deviating from the standards.

One more way to eliminate interference is to use gears with shorter teeth. If the addendum distance is reduced, then points *C* and *D* move inward. One way to do this is to purchase standard gears and then grind the top lands of the teeth to a new addendum distance. This, of course, makes the gears nonstandard and causes concern about repair or replacement, but it can be effective in eliminating interference. Again, careful study of Table 7.2 indicates that this is also possible by use of standard 20° *stub tooth* gears.

7.8 CONTACT RATIO

The zone of action of meshing gear teeth is shown in Fig. 7.19, where tooth contact begins and ends at the intersections of the two addendum circles with the line of action. As always, initial contact occurs at *C*, and final contact occurs at *D*. Tooth profiles drawn through

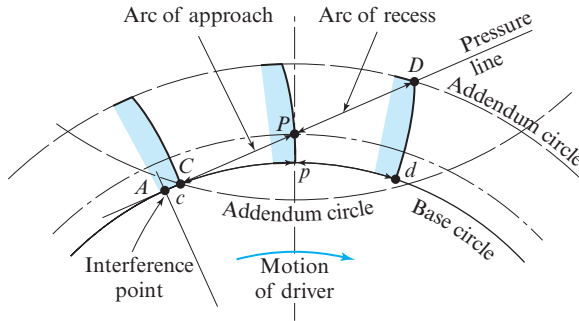


Figure 7.19

these points intersect the base circle at points c and d . Thinking back to our analogy of the unwrapping cord of Fig. 7.6, the linear distance CD , measured along the line of action, is equal to the arc length cd , measured along the base circle.

Consider a situation in which the arc length cd , or distance CD , is exactly equal to the base pitch, p_b , of Eq. (7.6). This means that one tooth and its tooth space spans the entire arc cd . In other words, when a tooth is just beginning contact at C , the tooth ahead of it is just ending its contact at D . Therefore, during the tooth action from C to D , there is exactly one pair of teeth in contact at all times.

Next, consider a situation for which the arc length cd , or distance CD , is greater than the base pitch, but not much greater, say $cd = 1.1p_b$. This means that when one pair of teeth is just entering contact at C , the previous pair, already in contact, has not yet reached D . Thus, for a short time, there are two pairs of teeth in contact, one in the vicinity of C , and the other nearing D . As meshing proceeds, the previous pair reaches D and ceases contact, leaving only a single pair of teeth in contact again, until the situation repeats itself with the next pair of teeth.

Because of the nature of this tooth action, with one, two, or even more pairs of teeth in contact simultaneously, it is convenient to define the term contact ratio as

$$m_c = \frac{CD}{p_b}. \quad (7.9)$$

This is a value for which the next lower integer indicates the average number of pairs of teeth in contact. Thus, a contact ratio of $m = 1.35$, for example, implies that there is always at least one tooth in contact, and there are two teeth in contact 35% of the time.

The minimum acceptable value of the contact ratio for smooth operation of meshing gears is $1.2 \leq m_c \leq 1.4$, and the recommended range of the contact ratio for most spur gearsets is $m_c > 1.4$.

The distance CD is quite convenient to measure if we are working graphically by making a drawing like Fig. 7.20 or Fig. 7.9. However, distances CP and PD can also be found analytically. From triangles O_3BC and O_3BP , we can write

$$CP = \sqrt{(R_3 + a)^2 - (R_3 \cos \phi)^2} - R_3 \sin \phi. \quad (7.10)$$

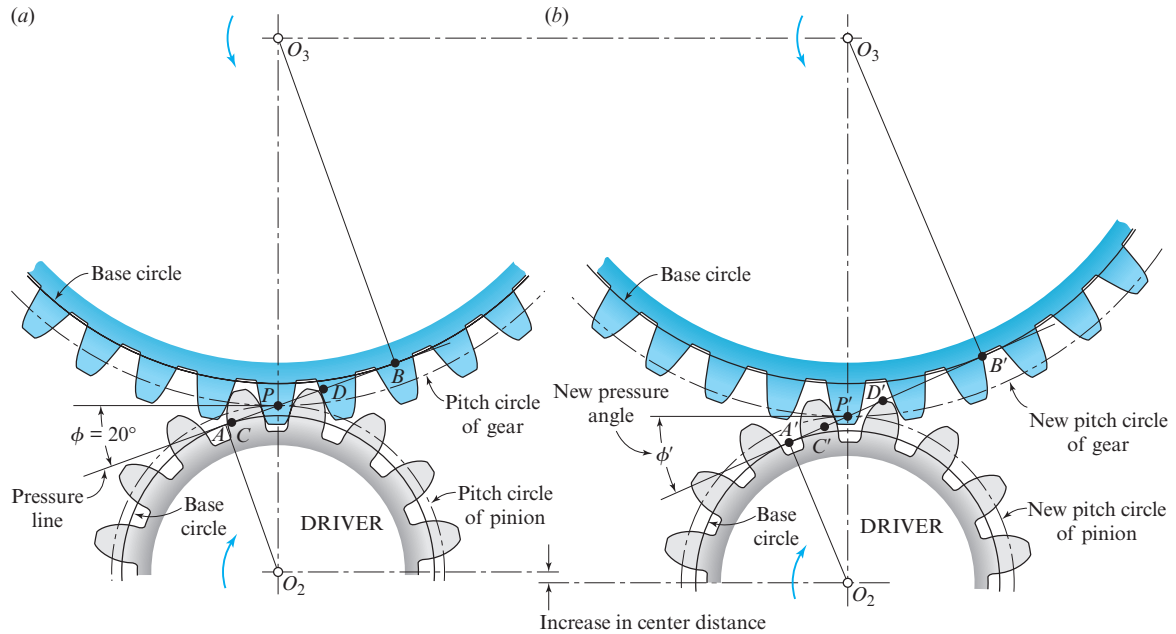


Figure 7.21 Effect of increased center distance on the action of involute gearing; mounting the gears at (a) normal center distance; (b) increased center distance.

When the center distance is increased, the base circles of the two gears do not change; they are fundamental to the shapes of the gears, once manufactured. However, review of Fig. 7.6 indicates that the same involute tooth shapes still touch as conjugate curves, and the fundamental law of gearing is still satisfied. However, the larger center distance results in an increase of the pressure angle and larger pitch circles passing through a new adjusted pitch point.

In Fig. 7.21b we see that triangles $O_2A'P'$ and $O_3B'P'$ are still similar to each other, although they are both modified by the change in pressure angle. Also, distances O_2A' and O_3B' are the base circle radii and have not changed. Therefore, the ratio of the new pitch radii, O_2P' and O_3P' , and the new velocity ratio remain the same as in the original design.

Another effect of increasing the center distance, observable in Fig. 7.21, is the shortening of the path of contact. The original path of contact CD in Fig. 7.21a is shortened to $C'D'$ in Fig. 7.21b. The contact ratio, Eq. (7.9), is also reduced when the path of contact $C'D'$ is shortened. Since a contact ratio of less than unity would imply periods during which no teeth would be in contact at all, the center distance must never be increased larger than that corresponding to a contact ratio of unity ($C'D' = p_b$).

7.10 INVOLUTOMETRY

Study of the geometry of the involute curve is called *involutometry*. In Fig. 7.22, a base circle with center at O is used to generate involute BC . AT is the generating line, ρ is the instantaneous radius of curvature of the involute, and r is the radius to point T on the

curve. If we designate the radius of the base circle R_b , line AT has the same length as the arc distance AB and so

$$\rho = r_b (\alpha + \varphi), \quad (a)$$

where α is the angle between the radius OT and the radius to the origin of the involute OB , and φ is the angle between the radius of the base circle OA and the radius OT . Since OAT is a right triangle,

$$\rho = r_b \tan \varphi. \quad (7.13)$$

Solving Eqs. (a) and (7.13) simultaneously to eliminate ρ and r_b gives

$$\alpha = \tan \varphi - \varphi,$$

which can be written

$$\text{inv } \varphi = \tan \varphi - \varphi \quad (7.14)$$

and defines the *involute function*. The angle φ in this equation is the variable involute angle, given in radians. Once φ is known, $\text{inv } \varphi$ can readily be determined from Eq. (7.14). The inverse problem, when $\text{inv } \varphi$ is given and φ is to be found, is more difficult. One approach is to expand Eq. (7.14) in an infinite series and to employ the first several terms to obtain a numeric approximation. Another approach is to use a root-finding technique [2, Chap. 4]. Here, we refer to Table 6 of Appendix A, in which the value of the involute function is tabulated, and the angle φ can be determined directly, in degrees.

Referring again to Fig. 7.22, we see that

$$r = \frac{r_b}{\cos \varphi}. \quad (7.15)$$

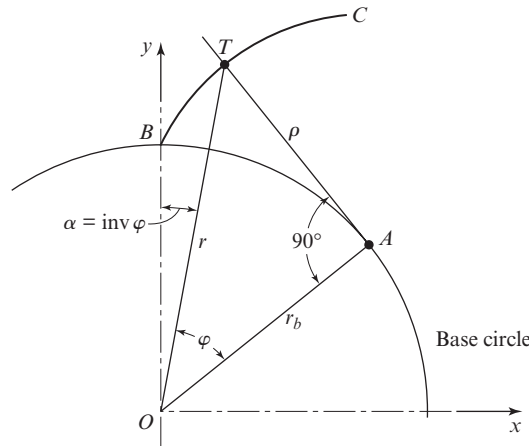


Figure 7.22

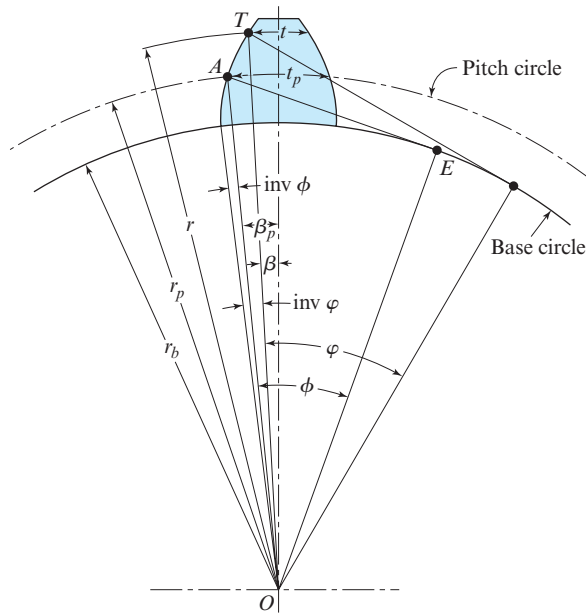


Figure 7.23

To illustrate the use of the relations obtained earlier, let us determine the tooth dimensions of Fig. 7.23. Here, only the portion of the tooth extending above the base circle has been drawn, and the thickness of the tooth, t_p , at the pitch circle (point A), equal to half of the circular pitch, is given. The problem is to determine the tooth thickness at some other point, say point T. The various quantities shown in Fig. 7.23 are identified as follows:

- r_b = radius of the base circle;
- r_p = radius of the pitch circle;
- r = radius at which the tooth thickness is to be determined;
- t_p = tooth thickness at the pitch circle;
- t = tooth thickness to be determined;
- ϕ = pressure angle corresponding to the pitch circle radius r_p ;
- φ = involute angle corresponding to point T;
- β_p = angular half-tooth thickness at the pitch circle; and
- β = angular half-tooth thickness at point T.

The half-tooth thicknesses at points A and T are

$$\frac{t_p}{2} = \beta_p r_p \quad \text{and} \quad \frac{t}{2} = \beta r, \quad (b)$$

so that

$$\beta_p = \frac{t_p}{2r_p} \quad \text{and} \quad \beta = \frac{t}{2r}. \quad (c)$$

From these, we can write

$$\text{inv } \phi - \text{inv } \phi = \beta_p - \beta = \frac{t_p}{2r_p} - \frac{t}{2r}. \quad (d)$$

The tooth thickness at point T is obtained by solving Eq. (d) for t :

$$t = 2r \left(\frac{t_p}{2r_p} + \text{inv } \phi - \text{inv } \phi \right). \quad (7.16)$$

EXAMPLE 7.2

A gear has 22 teeth cut full depth with pressure angle $\phi = 20^\circ$ and diametral pitch $P = 2$ teeth/in. Find the thickness of the teeth at the base circle and at the addendum circle.

SOLUTION

By the equations of Sec. 7.1 and Table 7.2, we find the radius of the pitch circle is $R_p = R = N/2P = 5.500$ in, the circular pitch is $p = 1.571$ in/tooth, the addendum is $a = 0.500$ in, and the dedendum is $d = 0.625$ in.

From right triangle $OE A$ in Fig. 7.23, the radius of the base circle can be written as

$$r_b = r_p \cos \phi = (5.500 \text{ in}) \cos 20^\circ = 5.168 \text{ in}.$$

The thickness of the tooth at the pitch circle is

$$t_p = \frac{p}{2} = \frac{1.571 \text{ in/tooth}}{2} = 0.7855 \text{ in}.$$

Converting the tooth pressure angle into radians gives $\phi = 20^\circ = 0.349066$ rad. Then, the involute function from Eq. (7.14) is

$$\text{inv } \phi = \tan 0.349066 - 0.349066 = 0.014904 \text{ rad}.$$

The involute angle at the base circle from Eq. (7.15), is $\phi_b = 0$. Therefore, the involute function is

$$\text{inv } \phi_b = 0.$$

Substituting these results into Eq. (7.16), the tooth thickness at the base circle is

$$t_b = 2r_b \left[\frac{t_p}{2r_p} + \text{inv } \phi - \text{inv } \phi_b \right] = 2(5.168 \text{ in}) \left[\frac{0.7855 \text{ in}}{2(5.500 \text{ in})} + 0.014904 - 0 \right] = 0.892 \text{ in}.$$

Ans.

The radius of the addendum circle is $r_a = r_p + a = 5.500 + 0.500 = 6.000$ in. Therefore, the involute pressure angle corresponding to this radius, from Eq. (7.15), is

$$\phi = \cos^{-1} \left(\frac{r_b}{r} \right) = \cos^{-1} \left(\frac{5.168 \text{ in}}{6.000 \text{ in}} \right) = 30.53^\circ = 0.532806 \text{ rad}.$$

Thus, the involute function is

$$\text{inv } \phi = \tan 0.532\ 806 - 0.532\ 806 = 0.056\ 887 \text{ rad.}$$

Substituting these results into Eq. (7.16), the tooth thickness at the addendum circle is

$$\begin{aligned} t_a &= 2r_a \left[\frac{t_p}{2r_p} + \text{inv } \phi - \text{inv } \phi \right] = 2(6.000 \text{ in}) \left[\frac{0.785\ 5 \text{ in}}{2(5.500 \text{ in})} + 0.014\ 904 - 0.056\ 887 \right] \\ &= 0.353 \text{ in.} \end{aligned} \quad \text{Ans.}$$

Note that the tooth thickness at the base circle is more than double the tooth thickness at the addendum circle.

7.11 NONSTANDARD GEAR TEETH

In this section, we will examine the effects obtained by deviating from the specified standards and modifying such things as pressure angle, tooth depth, addendum, or center distance. Some of these modifications do not eliminate interchangeability; all of them are discussed with the intent of obtaining improved performance. Still, making such modifications probably means increased cost, since modified gears are available and must be specially machined for the particular application. Of course, it is also necessary at the time of any future repair or design modification.

The designer is often under great pressure to produce a design using gears that are small and yet able to transmit a large amount of power. Consider, for example, a gearset that must have a 4:1 velocity ratio. If the smallest pinion that will carry the load has a pitch diameter of 2 in, the mating gear has a pitch diameter of 8 in, making the overall space required for the two gears more than 10 in. On the other hand, if the pitch diameter of the pinion can be reduced by only $\frac{1}{4}$ in, the pitch diameter of the gear is reduced by a full 1 in, and the overall size of the gearset is reduced by $1\frac{1}{4}$ in. This reduction assumes considerable importance when it is realized that the associated machine elements, such as shafts, bearings, and enclosure, are also reduced in size.

If a tooth of a certain pitch is required to carry the load, the only method of decreasing the pinion diameter is to use fewer teeth. However, we have already seen that problems involving interference, undercutting, and contact ratio are encountered when the tooth numbers are made too small. Thus, three principal reasons for employing nonstandard gears are: (a) to eliminate undercutting, (b) to prevent interference, and (c) to maintain a reasonable contact ratio. Note also that if a pair of gears is manufactured of the same material, the pinion is the weaker and is subject to greater wear, since each of its teeth are in contact a greater portion of the time. Therefore, any undercutting weakens the tooth that is already weaker. Thus, another objective of nonstandard gears is to gain a better balance of strength between the pinion and the gear.

As an involute curve is generated from its base circle, its radius of curvature becomes larger and larger. Near the base circle, the radius of curvature is quite small, being theoretically zero at the base circle. Contact near this region of sharp curvature should be avoided if possible because of the difficulty in obtaining good cutting accuracy in areas

of small curvature, and since contact stresses are likely to be very high. Nonstandard gears present the opportunity of designing to avoid these sensitive areas.

Clearance Modification A larger fillet radius at the root of the tooth increases the fatigue strength of the tooth and provides extra depth for shaving the tooth profile. Since interchangeability is not lost, the dedendum is sometimes increased to $1.300/P$ or $1.400/P$ to obtain space for a larger fillet radius.

Center-Distance Modification When gears of low tooth numbers are paired with each other or with larger gears, reduction in interference and improvement in the contact ratio can be obtained by increasing the center distance to greater than standard. Although such a system changes the tooth proportions and the pressure angle of the gears, the resulting tooth shapes can be generated with rack cutters (or hobs) of standard pressure angles or with standard pinion shapers. Before introducing this system, however, it will be of value to develop certain additional relations about the geometry of gears.

The first new relation is for finding the thickness of a tooth that is cut by a rack cutter (or hob) when the pitch line of the rack cutter is displaced or offset a distance, e , from the pitch circle of the gear being cut. What we are doing here is moving the rack cutter away from the center of the gear being cut. Stated another way, suppose the rack cutter does not cut as deeply into the gear blank, and the teeth are not cut to full depth. This produces teeth that are thicker than the standard, and this thickness will now be determined. Figure 7.24a shows the problem, and Fig. 7.24b shows the solution. The increase of tooth thickness at the pitch circle is $2e \tan \phi$, so that

$$t = 2e \tan \phi + \frac{p}{2}, \quad (7.17)$$

where ϕ is the pressure angle of the rack cutter and t is the thickness of the modified gear tooth measured on its modified pitch circle.

Now suppose that two gears of different tooth numbers have both been cut with the cutter offset from their pitch circles as in the previous paragraph. Since the teeth of both have been cut with offset cutters, they will mate at a modified pressure angle and with modified pitch circles and consequently modified center distances. The word *modified* is used here in the sense of being nonstandard. Our problem is to determine the radii of these modified pitch circles and the value of the modified pressure angle.

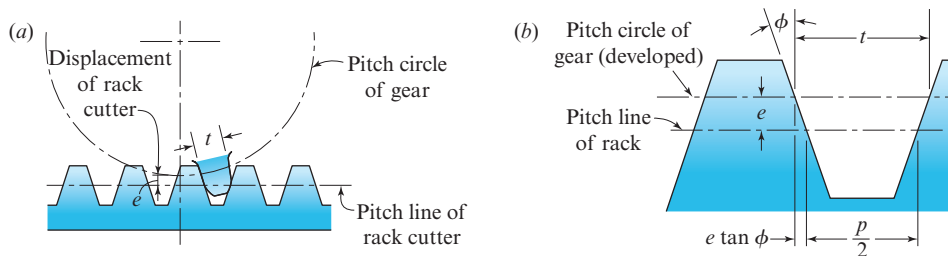


Figure 7.24

In the following notation, the word *standard* refers to values that would have been obtained had the usual, or standard, systems been employed to obtain the dimensions:

- ϕ = standard pressure angle of generating rack cutter;
- ϕ' = modified pressure angle at which gears will mate;
- R_2 = standard pitch radius of pinion;
- R'_2 = modified pitch radius of pinion when meshing with given gear;
- R_3 = standard pitch radius of gear;
- R'_3 = modified pitch radius of gear when meshing with given pinion;
- t_2 = thickness of pinion tooth at standard pitch radius R_2 ;
- t'_2 = thickness of pinion tooth at modified pitch radius R'_2 ;
- t_3 = thickness of gear tooth at standard pitch radius R_3 ; and
- t'_3 = thickness of gear tooth at modified pitch radius R'_3 .

From Eq. (7.16), the thickness of a gear tooth at the standard pitch radius and at the modified pitch radius can be written, respectively, as

$$t'_2 = 2R'_2 \left(\frac{t_2}{2R_2} + \text{inv } \phi - \text{inv } \phi' \right) \quad (a)$$

and

$$t'_3 = 2R'_3 \left(\frac{t_3}{2R_3} + \text{inv } \phi - \text{inv } \phi' \right). \quad (b)$$

Note that the sum of these two thicknesses must be the new circular pitch. Therefore, using Eq. (7.3), we can write

$$t'_2 + t'_3 = p' = \frac{2\pi R'_2}{N_2}. \quad (c)$$

Since the pitch diameters of a pair of mating gears are proportional to their tooth numbers, then

$$R_3 = \frac{N_3}{N_2} R_2 \quad \text{and} \quad R'_3 = \frac{N_3}{N_2} R'_2. \quad (d)$$

Substituting Eqs. (a), (b), and (d) into Eq. (c) and rearranging gives

$$\text{inv } \phi' = \frac{N_2 (t'_2 + t'_3) - 2\pi R_2}{2R_2 (N_2 + N_3)} + \text{inv } \phi. \quad (7.18)$$

This equation gives the modified pressure angle ϕ' , at which a pair of gears will operate when the tooth thicknesses on their standard pitch circles are modified to t'_2 and t'_3 .

Although the base circle of a gear is fundamental to its shape and fixed once the gear is generated, gears have no pitch circles until a pair of them are brought into contact. Bringing a pair of gears into contact creates a pair of pitch circles that are tangent to each other at the modified pitch point. Throughout this discussion, the idea of a pair of so-called standard pitch circles has been used to define a certain point on the involute curves. These standard pitch circles, as we have seen, are the ones that would come into existence when

the gears are paired *if the gears are not modified from the standard dimensions*. On the other hand, the base circles are fixed circles that are not changed by tooth modifications. The base circle remains the same whether the tooth dimensions are changed or not, so we can determine the base circle radius using either the standard pitch circle or the new pitch circle. Thus, from Eq. (7.15), we can write

$$R_2 \cos \phi = R'_2 \cos \phi'.$$

Therefore, the modified pitch radius of the pinion can be written as

$$R'_2 = \frac{R_2 \cos \phi}{\cos \phi'}. \quad (7.19)$$

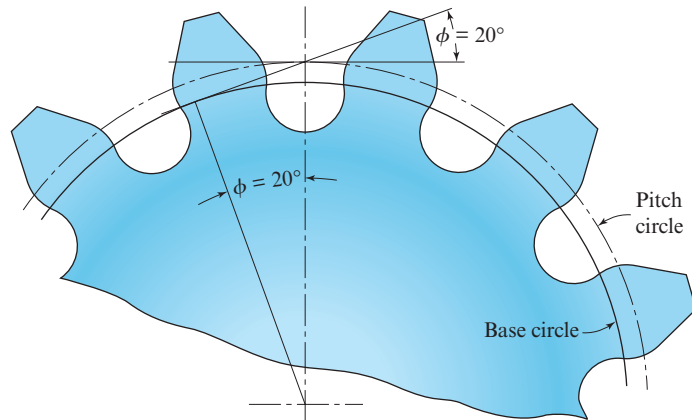
Similarly, the modified pitch radius of the gear can be written as

$$R'_3 = \frac{R_3 \cos \phi}{\cos \phi'}. \quad (7.20)$$

Equations (7.19) and (7.20) give the values of the actual pitch radii when the two gears with modified teeth are brought into mesh without backlash. The new center distance is, of course, the sum of these radii.

All the necessary relations have now been developed to create nonstandard gears with changes in the center distance. The use of these relations is now illustrated by an example.

Figure 7.25 is a drawing of a 20° pressure angle, 1-tooth/in diametral pitch, 12-tooth pinion generated with a rack cutter to full depth with a standard clearance of $0.250/P$. In the 20° full-depth system, interference is severe when the number of teeth is less than 14. The resulting undercutting is evident in the drawing.



 **Figure 7.25** Standard 20° pressure angle, 1-tooth/in diametral pitch, 12-tooth full-depth involute gear showing undercut.

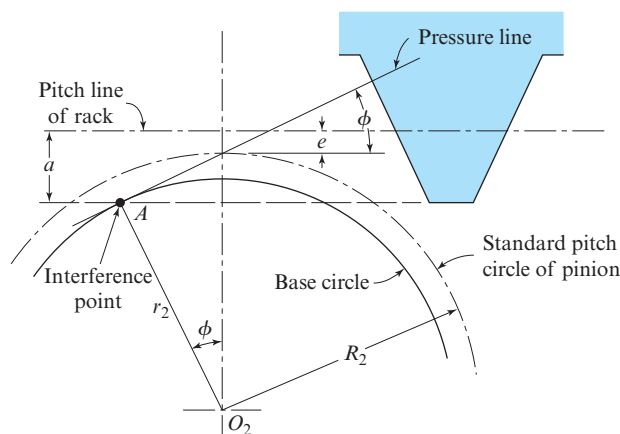


Figure 7.26 Offset of a rack cutter to cause its addendum line to pass through the interference point.

In an attempt to eliminate undercutting, improve the tooth action, and increase the contact ratio, suppose that this pinion were not cut to full depth; suppose instead that the rack cutter were only allowed to cut to a depth for which its addendum passes through the interference point A of the pinion being cut, that is—the point of tangency of the 20° line of action and the base circle—as shown in Fig. 7.26. From Eq. (7.15) we know that

$$r_2 = R_2 \cos \phi. \quad (e)$$

Then, from Fig. 7.26, the depth of the cut would be offset from the standard by

$$e = a + r_2 \cos \phi - R_2. \quad (f)$$

Substituting Eq. (e) into Eq. (f), the offset can be written as

$$e = a + R_2 \cos^2 \phi - R_2 = a - R_2 \sin^2 \phi. \quad (7.21)$$

If the offset is any less than this, then the rack will cut below the interference point, A , and will result in undercutting.

EXAMPLE 7.3

A 12-tooth pinion with pressure angle $\phi = 20^\circ$ and diametral pitch $P = 1$ tooth/in is to be mated with a standard 40-tooth gear. If the pinion were cut to full depth, then Eq. (7.9) demonstrates that the contact ratio would be 1.41, but there would be undercutting, as indicated in Fig. 7.25. Instead, let the 12-tooth pinion be cut from a larger blank using center-distance modifications. Determine the cutter offset, the modified pressure angle, the modified pitch radii of the pinion and the gear, the modified center distance, the modified outside radii of the pinion and gear, and the contact ratio. Has the contact ratio increased significantly?

SOLUTION

Designating the pinion as subscript 2 and the gear as 3, then with $P = 1$ tooth/in and $\phi = 20^\circ$, the following values are determined:

$$p = 3.142 \text{ in/tooth}, R_2 = 6 \text{ in}, R_3 = 20 \text{ in}, N_2 = 12 \text{ teeth}, \\ N_3 = 40 \text{ teeth}, \quad \text{and} \quad t_3 = 1.571 \text{ in}.$$

For a standard rack cutter, from Table 7.2, the addendum is $a = 1/P = 1.0$ in.

From Eq. (7.21), the rack cutter will be offset by

$$e = 1.0 - 6.0 \sin^2 20^\circ = 0.298 \text{ in.} \quad \text{Ans.}$$

Then, the thickness of the pinion tooth at the 6-in pitch circle, using Eq. (7.17), is

$$t'_2 = 2e \tan \phi + \frac{p}{2} = 2(0.298 \text{ in}) \tan 20^\circ + \frac{3.142 \text{ in}}{2} = 1.788 \text{ in}.$$

The pressure angle at which this (and only this) gearset will operate is determined from Eq. (7.18); that is,

$$\begin{aligned} \text{inv } \phi' &= \frac{N_2 (t'_2 + t'_3) - 2\pi R_2}{2R_2 (N_2 + N_3)} + \text{inv } \phi \\ &= \frac{12(1.788 \text{ in} + 1.571 \text{ in}) - 2\pi (6.0 \text{ in})}{2(6.0 \text{ in})(12 + 40)} + \text{inv } 20^\circ = 0.01908 \text{ rad.} \end{aligned}$$

From Appendix A, Table 6, we find that the new pressure angle is

$$\phi' = 21.65^\circ. \quad \text{Ans.}$$

Using Eqs. (7.19) and (7.20), the modified pitch radii are found to be

$$R'_2 = \frac{R_2 \cos \phi}{\cos \phi'} = \frac{(6.0 \text{ in}) \cos 20^\circ}{\cos 21.65^\circ} = 6.066 \text{ in}, \quad \text{Ans.}$$

$$R'_3 = \frac{R_3 \cos \phi}{\cos \phi'} = \frac{(20.0 \text{ in}) \cos 20^\circ}{\cos 21.65^\circ} = 20.220 \text{ in.} \quad \text{Ans.}$$

So, the modified center distance is

$$R'_2 + R'_3 = 6.066 + 20.220 = 26.286 \text{ in.} \quad \text{Ans.}$$

Note that the center distance has not increased as much as the offset of the rack cutter.

Standard clearance of $0.25/P$ results from the standard dedendums equal to $1.25/P$, as indicated in Table 7.2. So, the root radii of the two gears are

$$\text{Root radius of pinion} = 6.298 - 1.250 = 5.048 \text{ in},$$

$$\text{Root radius of gear} = 20.000 - 1.250 = 18.750 \text{ in},$$

$$\text{Sum of root radii} = 23.798 \text{ in}.$$

The difference between this sum and the center distance is the working depth plus twice the clearance. Since the clearance is 0.25 in for each gear, the working depth is

$$\text{Working depth} = 26.286 - 23.798 - 2(0.250) = 1.988 \text{ in.}$$

The outside radius of each gear is the sum of the root radius, the clearance, and the working depth; that is,

$$\text{Outside radius of pinion} = 5.048 + 0.250 + 1.988 = 7.286 \text{ in.;} \quad \text{Ans.}$$

$$\text{Outside radius of gear} = 18.750 + 0.250 + 1.988 = 20.988 \text{ in.} \quad \text{Ans.}$$

The result is shown in Fig. 7.27, and the pinion is seen to have a stronger-looking form than the one of Fig. 7.25. Undercutting has been completely eliminated.

The contact ratio can be obtained from Eqs. (7.9) through (7.11). The following quantities are needed:

$$\text{Outside radius of pinion} = R'_2 + a = 7.286 \text{ in,}$$

$$\text{Outside radius of gear} = R'_3 + a = 20.988 \text{ in,}$$

$$r_2 = R_2 \cos \phi = (6.000 \text{ in}) \cos 20^\circ = 5.638 \text{ in,}$$

$$r_3 = R_3 \cos \phi = (20.000 \text{ in}) \cos 20^\circ = 18.794 \text{ in,}$$

$$p_b = p \cos \phi = (3.1416 \text{ in/tooth}) \cos 20^\circ = 2.952 \text{ in/tooth.}$$

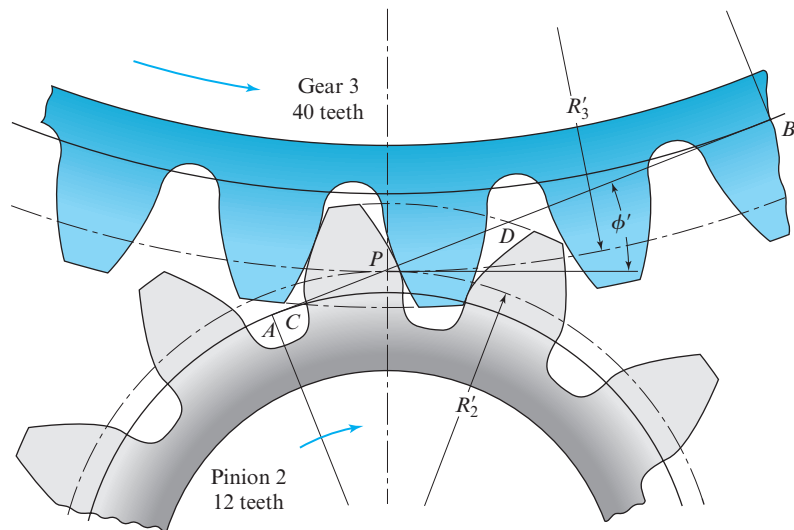


Figure 7.27

Therefore, for Eqs. (7.10) and (7.11), we have

$$\begin{aligned}
 CP &= \sqrt{(R'_3 + a)^2 - r_3^2} - R'_3 \sin \phi \\
 &= \sqrt{(20.988 \text{ in})^2 - (18.794 \text{ in})^2} - (20.220 \text{ in}) \sin 21.65^\circ = 1.883 \text{ in}, \\
 PD &= \sqrt{(R'_2 + a)^2 - r_2^2} - R'_2 \sin \phi \\
 &= \sqrt{(7.286 \text{ in})^2 - (5.638 \text{ in})^2} - (6.066 \text{ in}) \sin 21.65^\circ \\
 &= 2.377 \text{ in}.
 \end{aligned}$$

Finally, from Eq. (7.9), the contact ratio is

$$m_c = \frac{CP + PD}{p_b} = \frac{1.883 \text{ in} + 2.377 \text{ in}}{2.952 \text{ in/tooth}} = 1.443 \text{ teeth avg.} \quad \text{Ans.}$$

Therefore, the contact ratio has increased only slightly (approximately 2% increase). The modification, however, is justified because of the elimination of undercutting, which results in a substantial improvement in the strength of the teeth.

Long-and-Short-Addendum System It often happens in the design of machinery that the center distance between a pair of gears is fixed by some other design consideration or feature of the machine. In such a case, modifications to obtain improved performance cannot be made by varying the center distance.

In the previous section, we saw that improved action and tooth shape can be obtained by backing the rack cutter away from the gear blank during forming of the teeth. The effect of this withdrawal is to create the active tooth profile farther away from the base circle. Examination of Fig. 7.27 indicates that more dedendum could be used on the gear (not the pinion) before the interference point is reached. If the rack cutter is advanced into the gear blank by a distance equal to the withdrawal from the pinion blank, more of the gear dedendum is used, and at the same time, the center distance is not changed. This is called the *long-and-short-addendum system*.

In the long-and-short-addendum system, there are no changes in the pitch circles and consequently none in the pressure angle. The effect is to move the contact region away from the pinion center toward the gear center, thus shortening the approach action and lengthening the recess action.

The characteristics of the long-and-short-addendum system can be explained by reference to Fig. 7.28. Figure 7.28a shows a conventional (standard) set of gears having a dedendum equal to the addendum plus the clearance. Interference exists, and the tip of the gear tooth will have to be relieved as shown or the pinion will be undercut. This is indicated, since the addendum circle crosses the line of action at *C*, outside of the tangency or interference point *A*; hence, distance *AC* is a measure of the degree of interference.

To eliminate the undercutting or interference, the pinion addendum may be enlarged, as in Fig. 7.28b, until the addendum circle of the pinion passes through the interference point (point *B*) of the gear. In this manner, we shall be using all of the gear-tooth profile.

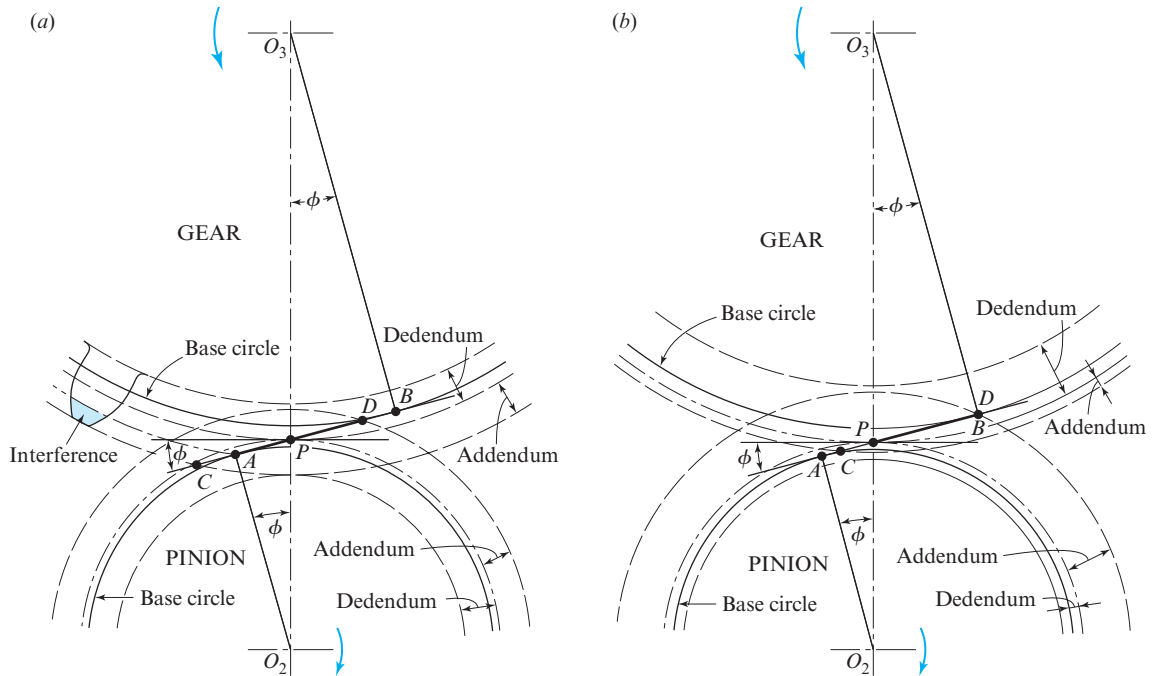


Figure 7.28 Comparison of standard gears and gears cut by the long-and-short-addendum system: (a) gear and pinion with standard addendum and dedendum; (b) gear and pinion with long-and-short addendum.

The same whole depth may be retained; hence, the dedendum of the pinion may be reduced by the same amount that the addendum is increased. This means that we must also lengthen the gear dedendum and shorten the dedendum of the mating pinion. With these changes, the path of contact is line CD of Fig. 7.28b. It is longer than path AD of Fig. 7.28a, and so the contact ratio is higher. Note, too, that the base circles, the pitch circles, the pressure angle, and the center distance have not changed. Both gears can be cut with standard cutters by advancing the cutter into the gear blank for this modification by a distance equal to the amount of withdrawal from the pinion blank. Finally, note that the blanks from which the pinion and gear are cut must now be of different diameters than the standard blanks.

The tooth dimensions for the long-and-short-addendum system can be determined using the equations developed in the previous sections.

A less obvious advantage of the long-and-short-addendum system is that more recess action than approach action is obtained. The approach action of gear teeth is analogous to pushing a piece of chalk across a blackboard; the chalk screeches. But, when the chalk is pulled across a blackboard, analogous to the recess action, it glides smoothly. Thus, recess action is always preferable because of the smoothness and the lower frictional forces.

7.12 PARALLEL-AXIS GEAR TRAINS

Mechanisms arranged in combinations so that the driven member of one mechanism is the driver for another mechanism are called *mechanism trains*. With certain exceptions, to be

explored here, the analysis of such trains can proceed in serial fashion by using the methods developed in the previous chapters.

In Chap. 3, we learned that the first-order *kinematic coefficient* is the term used to describe the ratio of the angular velocity of the driven member to that of the driving member. Thus, for example, in a four-bar linkage with link 2 as the driving or input member and link 4 as the driven or output member, we have

$$\theta'_{42} = \frac{d\theta_4}{d\theta_2} = \frac{d\theta_4/dt}{d\theta_2/dt} = \frac{\omega_4}{\omega_2}, \quad (a)$$

where it is noted that, as in Sec. 5.5 and after, we adopt the second subscript to explicitly indicate the number of the driving or input member. This second subscript is important in this and following sections, since many mechanism trains have more than one degree of freedom.

In this section, where we deal with serially connected gear trains, we prefer to write Eq. (a) as

$$\theta'_{LF} = \frac{d\theta_L}{d\theta_F} = \frac{d\theta_L/dt}{d\theta_F/dt} = \frac{\omega_L}{\omega_F}, \quad (7.22)$$

where ω_L is the angular velocity of the *last* gear and ω_F is the angular velocity of the *first* gear in the train, since, usually, the last gear is the output and is the driven gear, and the first is the input and driving gear.

The term θ'_{LF} in Eq. (7.22) is the first-order kinematic coefficient, called the *speed ratio* by some or the *train value* by others. Equation (7.22) is often written in the more convenient form:

$$\omega_L = \theta'_{LF} \omega_F. \quad (7.23)$$

Next, we consider pinion 2 driving gear 3. The speed of the driven gear is

$$\omega_3 = \pm \frac{R_2}{R_3} \omega_2 = \pm \frac{N_2}{N_3} \omega_2, \quad (b)$$

where, for each gear, R is the radius of the pitch circle, N is the number of teeth, and ω is either the angular velocity or the angular displacement completed during a chosen time interval.

For parallel-shaft gearing, the directions can be tracked by following the vector sense—that is, by specifying that angular velocity is positive when counterclockwise as seen from a chosen side. For parallel-shaft gearing, we shall use the following sign convention: If the last gear of a parallel-shaft gear train rotates with the same sense as the first gear, then θ'_{LF} is positive; if the last gear rotates in the opposite sense to the first gear, then θ'_{LF} is negative. This sign convention approach is not as easy, however, when the gear shafts are not parallel, as in bevel, crossed-helical, or worm gearing (Chap. 8). In such cases, it is often simpler to track the directions by visually inspecting a sketch of the train.

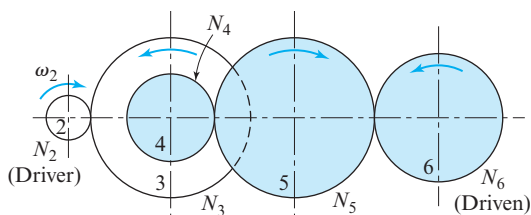


Figure 7.29

The gear train shown in Fig. 7.29 is made up of five gears in series. Applying Eq. (b) three times, we find the speed of gear 6 to be

$$\omega_6 = -\frac{R_5}{R_6} \frac{R_4}{R_5} \frac{R_2}{R_3} \omega_2 = -\frac{N_5}{N_6} \frac{N_4}{N_5} \frac{N_2}{N_3} \omega_2. \quad (c)$$

Here, we note that gear 5 is an *idler*; that is, its tooth numbers cancel in Eq. (c), and hence the only purpose served by gear 5 is to change the direction of rotation of gear 6. We further note that gears 5, 4, and 2 are drivers, whereas gears 6, 5, and 3 are driven members. Thus, Eq. (7.22) can also be written

$$\theta'_{LF} = \pm \frac{\text{product of driving tooth numbers}}{\text{product of driven tooth numbers}}. \quad (7.24)$$

Note also that, since they are proportional, pitch radii can be used in Eq. (7.24) just as well as tooth numbers.

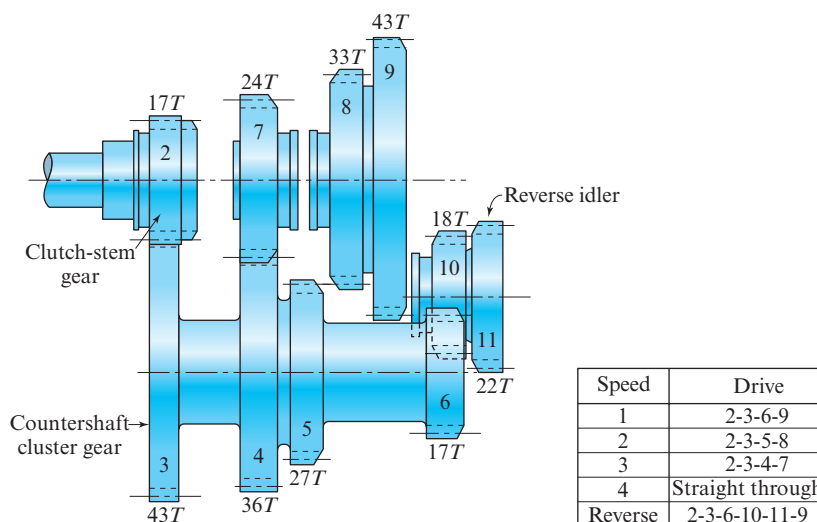


Figure 7.30 A truck transmission with gears having diametral pitch of 7 teeth/in and pressure angle of 22.5°.

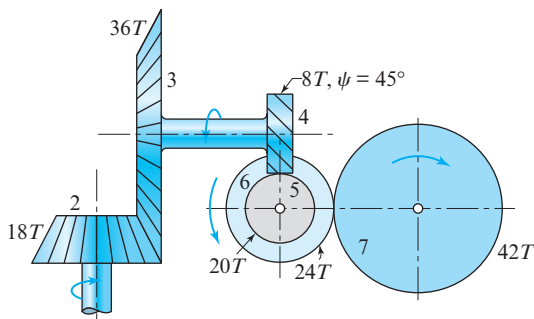


Figure 7.31 A gear train composed of bevel, crossed-helical, and spur gears.

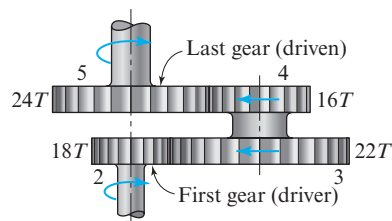


Figure 7.32 A reverted gear train.

In speaking of gear trains, it is convenient to describe a train having only one gear on each axis as a *simple* gear train. A *compound* gear train, then, is one that has two or more gears on one or more axes, such as the train shown in Fig. 7.29. Another example of a compound gear train is shown in Fig. 7.30. Figure 7.30 shows a transmission for a small- or medium-size truck that has four speeds forward and one in reverse.

The compound gear train shown in Fig. 7.31 is composed of bevel, helical (Chap. 8), and spur gears. The helical gears are crossed, so their direction of rotation depends upon their hand.

A *reverted* gear train is one in which the first and last gears have collinear axes of rotation, such as the one shown in Fig. 7.32. This produces a compact arrangement and is used in such applications as speed reducers, clocks (to connect the hour hand to the minute hand), and machine tools. As an exercise, it is suggested that you seek out a suitable set of diametral pitches for each pair of gears shown in Fig. 7.32 so that the first and last gears have the same axis of rotation with all gears properly engaged.

7.13 DETERMINING TOOTH NUMBERS

When notable power is transmitted through a speed reduction unit, the speed ratio of the last pair of meshing gears is usually chosen larger than that of the first gear pair, since the torque is greater at the low-speed end. In a given amount of space, more teeth can be used on gears of lesser pitch; hence, a greater speed reduction can be obtained at the high-speed end.

Without examining the problem of tooth strength, suppose we wish to use two pairs of gears in a train to obtain an overall kinematic coefficient of $\theta'_{LF} = \frac{1}{12}$. Let us also impose the restriction that the tooth numbers must not be less than 15 and that the reduction in the first pair of gears should be about twice that of the second pair. This means that the overall kinematic coefficient is

$$\theta'_{52} = \frac{N_4}{N_5} \frac{N_2}{N_3} = \frac{1}{12}, \quad (a)$$

where N_2/N_3 is the coefficient of the first gear pair, and N_4/N_5 is that of the second pair. Since the kinematic coefficient of the first pair should be half that of the second, Eq. (a) can be written as

$$\left(\frac{N_4}{N_5}\right) \left(\frac{N_4}{2N_5}\right) = \frac{1}{12} \quad (b)$$

or

$$\frac{N_4}{N_5} = \sqrt{\frac{1}{6}} = 0.408\,248 \quad (c)$$

to six decimal places. The following tooth numbers are seen to be close:

$$\frac{15}{37} \quad \frac{16}{39} \quad \frac{18}{44} \quad \frac{20}{49} \quad \frac{22}{54} \quad \frac{24}{59}.$$

Of these, $N_4/N_5 = 20/49$ is the closest approximation, but note that

$$\theta'_{52} = \left(\frac{N_4}{N_5}\right) \left(\frac{N_2}{N_3}\right) = \left(\frac{20}{49}\right) \left(\frac{20}{98}\right) = \frac{400}{4\,802} = \frac{1}{12.005},$$

which is very close to $1/12$. On the other hand, the choice of $N_4/N_5 = 18/44$ gives exactly

$$\theta'_{52} = \left(\frac{N_4}{N_5}\right) \left(\frac{N_2}{N_3}\right) = \left(\frac{18}{44}\right) \left(\frac{22}{108}\right) = \frac{396}{4\,752} = \frac{1}{12}.$$

In this case, the reduction in the first gear pair is not exactly twice the reduction in the second gear pair. However, this consideration is usually of only minor importance.

The problem of specifying tooth numbers and the number of pairs of gears to give a kinematic coefficient with a specified degree of accuracy has interested many people, but has no exact solution. Consider, for instance, the problem of specifying a set of gears to have a kinematic coefficient of $\theta'_{LF} = \pi/10$ accurate to eight decimal places, while we know that π is an irrational number and cannot be expressed as a ratio of integers.

7.14 EPICYCLIC GEAR TRAINS

Figure 7.33 shows an elementary epicyclic gear train together with its schematic diagram, as suggested by Lévai.* The train consists of a *central gear*, 2, and an *epicyclic gear*, 4, which produces epicyclic motion for its points by rolling around the periphery of the central gear. An *arm* 3 contains the bearing for the epicyclic gear to maintain the gears in mesh.

* Literature devoted to epicyclic gear trains is rather scarce; however, see [3]. For a comprehensive study in the English language, see [1]; this book lists 104 references.

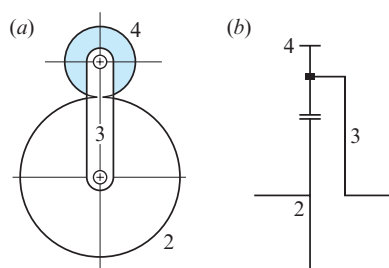


Figure 7.33 (a) Elementary epicyclic gear train; (b) its schematic diagram.

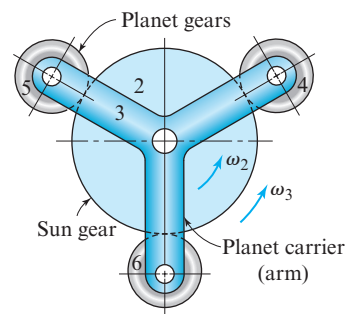


Figure 7.34 A planetary gearset.

Epicyclic trains are also called *planetary* or *sun-and-planet* gear trains. In this nomenclature, gear 2 of Fig. 7.33 is called the *sun gear*, gear 4 is called the *planet gear*, and crank 3 is called the *planet carrier*. Figure 7.34 shows the train of Fig. 7.33 with two redundant planet gears added. This produces better force balance; also, adding more planet gears allows lower forces by more load sharing. However, these additional planet gears do not change the kinematic characteristics at all. For this reason, we generally indicate only a single planet in the illustrations and problems in Chap. 7, although an actual machine will probably be designed with planets in trios.

The simple epicyclic gear train together with its schematic designation in Fig. 7.35 shows how the motion of the planet gear can be transmitted to another central gear. The second central gear in this case is gear 5, an internal gear. In Fig. 7.35a, internal gear 5 is stationary, but this is not a requirement, as shown in Fig. 7.35b. Figure 7.36 shows a similar arrangement, with the difference that both central gears are external gears. Note, in Fig. 7.36, that the double planet gears are mounted on a single planet shaft and that each planet gear is in mesh with a separate sun gear rotating at a different speed.

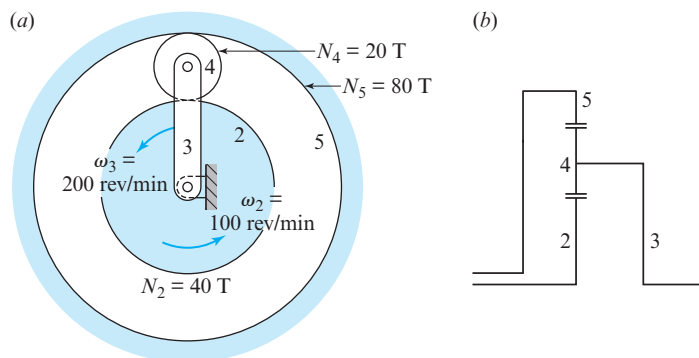


Figure 7.35 (a) A simple epicyclic gear train; (b) its schematic diagram.

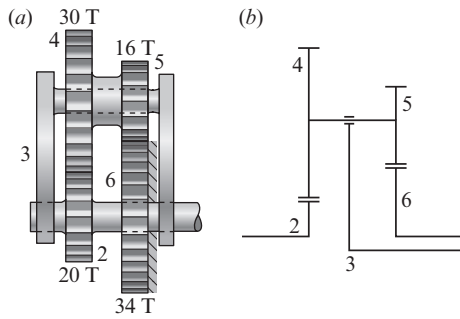


Figure 7.36 A simple epicyclic gear train with double planet gears.

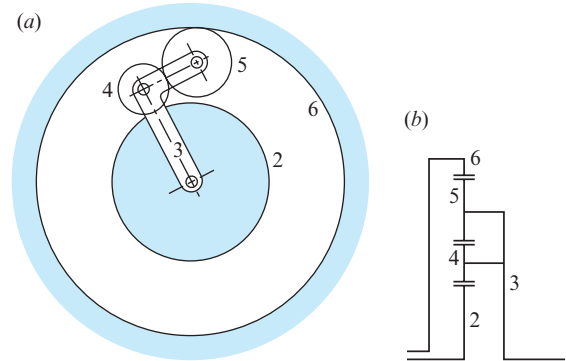


Figure 7.37 An epicyclic gear train with two planet gears.

In any case, no matter how many planets are used, only one planet carrier or arm may be used. This principle is shown in Fig. 7.34, in which redundant planets are used, and in Fig. 7.37, where two planets are used to alter the kinematic performance.

According to Lévai, only twelve variations of epicyclic gear trains are possible; they are all shown in schematic form in Fig. 7.38 as Lévai arranged them. In all variations, the arm (the planet carrier) is shown as link number 3. The trains in Figs. 7.38a and 7.38c are simple trains in which the planet gears mesh with both sun gears. The trains shown in Figs. 7.38b and 7.38d have planet gear pairs that are partly in mesh with each other and partly in mesh with the sun gears.

7.15 ANALYSIS OF EPICYCLIC GEAR TRAINS BY FORMULA

Figure 7.39 shows a planetary gear train composed of a sun gear 2, an arm or planet carrier, 3, and planet gears 4 and 5. Using the apparent angular velocity equation, Eq. (3.10), we can write that the angular velocity of gear 2 as it would appear from a coordinate system fixed to arm 3 is

$$\omega_{2/3} = \omega_2 - \omega_3. \quad (a)$$

Also, the angular velocity of gear 5 as it would appear from the coordinate system fixed to arm 3 is

$$\omega_{5/3} = \omega_5 - \omega_3. \quad (b)$$

Dividing Eq. (b) by Eq. (a) gives

$$\frac{\omega_{5/3}}{\omega_{2/3}} = \frac{\omega_5 - \omega_3}{\omega_2 - \omega_3}. \quad (c)$$

Equation (c) expresses the ratio of the apparent angular velocity of gear 5 to that of gear 2, with both taken as they would appear from arm 3. This ratio, which is proportional to

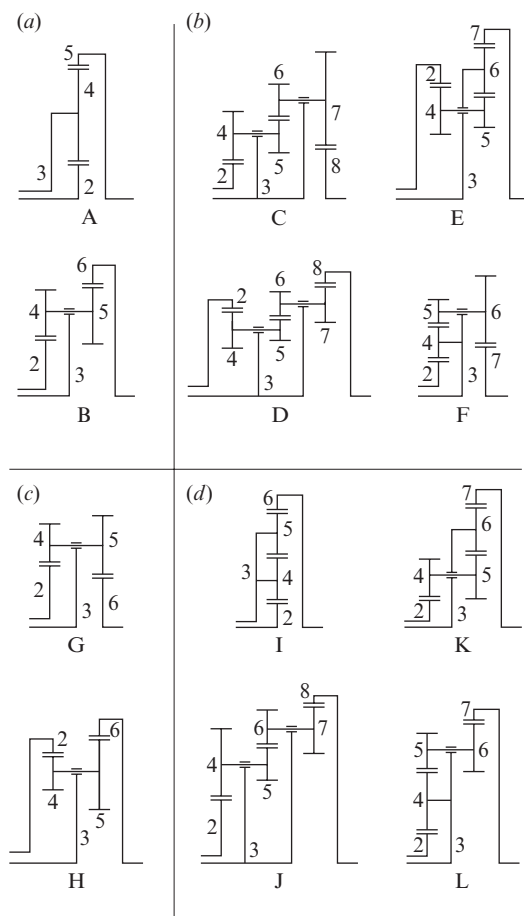
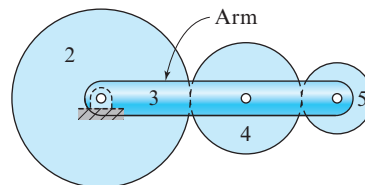


Figure 7.38 All twelve possible epicyclic gear train types according to Lévai [1]. (Reproduced with permission of the author.)

Figure 7.39



the tooth numbers, appears the same whether the arm is rotating or not; it is the first-order kinematic coefficient of the gear train. Therefore, the first-order kinematic coefficient can be expressed as

$$\theta'_{52/3} = \frac{\omega_5 - \omega_3}{\omega_2 - \omega_3}. \quad (d)$$

An equation similar to Eq. (d) is all that we need to find the angular velocities in a planetary gear train. It is convenient to express it in the form as it would appear from the arm:

$$\theta'_{LF/A} = \frac{\omega_L - \omega_A}{\omega_F - \omega_A}, \quad (7.25)$$

where

ω_F = angular velocity of the first gear in the train,
 ω_L = angular velocity of the last gear in the train, and
 ω_A = angular velocity of the arm.

Note that Eq. (d) and Eq. (7.25) can be expressed entirely in terms of kinematic coefficients; that is,

$$\theta'_{52/3} = \frac{\theta'_5 - \theta'_3}{\theta'_2 - \theta'_3} \quad (e)$$

and

$$\theta'_{LF/A} = \frac{\theta'_L - \theta'_A}{\theta'_F - \theta'_A}, \quad (7.26)$$

where

θ'_F = kinematic coefficient of the first gear in the train,
 θ'_L = kinematic coefficient of the last gear in the train, and
 θ'_A = kinematic coefficient of the arm.

The following examples help to illustrate the use of kinematic coefficients presented in Eqs. (7.25) and (7.26).

EXAMPLE 7.4

Figure 7.36 shows a reverted planetary gear train. Gear 2 is fastened to its shaft and is driven at 250 rev/min in a clockwise direction. Gears 4 and 5 are planet gears that are joined but are free to turn on the shaft carried by the arm. Gear 6 is stationary. Find the speed and direction of rotation of the arm.

SOLUTION

We must first decide which gears to designate the first and last members of the train. Since the speeds of gears 2 and 6 are both given, then either gear may be chosen as the first. The choice makes no difference to the results, but once the decision is made, it may not be changed. Here we choose gear 2 as first; therefore, gear 6 is last. Thus, choosing counterclockwise as positive gives

$$\omega_F = \omega_2 = -250 \text{ rev/min} \quad \text{and} \quad \omega_L = \omega_6 = 0 \text{ rev/min},$$

and, according to Eq. (7.24), the first-order kinematic coefficient is

$$\theta'_{LF} = \theta'_{62} = \left(-\frac{16}{34}\right) \left(-\frac{20}{30}\right) = \frac{16}{51},$$

where the positive sign results, since there are two external contacts.

Substituting the values into Eq. (7.25) gives the first-order kinematic coefficient:

$$\theta'_{LF/A} = \frac{16}{51} = \frac{0 - \omega_3}{-250 \text{ rev/min} - \omega_3}.$$

Rearranging this equation gives the angular velocity of arm A:

$$\omega_A = \omega_3 = 114.3 \text{ rev/min ccw.} \quad \text{Ans.}$$

EXAMPLE 7.5

Consider the simple planetary gear train shown in Fig. 7.40 in which the ring gear h is fixed (that is, the angular velocity of the ring gear is $\omega_h = 0$) and the input is the planet carrier (the arm). The dimensions of ring gear h , planet gear j , sun gear k , and arm A are $R_h = 200 \text{ mm}$, $R_j = 50 \text{ mm}$, $R_k = 100 \text{ mm}$, and $R_A = 150 \text{ mm}$, respectively. If the input angular velocity is $\omega_A = \omega_i = 10 \text{ rad/s ccw}$, determine the angular velocities of the planet gear, j , and the sun gear, k .

SOLUTION

Using Eq. (7.25), the first-order kinematic coefficient of planet gear j , in rolling contact with ring gear h , can be written as

$$\theta'_{jh/A} = \pm \frac{R_h}{R_j} = \frac{\omega_j - \omega_A}{\omega_h - \omega_A}. \quad (1a)$$

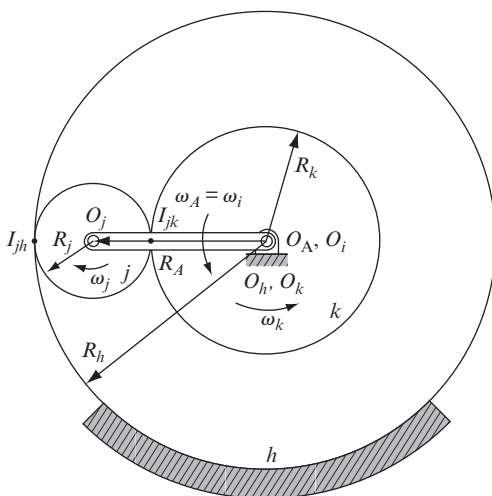


Figure 7.40 A simple planetary gear train with the ring gear fixed.

Similarly, the first-order kinematic coefficient of planet gear j , in rolling contact with sun gear k , can be written as

$$\theta'_{jk/A} = \pm \frac{R_k}{R_j} = \frac{\omega_j - \omega_A}{\omega_k - \omega_A}. \quad (1b)$$

Note that if the ring gear is removed, and the arm is fixed (that is, $\omega_A = 0$), then the planetary gear train reduces to an ordinary gear train, that is, Eq. (1b) reduces to

$$\theta'_{jk/A} = \pm \frac{R_k}{R_j} = \frac{\omega_j}{\omega_k}. \quad (2)$$

Since there is internal contact between the planet gear and the fixed ring gear, then the positive sign is used in Eq. (1a). Then, rearranging the equation and substituting $\omega_h = 0$ gives

$$R_j \omega_j = -(R_h - R_j) \omega_A = -R_A \omega_A. \quad (3)$$

Therefore, the angular velocity of the planet gear is

$$\omega_j = -\frac{R_A \omega_A}{R_j} = -\frac{(150 \text{ mm})(10 \text{ rad/s})}{50 \text{ mm}} = -30 \text{ rad/s.} \quad \text{Ans.}$$

The negative sign indicates that the direction of the angular velocity of the planet gear is opposite to the direction of the arm and, therefore, clockwise.

Similarly, since there is external contact between the planet gear and the sun gear, the negative sign is used in Eq. (1b), and the equation can be written as

$$R_k \omega_k = -R_j \omega_j + (R_k + R_j) \omega_A. \quad (4a)$$

The angular velocity of the sun gear can be expressed in terms of the input angular velocity of the arm by substituting Eq. (3) into Eq. (4a) and simplifying; that is,

$$R_k \omega_k = (R_k + R_h) \omega_A = 2R_A \omega_A. \quad (4b)$$

Therefore, the angular velocity of the sun gear is

$$\omega_k = \frac{2R_A \omega_A}{R_k} = \frac{2(150 \text{ mm})(10 \text{ rad/s})}{100 \text{ mm}} = 30 \text{ rad/s.} \quad \text{Ans.}$$

The positive result indicates that the direction of the angular velocity of the sun gear is the same as the arm and is, therefore, counterclockwise.

EXAMPLE 7.6

Consider the compound planetary gear train that is commonly used in an electric drill or screwdriver, shown in the schematic diagram in Fig. 7.41. The gear and the arm dimensions are $R_1 = 150 \text{ mm}$, $R_{2A} = R_{4A} = 100 \text{ mm}$, and $R_3 = R_4 = R_5 = R_6 = 50 \text{ mm}$.

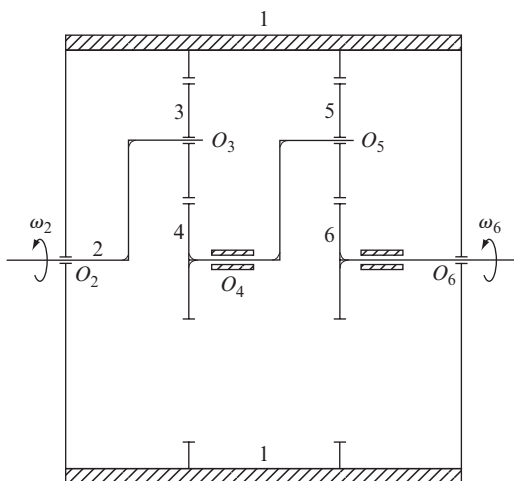


Figure 7.41 Schematic diagram of a compound planetary gear train.

The angular velocity of the input shaft (and arm 2) is $\omega_{2A} = 10$ rad/s ccw (looking from the left), and ring gear 1 is fixed—that is, the angular velocity of the ring gear is $\omega_1 = 0$. Determine the angular velocities of gears 3, 4, 5, and 6.

SOLUTION

This gear train is composed of two simple planetary gear trains in series; that is, the first planetary gear train is composed of ring gear 1, input arm 2, planet gear 3, and sun gear 4, and the second planetary gear train is composed of ring gear 1, arm 4, planet gear 5, and output sun gear 6. Note that the angular velocities of gear 4 and arm 4 are equal in magnitude and direction, since they are rigidly attached. Also, note that gear 4 acts as both the output of the first planetary gear train and the input to the second planetary gear train. Using Eq. (7.25), the first-order kinematic coefficient of gear 4 with respect to ring gear 1 can be written as

$$\theta'_{41/2A} = \left(-\frac{R_3}{R_4}\right) \left(\frac{R_1}{R_3}\right) = -\frac{R_1}{R_4} = \frac{\omega_4 - \omega_{2A}}{\omega_1 - \omega_{2A}}. \quad (1a)$$

The correct sign is negative, since gears 4 and 3 are in external contact, and gears 3 and 1 are in internal contact. Substituting the given data into Eq. (1a) gives

$$-\frac{150 \text{ mm}}{50 \text{ mm}} = \frac{\omega_4 - 10 \text{ rad/s}}{0 - 10 \text{ rad/s}}. \quad (1b)$$

Therefore, the angular velocity of gear 4 is

$$\omega_4 = 40 \text{ rad/s}, \quad \text{Ans.}$$

where the positive result implies that gear 4 is rotating in the same direction as the input gear 2—that is, counterclockwise.

Using Eq. (7.25), the first-order kinematic coefficient of gear 3 with respect to gear 4 can be written as

$$\theta'_{34/2A} = -\frac{R_4}{R_3} = \frac{\omega_3 - \omega_{2A}}{\omega_4 - \omega_{2A}}, \quad (2a)$$

where the negative sign denotes external contact between gears 3 and 4. Therefore, the first-order kinematic coefficient is

$$\theta'_{34/2A} = -\frac{50 \text{ mm}}{50 \text{ mm}} = \frac{\omega_3 - 10 \text{ rad/s}}{40 \text{ rad/s} - 10 \text{ rad/s}}. \quad (2b)$$

Therefore, the angular velocity of gear 3 is

$$\omega_3 = -20 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign indicates that gear 3 is rotating in the direction opposite to the input gear 2—that is, clockwise.

Using Eq. (7.25), the first-order kinematic coefficient of gear 6 with respect to the ring gear 1 can be written as

$$\theta'_{61/4A} = \left(-\frac{R_5}{R_6}\right) \left(\frac{R_1}{R_5}\right) = -\frac{R_1}{R_6} = \frac{\omega_6 - \omega_{4A}}{\omega_1 - \omega_{4A}}. \quad (3a)$$

The correct sign is negative, since gears 6 and 5 are in external contact, and gears 5 and 1 are in internal contact. Substituting the given data into Eq. (3a) gives

$$-\frac{150 \text{ mm}}{50 \text{ mm}} = \frac{\omega_6 - 40 \text{ rad/s}}{0 - 40 \text{ rad/s}}. \quad (3b)$$

Therefore, the angular velocity of gear 6 is

$$\omega_6 = 160 \text{ rad/s}, \quad \text{Ans.}$$

where the positive result implies that gear 4 is rotating in the same direction as the input gear 2—that is, counterclockwise.

Using Eq. (7.25), the first-order kinematic coefficient of gear 5 with respect to gear 6 can be written as

$$\theta'_{56/4A} = -\frac{R_6}{R_5} = \frac{\omega_5 - \omega_{4A}}{\omega_6 - \omega_{4A}}, \quad (4a)$$

where the negative sign denotes external contact between gears 5 and 6. Substituting the known data, the first-order kinematic coefficient is

$$\theta'_{56/4A} = -\frac{50 \text{ mm}}{50 \text{ mm}} = \frac{\omega_5 - 40 \text{ rad/s}}{160 \text{ rad/s} - 40 \text{ rad/s}}. \quad (4b)$$

Therefore, the angular velocity of gear 5 is

$$\omega_5 = -80 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign implies that gear 5 is rotating in the direction opposite to the input gear 2—that is, clockwise.

ALTERNATE SOLUTION

The rolling contact constraint between planet gear 3 and ring gear 1 can be written in terms of kinematic coefficients as

$$\frac{R_3}{R_1} = \frac{50 \text{ mm}}{150 \text{ mm}} = \frac{1}{3} = \frac{\theta'_{12} - \theta'_{22}}{\theta'_{32} - \theta'_{22}}, \quad (1)$$

where the positive sign is chosen, since planet gear 3 has internal contact with ring gear 1.

Since ring gear 1 is fixed, the kinematic coefficient of the ring gear is $\theta'_{12} = 0$ and, by definition, the kinematic coefficient of input arm 2 is $\theta'_{22} = 1$. Substituting these into Eq. (1), the kinematic coefficient of planet gear 3 with respect to input gear 2 is

$$\theta'_{32} = -2 \text{ rad/rad}, \quad (2)$$

where the negative sign indicates that gear 3 is rotating in the direction opposite to input gear 2—that is, clockwise. Therefore, the angular velocity of gear 3 is

$$\omega_3 = \theta'_{32}\omega_2 = -20 \text{ rad/s.} \quad \text{Ans.}$$

The rolling contact constraint between sun gear 4 and planet gear 3 can be written from Eq. (1) as

$$-\frac{R_4}{R_3} = -1 = \frac{\theta'_{32} - \theta'_{22}}{\theta'_{42} - \theta'_{22}}, \quad (3)$$

where the negative sign is chosen, since planet gear 3 has external contact with sun gear 4.

Substituting Eq. (2) and $\theta'_{22} = 1$, the known geometry, into Eq. (3), the kinematic coefficient of sun gear 4 with respect to gear 2 is

$$\theta'_{42} = 4 \text{ rad/rad}, \quad (4)$$

where the positive result implies that gear 4 is rotating in the same direction as input gear 2—that is, counterclockwise. Therefore, the angular velocity of gear 4 is

$$\omega_4 = \theta'_{42}\omega_2 = 40 \text{ rad/s ccw.} \quad \text{Ans.}$$

The rolling contact constraint between the planet gear 5 and ring gear 1 can be written from Eq. (1) as

$$\frac{R_5}{R_1} = \frac{50 \text{ mm}}{150 \text{ mm}} = \frac{1}{3} = \frac{\theta'_{12} - \theta'_{42}}{\theta'_{52} - \theta'_{42}}, \quad (5)$$

where the positive sign is used, since planet gear 5 has internal contact with ring gear 1.

Substituting Eq. (4) and $\theta'_{12} = 0$, the kinematic coefficient of planet gear 5 with respect to gear 2 is

$$\theta'_{52} = -8 \text{ rad/rad}, \quad (6)$$

where the negative sign indicates that gear 5 is rotating in the direction opposite to the input gear 2—that is, clockwise. Therefore, the angular velocity of gear 5 is

$$\omega_5 = \theta'_{52}\omega_2 = -80 \text{ rad/s (cw)}. \quad \text{Ans.}$$

The rolling contact constraint between the sun gear 6 and planet gear 5 can be written from Eq. (5) as

$$-\frac{R_6}{R_5} = -1 = \frac{\theta'_{52} - \theta'_{42}}{\theta'_{62} - \theta'_{42}}, \quad (7)$$

where the negative sign is used because of the external contact between planet gear 5 and sun gear 6.

Substituting Eqs. (4) and (6) and the known geometry into Eq. (7), the kinematic coefficient of sun gear 6 with respect to gear 2 is

$$\theta'_{62} = 16 \text{ rad/rad}, \quad (8)$$

where the positive result implies that gear 6 is rotating in the same direction as input gear 2. Therefore, the angular velocity of gear 6 is

$$\omega_6 = \theta'_{62}\omega_2 = 160 \text{ rad/s ccw}. \quad \text{Ans.}$$

EXAMPLE 7.7

Consider the planetary gear train shown in the schematic diagram in Fig. 7.42. The radius of the fixed sun gear is $R_1 = 100 \text{ mm}$, the radius of input gear 2 is $R_2 = 100 \text{ mm}$, the radius of ring gear 4 is $R_4 = 200 \text{ mm}$, and the radii of gears 3 and 5 are $R_3 = R_5 = 50 \text{ mm}$. The angular velocity of input shaft 2 is $\omega_2 = 10 \text{ rad/s ccw}$ (looking from the left). Determine the angular velocities of gears 3, and 4, and 5.

SOLUTION

The rolling contact constraint between input gear 2 and gear 5 can be written as

$$-\frac{R_2}{R_5} = -\frac{100 \text{ mm}}{50 \text{ mm}} = -2 = \frac{\theta'_{52}}{\theta'_{22}}. \quad (1)$$

Since the kinematic coefficient of input shaft 2 is $\theta'_{22} = 1 \text{ rad/rad}$, the kinematic coefficient of gear 5 is

$$\theta'_{52} = -2 \text{ rad/rad}, \quad (2)$$

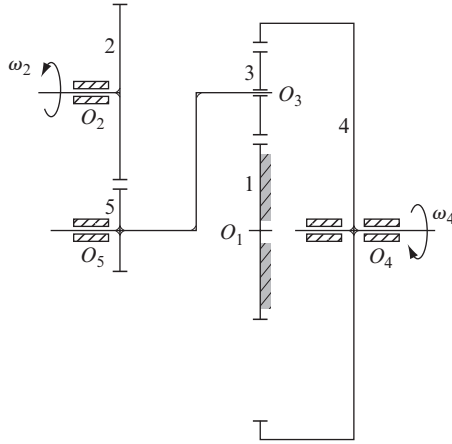


Figure 7.42 Schematic diagram of a planetary gear train.

where the negative sign implies that gear 5 is rotating in the opposite direction to input gear 2—that is, clockwise. Therefore, the angular velocity of gear 5 is

$$\omega_5 = \theta'_{52}\omega_2 = -20 \text{ rad/s (cw)}. \quad \text{Ans.}$$

The rolling contact constraint between planet gear 3 and fixed sun gear 1 can be written as

$$-\frac{R_3}{R_1} = \frac{\theta'_{12} - \theta'_{5A2}}{\theta'_{32} - \theta'_{5A2}}, \quad (3)$$

which, for the given conditions, becomes

$$-\frac{50 \text{ mm}}{100 \text{ mm}} = \frac{0 + 2 \text{ rad/rad}}{\theta'_{32} + 2 \text{ rad/rad}}.$$

Therefore, the kinematic coefficient of gear 3 with respect to fixed sun gear 1 is

$$\theta'_{32} = -6 \text{ rad/rad} \quad (4)$$

and the angular velocity of gear 3 is

$$\omega_3 = \theta'_{32}\omega_2 = -60 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign indicates that the direction of ω_3 is clockwise.

The rolling contact constraint between ring gear 4 and planet gear 3 can be written as

$$\frac{R_3}{R_4} = \frac{\theta'_{42} - \theta'_{5A2}}{\theta'_{32} - \theta'_{5A2}}, \quad (5)$$

which, using Eqs. (2) and (4) and the given dimensions, can be written as

$$\frac{50 \text{ mm}}{200 \text{ mm}} = \frac{\theta'_{42} + 2 \text{ rad/rad}}{-6 \text{ rad/rad} + 2 \text{ rad/rad}}.$$

Therefore, the kinematic coefficient of gear 4 with respect to fixed sun gear 1 is

$$\theta'_{42} = -3 \text{ rad/rad} \quad (6)$$

and the angular velocity of gear 4 is

$$\omega_4 = \theta'_{42} \omega_2 = -30 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign indicates that the direction of ω_4 is clockwise.

The kinematic coefficients in Eqs. (2), (4), and (6) define the geometry of the gear train and provide the capability for obtaining the angular velocities (magnitudes and directions) in terms of the input angular velocity, ω_2 .

7.16 TABULAR ANALYSIS OF EPICYCLIC GEAR TRAINS

Another method of determining the rotational speeds of epicyclic gear trains uses the principle of superposition. The total analysis is carried out by finding the apparent rotations of the components with respect to the arm or planet carrier and then summing with the rotation of the components as if all the gears are fixed to the arm. The process is easily carried out in a tabular procedure.

Figure 7.35 shows a planetary gear train composed of sun gear 2, planet carrier (arm) 3, planet gear 4, and internal gear 5 that is in mesh with the planet gear. Since this gear train has two degrees of freedom, we might reasonably specify the angular velocities of both the sun gear and the arm, and wish to determine the angular velocity of the internal gear.

The analysis can be carried out in the following three steps:

1. Consider all gears (including the fixed gear, if any) to be locked to the arm, and allow the arm to rotate with angular velocity ω_A . Tabulate the angular velocities of all components under this condition as also equal to ω_A .
2. Free all constraints of step 1, fix the arm, and allow some other gear B (such as the sun gear, for example) to rotate with angular velocity $\omega_{B/A}$. Tabulate the apparent angular velocities of all other gears with respect to the arm as multiples of $\omega_{B/A}$.
3. Add the angular velocities of each gear from steps 1 and 2, and apply the given input velocities in order to find numeric values for ω_A and $\omega_{B/A}$.

A few examples will clarify how this can be done in a convenient tabular procedure.

EXAMPLE 7.8

Let us assume the tooth numbers shown in Fig. 7.35, and let the angular velocities of the sun gear and the arm be $\omega_2 = 100 \text{ rev/min}$ and $\omega_3 = 200 \text{ rev/min}$, respectively, both in the

counterclockwise direction, chosen positive. What is the angular velocity of internal ring gear 5?

SOLUTION

The solution process described by the three steps enumerated above is demonstrated in Table 7.3.

Table 7.3 Tabular Analysis for Examples 7.8 and 7.9

Step Number	Gear 2	Arm 3	Gear 4	Gear 5
1. Gears fixed to arm	ω_3	ω_3	ω_3	ω_3
2. Arm fixed	$\omega_{2/3}$	0	$(-40/20)\omega_{2/3}$	$(20/80)(-40/20)\omega_{2/3}$
3. Total	$\omega_3 + \omega_{2/3}$	ω_3	$\omega_3 + (-40/20)\omega_{2/3}$	$\omega_3 + (20/80)(-40/20)\omega_{2/3}$

Next, comparing the given input velocities with the bottom row of columns 2 and 3, we see that, with $\omega_2 = \omega_3 + \omega_{2/3} = 100$ rev/min and $\omega_3 = 200$ rev/min, then $\omega_{2/3} = -100$ rev/min. Therefore, from the bottom row of column 5, the angular velocity of internal ring gear 5 can be written as

$$\omega_5 = \omega_3 + (20/80)(-40/20)\omega_{2/3};$$

that is,

$$\omega_5 = (200 \text{ rev/min}) + (-40/80)(-100 \text{ rev/min}) = 250 \text{ rev/min},$$

where the positive result implies that gear 5 is rotating in the same direction as the positive inputs—that is, counterclockwise.

EXAMPLE 7.9

What is the angular velocity of internal gear 5 of Fig. 7.35 if gear 2 rotates at 100 rev/min cw while arm 3 rotates at 200 rev/min ccw?

SOLUTION

The analysis is identical to that performed in Table 7.3 for Example 7.8. However, the input velocities have now changed. Still taking counterclockwise as positive, the bottom row of columns 2 and 3 indicate that, since

$$\omega_2 = \omega_3 + \omega_{2/3} = -100 \text{ rev/min} \quad \text{and} \quad \omega_3 = 200 \text{ rev/min},$$

then $\omega_{2/3} = -300 \text{ rev/min}$. Therefore, the bottom row of column 5 indicates

$$\begin{aligned}\omega_5 &= \omega_3 + (20/80)(-40/20)\omega_{2/3} = (200 \text{ rev/min}) + (-40/80)(-300 \text{ rev/min}) \\ &= 350 \text{ rev/min ccw.}\end{aligned}$$

Ans.

EXAMPLE 7.10

The planetary gear train shown in Fig. 7.43 is called *Ferguson's paradox*.^{*} Gear 2 is fixed to the frame. Arm 3 and gears 4, 5, and 6 are free to turn upon their shafts. Gears 2, 4, and 5 have tooth numbers of 100, 101, and 99, respectively, all cut with the same circular pitch (but with slightly different pitch circle radii) so that planet gear 6 meshes with all of them. Find the angular rotations of gears 4 and 5 when arm 3 is given one counterclockwise turn.

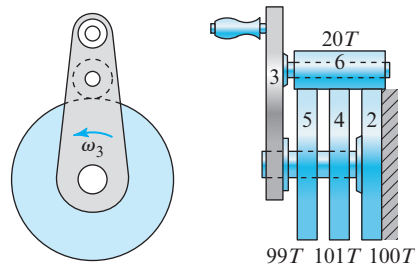


Figure 7.43 Ferguson's paradox.

SOLUTION

The solution process is indicated in Table 7.4.

Table 7.4 Tabular Analysis for Example 7.10

Step Number	Gear 2	Arm 3	Gear 4	Gear 5	Gear 6
1. Gears fixed to arm	$\Delta\theta_3$	$\Delta\theta_3$	$\Delta\theta_3$	$\Delta\theta_3$	$\Delta\theta_3$
2. Arm fixed	$\Delta\theta_{2/3}$	0	$(-20/101)(-100/20)\Delta\theta_{2/3}$	$(-20/99)(-100/20)\Delta\theta_{2/3}$	$(-100/20)\Delta\theta_{2/3}$
3. Total	$\Delta\theta_3 + \Delta\theta_{2/3}$	$\Delta\theta_3$	$\Delta\theta_3 + (-20/101)(-100/20)\Delta\theta_{2/3}$	$\Delta\theta_3 + (-20/99)(-100/20)\Delta\theta_{2/3}$	$\Delta\theta_3 + (-100/20)\Delta\theta_{2/3}$

It should be noted that angular displacements are indicated in Table 7.4 instead of angular velocities. This is caused by the nature of the question asked and recognizing that, during a chosen time interval, $\Delta\theta = \omega\Delta t$ for each of the elements. According to the problem statement, we choose the time interval Δt such that for the arm $\Delta\theta_3 = 1$

^{*}James Ferguson (1710–1776), Scottish physicist and astronomer, first published this device under the title *The Description and Use of a New Machine Called the Mechanical Paradox*, London, 1764.

rev ccw. Then, for gear 2 to remain stationary, column 2 indicates that $\Delta\theta_3 + \Delta\theta_{2/3} = (1 \text{ rev}) + \Delta\theta_{2/3} = 0$, and, therefore, $\Delta\theta_{2/3} = -1 \text{ rev}$. Finally, from the bottom row of columns 4 and 5, we determine that

$$\Delta\theta_4 = \Delta\theta_3 + (-20/101)(-100/20)\Delta\theta_{2/3} = (1 \text{ rev}) + (100/101)(-1 \text{ rev}) = 1/101 \text{ rev}$$

and

$$\Delta\theta_5 = \Delta\theta_3 + (-20/99)(-100/20)\Delta\theta_{2/3} = (1 \text{ rev}) + (100/99)(-1 \text{ rev}) = -1/99 \text{ rev}.$$

Thus, when arm 3 is turned 1 rev ccw, gear 4 rotates 1/101 rev ccw, whereas gear 5 turns 1/99 rev cw. *Ans.*

EXAMPLE 7.11

The overdrive unit shown in Fig. 7.44 is sometimes used following a standard automotive transmission to further reduce engine speed. The engine speed (after the transmission) corresponds to the speed of planet carrier 3, and the drive shaft speed corresponds to that of gear 5; sun gear 2 is held stationary. Determine the percentage additional reduction in engine speed obtained when the overdrive is active.

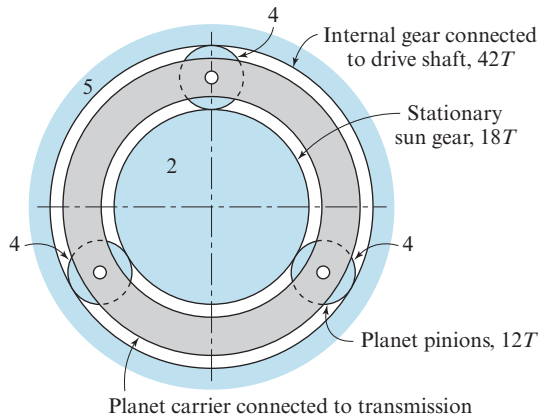


Figure 7.44 Overdrive unit.

SOLUTION

The analysis for this problem is demonstrated in Table 7.5. For gear 2 to remain stationary, column 2 indicates that $\omega_3 + \omega_{2/3} = 0$; therefore $\omega_{2/3} = -\omega_3$. Putting this into column 5 indicates that

$$\omega_5 = \omega_3 + (12/42)(-18/12)\omega_{2/3} = \omega_3 + (-18/42)(-\omega_3) = 1.429\omega_3.$$

Therefore, the percentage reduction in engine speed is

$$(1.429\omega_3 - 1.0\omega_3)/(1.429\omega_3) = 0.300 = 30\%.$$

Ans.

Table 7.5 Tabular Analysis for Example 7.11

Step Number	Gear 2	Arm 3	Gear 4	Gear 5
1. Gears fixed to arm	ω_3	ω_3	ω_3	ω_3
2. Arm fixed	$\omega_{2/3}$	0	$(-18/12)\omega_{2/3}$	$(12/42)(-18/12)\omega_{2/3}$
3. Total	$\omega_3 + \omega_{2/3}$	ω_3	$\omega_3 + (-18/12)\omega_{2/3}$	$\omega_3 + (12/42)(-18/12)\omega_{2/3}$

7.17 REFERENCES

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- [4] Shigley, J. E., and C. R. Mischke, 2001. *Mechanical Engineering Design*, 6th ed., Boston: McGraw-Hill.

PROBLEMS

- 7.1** Find the diametral pitch of a pair of gears having 32 and 84 teeth, respectively, whose center distance is 3.625 in.
- 7.2** Find the number of teeth and the circular pitch of a 6-in pitch-diameter gear whose diametral pitch is 9 teeth/in.
- 7.3** Determine the module of a pair of gears having 18 and 40 teeth, respectively, whose center distance is 58 mm.
- 7.4** Find the number of teeth and the circular pitch of a gear whose pitch diameter is 200 mm if the module is 8 mm/tooth.
- 7.5** Find the diametral pitch and the pitch diameter of a 40-tooth gear whose circular pitch is 3.50 in/tooth.
- 7.6** The pitch diameters of a pair of mating gears are 3.50 in and 8.25 in, respectively. If the diametral pitch is 16 teeth/in, how many teeth are there on each gear?
- 7.7** Find the module and the pitch diameter of a gear whose circular pitch is 40 mm/tooth if the gear has 36 teeth.
- 7.8** The pitch diameters of a pair of gears are 60 mm and 100 mm, respectively. If their module is 2.5 mm/tooth, how many teeth are there on each gear?
- 7.9** What is the diameter of a 33-tooth gear if its circular pitch is 0.875 in/tooth?
- 7.10** A shaft carries a 30-tooth, 3-teeth/in diametral pitch gear that drives another gear at a speed of 480 rev/min. How fast does the 30-tooth gear rotate if the shaft center distance is 9 in?
- 7.11** Two gears having an angular velocity ratio of 3:1 are mounted on shafts whose centers are 136 mm apart. If the module of the gears is 4 mm/tooth, how many teeth are there on each gear?
- 7.12** A gear having a module of 4 mm/tooth and 21 teeth drives another gear at a speed of 240 rev/min. How fast is the 21-tooth gear rotating if the shaft center distance is 156 mm?
- 7.13** A 4-tooth/in diametral pitch, 24-tooth pinion is to drive a 36-tooth gear. The gears are cut on the 20° full-depth involute system. Find and tabulate the addendum, dedendum, clearance, circular pitch, base pitch, tooth thickness, pitch circle radii, base circle radii, lengths of paths of approach and recess, and contact ratio.
- 7.14** A 5-tooth/in diametral pitch, 15-tooth pinion is to mate with a 30-tooth internal gear. The gears are 20° full-depth involute. Make a drawing of the gears showing several teeth on each gear. Can these gears be assembled in a radial direction? If not, what remedy should be used?

- 7.15** A $2\frac{1}{2}$ -teeth/in diametral pitch 17-tooth pinion and a 50-tooth gear are paired. The gears are cut on the 20° full-depth involute system. Find the angles of approach and recess of each gear, and the contact ratio.
- 7.16** A gearset with a module of 5 mm/tooth has involute teeth with $22\frac{1}{2}^\circ$ pressure angle, and 19 and 31 teeth, respectively. They have 1.0 m for the addendum and 1.25 m for the dedendum.* Tabulate the addendum, dedendum, clearance, circular pitch, base pitch, tooth thickness, base circle radii, and contact ratio.
- 7.17** A gear with a module of 8 mm/tooth and 22 teeth is in mesh with a rack; the pressure angle is 25° . The addendum and dedendum are 1.0 m and 1.25 m, respectively.* Find the lengths of the paths of approach and recess, and determine the contact ratio.
- 7.18** Repeat Problem 7.15 using the 25° full-depth system.
- 7.19** Draw a 2-tooth/in diametral pitch, 26-tooth, 20° full-depth involute gear in mesh with a rack. (a) Find the lengths of the paths of approach and recess, and the contact ratio. (b) Draw a second rack in mesh with the same gear but offset $1/8$ in farther away from the gear center. Determine the new contact ratio. Has the pressure angle changed?
- 7.20 to 7.24** Shaper gear cutters have the advantage that they can be used for either external or internal gears and also that only a small amount of runout is necessary at the end of the stroke. The generating action of a pinion shaper cutter can easily be simulated by employing a sheet of clear plastic. Figure P 7.20 shows 1 tooth of a 16-tooth pinion cutter with 20° pressure angle as it can be cut

from a plastic sheet. To construct the cutter, lay out the tooth on a sheet of drawing paper. Be sure to include the clearance at the top of the tooth. Draw radial lines through the pitch circle spaced at distances equal to one fourth of the tooth thickness, as shown in the figure. Next, fasten the sheet of plastic to the drawing and scribe the cutout, the pitch circle, and the radial lines onto the sheet. Then, remove the sheet and trim the tooth outline with a razor blade. Next, use a small piece of fine sandpaper to remove any burrs.

To generate a gear with the cutter, only the pitch circle and the addendum circle need be drawn. Divide the pitch circle into spaces equal to those used on the template and construct radial lines through them. The tooth outlines are then obtained by rolling the template pitch circle upon that of the gear and drawing the cutter tooth lightly for each position. The resulting generated tooth upon the gear will be evident. The following problems all employ a standard 1-tooth/in diametral pitch 20° full-depth template constructed as described earlier. In each, case, you should generate a few teeth and estimate the amount of undercutting.

Table P7.20 to P7.24

Problem no.	P7.20	P7.21	P7.22	P7.23	P7.24
No. of teeth	10	12	14	20	36

- 7.25** A 10-mm/tooth module gear has 17 teeth, a 20° pressure angle, an addendum of 1.0 m, and a dedendum of 1.25 m.* Find the thickness of the teeth at the base circle and at the addendum circle. What is the pressure angle corresponding to the addendum circle?

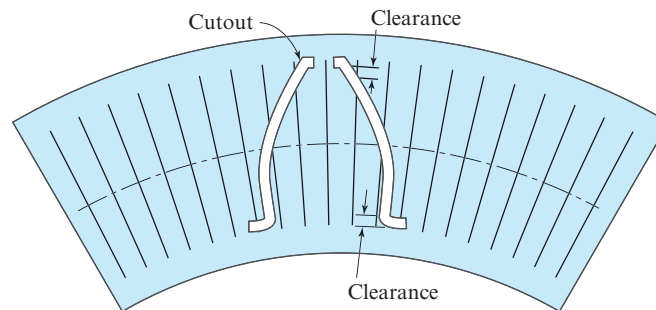


Figure P7.20

* In SI, tooth sizes are given in modules, m, and $a = 1.0$ m means 1 module, not 1 meter.

- 7.26** A 15-tooth pinion has $1\frac{1}{2}$ -tooth/in diametral pitch, 20° full-depth involute teeth. Calculate the thickness of the teeth at the base circle. What are the tooth thickness and the pressure angle at the addendum circle?
- 7.27** A tooth is 0.785 in thick at a pitch circle radius of 8 in and has a pressure angle of 25° . What is the thickness at the base circle?
- 7.28** A tooth is 1.571 in thick at the pitch radius of 16 in and has a pressure angle of 20° . At what radius does the tooth become pointed?
- 7.29** A 25° full-depth involute, 12-tooth/in diametral pitch pinion has 18 teeth. Calculate the tooth thickness at the base circle. What are the tooth thickness and pressure angle at the addendum circle?
- 7.30** A nonstandard 10-tooth 8-tooth/in diametral pitch involute pinion is to be cut with a $22\frac{1}{2}^\circ$ pressure angle. What maximum addendum can be used before the teeth become pointed?
- 7.31** The accuracy of cutting gear teeth can be measured by fitting hardened and ground pins in diametrically opposite tooth spaces and measuring the distance over the pins. For a 10-tooth/in diametral pitch, 20° full-depth involute system 96-tooth gear:
- Calculate the pin diameter that will contact the teeth at the pitch lines if there is to be no backlash.
 - What should be the distance measured over the pins if the gears are cut accurately?
- 7.32** A set of interchangeable gears with 4-tooth/in diametral pitch is cut on the 20° full-depth involute system. The gears have tooth numbers of 24, 32, 48, and 96. For each gear, calculate the radius of curvature of the tooth profile at the pitch circle and at the addendum circle.
- 7.33** Calculate the contact ratio of a 17-tooth pinion that drives a 73-tooth gear. The gears are 96-tooth/in diametral pitch and cut on the 20° full-depth involute system.
- 7.34** A 25° pressure angle 11-tooth pinion is to drive a 23-tooth gear. The gears have a diametral pitch of 8 teeth/in and have involute stub teeth. What is the contact ratio?
- 7.35** A 22-tooth pinion mates with a 42-tooth gear. The gears have full-depth involute teeth, have a diametral pitch of 16 teeth/in, and are cut with a $17\frac{1}{2}^\circ$ pressure angle.* Find the contact ratio.
- 7.36** The center distance of two 24-tooth, 20° pressure angle, full-depth involute spur gears with diametral pitch of 2 teeth/in is increased by 0.125 in over the standard distance. At what pressure angle do the gears operate?
- 7.37** The center distance of two 18-tooth, 25° pressure angle, full-depth involute spur gears with diametral pitch of 3 teeth/in is increased by 0.062 5 in over the standard distance. At what pressure angle do the gears operate?
- 7.38** A pair of mating gears have 24 teeth/in diametral pitch and are generated on the 20° full-depth involute system. If the tooth numbers are 15 and 50, what maximum addendums may they have if interference is not to occur?
- 7.39** A set of gears is cut with a $4\frac{1}{2}$ -in/tooth circular pitch and a $17\frac{1}{2}^\circ$ pressure angle.* The pinion has 20 full-depth teeth. If the gear has 240 teeth, what maximum addendum may it have to avoid interference?
- 7.40** Using the method described for Problems 7.20 to 7.24, cut a 1-tooth/in diametral pitch, 20° pressure angle full-depth involute rack tooth from a sheet of clear plastic. Use a nonstandard clearance of $0.35/P$ in order to obtain a stronger fillet. This template can be used to simulate the generating action of a hob. Now, using the variable-center-distance system, generate an 11-tooth pinion to mesh with a 25-tooth gear without interference. Record the values found for center distance, pitch radii, pressure angle, gear blank diameters, cutter offset, and contact ratio. Note that more than one satisfactory solution exists.
- 7.41** Using the template cut in Problem 7.40, generate an 11-tooth pinion to mesh with a 44-tooth gear with the long-and-short-addendum system. Determine and record suitable values for gear and pinion addendum and dedendum, and for the cutter offset and contact ratio. Compare the contact ratio with that of standard gears.
- 7.42** A pair of involute spur gears with 9 and 36 teeth are to be cut with a 20° full-depth cutter with diametral pitch of 3 teeth/in.
- Determine the amount that the addendum of the gear must be decreased to avoid interference.
 - If the addendum of the pinion is increased by the same amount, determine the contact ratio.

* Such gears came from an older standard and are now obsolete.

7.43 A standard 20° pressure angle full-depth involute 1-tooth/in diametral pitch 20-tooth pinion drives a 48-tooth gear. The speed of the pinion is 500 rev/min. Using the position of the point of contact along the line of action as the abscissa, plot a curve indicating the sliding velocity at all points of contact. Note that the sliding velocity changes sign when the point of contact passes through the pitch point.

7.44 Find the first-order kinematic coefficient of the gear train. What are the speed and direction of rotation of gear 8?

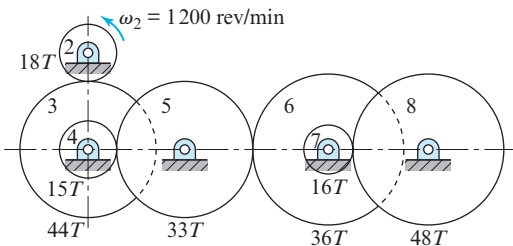


Figure P7.44

7.45 For the given pitch diameters of a set of spur gears forming a train, compute the first-order kinematic coefficient of the train. Determine the speeds and directions of rotation of gears 5 and 7.

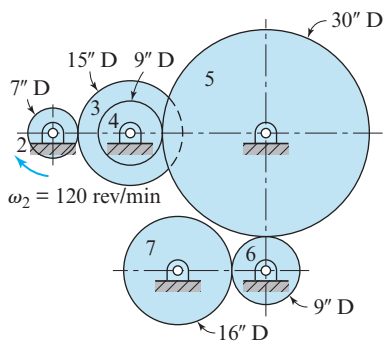


Figure P7.45

7.46 Use the truck transmission of Fig. 7.30 and an input speed of 3 000 rev/min to find the drive shaft speed for each forward gear and for the reverse gear.

7.47 Consider the gears in a speed-change gearbox used in machine tool applications. By sliding the cluster gears on shafts *B* and *C*, nine speed changes can be obtained. The problem of the machine tool designer is to select tooth numbers for the various gears

to produce a reasonable distribution of speeds for the output shaft. The smallest and largest gears are gears 2 and 9, respectively. Using 20 and 45 teeth for these gears, determine a set of suitable tooth numbers for the remaining gears. What are the corresponding speeds of the output shaft? Note that the problem has many solutions.

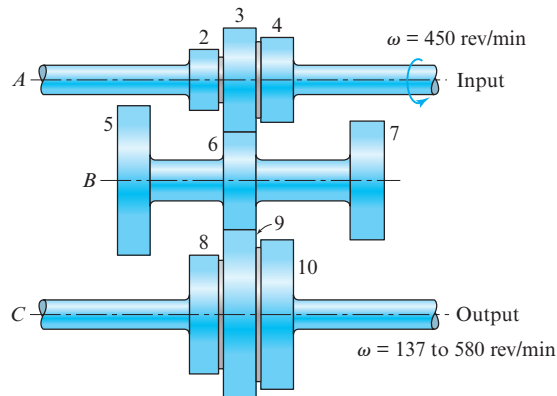


Figure P7.47

7.48 If internal gear 7 rotates at 60 rev/min ccw, determine the speed and direction of rotation of arm 3.

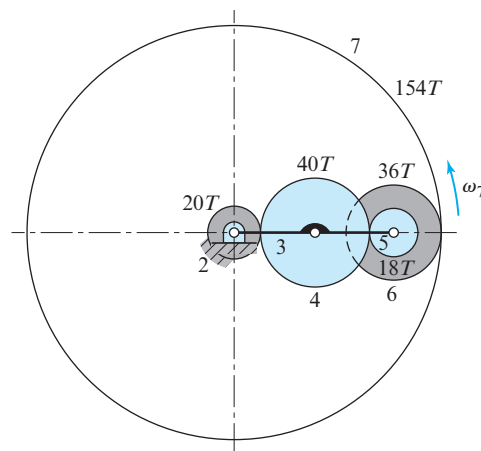


Figure P7.48

7.49 If the arm in Fig. P7.48 rotates at 300 rev/min ccw, find the speed and direction of rotation of internal gear 7.

7.50 If shaft *C* is stationary and gear 2 rotates at 800 rev/min ccw, determine the speed and direction of rotation of shaft *B*.

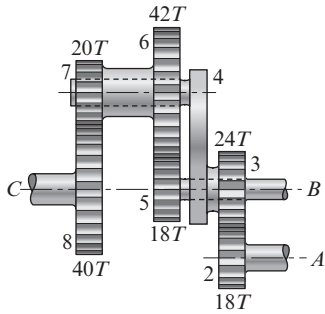


Figure P7.50

- 7.51** In Fig. P7.50, shaft B is stationary and shaft C is driven at 380 rev/min ccw. Determine the speed and direction of rotation of shaft A .
- 7.52** In Fig. P7.50, determine the speed and direction of rotation of shaft C if: (a) shafts A and B both rotate at 360 rev/min ccw; and (b) shaft A rotates at 360 rev/min cw and shaft B rotates at 360 rev/min ccw.
- 7.53** Gear 2 is connected to the input shaft, and arm 3 is connected to the output shaft. Determine the speed reduction. What is the sense of rotation of the output shaft? What changes could be made in the train to produce the opposite sense of rotation for the output shaft?

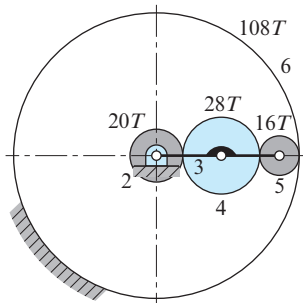


Figure P7.53

- 7.54** The Lévi type-L train shown in Fig. 7.38 has $N_2 = 16T$, $N_4 = 19T$, $N_5 = 17T$, $N_6 = 24T$, and $N_7 = 95T$. Internal gear 7 is fixed. Find the speed and direction of rotation of the arm if gear 2 is driven at 100 rev/min cw.
- 7.55** The Lévi type-A train of Fig. 7.38 has $N_2 = 20T$ and $N_4 = 32T$.
- (a) If the module is $m = 6$ mm/tooth, find the number of teeth on gear 5 and the crank arm radius.

- (b) If gear 2 is fixed and internal gear 5 rotates at 10 rev/min ccw, find the speed and direction of rotation of the arm.

7.56 The figure shows a possible arrangement of gears in a lathe headstock. Shaft A is driven by a motor at a speed of 720 rev/min. The three pinions can slide along shaft A to yield the meshes 2 with 5, 3 with 6, or 4 with 8. The gears on shaft C can also slide to mesh either 7 with 9 or 8 with 10. Shaft C is the mandrel shaft.

- (a) Make a table demonstrating all possible gear arrangements, beginning with the slowest speed for shaft C and ending with the highest, and enter in this table the speeds of shafts B and C .
- (b) If the gears all have a module of $m = 5$ mm/tooth, what must be the shaft center distances?

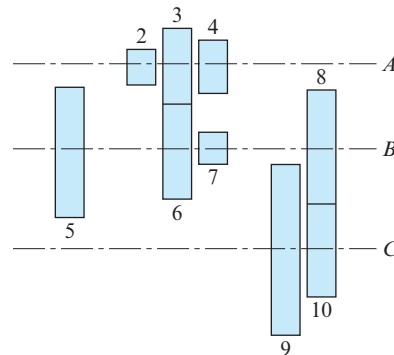


Figure P7.56 $N_2 = 16T$, $N_3 = 36T$, $N_4 = 25T$, $N_5 = 64T$, $N_6 = 66T$, $N_7 = 17T$, $N_8 = 55T$, $N_9 = 79T$, and $N_{10} = 41T$.

- 7.57** If shaft A is the output connected to the arm, and shaft B is the input driving gear 2, determine the speed ratio. Can you identify the Lévi type for this train?

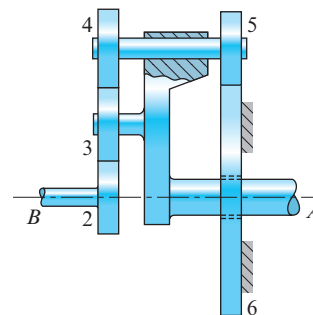


Figure P7.57 $N_2 = 16T$, $N_3 = 18T$, $N_4 = 16T$, $N_5 = 18T$, and $N_6 = 50T$.

7.58 In Problem 7.57, shaft B rotates at 100 rev/min cw. Find the speed of shaft A and of gears 3 and 4 about their own axes.

7.59 In the clock mechanism, a pendulum on shaft A drives an anchor (Fig. 1.12c). The pendulum period is such that 1 tooth of the 30T escapement wheel on shaft B is released every 2 s, causing shaft B to rotate once every minute. Note that the second (to the right) 64T gear is pivoted loosely on shaft D and is connected by a tubular shaft to the hour hand. (a) Show that the train values are such that the minute hand rotates once every hour and that the hour hand rotates once every 12 hours. (b) How many turns does the drum on shaft F make every day?

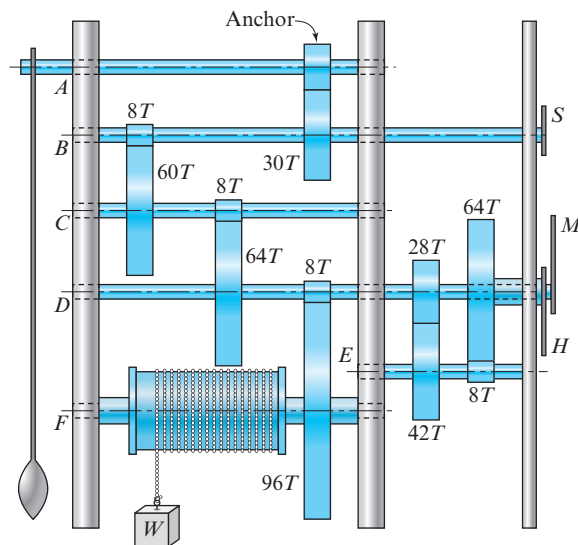


Figure P7.59 Clockwork mechanism.

8

Helical Gears, Bevel Gears, Worms, and Worm Gears



When rotational motion is to be transmitted between parallel shafts, engineers often prefer to use spur gears, since they are easy to design and very economical to manufacture. However, sometimes the design requirements are such that helical gears are a better choice. This is especially true when the loads are heavy, the speeds are high, or the noise level must be kept low.

When motion is to be transmitted between shafts that are not parallel, spur gears cannot be used; the designer must then choose between crossed-helical, bevel, hypoid, or worm gears. Bevel gears have straight teeth, line contact, and high efficiency. Crossed-helical and worm gears have a much lower efficiency because of their increased sliding action; however, if good engineering is used, crossed-helical and worm gears may be designed with quite acceptable values of efficiency. Bevel and hypoid gears are used for similar applications, and, although hypoid gears have inherently stronger teeth, their efficiency is often much less. Worm gears are used when a very small velocity ratio (first-order kinematic coefficient) is required.

8.1 PARALLEL-AXIS HELICAL GEARS

The shape of the tooth of a helical gear is shown in Fig. 8.1. If a piece of paper is cut into the shape of a parallelogram and wrapped around a cylinder, the angular edge of the paper wraps into a helix. The cylinder plays the same role as the base cylinder of a spur gear in Chap. 7. If the paper is unwound, each point on the angular edge generates an involute curve, as indicated in Sec. 7.3 for spur gears. The surface obtained when every point on the angular edge of the paper generates an involute is called an *involute helicoid*. If we imagine the strip of paper as unwrapping from a base cylinder on one gear and wrapping

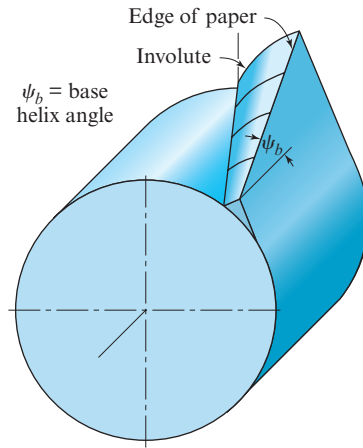


Figure 8.1 An involute helicoid.

up onto the base cylinder of another, then a slanted line on this strip of paper generates two involute helicoids meshing as two tangent tooth shapes.

The initial contact of spur gear teeth, as we saw in Chap. 7, is a line extending across the face of the tooth. The initial contact of helical gear teeth starts as a point and changes into a line as the teeth come into more engagement; in helical gears, however, the line is diagonal across the face of the tooth. It is this gradual engagement of the teeth and the smooth transfer of load from one tooth to another that give helical gears their ability to quietly transmit heavy loads at high speeds.

8.2 HELICAL GEAR TOOTH RELATIONS

As shown in Fig. 8.2, two parallel shaft helical gears must have equal pitches and equal helix angles to mesh properly, but must be of opposite hand. Helical gears with the same hand, a right-hand driver and a right-hand driven gear, for example, can be meshed with their axes skewed and are referred to as *crossed-axis gears* (Sec. 8.7).

Figure 8.3 represents a portion of the top view of a helical rack. Lines AB and CD are the centerlines of two adjacent helical teeth taken on the pitch plane. The angle ψ is the helix angle and is measured at the pitch diameter unless otherwise specified. The distance AC in the plane of rotation of the gear is the *transverse circular pitch*, p_t . The distance AE is the *normal circular pitch*, p_n , and is related to the transverse circular pitch as follows:

$$p_n = p_t \cos \psi. \quad (8.1)$$

Distance AD is called the *axial pitch*, p_x , and can be written as

$$p_x = \frac{p_t}{\tan \psi}. \quad (8.2)$$



Figure 8.2 Pair of helical gears in mesh. Note the opposite hand of the two gears. (Courtesy of Gleason Works, Rochester, NY.)

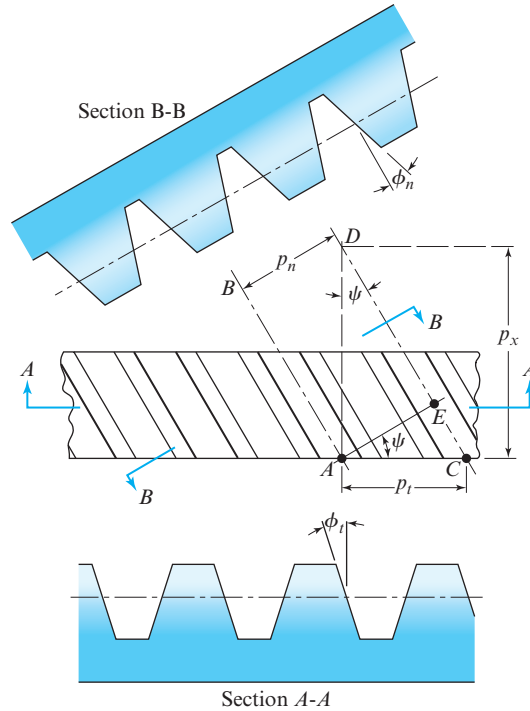


Figure 8.3 Helical gear tooth relations.

The *normal diametral pitch* P_n can be written as

$$P_n = \frac{\pi}{p_n} = \frac{\pi}{p_t \cos \psi} = \frac{P_t}{\cos \psi}, \quad (8.3)$$

where P_t is the *transverse diametral pitch*.

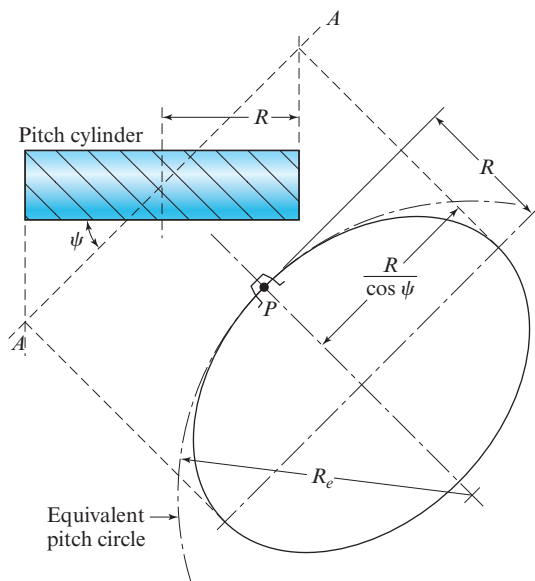
Because of the angularity of the teeth, we must define two different pressure angles, the *normal pressure angle* ϕ_n and the *transverse pressure angle*, ϕ_t , both shown in Fig. 8.3. The two pressure angles are related by

$$\tan \phi_n = \tan \phi_t \cos \psi. \quad (8.4)$$

In applying these equations, it is convenient to remember that all equations and relations that are valid for a spur gear apply equally for the transverse plane of a helical gear.

A better picture of the tooth relations can be obtained by an examination of Fig. 8.4. To obtain the geometric relations, a helical gear has been cut by the oblique plane AA at an angle ψ to a right section. For convenience, only the pitch cylinder of radius R is given. Figure 8.4 shows that the intersection of the AA plane and the pitch cylinder is an ellipse whose radius at the pitch point P is R_e . This is called the *equivalent pitch radius*, and it is the radius of curvature of the pitch surface in the normal cross section. For the condition

Figure 8.4



that $\psi = 0$, this radius of curvature is $R_e = R$. If we imagine angle ψ to be gradually increased from 0 to 90° , we see that R_e begins at a value of $R_e = R$ and increases until, when $\psi = 90^\circ$, the value of $R_e = \infty$.

It is demonstrated in a note at the end of this chapter (Sec. 8.17) that

$$R_e = \frac{R}{\cos^2 \psi}, \quad (8.5)$$

where R is the pitch radius of the helical gear and R_e is the pitch radius of an equivalent spur gear. This equivalence is taken on the normal section of the helical gear.

Let us define the number of teeth on the helical gear as N and that on the equivalent spur gear as N_e . Then,

$$N_e = 2R_e P_n. \quad (d)$$

Using Eqs. (8.3) and (8.5) we can write this as

$$N_e = 2 \frac{R}{\cos^2 \psi} \frac{P_t}{\cos \psi} = \frac{N}{\cos^3 \psi}. \quad (8.6)$$

8.3 HELICAL GEAR TOOTH PROPORTIONS

Except for fine pitch (normal diametral pitch of 20 teeth/in and finer), there is no generally accepted standard for the proportions of helical gear teeth.

In determining the tooth proportions for helical gears, it is necessary to consider the manner in which the teeth are formed. If the helical gear is hobbled, then tooth proportions

are calculated in a plane normal to the tooth. As a general guide, tooth proportions are then often based on a normal pressure angle of $\phi_n = 20^\circ$. Most of the proportions used for spur gears, given in Table 7.2, can then be used. The tooth proportions are calculated by using the normal diametral pitch, P_n . These proportions are suitable for helix angles from 0° to 30° , and all helix angles can be cut with the same hob. Of course, the normal diametral pitch of the hob and the gear are the same.

If the gear is cut by a shaper, an alternative set of tooth proportions is used, based on a transverse pressure angle of $\phi_t = 20^\circ$ and the transverse diametral pitch, P_t . For these gears, the helix angles are generally restricted to 15° , 23° , 30° , or 45° ; helix angles greater than 45° are not recommended. The normal diametral pitch, P_n , must be used to compute the tooth dimensions; the proportions given in Table 7.2 are usually satisfactory. If the shaper method is used, however, the same cutter cannot be used to cut both spur and helical gears.

8.4 CONTACT OF HELICAL GEAR TEETH

For spur gears, contact between meshing teeth occurs along a line that is parallel to their axes of rotation. As shown in Fig. 8.5, contact between meshing helical gear teeth occurs along a diagonal line. When contact of another tooth is just beginning at A , contact at the other end of the tooth may have already progressed from B to C .

Several kinds of contact ratios are used in evaluating the performance of helical gearsets. The *transverse contact ratio* is designated by m_t and is the contact ratio in the transverse plane. It is obtained exactly as was m_c for spur gears.

The *normal contact ratio*, m_n , is the contact ratio in the normal section. It is also determined exactly as was the contact ratio m_c for spur gears, but the dimensions of equivalent spur gears must be used in the determination. The base helix angle and the pitch helix angle for helical gears, are related by

$$\tan \psi_b = \tan \psi \cos \phi_t. \quad (8.7)$$

Then the transverse and normal contact ratios are related by

$$m_n = \frac{m_t}{\cos^2 \psi_b}. \quad (8.8)$$

The *axial contact ratio*, m_x , also called the *face contact ratio*, is the ratio of the face width of the gear to the axial pitch, determined from Eq. (8.2). It is given by

$$m_x = \frac{F}{p_x} = \frac{F \tan \psi}{p_t}, \quad (8.9)$$

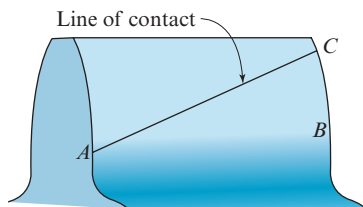


Figure 8.5 Line of contact on a helical gear tooth.

where F is the face width of the helical gear. Figure 8.5 shows that this contact ratio is greater than unity when another tooth is beginning contact, solely because of the helix angle of the teeth before the previous tooth contact has finished. Note that this face contact ratio, also called *overlap*, has no parallel for spur gears, and note that, because of the helix angle, this face contact ratio can be made greater than unity for helical gears by the choice of face width despite the choice of tooth size. If the face width is made greater than the axial pitch, continuous contact of at least one tooth is assured. This means that fewer teeth may be used on helical pinions than on spur pinions. The overlapping action also results in smoother operation of the gears. Note also that the face contact ratio depends solely on the geometry of a single gear, whereas the transverse and normal contact ratios depend upon the geometry of a pair of mating gears.

The *total contact ratio* is the sum of the face contact ratio m_x and the transverse contact ratio m_t . In a sense, this sum gives the average total number of teeth in contact.

8.5 REPLACING SPUR GEARS WITH HELICAL GEARS

Because of their ability to carry heavy loads at high speed with little noise, it is sometimes desirable to replace a pair of spur gears by parallel shaft helical gears, although the cost may be slightly higher. An example illustrates the calculations.

EXAMPLE 8.1

A pair of 20° full-depth involute spur gears with 32 and 80 teeth, diametral pitch of 16 teeth/in, and face width of 0.75 in are to be replaced by helical gears. The same hob used for the spur gears is to be used for the helical gears. The shaft center distance and the angular velocity ratio must remain the same. The helix angle is to be as small as possible, and the overlap is to be 1.5 or greater. Determine the helix angle, the numbers of teeth, and the face width of the new helical gears.

SOLUTION

From the spur gear data and Eq. (7.1), the center distance is

$$R_2 + R_3 = \frac{N_2 + N_3}{2P} = \frac{32 \text{ teeth} + 80 \text{ teeth}}{2(16 \text{ teeth/in})} = 3.5 \text{ in.} \quad (1)$$

From Eq. (7.5), the first-order kinematic coefficient, the angular velocity ratio, is

$$\left| \theta'_{3/2} \right| = \left| \frac{\omega_3}{\omega_2} \right| = \frac{R_2}{R_3} = \frac{N_2}{N_3} = \frac{32 \text{ teeth}}{80 \text{ teeth}} = 0.4. \quad (2)$$

Since the same hob is to be used, the normal diametral pitch, P_n , for the helical gears must also be 16 teeth/in. Since the shaft center distance must remain the same that is,

$$R_2 + R_3 = \frac{N_2 + N_3}{2P_n \cos \psi} = \frac{N_2 + N_3}{2(16 \text{ teeth/in}) \cos \psi} = 3.5 \text{ in}$$

or

$$\cos \psi = \frac{N_2 + N_3}{112 \text{ teeth}} = \frac{(N_2/N_3)N_3 + N_3}{112 \text{ teeth}} = \frac{1.4N_3}{112 \text{ teeth}} = \frac{N_3}{80 \text{ teeth}}. \quad (3)$$

This implies that N_3 must be less than 80 teeth, while Eq. (2) requires that the ratio N_2/N_3 must remain 0.4, or $N_2 = 0.4N_3$.

Since $N_3 = 79$ teeth does not give an integer solution for N_2 , the next smallest integer solution (which gives the smallest nonzero helix angle, ψ) is $N_3 = 75$ teeth and $N_2 = 30$ teeth, giving a helix angle of $\psi = 20.364^\circ$. The transverse circular pitch is

$$p_t = \frac{\pi}{P_n \cos \psi} = \frac{\pi}{(16 \text{ teeth/in}) \cos 20.364^\circ} = 0.209 \text{ in/tooth},$$

for which Eq. (8.9) indicates a face width of $F \geq 0.845$ in. Unfortunately, the space available does not allow this increase in face width. Therefore, this solution is not acceptable.

The next integer solution for tooth numbers is $N_3 = 70$ teeth and $N_2 = 28$ teeth, giving a helix angle of $\psi = 28.955^\circ$. The transverse circular pitch is $p_t = \pi / (P_n \cos \psi) = 0.224$ in/tooth, and the face width is $F \geq 0.607$ in. This is an acceptable solution with face width of less than the original spur gears. The same face width of $F = 0.750$ in can still be used if desired.

8.6 HERRINGBONE GEARS

Double-helical or *herringbone gears* comprise teeth having both a right- and a left-hand helix cut on the same gear blank, as shown in Fig. 8.6. One of the primary disadvantages of the single-helical gear is the axial thrust loads that must be accounted for in the design of the bearings. In addition, the desire to obtain a good overlap without an excessively large face width may lead to the use of a comparatively larger helix angle,



Figure 8.6 A herringbone gearset. (Courtesy of Matzner Photography, Madison, WI.)

thus producing even higher axial thrust loads. These thrust loads are eliminated by the herringbone configuration, since the axial force of the right-hand half is balanced by that of the left-hand half. Thus, with the absence of thrust reactions, helix angles are usually larger for herringbone gears than for single-helical gears. However, one of the members of a herringbone gearset should always be mounted with some axial play or float to accommodate slight tooth errors and mounting tolerances.

For the efficient transmission of large amounts of power at high speeds, herringbone gears are almost universally employed.

8.7 CROSSED-AXIS HELICAL GEARS

Crossed-axis helical or *spiral gears* are sometimes used when the shaft centerlines are neither parallel nor intersecting. These are essentially non-enveloping worm gears (Sec. 8.14), since the gear blanks have a cylindrical form with the two cylinder axes skew to each other.

The tooth action of crossed-axis helical gears is quite different from that of parallel-axis helical gears. The teeth of crossed-axis helical gears have only *point contact*. In addition, there is much greater sliding action along the tooth surfaces than for parallel-axis helical gears. For these reasons, they are chosen only to carry small loads. Because of the point contact, however, they need not be mounted accurately; either the center distance or the shaft angle may also vary slightly without affecting the amount of contact.

There is no difference between crossed-axis helical gears and other helical gears until they are mounted in mesh. They are manufactured in the same way. Two meshing crossed-axis helical gears usually have the same hand; that is, a right-hand driver goes with a right-hand driven gear. The relation between thrust, hand, and rotation for crossed-axis helical gears is shown in Fig. 8.7.

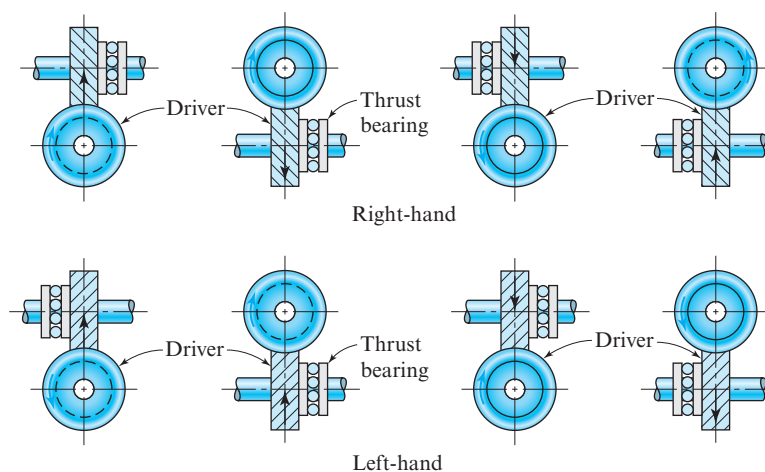


Figure 8.7 Thrust, rotation, and hand relations for crossed-axis helical gearing. (Courtesy of Boston Gear Works, Inc., North Quincy, MA.)

For crossed-axis helical gears to mesh properly, they must share the same normal pitch. When tooth sizes are specified, the normal pitch should always be used. The reason for this is that when different helix angles are used for the driver and the driven gear, the transverse pitches are not the same. The relation between the shaft and helix angles is

$$\Sigma = \psi_2 \pm \psi_3. \quad (8.10)$$

The positive sign is used when both helix angles are of the same hand, and the negative sign is used when they are of opposite hand. Opposite-hand crossed-axis helical gears are used when the shaft angle Σ is small. The first-order kinematic coefficient, the angular velocity ratio between the shafts, is

$$\left| \theta'_{3/2} \right| = \left| \frac{\omega_3}{\omega_2} \right| = \frac{N_2}{N_3} = \frac{R_2 \cos \psi_2}{R_3 \cos \psi_3}. \quad (8.11)$$

Crossed-axis helical gears have the least sliding at the point of contact when the two helix angles are equal. If the two helix angles are not equal, the larger helix angle should be used with the driver if both gears have the same hand.

There is no widely accepted standard for crossed-axis helical gear tooth proportions. Many different combinations of proportions give good tooth action. Since the teeth are in point contact, an effort should be made to obtain a contact ratio of 2 or more. For this reason, crossed-axis helical gears are usually cut with a low pressure angle and a deep tooth. The tooth proportions given in Table 8.1 are representative of good design. The driver tooth numbers indicated are the minimum required to avoid undercut. The driven gear should have 20 teeth or more if a contact ratio of 2 is to be obtained.

Table 8.1 Tooth Proportions for Crossed-Axis Helical Gears

Driver		Driven	Both
Helix angle	Minimum tooth number	Helix angle	Normal pressure angle
ψ_2	N_2	ψ_3	ϕ_n
45°	20	45°	14½°
60°	9	30°	17½°
75°	4	15°	19½°
86°	1	4°	20°

Normal diametral pitch $P_n = 1$ teeth/in; working depth = 2.400 in; whole depth = 2.650 in; addendum $a = 1.200$ in.

To illustrate the calculations for a pair of crossed-axis helical gears, consider the following example.

EXAMPLE 8.2

Two shafts at an angle of 60° are to have a velocity ratio of $1/1.5$. The center distance between the shafts is 8.63 in. Design a pair of crossed-axis helical gears for this application.

SOLUTION

Choosing $\psi_2 = 35^\circ$ for the pinion, Eq. (8.10) then gives $\psi_3 = 25^\circ$ for the gear. Substituting these angles into Eq. (8.11), the first-order kinematic coefficient can be written as

$$\left| \theta'_{3/2} \right| = \left| \frac{\omega_3}{\omega_2} \right| = \frac{R_2 \cos 35^\circ}{R_3 \cos 25^\circ} = \frac{1}{1.5}.$$

Therefore, the pitch radius of the pinion is

$$R_2 = 0.7376R_3.$$

This, along with the given shaft center distance, $R_2 + R_3 = 8.63$ in, gives the pitch radius of the pinion, $R_2 = 3.663$ in, and the pitch radius of the gear, $R_3 = 4.967$ in. Choosing a normal diametral pitch of $P_n = 6$ teeth/in, the numbers of teeth on the pinion and the gear, respectively, are

$$N_2 = 2P_n R_2 \cos \psi_2 = 2(6 \text{ teeth/in})(3.663 \text{ in}) \cos 35^\circ = 36 \text{ teeth} \quad \text{Ans.}$$

and

$$N_3 = 2P_n R_3 \cos \psi_3 = 2(6 \text{ teeth/in})(4.967 \text{ in}) \cos 25^\circ = 54 \text{ teeth.} \quad \text{Ans.}$$

8.8 STRAIGHT-TOOTH BEVEL GEARS

When rotational motion is transmitted between shafts whose axes intersect, some form of bevel gears is usually used. Bevel gears have pitch surfaces that are cones, with their cone axes matching the two shaft rotation axes, as shown in Fig. 8.8. The gears are mounted so that the apexes of the two pitch cones are coincident with the point of intersection of the shaft axes. These pitch cones roll together without slipping.

Although bevel gears are often made for an angle of 90° between the shafts, they can be designed for almost any angle. When the shaft intersection angle is other than 90° , the gears are called *angular bevel gears*. For the special case where the shaft intersection angle is 90° and both gears are of equal size, such bevel gears are called *miter gears*. A pair of miter gears is shown in Fig. 8.9.

For straight-tooth bevel gears, the true shape of a tooth is obtained by taking a spheric section through the tooth, where the center of the sphere is at the common apex, as shown in Fig. 8.8. As the radius of the sphere increases, the same number of teeth is projected

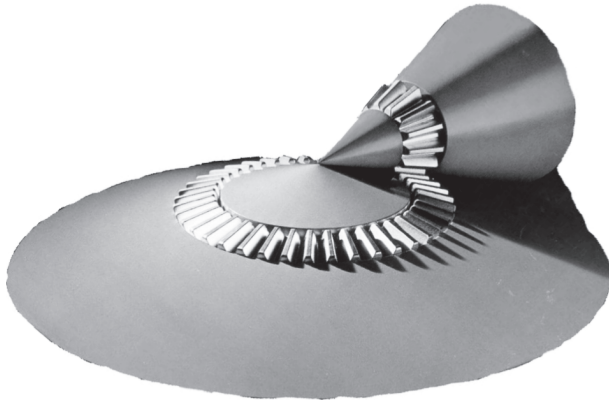


Figure 8.8 The pitch surfaces of bevel gears are cones that have only rolling contact. (Courtesy of Gleason Works, Rochester, NY.)



Figure 8.9 A pair of miter gears in mesh. (Courtesy of Gleason Works, Rochester, NY.)

onto a larger surface; therefore, the size of the teeth increases as larger spheric sections are taken. We have seen that the action and contact conditions for spur gear teeth may be viewed in a plane taken at right angles to the axes of the spur gears. For bevel gear teeth, the action and contact conditions should properly be viewed on a spheric surface (instead of a plane). We can even think of spur gears as a special case of bevel gears in which



Figure 8.10 A pair of straight-tooth bevel gears. (Courtesy of Gleason Works, Rochester, NY.)

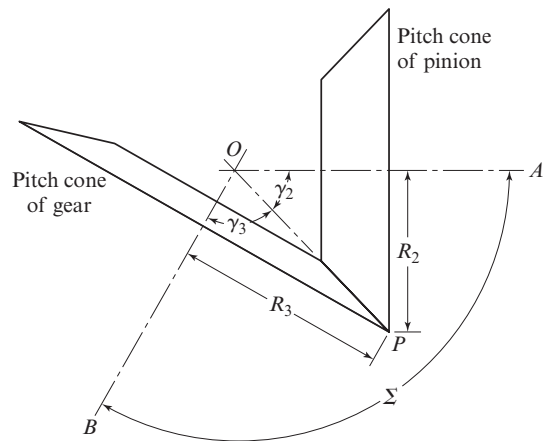


Figure 8.11 Pitch cones of bevel gears.

the spheric radius is infinite, thus producing a plane surface on which the tooth action is viewed. Figure 8.10 is typical of many straight-tooth bevel gearsets.

It is standard practice to specify the pitch diameter of a bevel gear at the large end of the teeth. In Fig. 8.11, the pitch cones of a pair of bevel gears are drawn, and the pitch radii are given as R_2 and R_3 , respectively, for the pinion and the gear. The cone angles γ_2 and γ_3 are defined as the pitch angles, and their sum is equal to the shaft intersection angle, Σ ; that is

$$\Sigma = \gamma_2 + \gamma_3.$$

The first-order kinematic coefficient, the angular velocity ratio between the shafts, is obtained in the same manner as for spur gears and is

$$\left| \theta'_{3/2} \right| = \left| \frac{\omega_3}{\omega_2} \right| = \frac{R_2}{R_3} = \frac{N_2}{N_3}. \quad (8.12)$$

In the kinematic design of bevel gears, the tooth numbers of the two gears and the shaft angle are usually given, and the corresponding pitch angles are to be determined. Although they can be determined graphically, the analytic approach gives more precise values. From Fig. 8.11, distance OP may be written as

$$OP = \frac{R_2}{\sin \gamma_2} = \frac{R_3}{\sin \gamma_3} \quad (a)$$

so that

$$\sin \gamma_2 = \frac{R_2}{R_3} \sin \gamma_3 = \frac{R_2}{R_3} \sin (\Sigma - \gamma_2) \quad (b)$$

or

$$\sin \gamma_2 = \frac{R_2}{R_3} (\sin \Sigma \cos \gamma_2 - \cos \Sigma \sin \gamma_2). \quad (c)$$

Dividing both sides of this equation by $\cos \gamma_2$ and rearranging gives

$$\tan \gamma_2 = \frac{R_2}{R_3} (\sin \Sigma - \cos \Sigma \tan \gamma_2).$$

Then, rearranging this equation gives

$$\tan \gamma_2 = \frac{\sin \Sigma}{(R_3/R_2) + \cos \Sigma} = \frac{\sin \Sigma}{(N_3/N_2) + \cos \Sigma}. \quad (8.13)$$

Similarly,

$$\tan \gamma_3 = \frac{\sin \Sigma}{(N_2/N_3) + \cos \Sigma}. \quad (8.14)$$

For a shaft angle of $\Sigma = 90^\circ$, the previous expressions reduce to

$$\tan \gamma_2 = \frac{N_2}{N_3} \quad (8.15)$$

and

$$\tan \gamma_3 = \frac{N_3}{N_2}. \quad (8.16)$$

The projection of bevel gear teeth onto the surface of a sphere would indeed be a difficult and time-consuming task. Fortunately, an approximation that reduces the problem to that of ordinary spur gears is common. This approximation is called *Tredgold's approximation*, and, as long as the gear has eight or more teeth, it is accurate enough for practical purposes. It is in almost universal use, and the terminology of bevel gear teeth has evolved around it.

In Tredgold's method, a *back cone* is formed of elements that are perpendicular to the elements of the pitch cone at the large end of the teeth, as shown in Fig. 8.12. The length of a back-cone element is called the *back-cone radius*. Now an equivalent spur gear is constructed whose pitch radius, R_e , is equal to the back-cone radius. Thus, from a pair of bevel gears, using Tredgold's approximation, we can obtain a pair of equivalent spur gears that are then used to define the tooth profiles. They can also be used to determine the tooth action and the contact conditions, just as for ordinary spur gears, and the results correspond closely to those for the bevel gears.

From the geometry of Fig. 8.12, the equivalent pitch radii are

$$R_{e2} = \frac{R_2}{\cos \gamma_2}, \quad R_{e3} = \frac{R_3}{\cos \gamma_3}. \quad (8.17)$$

The number of teeth on each of the equivalent spur gears is

$$N_e = \frac{2\pi R_e}{p}, \quad (8.18)$$

where p is the circular pitch of the bevel gear measured at the large end of the teeth. Usually, the equivalent spur gears do *not* have integral numbers of teeth.

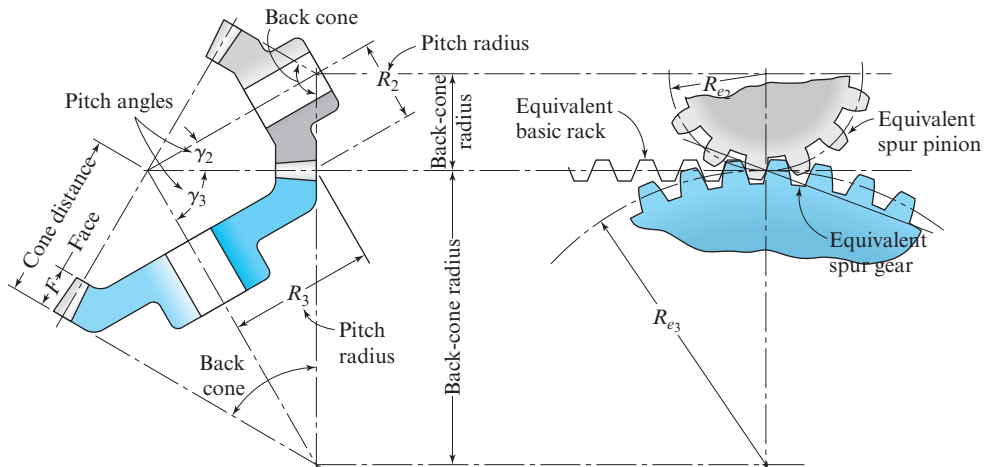


Figure 8.12 Tredgold's approximation.

8.9 TOOTH PROPORTIONS FOR BEVEL GEARS

Practically all straight-tooth bevel gears manufactured today use a 20° pressure angle. It is not necessary to use an interchangeable tooth form, since bevel gears cannot be interchanged. For this reason, the long-and-short-addendum system, described in Sec. 7.11, is used. The proportions are tabulated in Table 8.2.

Bevel gears are usually mounted on the outboard side of the bearings, since the shaft axes intersect, which means that the effect of shaft deflection is to pull the small end of the teeth away from mesh, causing the larger end to take more of the load. Thus, the load across the tooth is variable; for this reason, it is desirable to design a fairly short tooth. As indicated in Table 8.2, the face width is usually limited to about one third of the cone distance. We note also that a short face width simplifies the tooling problems in cutting bevel gear teeth.

Figure 8.13 defines additional terminology characteristic of bevel gears. Note that a constant clearance is maintained by making the elements of the face cone parallel to the elements of the root cone of the mating gear. This explains why the face cone apex is not coincident with the pitch cone apex in Fig. 8.13. This permits a larger fillet than would otherwise be possible.

8.10 BEVEL GEAR EPICYCLIC TRAINS

The bevel gear train shown in Fig. 8.14 is called *Humpage's reduction gear*. Bevel gear epicyclic trains are used quite frequently, and they are similar to spur gear epicyclic trains except that their axes of rotation are not all on parallel shafts. The train of Fig. 8.14

Table 8.2 Tooth Proportions for 20° Straight-tooth Bevel Gears

Item	Formula			
Working depth	$h_k = 2.0/P$			
Clearance	$c = 0.188/P + 0.002 \text{ in.}$			
Addendum of gear	$a_G = \frac{0.540}{P} + \frac{0.460}{P(m_{90})^2}$			
Gear ratio	$m_G = N_G/N_P$			
Equivalent 90° ratio	$m_{90} = \begin{cases} m_G & \text{when } \Sigma = 90^\circ \\ \sqrt{m_G \frac{\cos \gamma_P}{\cos \gamma_G}} & \text{when } \Sigma \neq 90^\circ \end{cases}$			
Face width	$F = \frac{1}{3}$ or $F = \frac{10}{P}$, whichever is smaller			
Minimum number of teeth	Pinion	16	15	14
	Gear	16	17	20

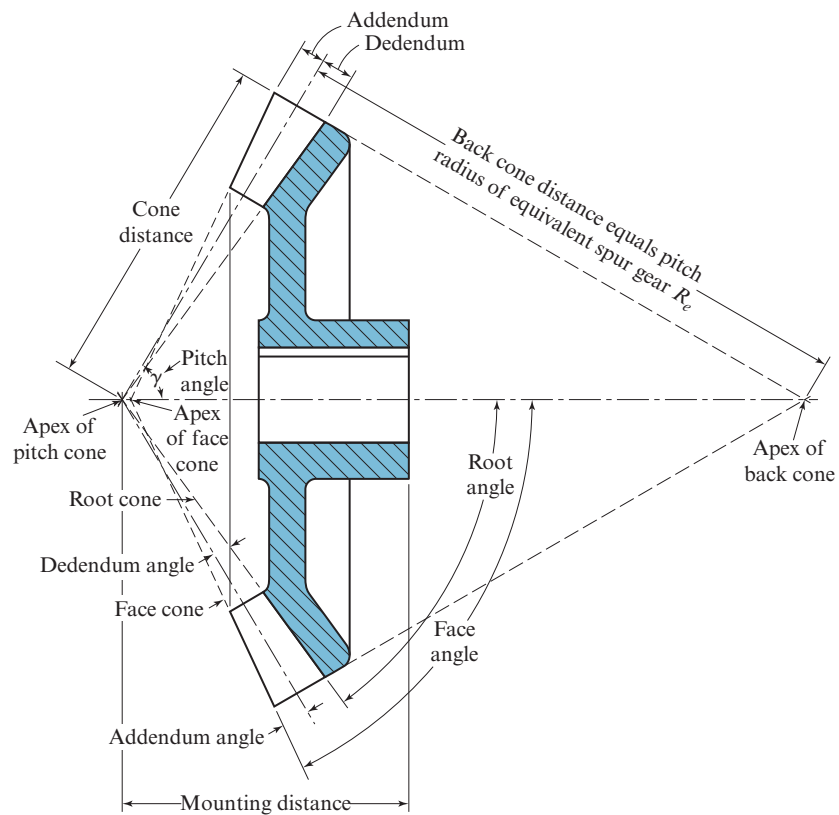


Figure 8.13

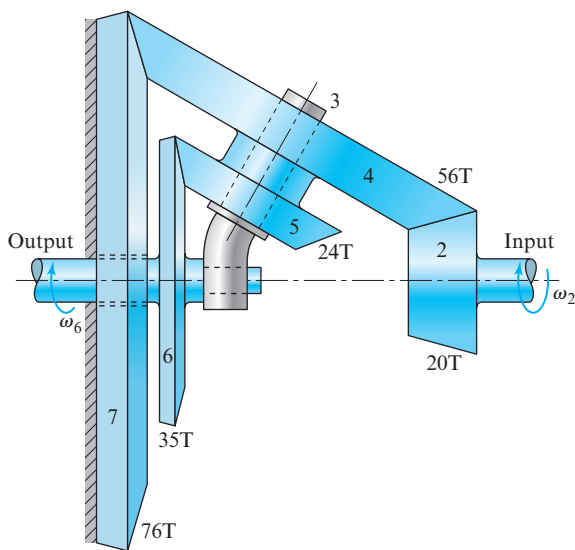


Figure 8.14 Humpage's reduction gear.

is, in fact, a double epicyclic train, and the spur gear counterpart of each can be found in Fig. 7.38. The analysis of such trains can be done by formulae, the same as for spur gear trains, as shown by the following example.

EXAMPLE 8.3

In the bevel gear train in Fig. 8.14, the input is gear 2, and the output is gear 6, which is connected to the output shaft. Arm 3 turns freely on the output shaft and carries planets 4 and 5. Gear 7 is fixed to the frame. What is the output shaft speed if gear 2 rotates at 2 000 rev/min?

SOLUTION

The problem is solved in two steps. In the first step we consider the train to be made up of gears 2, 4, and 7, and calculate the rotational speed of the arm. Thus,

$$\omega_F = \omega_2 = 2\,000 \text{ rev/min} \quad \text{and} \quad \omega_L = \omega_7 = 0 \text{ rev/min},$$

and, according to Eq. (7.22),

$$\theta'_{LF} = \theta'_{72} = -\left(\frac{56}{76}\right)\left(\frac{20}{56}\right) = -\frac{5}{19}.$$

The negative sign is chosen because, if gear 7 were not fixed, it would appear to rotate in the direction opposite to that of gear 2, when viewed from a coordinate system fixed to arm 3.

Substituting into Eq. (7.25) and solving for the angular velocity of arm 3 gives

$$\theta'_{LF/A} = -\frac{5}{19} = \frac{0 - \omega_3}{2\,000 \text{ rev/min} - \omega_3},$$

$$\omega_A = \omega_3 = 416.67 \text{ rev/min}.$$

Next, we consider the train as composed of gears 2, 4, 5, and 6. Then $\omega_F = \omega_2 = 2\,000 \text{ rev/min}$, as before, and $\omega_L = \omega_6$, which is to be found. The first-order kinematic coefficient of the train is

$$\theta'_{LF} = \theta'_{62} = \left(\frac{24}{35}\right)\left(-\frac{20}{56}\right) = -\frac{12}{49}.$$

Again, the negative sign is chosen because gear 6 would appear to rotate in the direction opposite to that of gear 2, when viewed from a coordinate system fixed to arm 3.

Substituting into Eq. (7.25) again and solving for ω_6 , with ω_3 known above, now gives

$$\theta'_{LF/A} = -\frac{12}{49} = \frac{\omega_L - 416.67 \text{ rev/min}}{2\,000 \text{ rev/min} - 416.67 \text{ rev/min}}.$$

Rearranging this equation, the speed of gear 6 (and the output shaft) is

$$\omega_L = \omega_6 = 28.91 \text{ rev/min}.$$

Ans.

Because the result is positive, we conclude that the output shaft rotates in the same direction as input shaft 2, with a speed reduction of 2 000:28.91 or 69.18:1.

8.11 CROWN AND FACE GEARS

If the pitch angle of one of a pair of bevel gears is made equal to 90° , the pitch cone becomes a flat surface and the resulting gear is called a *crown gear*. Figure 8.15 shows a crown gear in mesh with a bevel pinion. Notice that a crown gear is the counterpart of a rack in spur gearing. The back cone of a crown gear is a cylinder, and the resulting involute teeth have straight sides, as indicated in Fig. 8.12.

A pseudo-bevel gearset can be obtained using a cylindric spur gear for a pinion in mesh with a gear having a planar pitch surface (similar to a crown gear) called a *face gear*. When the axes of the pinion and gear intersect, the face gear is called *on center*; when the axes do not intersect, the face gear is called *off center*.

To understand how a spur pinion, with a cylindric rather than conic pitch surface, can properly mesh with a face gear, we must consider how the face gear is formed; it is generated by a reciprocating cutter that is a replica of the spur pinion. Since the cutter and the gear blank are rotated as if in mesh, the resulting face gear is conjugate to the cutter and, therefore, to the spur pinion. The face width of the teeth on the face gear must be held quite short, however; otherwise the top land will become pointed.

Face gears are not capable of carrying heavy loads, but since the axial mounting posture of the pinion is not critical, they are sometimes more suitable for angular drives than bevel gears.

8.12 SPIRAL BEVEL GEARS

Straight-tooth bevel gears are easy to design, simple to manufacture, and give very good results in service if they are mounted accurately and positively. As in the case of spur gears, however, they become noisy at high pitch-line velocities. In such cases, it is often

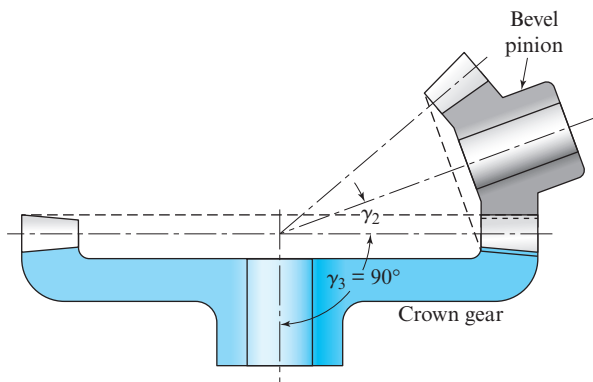


Figure 8.15 A crown gear and bevel pinion.



Figure 8.16 Spiral bevel gears.
(Courtesy of Gleason Works,
Rochester, NY.)

good design practice to use *spiral bevel gears*, which are the bevel counterparts of helical gears. A mating pair of spiral bevel gears is shown in Fig. 8.16. The pitch surfaces and the nature of contact are the same as for straight-tooth bevel gears except for the differences brought about by the spiral-shaped teeth.

Spiral bevel gear teeth are conjugate to a basic crown rack, which can be generated as shown in Fig. 8.17 using a circular cutter. The spiral angle ψ is measured at the mean radius of the gear. As with helical gears, spiral bevel gears give much smoother tooth action than straight-tooth bevel gears and hence are useful where high speeds are required. To obtain true spiral tooth action, the face contact ratio should be at least 1.25.

Pressure angles used with spiral bevel gears are generally $14\frac{1}{2}^\circ$ to 20° , whereas the spiral angle is about 30° or 35° . As far as tooth action is concerned, the spiral may be either right- or left-handed; it makes no difference. However, looseness in the bearings might result in jamming or separating of the teeth, depending on the direction of rotation and the hand of the spiral. Since jamming of the teeth would do the most damage, the hand of the spiral should be such that the teeth tend to separate.

Zerol Bevel Gears The Zerol bevel gear is a patented gear that has curved teeth but a 0° spiral angle. An example is shown in Fig. 8.18. It has no advantage in tooth action over the straight-tooth bevel gear and is designed simply to take advantage of the cutting machinery used for cutting spiral bevel gears.

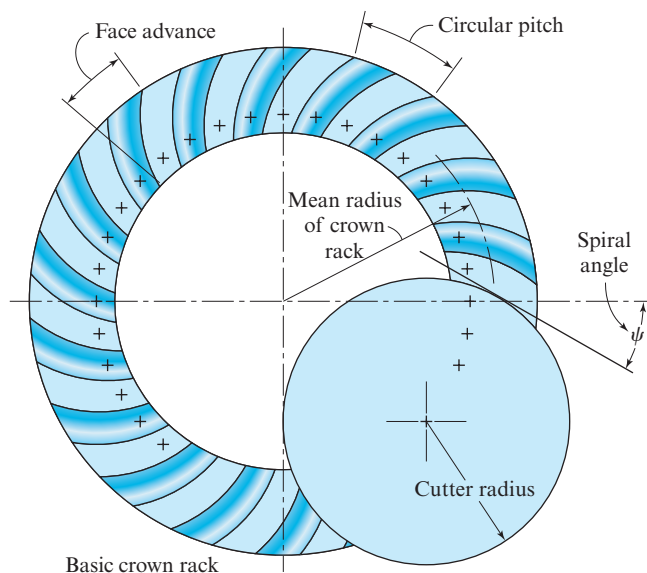


Figure 8.17 Cutting spiral bevel gear teeth on a basic crown rack.



Figure 8.18 Zerol bevel gears. (Courtesy of Gleason Works, Rochester, NY.)

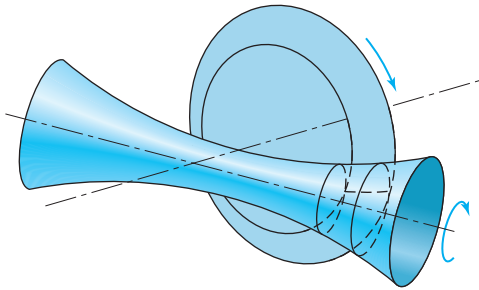


Figure 8.19 The pitch surfaces for hypoid gears are hyperboloids of revolution.



Figure 8.20 Hypoid gears. (Courtesy of Gleason Works, Rochester, NY.)

8.13 HYPOID GEARS

It is frequently desirable, as in the case of rear-wheel drive automotive differentials, to have a gearset similar to bevel gears but where the shafts do not intersect. Such gears are called *hypoid gears*, since, as shown in Fig. 8.19, their pitch surfaces are hyperboloids of revolution. Figure 8.20 shows a pair of hypoid gears in mesh. The tooth action between these gears is a combination of rolling and sliding along a straight line and has much in common with that of worm gears (Sec. 8.14).

8.14 WORMS AND WORM GEARS

A *worm* is a machine element having a screw-like thread, and worm teeth are frequently spoken of as *threads*. A worm meshes with a conjugate gear-like member called a *worm gear* or a *worm wheel*. Figure 8.21 shows a worm and worm gear in an application. These gears are used with nonintersecting shafts that are usually at a shaft angle of 90° , but there is no reason why shaft angles other than 90° cannot be used if a design demands it.

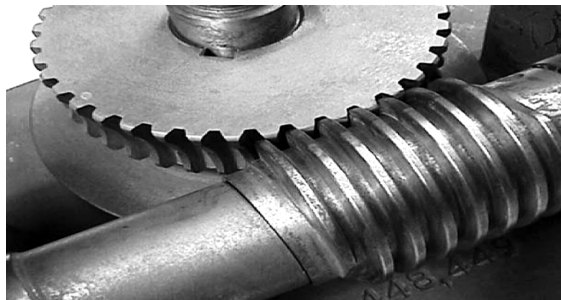


Figure 8.21

A single-enveloping worm and worm gearset. (Courtesy of Gleason Works, Rochester, NY.)

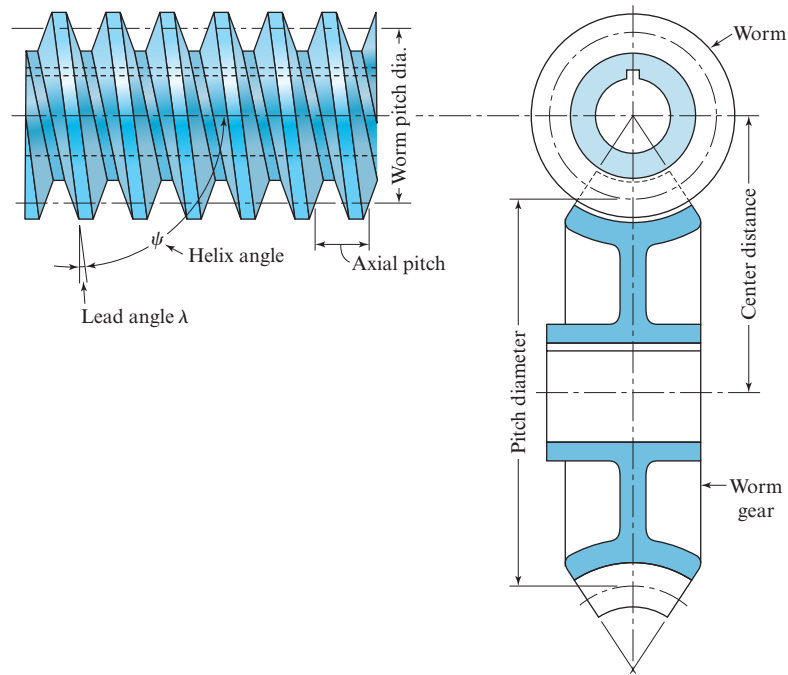


Figure 8.22 Nomenclature of a single-enveloping worm and worm gear set.

Worms in common use have from one to four teeth and are said to be *single threaded*, *double threaded*, and so on. As we will see, there is no definite relation between the number of teeth and the pitch diameter of a worm. The number of teeth on a worm gear is usually much higher, and, therefore, the angular velocity of the worm gear is usually much lower than that of the worm. In fact, often, one primary application for a worm and worm gear is to obtain a very large angular velocity reduction—that is, a very low first-order kinematic coefficient. In keeping with this low velocity ratio, the worm gear is usually the driven member of the pair, and the worm is usually the driving member.

A worm gear, unlike a spur or helical gear, has a face that is made concave so that it partially wraps around, or envelopes, the worm, as shown in Fig. 8.22. Worms are sometimes designed with a cylindric pitch surface, or they may have an hourglass shape, such that the worm also wraps around or partially encloses the worm gear. If an enveloping worm gear is mated with a cylindric worm, the set is said to be *single enveloping*. When the worm is hourglass shaped, the worm and worm gearset is said to be *double enveloping*, since each member partially wraps around the other; such a worm is sometimes called a *Hindley worm*. The nomenclature of a single-enveloping worm and worm gearset is shown in Fig. 8.22.

A worm and worm gear combination is similar to a pair of mating crossed helical gears except that the worm gear partially envelops the worm. For this reason, they have line contact instead of the point contact found in crossed helical gears and are thus able to transmit more power. When a double-enveloping worm and worm gearset is used, even

more power can be transmitted, at least in theory, since contact is distributed over an area on both tooth surfaces.

In a single-enveloping worm and worm gearset, it makes no difference whether the worm rotates on its own axis and drives the gear by a screwing action or whether the worm is translating along its axis and drives the worm gear through rack action. The resulting motion and contact are the same. For this reason, a single-enveloping worm need not be accurately mounted along its axis. However, the worm gear should be accurately mounted along its rotation axis; otherwise, its pitch surface is not properly aligned with the worm axis. In a double-enveloping worm and worm gearset, both members are throated, and therefore both must be accurately mounted in all directions to obtain correct contact.

A mating worm and worm gear with a 90° shaft angle have the same hand of helix, but the helix angles are usually very different. The helix angle on the worm is usually quite large (at least for one or two teeth) and quite small on the worm gear. On the worm, the *lead angle* is the complement of the helix angle, as shown in Fig. 8.22. Because of this, it is customary to specify the lead angle for the worm and specify the helix angle for the worm gear. This is convenient, since these two angles are equal for a 90° shaft angle.

In specifying the pitch of a worm and worm gearset, it is usual to specify the axial pitch of the worm and the circular pitch of the worm gear. These are equal if the shaft angle is 90° . It is common to employ simple fractions, such as $1/4$, $3/8$, $1/2$, $3/4$, 1, and $1 1/4$ in/tooth, and so on, for the circular pitch of the worm gear; there is no reason, however, why the AGMA standard diametral pitches used for spur gears (Table 7.1) should not also be used for worm gears.

The pitch radius of a worm gear is determined in the same manner as that of a spur gear; that is,

$$R_3 = \frac{N_3 p}{2\pi}, \quad (8.19)$$

where all values are defined in the same manner as for spur gears, but refer to the parameters of the worm gear.

The pitch radius of the worm may have any value, but it should be the same as that of the hob used to cut the worm gear teeth. The relation between the pitch radius of the worm and the center distance, as recommended by AGMA, is

$$R_2 = \frac{(R_2 + R_3)^{0.875}}{4.4}, \quad (8.20)$$

where the quantity $(R_2 + R_3)$ is the center distance in inches. This equation gives proportions that result in good power capacity. The AGMA standard also states that the denominator of Eq. (8.20) may vary from 3.4 to 6.0 without appreciably affecting the power capacity. Equation (8.20) is not required, however; other proportions will also serve well, and, in fact, power capacity may not always be the primary consideration. However, there are a lot of variables in worm gear design, and the equation is helpful in obtaining trial dimensions.

The *lead* of a worm has the same meaning as for a screw thread and is the axial distance through which a point on the helix will move when the worm is turned through one revolution. Thus, in equation form, the lead of the worm is given by

$$l = p_x N_2, \quad (8.21)$$

where p_x is the axial pitch, and N_2 is the number of teeth (threads) on the worm. The lead and the *lead angle* are related as follows:

$$\lambda = \tan^{-1} \left(\frac{l}{2\pi R_2} \right), \quad (8.22)$$

where λ is the lead angle, as shown in Fig. 8.22.

The teeth on a worm are usually cut in a milling machine or on a lathe. Worm gear teeth are most often produced by hobbing. Except for clearance at the top of the hob teeth, the worm should be an exact duplicate of the hob in order to obtain conjugate action. This also means that, where possible, the worm should be designed using the dimensions of existing hobs.

The pressure angles used on worms and worm gearsets vary widely and should depend approximately on the value of the lead angle. Good tooth action is obtained if the pressure angle is made large enough to eliminate undercutting of the worm gear tooth on the side at which the contact ends. Recommended values are given in Table 8.3.

A satisfactory tooth depth that has about the right relation to the lead angle is obtained by making the depth a proportion of the normal circular pitch. Using an addendum of $1/P = p_n/\pi$, as for full-depth spur gears, we obtain the following proportions for worms and worm gears:

$$\text{Addendum} = 1.000/P = 0.318 \, 3p_n,$$

$$\text{Whole depth} = 2.000/P = 0.636 \, 6p_n,$$

$$\text{Clearance} = 0.157/P = 0.050 \, 7p_n.$$

The face width of the worm gear should be obtained as shown in Fig. 8.23. This makes the face of the worm gear equal to the length of a tangent to the worm pitch circle between its points of intersection with the addendum circle.

Table 8.3 Recommended Pressure Angles for Worm and Worm Gear Sets

Lead angle λ	Pressure angle ϕ
0° – 16°	$14 \, 1/2^\circ$
16° – 25°	20°
25° – 35°	25°
35° – 45°	30°

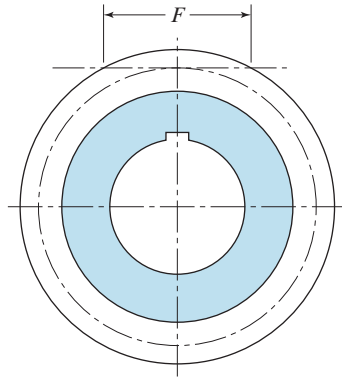


Figure 8.23 Face width of a worm gear.

8.15 SUMMERS AND DIFFERENTIALS

Figure 8.24 shows a variety of mechanisms used as computing devices. Since each of these is a two-degree-of-freedom mechanism, two input motion variables must be defined so that the motions of the remaining elements of the system are determined. The equation below each of these mechanisms indicates that the output parameter is a direct measure of the sum of the two input parameters. For this reason, such a mechanism is referred to as an *adder* or a *summer*.

The spur-gear differential of Fig. 8.25 helps in visualizing its action. If planet carrier 2 is held stationary and gear 3 is turned by some amount $\Delta\theta_3$, then gear 8 turns in the opposite direction by an amount $\Delta\theta_8 = -\Delta\theta_3$. If the planet carrier is also allowed to turn, then $\Delta\theta_8 = \Delta\theta_2 - \Delta\theta_3$. It is for this reason that this two-degree-of-freedom mechanism is called a *differential*. Of course, a better force balance is obtained by employing several sets of planets equally spaced about the sun gears; three sets are usual. Also, you will note in Fig. 8.25 that planets 4, 5, 6, and 7 are identical; by making planets 4 and 7 longer (thicker), they meet with each other and eliminate the need for planets 5 and 6.

If a spur-gear differential were used on the driving axles of an automobile, then shafts A and B of Fig. 8.25 would drive the right and left wheels, respectively, whereas arm 2 receives power from a main drive shaft connected to the transmission.

It is interesting that a differential was used in China, long before the invention of the magnetic compass, to indicate geographic direction. In Fig. 8.26, each wheel of a carriage drives a vertical shaft through pin wheels. The right-hand shaft drives the upper pin wheel and the left-hand shaft drives the lower pin wheel of the differential shown in Fig. 8.27. When the cart is driven in a straight line, the upper and lower pin wheels rotate at the same speed but in opposite directions. Thus, the planet gear turns about its own center, but the axle to which it is mounted remains stationary, and so the figure continues to point in a constant direction. However, when the cart makes a turn, one of the pin wheels rotates faster than the other, causing the planet axle to turn just enough to cause the figure to continue to point in the same geographic direction as before.

Figure 8.28 is a drawing of the ordinary bevel-gear automotive differential. The drive shaft pinion and the ring gear are normally hypoid gears (Sec. 8.13). The ring gear acts as the planet carrier, and its speed can be calculated as for a simple gear train when the speed

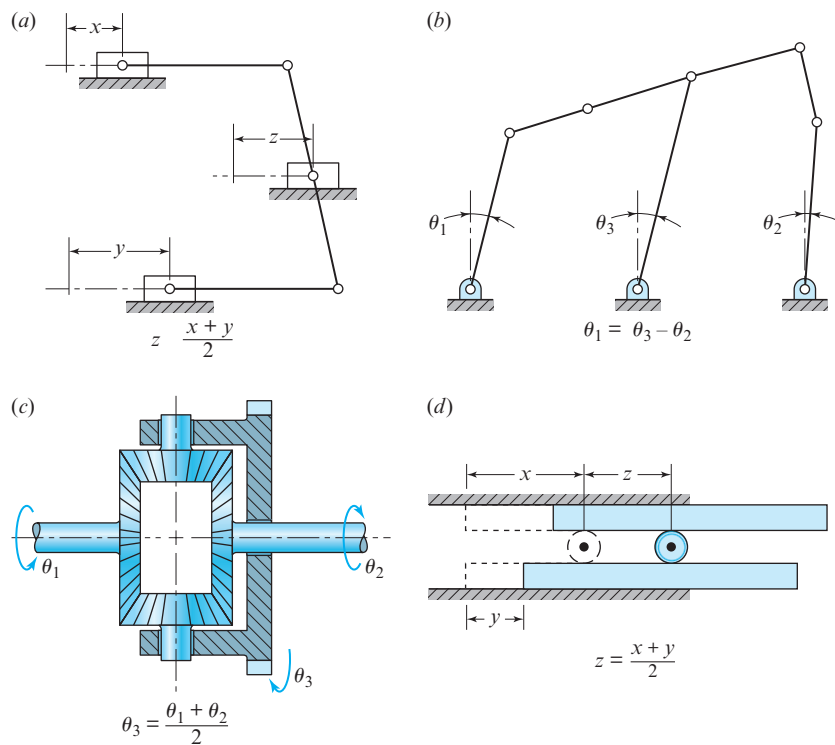


Figure 8.24 Differential mechanisms used for: (a, c, d) adding; (b) subtracting; (a, c, d) averaging two quantities.

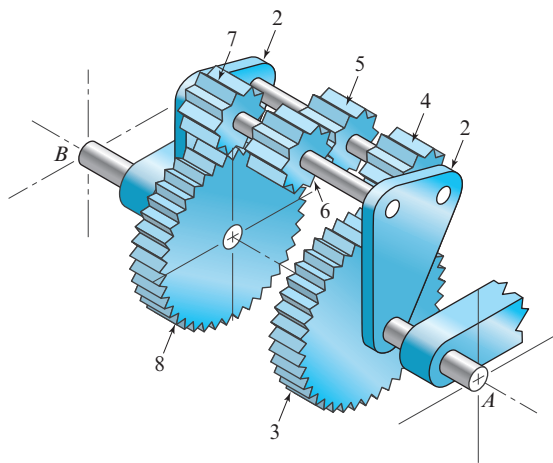


Figure 8.25 A spur-gear differential.

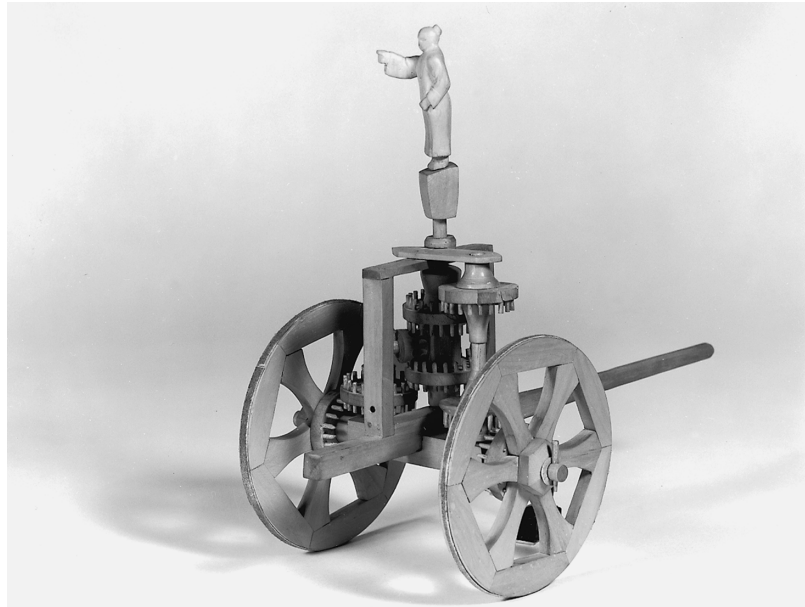


Figure 8.26 The image on this chariot continues to point in a constant geographic direction. (Smithsonian Institute, photo P63158-B.)

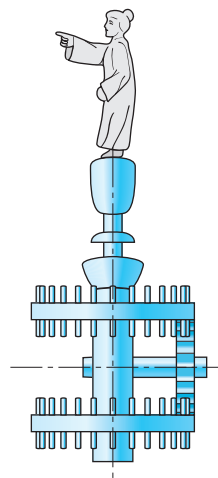


Figure 8.27 The Chinese differential.

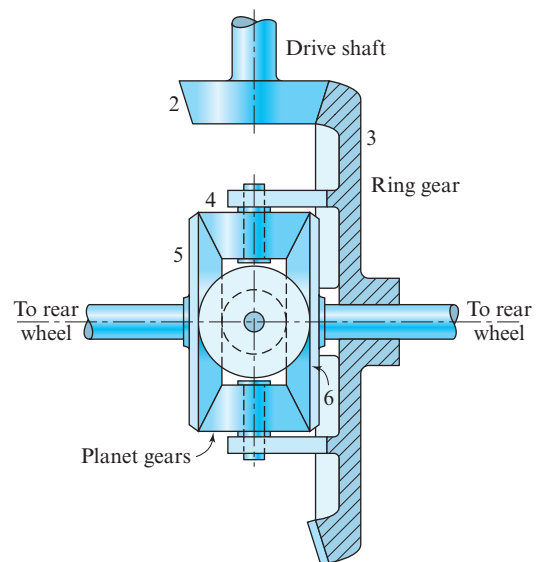


Figure 8.28 Schematic drawing of a bevel-gear automotive rear-axle differential.

of the drive shaft is given. Gears 5 and 6 are connected, respectively, to each of the rear wheels, and, when the car is traveling in a straight line, these two gears rotate in the same direction with the same speed. Thus, for straight-line motion of the car, there is no relative motion between the planet gears and gears 5 and 6. The planet gears, in effect, serve only as keys to transmit motion from the planet carrier to both wheels.

When the vehicle is making a turn, the wheel on the inside of the curve makes fewer revolutions than the wheel with the larger turning radius. Unless this difference in speed is accommodated in some manner, one or both of the tires must slip to make the turn. The differential permits the two wheels to rotate at different angular velocities while, at the same time, delivering power to both. During a turn, the planet gears rotate about their own axes, thus permitting gears 5 and 6 to revolve at different angular velocities.

The purpose of the differential is to allow different speeds for the two driving wheels. In the usual differential of a rear-wheel-drive passenger car, the torque is divided approximately equally whether the car is traveling in a straight line or on a curve. Sometimes, however, the road conditions are such that the tractive effort developed by the two wheels is unequal. In such a case, the total tractive effort is only twice that at the wheel having the least traction, since the differential divides the torque equally. If one wheel happens to be traveling on snow or ice, the total tractive effort possible at that wheel is very small, since only a small torque is required to cause the wheel to slip. Thus, the car remains stationary with one wheel spinning and the other having only trivial tractive effort. If the car is in motion and encounters a slippery surface, then all traction as well as control is lost!

Limited-Slip Differential It is possible to overcome this disadvantage of the simple bevel-gear differential by adding a coupling unit that is sensitive to wheel speeds. The objective of such a unit is to cause more of the torque to be directed to the slower-moving wheel. Such a combination is then called a *limited-slip differential*.

Mitsubishi, for example, utilizes a viscous coupling unit, called a *VCU*, which is torque sensitive to wheel speeds. A difference in wheel speeds causes more torque to be delivered to the slower-moving wheel. A large difference, perhaps caused by the spinning of one wheel on ice, causes a much larger share of the torque to be delivered to the non-spinning wheel. The arrangement, as used on the rear axle of an automobile, is shown in Fig. 8.29.

Another approach is to employ Coulomb friction, or clutching action, in the coupling. Such a unit, as with the VCU, is engaged whenever a significant difference in wheel speeds occurs.

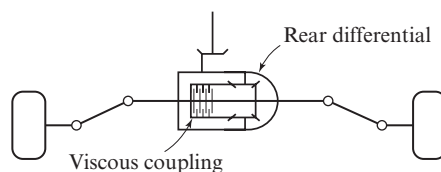


Figure 8.29 Viscous coupling used on the rear axle of the Mitsubishi Galant and Eclipse GSX automobiles.

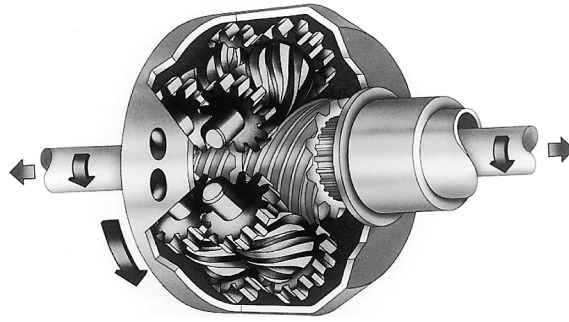


Figure 8.30 The TORSSEN differential used on the drive shaft of Audi automobiles. (Courtesy of Audi of America, Inc., Troy, MI.)

Of course, it is also possible to design a bevel-gear differential that is capable of being locked by the driver whenever dangerous road conditions are encountered. This is equivalent to a solid axle and forces both wheels to move at the same speed. It seems obvious that such a differential should not be locked when the tires are on dry pavement, because of excessive wear caused by tire slipping.

Worm-Gear Differential If gears 3 and 8 in Fig. 8.25 were replaced with worm gears, and planet gears 4 and 7 were replaced with mating worm wheels, then the result is a *worm-gear differential*. Of course, planet carrier 2 would have to rotate about a new axis perpendicular to axle *AB*, since worm and worm wheel axes act at right angles to each other. Such an arrangement can provide the traction of a locked differential or solid axle without the penalty of restricting small differential movement between the wheels.

The worm-gear differential was invented by Mr. Vernon Gleasman of Pittsford, NY, in 1958 and was developed by Gleason Works as the TORSSEN differential, a registered trademark now owned by JTEKT Torsen North America, Inc. The word TORSSEN is an acronym for the words “torque sensing” since the differential can be designed to provide any desired locking value by varying the lead angle of the worm. Figure 8.30 shows a TORSSEN differential as used on Audi automobiles.*

8.16 ALL-WHEEL DRIVE TRAIN

As shown in Fig. 8.31, an all-wheel drive train consists of a center differential, geared to the transmission, driving the ring gears on both the front- and rear-axle differentials. Dividing the thrusting force between all four wheels instead of only two is itself an advantage, since it allows easier handling on curves and in crosswinds.

Dr. Herbert H. Dobbs, Colonel (Ret.), an automotive engineer, states:

One of the major improvements in automotive design is antilocking braking. This provides stability and directional control when stopping by ensuring a balanced transfer of momentum from the car to the road through all wheels. As too many have

*A very nice animation of a TORSSEN worm-gear differential can be viewed at http://www.outube.ycom/watch?v=Z9iPqIQ_8iM.

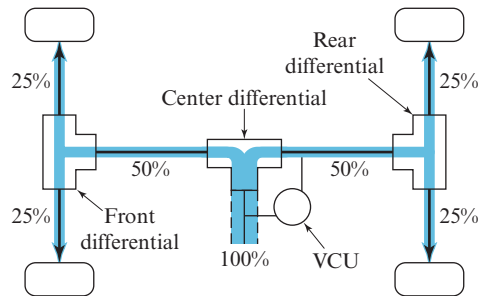


Figure 8.31 All-wheel drive (AWD) system used on the Mitsubishi Galant, demonstrating the power distribution for straight-ahead operation.

found out, loss of traction at one of the wheels when braking produces unbalanced forces on the car, which can throw it out of control.

It is equally important to provide such control during starting and acceleration. As with braking, this is not a problem when a car is driven prudently and driving conditions are good. When driving conditions are not good, the problems quickly become manifold. The antilock braking system provides the answer for the deceleration portion of the driving cycle, but the devices generally available to help during the remainder of the cycle are much less satisfactory.*

An early solution to this problem, used by Audi, is to electrically lock the center or the rear differential, or both, when driving conditions deteriorate. Locking only the center differential causes one half of the power to be delivered to the rear wheels and one half to the front wheels. If one of the rear wheels, say, rests on a slippery surface such as ice, the other rear wheel has no traction. But the front wheels still provide traction. So the car has two-wheel drive. If the rear differential is then also locked, the car has three-wheel drive, since the rear-wheel drive is then 50-50 distributed.

Another solution is to use a limited-slip differential as the center differential on an all-wheel drive (AWD) vehicle. This then has the effect of distributing most of the driving torque to the front or rear axle, depending on which is moving slower. An even better solution is to use limited-slip differentials on both the center and the rear differentials.

Unfortunately, both locking differentials and limited-slip differentials interfere with antilock braking systems. However, they are quite effective during low-speed winter operation.

The most effective solution seems to be the use of TORSEN differentials in an AWD vehicle. Here is what Dr. Dobbs has to say about their use:

If they are cut to preclude any slip, the TORSEN distributes torque proportional to available traction at the driven wheels under all conditions just like a solid axle does, but it never locks up under any circumstances. Both of the driven wheels are always free to follow the separate paths dictated for them by the vehicle's motion, but are constrained by the balancing gears to stay synchronized with each other. All this adds up to a true "TORque SENsing and proportioning" differential, which of course is where the name came from.

*Dr. Herbert H. Dobbs, Rochester Hills, MI, personal communication.

The result, particularly with a high-performance front-wheel-drive vehicle, is remarkable, to say the least. The Army has TORSSENS in the High-Mobility Multipurpose Wheeled Vehicle (HMMWV) or “Hummer,” which has replaced the Jeep. The only machines in production more mobile off-road than this one have tracks, and it is very capable on highways as well. It is fun to drive. The troops love it. Beyond that, Teledyne has an experimental “FAst Attack Vehicle” with TORSSENS front, center, and rear. I’ve driven that machine over 50 mi/h on loose sand washes at Hank Hodges’ Nevada Automotive Center, and it handled like it was running on dry pavement. The constant redistribution of torque to where traction was available kept all wheels driving and none digging!

8.17 NOTE

The equation of an ellipse with its center at the origin of an xy coordinate system with a and b as its semi-major and semi-minor axes, respectively, is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad (a)$$

Also, the formula for radius of curvature is

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2}. \quad (b)$$

Using these two equations, it is not difficult to find the radius of curvature corresponding to $x = 0$, $y = b$. The result is

$$\rho = a^2/b. \quad (c)$$

Then, referring to Fig. 8.3, we substitute $\rho = R_e$, $a = R/\cos\psi$, and $b = R$ into Eq. (c) and obtain Eq. (8.5).

PROBLEMS

- 8.1** A pair of parallel-axis helical gears has $14^{1/2}^\circ$ normal pressure angle, diametral pitch of 6 teeth/in, and 45° helix angle. The pinion has 15 teeth, and the gear has 24 teeth. Calculate the transverse and normal circular pitches, the normal diametral pitch, the pitch radii, and the equivalent tooth numbers.
- 8.2** A pair of parallel-axis helical gears are cut with a 20° normal pressure angle and a 30° helix angle. They have diametral pitch of 16 teeth/in and have 16 and 40 teeth, respectively. Find the transverse pressure angle, the normal circular pitch, the axial pitch, and the pitch radii of the equivalent spur gears.
- 8.3** A parallel-axis helical gearset is made with a 20° transverse pressure angle and a 35° helix angle. The gears have diametral pitch of 10 teeth/in and have 15 and 25 teeth, respectively. If the face width is 0.75 in, calculate the base helix angle and the axial contact ratio.
- 8.4** A pair of helical gears is to be cut for parallel shafts whose center distance is to be about 3.5 in to give a velocity ratio of approximately 1.8. The gears are to be cut with a standard 20° pressure angle hob whose diametral pitch is 8 teeth/in. Using a helix angle of 30° , determine the transverse values

of the diametral and circular pitches and the tooth numbers, pitch radii, and center distance.

- 8.5** A 16-tooth helical pinion is to run at 1 800 rev/min and drive a helical gear on a parallel shaft at 400 rev/min. The centers of the shafts are to be spaced 11 in apart. Using a helix angle of 23° and a pressure angle of 20° , determine the values for the tooth numbers, pitch radii, normal circular pitch and diametral pitch, and the face width.
- 8.6** The catalog description of a pair of helical gears is as follows: $14\frac{1}{2}^\circ$ normal pressure angle, 45° helix angle, diametral pitch of 8 teeth/in, 1.0-in face width, and normal diametral pitch of 11.31 teeth/in. The pinion has 12 teeth and a 1.500-in pitch diameter, and the gear has 32 teeth and a 4.000-in pitch diameter. Both gears have full-depth teeth, and they may be purchased either right- or left-handed. If a right-hand pinion and left-hand gear are placed in mesh, find the transverse contact ratio, the normal contact ratio, the axial contact ratio, and the total contact ratio.
- 8.7** In a medium-size truck transmission, a 22-tooth clutch-stem gear meshes continuously with a 41-tooth countershaft gear. The data indicate normal diametral pitch of 7.6 teeth/in, $18\frac{1}{2}^\circ$ normal pressure angle, $23\frac{1}{2}^\circ$ helix angle, and a 1.12-in face width. The clutch-stem gear is cut with a left-hand helix, and the countershaft gear is cut with a right-hand helix. Determine the normal and total contact ratios if the teeth are cut full-depth with respect to the normal diametral pitch.
- 8.8** A helical pinion is right-handed, has 12 teeth, has a 60° helix angle, and is to drive another gear at a velocity ratio of 3.0. The shafts are at a 90° angle, and the normal diametral pitch of the gears is

8 teeth/in. Find the helix angle and the number of teeth on the mating gear. What is the shaft center distance?

- 8.9** A right-hand helical pinion is to drive a gear at a shaft angle of 90° . The pinion has 6 teeth and a 75° helix angle and is to drive the gear at a velocity ratio of 6.5. The normal diametral pitch of the gear is 12 teeth/in. Calculate the helix angle and the number of teeth on the mating gear. Also determine the pitch radius of each gear.
- 8.10** In Fig P8.10 gear 2 rotates clockwise and drives gear 3 counterclockwise at a velocity ratio of 2:1. Use a normal diametral pitch of 5 teeth/in, a shaft center distance of about 10 in, and the same helix angle for both gears. Find the tooth numbers, the helix angles, and the exact shaft center distance.
- 8.11** A pair of straight-tooth bevel gears is to be manufactured for a shaft angle of 90° . If the driver is to have 18 teeth, and the velocity ratio is to be 3:1, what are the pitch angles?
- 8.12** A pair of straight-tooth bevel gears has a velocity ratio of 1.5 and a shaft angle of 75° . What are the pitch angles?
- 8.13** A pair of straight-tooth bevel gears is to be mounted at a shaft angle of 120° . The pinion and gear are to have 15 and 33 teeth, respectively. What are the pitch angles?
- 8.14** A pair of straight-tooth bevel gears with diametral pitch of 2 teeth/in have 19 teeth and 28 teeth, respectively. The shaft angle is 90° . Determine the pitch diameters, pitch angles, addendum, dedendum, face width, and pitch diameters of the equivalent spur gears.

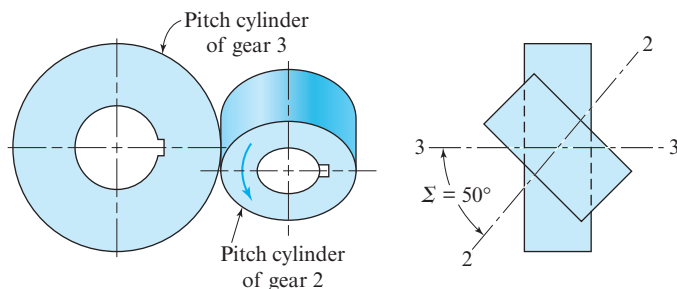


Figure P8.10

- 8.15** A pair of straight-tooth bevel gears with diametral pitch of 8 teeth/in has 17 teeth and 28 teeth, respectively, and a shaft angle of 105° . For each gear, calculate the pitch radius, pitch angle, addendum, dedendum, face width, and equivalent number of teeth. Make a sketch of the two gears in mesh. Use standard tooth proportions as for a 90° shaft angle.
- 8.16** A worm having 4 teeth and a lead of 1.0 in drives a worm gear at a velocity ratio of 7.5. Determine the pitch diameters of the worm and worm gear for a center distance of 1.75 in.
- 8.17** Specify a suitable worm and worm-gear combination for a velocity ratio of 60 and a center distance of 6.50 in. Use an axial pitch of 0.500 in/tooth.
- 8.18** A triple-threaded worm drives a worm gear having 40 teeth. The axial pitch is 1.25 in, and the pitch diameter of the worm is 1.75 in. Calculate the lead and lead angle of the worm. Find the helix angle and pitch diameter of the worm gear.
- 8.19** A triple-threaded worm with a lead angle of 20° and an axial pitch of 0.400 in/tooth drives a worm gear with a velocity reduction of 15 to 1. Determine the following for the worm gear: (a) the number of teeth, (b) the pitch radius, (c) the helix angle, (d) the pitch radius of the worm, and (e) the center distance.
- 8.20** The gear train shown in Figure P8.20 consists of bevel gears, spur gears, and a worm and worm gear.

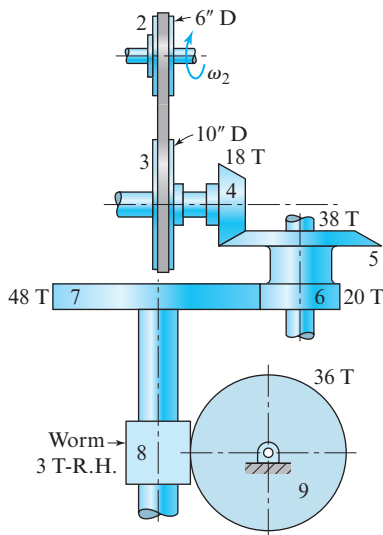
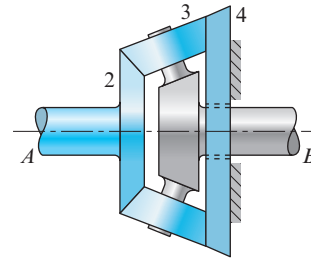


Figure P8.20

The bevel pinion is mounted on a shaft that is driven by a V-belt on pulleys. If pulley 2 rotates at 1 200 rev/min in the direction indicated, find the speed and direction of rotation of gear 9.

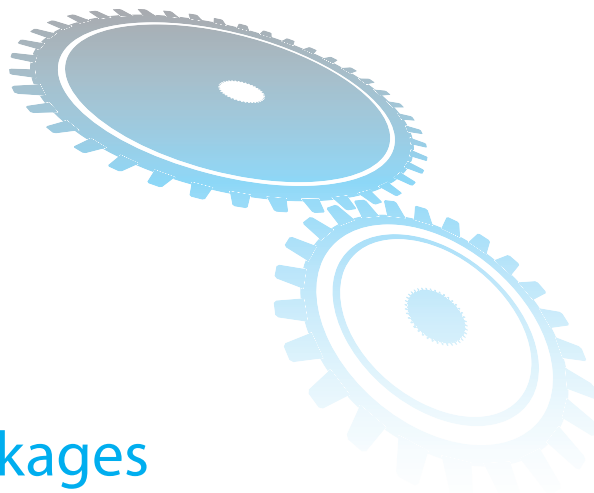
- 8.21** The marine reduction differential shown in Figure P8.21 has bevel gear 2 driven by engine shaft A. Bevel planets 3 mesh with fixed crown gear 4 and are pivoted on the spider (arm), which is connected to propeller shaft B. Find the percent speed reduction.

Figure P8.21 $N_2 = 36T$, $N_3 = 21T$, $N_4 = 52T$.

- 8.22** The tooth numbers for the automotive differential shown in Fig. 8.28 are $N_2 = 17T$, $N_3 = 54T$, $N_4 = 11T$, and $N_5 = N_6 = 16T$. The drive shaft turns at 1 200 rev/min. What is the speed of the right wheel if it is jacked up and the left wheel is resting on the road surface?
- 8.23** A vehicle using the differential shown in Fig. 8.28 turns to the right at a speed of 30 mi/h on a curve of 80-ft radius. Use the same tooth numbers as in Problem 8.22. The tire diameter is 15 in. Use 60 in as the distance between treads. Calculate the speed of each rear wheel and the speed of the ring gear.

9

Synthesis of Linkages



In previous chapters, we have concentrated on the analysis of mechanisms, where the dimensions of the links are known. By kinematic *synthesis* we mean the design or creation of a new mechanism to yield a desired set of motion characteristics. Because of the very large number of techniques available, this chapter presents only a few of the more useful approaches to illustrate applications of the planar theory.*

9.1 TYPE, NUMBER, AND DIMENSIONAL SYNTHESIS

There are three general stages in the design of a new mechanism: type synthesis, number synthesis, and dimensional synthesis. Although these three steps may not be consciously followed, they are always present in the creation of any new device.

Type synthesis refers to the choice of the kind of mechanism to be used; it might be a linkage, a geared system, a cam system, or even belts and pulleys. This beginning stage of the total design process usually involves the consideration of design factors, such as manufacturing processes, materials, space, safety, and economics. The study of kinematics is usually only slightly involved in type synthesis.

Number synthesis deals with finding a satisfactory number of links and number of joints to obtain the desired mobility (Sec. 1.6). Number synthesis is the second stage in the design process and follows type synthesis.

The third stage in design, determining the detailed dimensions of the individual links, is called *dimensional synthesis*. This is the subject of the balance of this chapter.

* Some of the most useful works on planar kinematic synthesis in the English language are included in the References at the end of this chapter. Extensive bibliographies may be found in [4] and [8].

9.2 FUNCTION GENERATION, PATH GENERATION, AND BODY GUIDANCE

A frequent requirement in mechanism design is that of causing an output body to rotate, oscillate, or reciprocate according to a specified function of time or function of the input motion. This is the first type of synthesis and is called *function generation*. A simple example is that of synthesizing a four-bar linkage to generate the function $y = f(x)$. In this case, x represents the motion (crank angle) of the input crank, and the linkage is to be designed so that the motion (angle) of the output rocker approximates the desired function y . A few other examples of function generation are as follows:

1. In a conveyor line the output body of a mechanism must move at the constant velocity of the conveyer while some operation is performed—for example, capping a bottle—and then it must return, pick up the next cap, and repeat.
2. The output body of a mechanism must pause or stop during its motion cycle to provide time for another event. The second event might be a sealing, stapling, or fastening operation of some kind.
3. The output body of a mechanism must rotate according to a specified nonuniform velocity function, since it must be synchronized with another mechanism that requires such a rotating motion.

Recall that in Chap. 2 we have defined the location of a point to be described by the term *position*. However, the location and orientation of a rigid body and the arrangement of a mechanism or system of rigid bodies are both described by the term *posture*. This terminology continues consistently throughout this chapter.

A second type of synthesis problem is called *path generation*. This refers to a problem in which the position of a coupler point must follow a path having a prescribed shape. Common requirements are that a portion of the path be circular, elliptic, or a straight line. Sometimes it is required that the path cross over itself, as in a figure eight.

The third general class of synthesis problems is called *body guidance*. Here we are interested in moving a rigid body from one posture to another. The problem may call for a simple translation or a combination of translation and rotation. In the construction industry, for example, a heavy part such as a scoop or a bulldozer blade must be moved through a series of prescribed postures.

The general problem of dimensional synthesis is to design a mechanism that will guide a moving rigid body through N finitely separated postures in a single plane (where $N = 2, 3, 4, \dots$). In general, the N postures are specified; that is, the design problem dictates the N postures to which the rigid body must move. The desire is to synthesize a planar, single-degree-of-freedom mechanism that will achieve the locations and orientations of the rigid body for each of the N postures. For example, the synthesis of a planar four-bar linkage might be the first attempt. If the rigid body that is to be guided through the N postures is the coupler link, then the synthesis problem is one of body guidance. If the rigid body is the input or output crank, then the synthesis problem is one of function generation.

In this chapter, we present a general geometric approach that can be used to synthesize a four-bar linkage for $N = 2, 3, 4$, or 5 finitely separated postures of the coupler link. We will see that the number of four-bar linkages that can be found to guide a rigid body through two finitely separated postures is ∞^6 ; for three finitely separated postures this number

is ∞^4 ; for four finitely separated postures there are ∞^1 solutions; and through five finitely separated postures, there are only a small finite number of solutions. In general, it is not possible to synthesize a four-bar linkage for N greater than five finitely separated postures of a rigid body unless the postures share special geometric relationships.

9.3 TWO FINITELY SEPARATED POSTURES OF A RIGID BODY ($N = 2$)

Let us start with the study of simple two-posture synthesis methods.

Two-Posture Synthesis of a Slider-Crank Linkage The central slider-crank linkage of Fig. 9.1a has a stroke, B_1B_2 , equal to twice the crank radius, r_2 . As shown, the extreme positions of points B_1 and B_2 , also called limit postures of the slider, are determined by constructing circular arcs around O_2 of length $r_3 - r_2$ and $r_3 + r_2$, respectively. The two dimensions r_2 and r_3 can be found from these two measured lengths.

In general, the central slider-crank linkage must have r_3 larger than r_2 . However, the special case of $r_3 = r_2$ results in the *isosceles slider-crank linkage*, in which the slider reciprocates along an axis through O_2 , and the stroke is four times the crank radius. All points on the coupler of the isosceles slider-crank linkage generate elliptic paths. The paths generated by points on the coupler of the central slider-crank linkage of Fig. 9.1a, which is not isosceles, are not elliptic; however, they are always symmetric about the sliding axis O_2B .

The linkage of Fig. 9.1b is called the *general* or *offset slider-crank linkage*. Note that the limit postures of the slider can be found in the same manner as explained above for the central slider-crank linkage. Certain special effects can be obtained by changing the offset distance, e . For example, stroke B_1B_2 is always greater than twice the crank radius. Also, the crank angle required to execute the forward stroke is different from that for the return stroke. This feature can be used to synthesize quick-return mechanisms where a slower working stroke may be desired to reduce power requirements (Sec. 1.7).

Two-Posture Synthesis of a Crank-Rocker Linkage The limit postures of the rocker in a crank-rocker linkage are shown by points B_1 and B_2 in Fig. 9.2. Note that the crank and the coupler form a single straight line at each of the limit postures. Also, note that

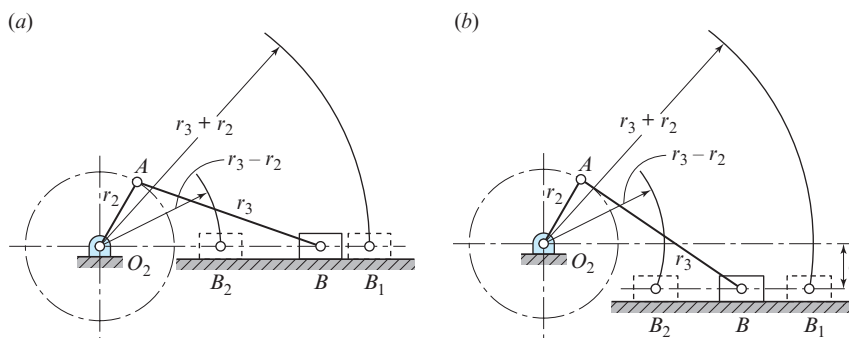


Figure 9.1 (a) Central slider-crank linkage; (b) general or offset slider-crank linkage.

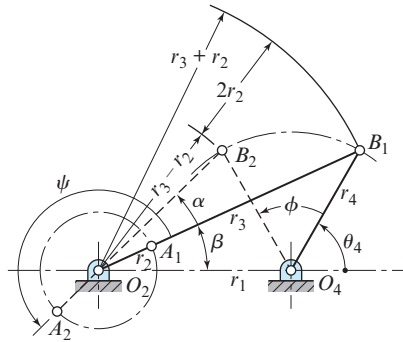


Figure 9.2 Two limiting postures of the crank-rocker linkage.

the two dimensions r_2 and r_3 can be determined from measurements made in these two postures in the same way as for the slider-crank linkage.

In this case, the crank executes angle ψ while the rocker moves from B_1 to B_2 through angle ϕ . Note on the return stroke that the rocker swings from B_2 back to B_1 through the same angle, $-\phi$, but the crank moves through the angle $360^\circ - \psi$.

There are many cases in which a crank-rocker linkage is superior to a cam-and-follower system. Among the advantages over cam systems are the smaller forces involved, the elimination of the retaining spring, and the smaller clearances obtainable by the use of revolute joints.

If the direction of rotation of the input crank is chosen so that $\psi > 180^\circ$ in Fig. 9.2, then we can define $\alpha = \psi - 180^\circ$, and an equation for the advance-to-return ratio (Sec. 1.7) can be written as

$$Q = \frac{180^\circ + \alpha}{180^\circ - \alpha}. \quad (9.1)$$

A problem that frequently arises in the synthesis of crank-rocker linkages is to obtain the dimensions or geometry that will cause the linkage to generate a specified output displacement, ϕ , when the advance-to-return ratio Q is specified.*

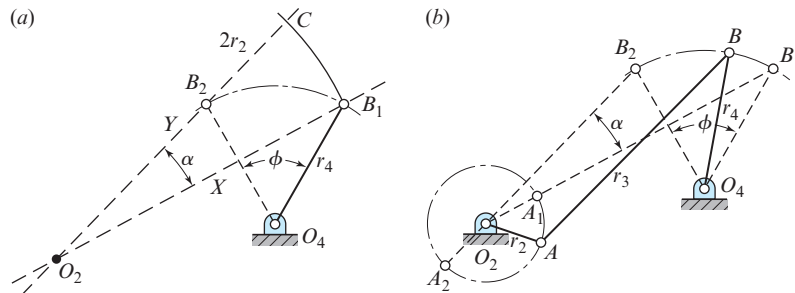


Figure 9.3 Synthesis of a four-bar linkage to generate a specified rocker displacement, ϕ , with a specified advance-to-return ratio, Q .

* The method described here appears in [9, p. 33] and in [16, p. 257]. Both [8, p. 317] and [18, p. 241] describe another method that yields different results.

To synthesize a crank-rocker linkage for specified values of ϕ and α , we locate an arbitrary point, O_4 , in Fig. 9.3a and choose any desired rocker length, r_4 ; we note that this choice does not change the solution but only sets the scale of the figure. Then, we draw the two postures of link 4, that is, O_4B_1 and O_4B_2 , separated by the given angle, ϕ . Through B_1 , we construct any line X . Then, through B_2 , we construct line Y at the given angle, α , to line X . The intersection of these two lines defines the location of the crank pivot, O_2 . Since the orientation of line X was chosen arbitrarily, there are an infinite number of solutions to this problem.

Next, as shown in Fig. 9.3a, distance B_2C is $2r_2$, or twice the crank length. So, we can bisect this distance to find r_2 . Then, the coupler length is found from $r_3 = O_2B_1 - r_2$. The completed linkage is shown in Fig. 9.3b.

Pole of a Finite Displacement For two finitely separated postures of a rigid body in planar motion, the moving body can be represented by a line AB of constant length, as shown in Fig. 9.4. In the first posture, the body is denoted as A_1B_1 , and in the second posture, the body is denoted as A_2B_2 . The displacement of the body from posture 1 to posture 2 can be described by a rotation, θ_{12} , and a translation, d_{12} , as shown in Fig. 9.4.

Note that a unique point can be located as the center of rotation of the rigid body displacement. The center of rotation is the point of intersection of the perpendicular bisector of the line connecting A_1 to A_2 and the perpendicular bisector of the line connecting B_1 to B_2 . The center of rotation will henceforth be referred to as the *pole* for the finite displacement and denoted as P_{12} as shown in Fig. 9.5.

The angle of rotation of body AB is denoted

$$\angle A_1P_{12}A_2 = \angle B_1P_{12}B_2 = \theta_{12} = 2\phi_{12}. \quad (9.2)$$

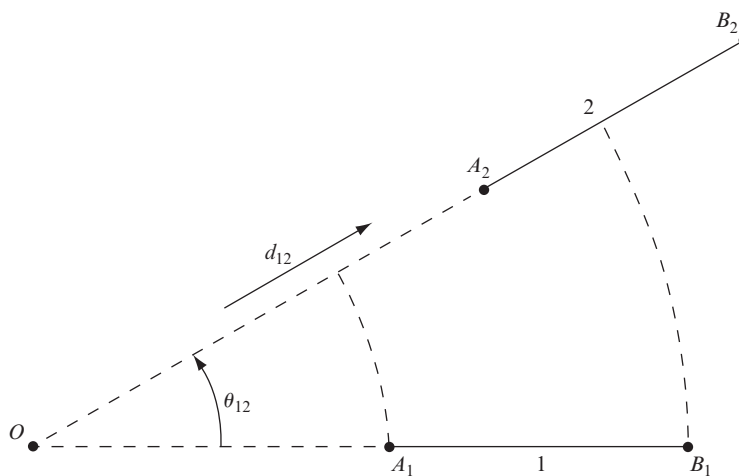


Figure 9.4 Two finitely separated postures of a rigid body.

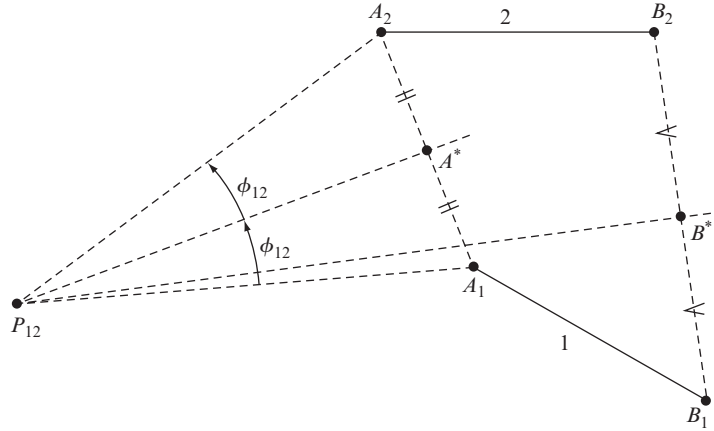


Figure 9.5 A single pivot to guide the rigid body through a finite displacement.

Therefore, the angles

$$\begin{aligned}\angle A_1 P_{12} A^* &= \angle A^* P_{12} A_2 = \phi_{12}, \\ \angle B_1 P_{12} B^* &= \angle B^* P_{12} B_2 = \phi_{12},\end{aligned}\quad (9.3)$$

where A^* and B^* are the midpoints of lines $A_1 A_2$ and $B_1 B_2$, respectively. Note that the pole P_{ij} for a finite displacement of the body from posture i to posture j is analogous to the instantaneous center of velocity for an infinitesimal displacement of a rigid body (Sec. 3.12). The following two statements are important: (a) the pole for a finite displacement is located at the point of the moving body having zero displacement; and (b) for the purpose of calculating displacements, the finite displacement of the body can be considered a rotation around the pole.

A better solution than a single pivot to guide the moving body between two postures is to use a four-bar linkage, since a four-bar linkage is more stable and can support a larger load than a single pivot. A possible four-bar linkage solution is shown in Fig. 9.6. The body AB would be attached to the coupler link; the previous or new points A and B could be used as the two coupler pivots; ground pivot O_A can be chosen as any point on the perpendicular bisector $A^* P_{12}$, and ground pivot O_B can be chosen as any point on the perpendicular bisector $B^* P_{12}$.

The total number of possible four-bar linkages that can guide the body through two finitely separated postures can be obtained from the fact that there are ∞^2 locations for pin A , ∞^2 locations for pin B , ∞^1 choices for ground pivot O_A , and ∞^1 choices for ground pivot O_B along the perpendicular bisector lines. Therefore, there are a total of ∞^6 possible four-bar linkages that can guide the body through the two specified finitely separated postures.

It is important to note the relationships between the signs of the rotation angles and the order of their subscripts. The rotation angle through which the body turns in moving from posture 1 to posture 2 is denoted $2\phi_{12}$ [Eq. (9.2)]. Similarly, the rotation angle

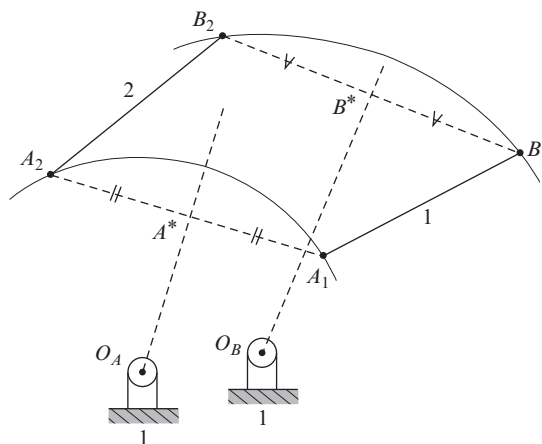


Figure 9.6 A possible four-bar linkage.

through which the body turns in moving from posture 2 back to posture 1 is denoted $2\phi_{21}$. Therefore, we can write that

$$2\phi_{12} + 2\phi_{21} = 0^\circ, 360^\circ$$

or

$$\phi_{12} + \phi_{21} = 0^\circ, 180^\circ. \quad (9.4)$$

Note that Eqs. (9.4) are valid when the two angles are measured in the same direction. If the angles are measured in opposite directions, then

$$\phi_{21} = -\phi_{12}. \quad (9.5)$$

An expression for the average velocity of an arbitrary point fixed in the body, say point A, can be written as

$$(V_A)_{\text{avg}} = \frac{\Delta s_A}{\Delta t}, \quad (9.6)$$

where the arc distance is $\Delta s_A = A_1A_2$. The relationship between the distance, Δs_A , and the angle, ϕ_{12} , can be obtained as follows. Consider the triangle $A_1P_{12}A^*$ of Fig. 9.5; then,

$$\sin \phi_{12} = \frac{A_1A^*}{P_{12}A_1},$$

which can be written as

$$A_1A^* = (P_{12}A_1) \sin \phi_{12}. \quad (9.7)$$

Note that

$$A_1A^* = \frac{A_1A_2}{2} = \frac{1}{2} \Delta s_A. \quad (9.8)$$

Setting Eq. (9.8) equal to Eq. (9.7) and rearranging, the displacement can be written as

$$\Delta s_A = 2A_1A^* = 2(P_{12}A_1) \sin \phi_{12}. \quad (9.9)$$

Then, substituting Eq. (9.9) into Eq. (9.6) and rearranging, the average velocity of point A can be written as

$$(V_A)_{\text{avg}} = \frac{2 \sin \phi_{12}}{\Delta t} (P_{12}A_1). \quad (9.10)$$

From this equation, the magnitude of the instantaneous velocity of point A in the body can be written as

$$V_A = \lim_{\Delta t \rightarrow 0} \left(\frac{2 \sin \phi_{12}}{\Delta t} \right) P_{12}A_1. \quad (9.11)$$

Recall that the magnitude of the instantaneous velocity of point A (Chap. 3) can also be written as

$$V_A = \omega R, \quad (9.12)$$

where ω is the angular velocity of the body, and $R = P_{12}A_1$ is the distance of point A from the pole (center of rotation). Comparing Eqs. (9.11) and (9.12), we note that the angular velocity of the body can be written as

$$\omega = \lim_{\Delta t \rightarrow 0} \left(\frac{2 \sin \phi_{12}}{\Delta t} \right). \quad (9.13)$$

9.4 THREE FINITELY SEPARATED POSTURES OF A RIGID BODY ($N = 3$)

Three finitely separated postures of a body, denoted 1, 2, and 3, can be specified by: (a) the poles P_{12} , P_{23} , and P_{31} ; (b) the rotation angle from posture 1 to posture 2 denoted $2\phi_{12}$, the rotation angle from posture 2 to posture 3 denoted $2\phi_{23}$, and the rotation angle from posture 3 to posture 1 denoted $2\phi_{31}$.

We can obtain the locations of the three poles in the same manner as before. We choose an arbitrary line in the body, say line DE shown in Fig. 9.7. For the body in its first posture, the line is denoted D_1E_1 ; in the second posture, the line is denoted D_2E_2 ; and in the third posture, the line is denoted D_3E_3 . The intersection of the perpendicular bisectors of the lines D_1D_2 and E_1E_2 is the pole P_{12} . The intersection of the perpendicular bisectors of lines D_2D_3 and E_2E_3 is pole P_{23} . Finally, the intersection of the perpendicular bisectors of lines D_3D_1 and E_3E_1 is pole P_{31} . The three poles for the three finitely separated postures of the body are shown in Fig. 9.7.

We connect poles P_{12} and P_{23} by a straight line; we connect poles P_{23} and P_{31} by a straight line; and we connect poles P_{31} and P_{12} by a straight line. These three lines form a triangle called the *pole triangle*. An understanding of the geometry of the pole triangle (for example, the interior angles and the exterior angles) is essential for synthesizing a mechanism that can guide the body through the three finitely separated postures.

For convenience, the finite displacements of the rigid body in Fig. 9.8a are shown counterclockwise. For the pole triangle shown, the sides are labeled so that side 1 is the

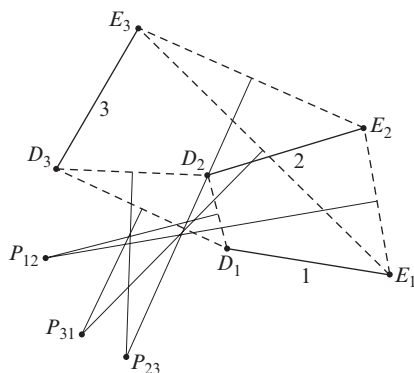


Figure 9.7 Three poles for three postures of rigid body DE .

side with the common subscript 1 for the two poles ($P_{31}P_{12}$), side 2 is the side with the common subscript 2 for the two poles ($P_{12}P_{23}$), and side 3 is the side with the common subscript 3 for the two poles ($P_{23}P_{31}$).

Interior Angles of a Pole Triangle The order of the subscripts for the interior angles of a pole triangle are defined as follows: (a) The interior angle about pole P_{12} from side 1 to side 2 of the pole triangle is the angle ϕ_{12} (positive counterclockwise), (b) the interior angle about pole P_{23} from side 2 to side 3 of the pole triangle is angle ϕ_{23} (positive counterclockwise), and (c) the interior angle about pole P_{31} from side 3 to side 1 of the pole triangle is the angle ϕ_{31} (positive counterclockwise).

Recall that these interior angles are half the rotation angles for the body; that is,

$$\phi_{12} = \frac{\theta_{12}}{2}, \quad \phi_{23} = \frac{\theta_{23}}{2}, \quad \text{and} \quad \phi_{31} = \frac{\theta_{31}}{2}. \quad (9.14)$$

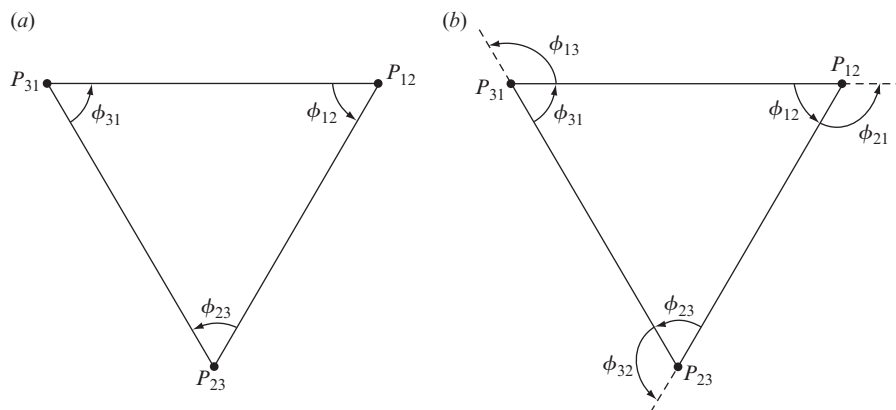


Figure 9.8 (a) Interior angles of a pole triangle; (b) exterior angles of a pole triangle.

Also note that, consistent with Eq. (9.14), the sum of the three interior angles of the pole triangle must be

$$\phi_{12} + \phi_{23} + \phi_{31} = 180^\circ. \quad (9.15)$$

Exterior Angles of a Pole Triangle Note the following relationships, again consistent with Eq. (9.14):

$$\phi_{12} + \phi_{21} = 180^\circ, \quad \phi_{23} + \phi_{32} = 180^\circ, \quad \text{and} \quad \phi_{31} + \phi_{13} = 180^\circ, \quad (9.16)$$

which are valid when the angles are measured in the same direction, as shown in Fig. 9.8*b*. Also, note the relationships

$$\phi_{12} = -\phi_{21}, \quad \phi_{23} = -\phi_{32}, \quad \text{and} \quad \phi_{31} = -\phi_{13}, \quad (9.17)$$

which are valid when the angles are measured in opposite directions, consistent with Eq. (9.5).

EXAMPLE 9.1

Given the pole triangle shown in Fig. 9.9, where the length of side 1 is $P_{31}P_{12} = 2$ in and the interior angles are $\phi_{12} = \theta_{12}/2 = 53.13^\circ$ ccw and $\phi_{31} = \theta_{31}/2 = 36.87^\circ$ ccw, and also given that the location of point D fixed in the moving body in posture 1, that is, point D_1 , is midway between poles P_{31} and P_{12} and 1.5 in vertically below this line, then determine the location of point D when the rigid body is in posture 2 and in posture 3; that is, find points D_2 and D_3 .

SOLUTION

Note that the interior angle $\phi_{23} = \theta_{23}/2 = 90^\circ$ ccw; therefore, the pole triangle is a 3:4:5 right-angled triangle, and the lengths of the other two sides are $P_{12}P_{23} = 1.2$ in for side 2 and $P_{23}P_{31} = 1.6$ in. for side 3.

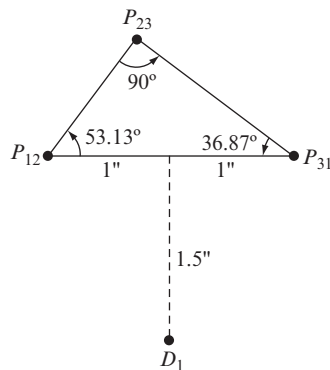


Figure 9.9 Pole triangle and point D_1 .